

# Qualifying Exam notes

Songyu Ye

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## Abstract

We begin preparation for the qualifying exam.

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## 1 Algebraic geometry

The primary reference is Hartshorne's *Algebraic Geometry*, and the secondary reference is Vakil's *The Rising Sea*.

### 1.1 Sheaves

#### 1.1.1 Presheaves

**Definition 1.1.** A **presheaf** of abelian groups  $\mathcal{F}$  on a space  $X$  is a contravariant functor from the poset category of open sets to the category of abelian groups.

$$\mathcal{F} : \text{Open}^{\text{op}}(X) \rightarrow \text{Ab}$$

**Definition 1.2.** A presheaf  $\mathcal{F}$  on a topological space  $X$  is a **sheaf** if it satisfies the following supplementary conditions:

**Separated** If  $U$  is an open set, if  $\{V_i\}$  is an open covering of  $U$ , and if  $s \in \mathcal{F}(U)$  is an element such that  $s|_{V_i} = 0$  for all  $i$ , then  $s = 0$ ;

**Gluing** If  $U$  is an open set, if  $\{V_i\}$  is an open covering of  $U$ , and if we have elements  $s_i \in \mathcal{F}(V_i)$  for each  $i$ , with the property that for each  $i, j$ ,  $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$ , then there is an element  $s \in \mathcal{F}(U)$  such that  $s|_{V_i} = s_i$  for each  $i$ .

**Example 1.3** (Presheaf which is not a sheaf). Let  $X$  be a topological space and  $A$  an abelian group. Let  $\mathcal{F}(U) = A$  for all open sets  $U$ . Then  $\mathcal{F}$  is a presheaf with the obvious restriction maps, however  $\mathcal{F}$  is not a sheaf because if  $U, V$  are disjoint open sets then  $A = \mathcal{F}(U \cup V) = \mathcal{F}(U) \times \mathcal{F}(V) = A \times A$ .

**Example 1.4** (Sheaf of functions). Let  $X$  be a topological space and for every open set  $U \subset X$  let  $C(U)$  denote the continuous functions  $U \rightarrow \mathbb{R}$ . Then  $C$  is a sheaf because every continuous function on  $U$  is completely determined by its restriction to any open cover  $U_i$ , and because continuous functions on  $U_i$  which agree on overlaps  $U_{ij}$  glue to continuous functions on  $U$ .

The sheaf axioms are laid out so that sheaves behave like continuous functions on a topological space. However, there are many examples of sheaves which are not the sheaves of functions on a space.

**Example 1.5** (Skyscraper sheaf). Let  $X$  be a topological space. For  $x \in X$  and  $A$  an

abelian group, the **skyscraper sheaf**  $i_x(A)$  has:

$$i_x(A)(U) = \begin{cases} A & \text{if } x \in U \\ 0 & \text{otherwise} \end{cases}$$

Skyscraper sheaves probe stalks, in the sense that they satisfy the following universal property: for any sheaf  $\mathcal{F}$  on  $X$ ,

$$\mathrm{Hom}(\mathcal{F}, i_x(A)) \cong \mathrm{Hom}(\mathcal{F}_x, A)$$

**Definition 1.6.** Let  $\mathcal{F}$  and  $\mathcal{G}$  be sheaves on a topological space  $X$ . A **morphism of sheaves**  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  is a collection of morphisms

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

for each open set  $U \subseteq X$ , such that:

1. The following diagram commutes for every pair of open sets  $V \subseteq U$ :

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \rho_V^U \downarrow & & \downarrow \rho_V^U \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

where  $\rho_V^U$  denotes the restriction maps of both sheaves.

2. Each  $\varphi_U$  is a morphism of abelian groups.

Equivalently, a sheaf morphism is a natural transformation between the sheaves viewed as functors from the category of open sets of  $X$  to some algebraic category.

**Definition 1.7.** The stalk of a sheaf  $\mathcal{F}$  at a point  $x \in X$  is the colimit

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U)$$

where the limit is taken over all open sets  $U$  containing  $x$ .

A morphism of sheaves  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  induces a morphism on stalks  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  for each  $x \in X$ .

### 1.1.2 Leaf space, exactness, and constant sheaves

One way to think about the category of sheaves of abelian groups is that it should be very similar to the category of abelian groups. In fact, one takes theorems about abelian groups and tries to prove them for sheaves, or in general for any abelian category. The first question one can ask is about exact sequences.

The naive guess which turns out to be wrong is to declare that a sequence of sheaves of abelian groups

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is exact if when evaluated for each open set  $U$ , the corresponding sequence of abelian groups is exact. This definition works for left exactness but turns out to give the wrong notion for surjectivity.

**Example 1.8.** Consider  $X = S^1$  and  $X_1 = X$  and  $X_2 =$  nontrivial double cover of  $X$ . Let  $F_1$  and  $F_2$  denote the corresponding sheaves of sections. Then the map  $X_2 \rightarrow X_1$  is surjective, but the map  $F_2(X) \rightarrow F_1(X)$  is not surjective. Indeed  $F_2(X) = \emptyset$  and  $F_1(X) = \{\text{id}_X\}$ .

Thus we encounter two notions of surjectivity. If we think about all sheaves as the sheaves of sections of some space living over  $X$ , which we will introduce as the leaf space (or etale space), then morphisms of sheaves correspond fully faithfully to morphisms of leaf spaces. Then there is a local notion of surjectivity, which corresponds to the morphism of leaf spaces being surjective, and a global notion of surjective, corresponding to evaluation on all open sets.

**Definition 1.9.** Let  $\mathcal{F}$  be a sheaf on a topological space  $X$ . The **leaf space**  $L(\mathcal{F})$  is the topological space defined by:

$$L(\mathcal{F}) = \bigsqcup_{x \in X} \mathcal{F}_x$$

where  $\mathcal{F}_x$  denotes the stalk of  $\mathcal{F}$  at  $x$ , equipped with the topology generated by the basis:

$$\{\langle s, U \rangle \mid U \subseteq X \text{ open}, s \in \mathcal{F}(U)\}$$

where  $\langle s, U \rangle = \{s_x \in \mathcal{F}_x \mid x \in U\}$  and  $s_x$  is the germ of  $s$  at  $x$ .

**Remark 1.10.** The projection map  $\pi : L(\mathcal{F}) \rightarrow X$  sending each germ  $s_x$  to  $x$  is a local homeomorphism. The original sheaf  $\mathcal{F}$  can be recovered as the sheaf of continuous sections of  $\pi$ .

The right definition of surjectivity to use is the local notion. By the definition of a leaf space, surjectivity of sheaves is equivalent to surjectivity on all stalks. Another clue that this is the right notion of surjectivity is that the local notion agrees with the notion of epimorphism in the category of sheaves of abelian groups.

Leaf spaces look very strange. For example, you might say that if  $X$  is a manifold and  $\mathcal{F}$  is the sheaf of smooth functions on  $X$ , then the leaf space of  $\mathcal{F}$  is locally homeomorphic to  $X$  and therefore is also a manifold. However the issue with this is that the leaf space is non-Hausdorff.

**Definition 1.11.** Let  $\mathcal{F}$  be a presheaf on a topological space  $X$ . The **sheafification** of  $\mathcal{F}$  is the sheaf of continuous sections of the leaf space  $L(\mathcal{F})$ . It comes equipped with

a morphism of presheaves  $\mathcal{F} \rightarrow \mathcal{F}^+$  such that for any sheaf  $\mathcal{G}$  and any morphism of presheaves  $\mathcal{F} \rightarrow \mathcal{G}$ , there is a unique morphism of sheaves  $\mathcal{F}^+ \rightarrow \mathcal{G}$  making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & \mathcal{F}^+ \\ & \searrow \exists! & \downarrow \\ & & \mathcal{G} \end{array}$$

**Definition 1.12.** A **constant sheaf** on a topological space  $X$  with value in an abelian group  $A$  is the sheafification of the constant presheaf defined by  $\mathcal{F}(U) = A$  for all open sets  $U \subseteq X$ .

**Exercise 1.13.** On a connected locally connected space,  $\underline{A}(U)$  is the set of locally constant functions  $U \rightarrow A$ , not literally just  $A$  for every open  $U$ .

**Solution 1.14.** Let  $\mathcal{L}_A$  be the sheaf given by

$$\mathcal{L}_A(U) = \{f : U \rightarrow A \mid f \text{ is locally constant}\}.$$

There is a natural morphism from the constant presheaf  $P(U) = A$  to  $\mathcal{L}_A$ , sending  $a \in A$  to the constant function with value  $a$ . By the universal property of sheafification, this induces a morphism

$$\Phi : \underline{A} = P^\# \rightarrow \mathcal{L}_A.$$

We claim that  $\Phi$  is an isomorphism. It suffices to check stalks. For any  $x \in X$ ,

$$(\underline{A})_x \cong P_x \cong A,$$

since sheafification does not change stalks and the stalk of the constant presheaf is the colimit of the constant system  $A$ . Also,

$$(\mathcal{L}_A)_x \cong A,$$

because the germ of a locally constant function is determined by its value at  $x$ . Under these identifications,  $\Phi_x$  is the identity on  $A$ , hence an isomorphism. Therefore  $\Phi$  is an isomorphism of sheaves.

Thus

$$\underline{A}(U) \cong \{f : U \rightarrow A \mid f \text{ is locally constant}\}.$$

If  $X$  is locally connected, then each connected component of  $U$  is open, and a function  $U \rightarrow A$  is locally constant if and only if it is constant on each connected component. Therefore

$$\underline{A}(U) \cong \prod_{C \in \pi_0(U)} A.$$

### 1.1.3 Hom, kernel, image, cokernel, $f_*$ and $f^{-1}$

The category of sheaves of abelian groups on a topological space  $X$  is an abelian category. In particular, given a morphism of sheaves

$$\varphi : \mathcal{F} \rightarrow \mathcal{G},$$

one can form its kernel, image, cokernel, and coimage, and these satisfy the usual formal properties familiar from modules and abelian groups. The point is that these constructions must be made sheaf-theoretically rather than merely sectionwise.

**Definition 1.15.** *Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of sheaves of abelian groups on  $X$ .*

1. The **kernel sheaf**  $\ker(\varphi)$  is the sheaf defined by

$$(\ker \varphi)(U) := \ker(\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

2. The **image presheaf**  $\mathrm{im}^{\mathrm{pre}}(\varphi)$  is the presheaf defined by

$$\mathrm{im}^{\mathrm{pre}}(\varphi)(U) := \mathrm{im}(\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

*Its associated sheaf is called the **image sheaf** and is denoted  $\mathrm{im}(\varphi)$ .*

3. The **cokernel sheaf**  $\mathrm{coker}(\varphi)$  is the sheaf associated to the presheaf

$$U \mapsto \mathcal{G}(U) / \varphi_U(\mathcal{F}(U)).$$

**Remark 1.16.** *The kernel is already a sheaf, because the sheaf axioms are checked inside  $\mathcal{F}$ . By contrast, the naive image and cokernel need not be sheaves, so one must sheafify.*

**Proposition 1.17.** *Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of sheaves of abelian groups.*

1. For every  $x \in X$ ,

$$(\ker \varphi)_x \cong \ker(\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x).$$

2. For every  $x \in X$ ,

$$(\mathrm{im} \varphi)_x \cong \mathrm{im}(\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x).$$

3. For every  $x \in X$ ,

$$(\mathrm{coker} \varphi)_x \cong \mathrm{coker}(\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x).$$

*Proof.* Stalks are filtered colimits over neighborhoods of  $x$ , and filtered colimits of abelian groups are exact. For the kernel this gives

$$(\ker \varphi)_x = \varinjlim_{x \in U} \ker(\varphi_U) \cong \ker\left(\varinjlim_{x \in U} \mathcal{F}(U) \rightarrow \varinjlim_{x \in U} \mathcal{G}(U)\right) = \ker(\varphi_x).$$

The statements for image and cokernel follow similarly, using exactness of filtered colimits and the fact that sheafification does not change stalks.  $\square$

The previous proposition explains why exactness for sheaves is tested on stalks.

**Definition 1.18.** *A sequence of sheaves of abelian groups*

$$\mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}''$$

*is **exact at**  $\mathcal{F}$  if*

$$\mathrm{im}(\alpha) = \ker(\beta)$$

*as subsheaves of  $\mathcal{F}$ . Equivalently, the sequence is exact at  $\mathcal{F}$  if and only if for every point  $x \in X$ , the induced sequence on stalks*

$$\mathcal{F}'_x \xrightarrow{\alpha_x} \mathcal{F}_x \xrightarrow{\beta_x} \mathcal{F}''_x$$

*is exact.*

**Example 1.19** (Image presheaf need not be a sheaf). *Let  $\mathcal{O}$  denote the sheaf of holomorphic functions on  $\mathbb{C}$ , and consider the morphism*

$$d : \mathcal{O} \rightarrow \mathcal{O}, \quad f \mapsto f'.$$

*On a simply connected open set  $U$ , every holomorphic function has a primitive, so  $d_U$  is surjective. But on  $U = \mathbb{C}^\times$ , the function  $1/z$  has no global primitive, so  $1/z \notin \mathrm{im}(d_U)$ . Thus the presheaf image is not determined by local data and is not a sheaf.*

*Its sheafification is the whole sheaf  $\mathcal{O}$ , because every holomorphic function is locally a derivative.*

This example shows that a morphism of sheaves may be surjective on stalks without being surjective on sections over a given open set.

**Definition 1.20.** *A morphism of sheaves  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  is called*

1. a **monomorphism** if  $\ker(\varphi) = 0$ ;
2. an **epimorphism** if  $\mathrm{coker}(\varphi) = 0$ ;
3. an **isomorphism** if it is both a monomorphism and an epimorphism.

Equivalently,  $\varphi$  is a monomorphism, epimorphism, or isomorphism if and only if each stalk map

$$\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$$

has the corresponding property for every  $x \in X$ .

**Proposition 1.21.** *Let  $f : X \rightarrow Y$  be a continuous map of topological spaces.*

1. *If  $\mathcal{F}, \mathcal{G}$  are sheaves on  $X$ , then the assignment*

$$U \longmapsto \mathrm{Hom}_{\mathcal{S}\widetilde{\sim}(U)}(\mathcal{F}|_U, \mathcal{G}|_U)$$

*defines a sheaf on  $X$ , called the **internal Hom sheaf** and denoted  $\mathcal{H}om(\mathcal{F}, \mathcal{G})$*



2. If  $\mathcal{F}$  is a sheaf on  $X$ , then the direct image sheaf  $f_*\mathcal{F}$  on  $Y$  is defined by

$$(f_*\mathcal{F})(V) := \mathcal{F}(f^{-1}(V))$$

for each open set  $V \subseteq Y$ .

3. If  $\mathcal{G}$  is a sheaf on  $Y$ , then the inverse image sheaf  $f^{-1}\mathcal{G}$  on  $X$  is the sheafification of the presheaf

$$U \longmapsto \varinjlim_{V \subseteq f^{-1}(U)} \mathcal{G}(V).$$

**Remark 1.22.** For sheaves of abelian groups,  $f^{-1}$  is exact, while  $f_*$  is left exact but usually not right exact. This is the sheaf-theoretic origin of higher direct images  $R^i f_*$ .

**Remark 1.23.** If one works with sheaves of rings and modules, then one often uses the notation  $f^*\mathcal{F} := \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{F}$  for the pullback of modules. For plain sheaves of abelian groups, the fundamental functor is  $f^{-1}$ .

**Exercise 1.24.** Show that a morphism of sheaves  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  is an isomorphism if and only if it is an isomorphism on stalks, i.e. if and only if  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is an isomorphism for every  $x \in X$ .

The forward direction is clear since stalks are functorial and an isomorphism of sheaves induces an isomorphism on stalks. **The converse direction is the nontrivial part, and it relies on the fact that if a sheaf has zero stalks everywhere, then it must be the zero sheaf.**

**Solution 1.25.** Assume that for every  $x \in X$ , the stalk map  $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is an isomorphism. Then for every  $x$ ,

$$(\ker \varphi)_x \cong \ker(\varphi_x) = 0, \quad (\operatorname{coker} \varphi)_x \cong \operatorname{coker}(\varphi_x) = 0.$$

Hence  $\ker(\varphi)$  and  $\operatorname{coker}(\varphi)$  have zero stalks everywhere.

We claim that if a sheaf  $\mathcal{H}$  has  $\mathcal{H}_x = 0$  for all  $x \in X$ , then  $\mathcal{H} = 0$ . Indeed, let  $U \subseteq X$  be open and let  $s \in \mathcal{H}(U)$ . For every  $x \in U$ , the germ  $s_x \in \mathcal{H}_x$  is zero, so there exists an open neighborhood  $V_x \subseteq U$  of  $x$  such that  $s|_{V_x} = 0$ . Thus  $s$  restricts to zero on the open cover  $\{V_x\}_{x \in U}$  of  $U$ , and by the separatedness axiom for sheaves we get  $s = 0$ . Therefore  $\mathcal{H}(U) = 0$  for all open  $U$ , so  $\mathcal{H} = 0$ .

Applying this to  $\ker(\varphi)$  and  $\operatorname{coker}(\varphi)$ , we get

$$\ker(\varphi) = 0, \quad \operatorname{coker}(\varphi) = 0.$$

Therefore  $\varphi$  is both a monomorphism and an epimorphism, hence an isomorphism.

**Exercise 1.26.** Construct a morphism of sheaves which is surjective on stalks but not surjective on sections over some open set.

Note that the corresponding statement for injectivity is false: if a morphism of sheaves is injective on stalks, then it is injective on sections over every open set. This is because the separatedness axiom for sheaves implies that if a section restricts to zero for every stalk, then it must be zero.

**Solution 1.27.** Consider the morphism of sheaves

$$d : \mathcal{O} \rightarrow \mathcal{O}, \quad f \mapsto f'$$

on the sheaf of holomorphic functions on  $\mathbb{C}$ . It is a morphism of sheaves because the derivative of a holomorphic function is holomorphic, and the restriction of a derivative is the derivative of the restriction. It is surjective on stalks because every germ of a holomorphic function has a primitive germ. However, it is not surjective on sections over  $\mathbb{C}^\times$ , since  $1/z$  has no global primitive on  $\mathbb{C}^\times$ .

**Exercise 1.28.** Let  $X$  be a locally compact topological space and let  $F$  be the presheaf of bounded continuous functions on  $X$  with values in  $\mathbb{R}$ . Describe the associated sheaf  $\tilde{F}$ .

**Solution 1.29.** The associated sheaf  $\tilde{F}$  is the sheaf of all locally bounded continuous functions on  $X$  with values in  $\mathbb{R}$ . Since  $X$  is locally compact, this is the same as the sheaf of all continuous functions on  $X$ .

Let  $\mathcal{L}$  be the sheaf of locally bounded continuous functions on  $X$ . To check that  $\mathcal{L}$  is a sheaf, it reduces to checking that the glued function is locally bounded, which is clear after passing to the appropriate open set. To see that  $\mathcal{L}$  is the sheafification, note that there is an obvious map  $\mathcal{F} \rightarrow \mathcal{L}$  which is an isomorphism on stalks, because every germ of a locally bounded continuous function is represented on some neighborhood by a bounded continuous function.

## 1.2 Schemes and Morphisms

### 1.2.1 Affine schemes and gluing

The basic objects in scheme theory are affine schemes.

**Definition 1.30.** Let  $A$  be a commutative ring. The **affine scheme** associated to  $A$  is the locally ringed space

$$\mathrm{Spec} A := (\mathrm{Spec} A, \mathcal{O}_{\mathrm{Spec} A})$$

where:

1. the underlying topological space  $\mathrm{Spec} A$  is the set of prime ideals of  $A$ ;
2. the closed sets are of the form

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec} A : I \subseteq \mathfrak{p}\}$$

for ideals  $I \subseteq A$ ;

3. the structure sheaf is characterized by

$$\mathcal{O}_{\mathrm{Spec} A}(D(f)) = A_f$$

on the basic distinguished opens

$$D(f) := \{\mathfrak{p} \in \mathrm{Spec} A : f \notin \mathfrak{p}\}.$$

The local ring at a point  $\mathfrak{p} \in \mathrm{Spec} A$  is

$$\mathcal{O}_{\mathrm{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}.$$

In particular, points of an affine scheme carry local algebraic information.

**Proposition 1.31.** *The distinguished opens  $D(f)$  form a basis for the Zariski topology on  $\mathrm{Spec} A$ , and for  $f, g \in A$  one has*

$$D(f) \cap D(g) = D(fg).$$

Moreover,

$$D(f) \cong \mathrm{Spec} A_f$$

as locally ringed spaces.

**Definition 1.32.** A **scheme** is a locally ringed space  $(X, \mathcal{O}_X)$  such that every point  $x \in X$  has an open neighborhood isomorphic, as a locally ringed space, to an affine scheme.

**Example 1.33** ( $\mathbb{A}_{\mathbb{R}}^1$  and  $\mathbb{P}_{\mathbb{R}}^1$ ). *The affine line over  $\mathbb{R}$  is*

$$\mathbb{A}_{\mathbb{R}}^1 = \mathrm{Spec} \mathbb{R}[x].$$

*Its points are the prime ideals of  $\mathbb{R}[x]$ . Since  $\mathbb{R}[x]$  is a PID, the nonzero prime ideals are generated by irreducible polynomials. Over  $\mathbb{R}$ , the irreducible polynomials are either linear polynomials  $(x - a)$  with  $a \in \mathbb{R}$ , or irreducible quadratics. Thus the points of  $\mathbb{A}_{\mathbb{R}}^1$  are:*

- the generic point  $(0)$ ;
- the closed points  $(x - a)$  for  $a \in \mathbb{R}$ ;
- the closed points  $(f)$  where  $f \in \mathbb{R}[x]$  is irreducible quadratic.

*So  $\mathbb{A}_{\mathbb{R}}^1$  is not just the set  $\mathbb{R}$ : it also has one closed point for each conjugate pair of nonreal complex numbers, together with the generic point.*

*The projective line over  $\mathbb{R}$  is*

$$\mathbb{P}_{\mathbb{R}}^1 = \mathrm{Proj} \mathbb{R}[s, t].$$

*It is covered by the two standard affine opens*

$$D_+(s) \cong \mathrm{Spec} \mathbb{R} \left[ \frac{t}{s} \right] \cong \mathbb{A}_{\mathbb{R}}^1, \quad D_+(t) \cong \mathrm{Spec} \mathbb{R} \left[ \frac{s}{t} \right] \cong \mathbb{A}_{\mathbb{R}}^1.$$

Thus  $\mathbb{P}_{\mathbb{R}}^1$  is obtained by gluing two copies of  $\mathbb{A}_{\mathbb{R}}^1$  along

$$\operatorname{Spec} \mathbb{R}[u, u^{-1}]$$

via the transition function  $u \mapsto u^{-1}$ .

Its closed points consist of:

- the real points  $[a : 1]$  for  $a \in \mathbb{R}$ ;
- the point at infinity  $[1 : 0]$ ;
- the points corresponding to irreducible quadratics, equivalently conjugate pairs  $\{z, \bar{z}\} \subset \mathbb{C} \setminus \mathbb{R}$ .

There is also the generic point of the integral scheme  $\mathbb{P}_{\mathbb{R}}^1$ .

So the set of  $\mathbb{R}$ -rational points is

$$\mathbb{P}^1(\mathbb{R}) = \mathbb{R} \cup \{\infty\},$$

but as a scheme  $\mathbb{P}_{\mathbb{R}}^1$  has additional closed points with residue field  $\mathbb{C}$ , as well as its generic point.

Thus schemes are obtained by gluing affine schemes along open subsets.

**Definition 1.34.** A **morphism of schemes**  $f : X \rightarrow Y$  is a morphism of locally ringed spaces, i.e. a continuous map  $f : X \rightarrow Y$  together with a morphism of sheaves of rings  $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$  such that for every  $x \in X$ , the induced morphism on local rings

$$f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$$

is a local homomorphism of local rings.

The data of the morphism of sheaves  $f^\#$  is saying how to pull back functions on  $Y$  to functions on  $X$  and in particular is extra data.

**Remark 1.35.** A morphism between affine schemes  $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$  is precisely a ring homomorphism  $A \rightarrow B$ . Given a morphism of affine schemes, the map on global sections is the desired ring homomorphism. Conversely, given a ring homomorphism  $\varphi : A \rightarrow B$ , the corresponding morphism of affine schemes is given by the induced map on prime ideals and the morphism of structure sheaves

$$f^\# : \mathcal{O}_{\operatorname{Spec} A} \rightarrow f_* \mathcal{O}_{\operatorname{Spec} B}$$

is defined on distinguished opens by

$$f_{D(f)}^\# : A_f \longrightarrow B_{\varphi(f)}, \quad \frac{a}{f^n} \mapsto \frac{\varphi(a)}{\varphi(f)^n},$$

**Proposition 1.36** (Gluing of schemes). *Let  $\{X_i\}_{i \in I}$  be schemes, and suppose for each  $i, j$  there are open subsets  $U_{ij} \subseteq X_i$  together with isomorphisms*

$$\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$$

*satisfying:*

1.  $\varphi_{ii} = \text{id}$ ;
2.  $\varphi_{ji} = \varphi_{ij}^{-1}$ ;
3.  $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$  on triple overlaps

*Then the  $X_i$  glue to a scheme  $X$  containing each  $X_i$  as an open subscheme.*

Recall that an affine algebraic set is the zero locus of a collection of polynomials in  $\mathbb{A}_k^n$ . There is a correspondence between the closed subsets of  $\mathbb{A}_k^n$  and radical ideals of  $k[x_1, \dots, x_n]$ , which is the content of the Nullstellensatz. This correspondence refines to a correspondence between the irreducible closed subsets of  $\mathbb{A}_k^n$  and prime ideals of  $k[x_1, \dots, x_n]$ , and further refines to a correspondence between the points of  $\mathbb{A}_k^n$  and maximal ideals of  $k[x_1, \dots, x_n]$ .

This motivates the definition of affine varieties as ringed spaces which are isomorphic irreducible closed subsets of  $\mathbb{A}_k^n$ . On the level of coordinate rings, this corresponds to the definition of affine varieties as integral finitely generated  $k$ -algebras. Clearly, the coordinate ring will be an integral domain since we have quotiented out by a prime ideal. Moreover, since the defining ideal is radical, the coordinate ring will be reduced. It is irreducible because the defining ideal is prime.

**Theorem 1.37.** *The category of affine varieties over an algebraically closed field  $k$  is equivalent to the opposite of the category of integral finitely generated  $k$ -algebras. In particular, it is a full subcategory of the category of affine schemes over  $k$ .*

**Exercise 1.38.**

1. Show that every irreducible locally closed subset of an affine scheme  $X = \text{Spec } A$  contains a unique generic point.
2. Show that every irreducible subset of  $X$  contains at most one generic point.
3. Let  $A$  be a principal ideal domain with infinitely many maximal ideals (e.g.  $A = \mathbb{Z}$  or  $A = k[T]$ , where  $k$  is a field). Show that any subset of  $\text{Spec } A$  that consists of infinitely many closed points is irreducible but does not contain a generic point.

**Solution 1.39.** Let  $X = \text{Spec } A$ . Recall that for a prime ideal  $\mathfrak{p} \subset A$ ,

$$V(\mathfrak{p}) = \overline{\{\mathfrak{p}\}},$$

and the irreducible closed subsets of  $X$  are exactly the sets  $V(\mathfrak{p})$  with  $\mathfrak{p}$  prime.

(a) Let  $Y \subset X$  be locally closed and irreducible. Then by definition  $Y = U \cap Z$  with  $U \subset X$  open and  $Z \subset X$  closed. Since  $Y$  is irreducible, its closure  $\bar{Y}$  is also irreducible. Since  $\bar{Y} \subset Z$  and  $Z$  is closed,  $\bar{Y}$  is an irreducible closed subset of  $X$ . Therefore

$$\bar{Y} = V(\mathfrak{p})$$

for a unique prime ideal  $\mathfrak{p}$ .

Now  $Y$  is open in  $\bar{Y}$ , since  $Y = U \cap \bar{Y}$ . Also  $Y \neq \emptyset$ , because an irreducible space is by convention nonempty. Since  $\mathfrak{p}$  is the generic point of  $\bar{Y}$ , every nonempty open subset of  $\bar{Y}$  contains  $\mathfrak{p}$ . In particular,  $Y$  contains  $\mathfrak{p}$ .

Thus  $\mathfrak{p} \in Y$ , and  $\overline{\{\mathfrak{p}\}} = \bar{Y}$ . Taking closure inside the subspace  $Y$ , we get

$$\overline{\{\mathfrak{p}\}}^Y = Y \cap \overline{\{\mathfrak{p}\}} = Y \cap \bar{Y} = Y$$

So  $\mathfrak{p}$  is a generic point of  $Y$ .

(b) Let  $Y \subset X$  be irreducible, and suppose  $y, y' \in Y$  are both generic points of  $Y$ . Then

$$\overline{\{y\}}^Y = Y = \overline{\{y'\}}^Y.$$

Taking closures in  $X$ , we obtain

$$\overline{\{y\}} = \bar{Y} = \overline{\{y'\}}.$$

Write  $y = \mathfrak{p}$  and  $y' = \mathfrak{q}$ . Then

$$V(\mathfrak{p}) = V(\mathfrak{q}).$$

But in  $\text{Spec } A$ , if  $\mathfrak{p}$  and  $\mathfrak{q}$  are prime and

$$V(\mathfrak{p}) = V(\mathfrak{q}),$$

then

$$\mathfrak{p} = \sqrt{\mathfrak{p}} = I(V(\mathfrak{p})) = I(V(\mathfrak{q})) = \sqrt{\mathfrak{q}} = \mathfrak{q}.$$

Hence  $y = y'$ .

So every irreducible subset of  $X$  contains at most one generic point.

(c) Now assume that  $A$  is a principal ideal domain with infinitely many maximal ideals, and let

$$S \subset \text{Spec } A$$

be any subset consisting of infinitely many closed points. Since closed points of  $\text{Spec } A$  are exactly maximal ideals,  $S$  is an infinite set of maximal ideals.

We must show that:

1.  $S$  is irreducible;
2.  $S$  has no generic point.

We first show that  $S$  is irreducible. It is enough to prove that any two nonempty open subsets of  $S$  intersect.

Let

$$S \cap U, \quad S \cap V$$

be two nonempty open subsets of  $S$ , where  $U, V \subset X$  are open. Since  $X = \operatorname{Spec} A$ , every open set is a union of distinguished opens. Therefore, after choosing a point in each of  $S \cap U$  and  $S \cap V$ , we may find  $f, g \in A$  such that

$$S \cap D(f) \neq \emptyset, \quad S \cap D(g) \neq \emptyset.$$

Now if  $h \in A$  is nonzero, then  $h$  is contained in only finitely many maximal ideals. Indeed, since  $A$  is a PID, we can write

$$h = up_1^{e_1} \cdots p_r^{e_r}$$

with  $u$  a unit and the  $p_i$  irreducible. The maximal ideals containing  $h$  are exactly

$$(p_1), \dots, (p_r).$$

Thus  $D(h)$  omits only finitely many closed points of  $\operatorname{Spec} A$ . Applying this to  $f$  and  $g$ , we see that

$$S \setminus D(f), \quad S \setminus D(g)$$

are finite subsets of  $S$ . Since  $S$  is infinite, it follows that

$$S \cap D(f) \cap D(g) = S \setminus ((S \setminus D(f)) \cup (S \setminus D(g)))$$

is nonempty. Hence

$$(S \cap U) \cap (S \cap V) \neq \emptyset.$$

So any two nonempty open subsets of  $S$  meet, and therefore  $S$  is irreducible.

Now we show that  $S$  has no generic point. Suppose  $x \in S$  were a generic point of  $S$ . Since  $S$  consists only of closed points of  $X$ , the point  $x$  is closed in  $X$ . Hence

$$\overline{\{x\}} = \{x\}$$

in  $X$ , and therefore in the subspace  $S$ ,

$$\overline{\{x\}}^S = \{x\}.$$

But if  $x$  were a generic point of  $S$ , we would need

$$\overline{\{x\}}^S = S.$$

This would force  $S = \{x\}$ , contradicting the assumption that  $S$  is infinite. Therefore  $S$  has no generic point.

### 1.2.2 Subschemes

#### Definition 1.40.

1. Let  $X$  be a scheme. A **subscheme** of  $X$  is a scheme  $(Y, \mathcal{O}_Y)$ , such that  $Y \subseteq X$  is a locally closed subset, and such that  $Y$  is a closed subscheme of the open subscheme  $U \subseteq X$ , where  $U$  is the largest open subset of  $X$  which contains  $Y$  and in which  $Y$  is closed (i.e.  $U$  is the complement of  $\overline{Y} \setminus Y$ ). We then have a natural morphism of schemes  $Y \rightarrow X$
2. An **immersion**  $i : Y \rightarrow X$  is a morphism of schemes whose underlying continuous map is a homeomorphism of  $Y$  onto a locally closed subset of  $X$ , and such that for all  $y \in Y$  the ring homomorphism

$$i_y^\# : \mathcal{O}_{X, i(y)} \rightarrow \mathcal{O}_{Y, y}$$

between the local rings is surjective.

Open and closed immersions are examples of immersions, and every immersion factors as a closed immersion followed by an open immersion.

**Definition 1.41.** An **open immersion** is a morphism of schemes  $i : Y \rightarrow X$  such that the underlying continuous map is an open embedding and such that for all  $y \in Y$  the ring homomorphism

$$i_y^\# : \mathcal{O}_{X, i(y)} \rightarrow \mathcal{O}_{Y, y}$$

between the local rings is an isomorphism.

**Definition 1.42.** Let  $X$  be a scheme.

1. A **closed subscheme** of  $X$  is given by a closed subset  $Z \subseteq X$  and an ideal sheaf  $\mathcal{I} \subseteq \mathcal{O}_X$  such that

$$Z = \{x \in X : (\mathcal{O}_X/\mathcal{I})_x \neq 0\}$$

and  $(Z, (\mathcal{O}_X/\mathcal{I})|_Z)$  is a scheme.

2. A morphism  $i : Z \rightarrow X$  of schemes is called a **closed immersion** if the underlying continuous map is a homeomorphism between  $Z$  and a closed subset of  $X$ , and the sheaf homomorphism

$$i^\flat : \mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$$

is surjective.

**Proposition 1.43.** Let  $X$  be a scheme. Then the data of a closed immersion  $i : Z \rightarrow X$  is equivalent to the data of a **quasi-coherent** ideal sheaf  $\mathcal{I} \subseteq \mathcal{O}_X$ . Under this correspondence,

$$i_*\mathcal{O}_Z \cong \mathcal{O}_X/\mathcal{I},$$

and  $Z$  is identified with the closed subscheme  $V(\mathcal{I}) \subseteq X$ .



Fiber products exist in the category of schemes.

**Theorem 1.44.** *Let  $X, Y, Z$  be schemes and let  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$  be morphisms of schemes. Then the fiber product  $X \times_Z Y$  exists in the category of schemes. It is locally modeled by the tensor product of rings: if  $X = \operatorname{Spec} A$ ,  $Y = \operatorname{Spec} B$ , and  $Z = \operatorname{Spec} C$  are affine schemes, and the morphisms  $f$  and  $g$  correspond to ring homomorphisms  $C \rightarrow A$  and  $C \rightarrow B$ , then*

$$X \times_Z Y \cong \operatorname{Spec}(A \otimes_C B).$$

We can use fiber products to define the fiber of a morphism over a point.

**Definition 1.45.** *The **fiber** of a morphism of schemes  $f : X \rightarrow Y$  over a point  $y \in Y$  is the fiber product*

$$X_y := X \times_Y \operatorname{Spec} k(y),$$

*where  $k(y)$  is the residue field of  $y$ , and the morphism  $\operatorname{Spec} k(y) \rightarrow Y$  is the natural morphism corresponding to the point  $y$ .*

### 1.2.3 Dimension

Recall that we say a topological space is **irreducible** if it cannot be written as the union of two proper closed subsets. An irreducible closed subset of a scheme is called an **irreducible component**. Every scheme can be written as a finite union of irreducible components, and this decomposition is unique up to reordering.

**Remark 1.46.** *Irreducible components and connected components are not the same. An irreducible component is necessarily connected, but connected does not imply irreducible. For example,  $X = \operatorname{Spec} k[x, y]/(xy)$  is the union of two affine lines crossing at the origin. It is connected, because the two components meet, but it is not irreducible because  $V(xy) = V(x) \cup V(y)$ .*

The dimension of a scheme sees only the underlying topological space and is defined as the supremum of the lengths of chains of irreducible closed subsets. For an affine scheme  $\operatorname{Spec} A$ , the dimension is equal to the Krull dimension of  $A$ , which is the supremum of the lengths of chains of prime ideals in  $A$ .

Dimension can be checked on an open cover, along irreducible components, or at local rings.

**Proposition 1.47.**

1. Suppose  $X = \bigcup_i U_i$  is a union of open subsets. Then

$$\dim X = \sup_i \dim U_i.$$

2. If  $X = \bigcup_i X_i$  is a union of irreducible components, then

$$\dim X = \sup_i \dim X_i.$$

3.

$$\dim X = \sup_{x \in X} \dim \mathcal{O}_{X,x}.$$

**Theorem 1.48.** *Let  $X$  be an irreducible  $k$ -scheme locally of finite type with generic point  $\eta$ .*

1. *If  $X$  is integral, then  $\kappa(\eta)$  is simply the function field of  $X$ .*

$$\dim X = \text{trdeg}_k \kappa(\eta).$$

2. *Let  $x \in X$  be any closed point. Then*

$$\dim \mathcal{O}_{X,x} = \dim X.$$

3. *Let  $f : Y \rightarrow X$  be a morphism of  $k$ -schemes of finite type such that  $f(Y)$  contains the generic point  $\eta$  of  $X$ . Then*

$$\dim Y \geq \dim X.$$

*In particular,*

$$\dim U = \dim X$$

*for any nonempty open subscheme  $U$  of  $X$ .*

4. *Let  $f : Y \rightarrow X$  be a morphism of  $k$ -schemes of finite type with finite fibers. Then*

$$\dim Y \leq \dim X.$$

**Definition 1.49.** *The codimension of an arbitrary subset  $Y \subseteq X$  is defined as*

$$\text{codim}(Y, X) = \inf_{y \in Y} \dim \mathcal{O}_{X,y}.$$

*If  $Y$  is an irreducible closed subset of  $X$ , then this coincides with the supremum of the lengths of chains of irreducible closed subsets containing  $Y$ .*

If  $X = \text{Spec } A$  is affine, then the codimension of an irreducible closed subset  $V(\mathfrak{p})$  is the height of the prime ideal  $\mathfrak{p}$ , which is the supremum of the lengths of chains of prime ideals contained in  $\mathfrak{p}$ .

### 1.2.4 Morphisms of finite type and finite morphisms

**Definition 1.50.** A morphism of affine schemes

$$\operatorname{Spec} B \rightarrow \operatorname{Spec} A$$

is of **finite type** if  $B$  is a finitely generated  $A$ -algebra.

**Definition 1.51.** A morphism of schemes  $f : X \rightarrow Y$  is **locally of finite type** if every point of  $X$  has an affine neighborhood  $\operatorname{Spec} B$  mapping into an affine open  $\operatorname{Spec} A \subseteq Y$  such that  $B$  is a finitely generated  $A$ -algebra. It is **of finite type** if in addition  $f$  is quasi-compact.

**Example 1.52.** The morphism

$$\mathbb{A}_A^n = \operatorname{Spec} A[x_1, \dots, x_n] \rightarrow \operatorname{Spec} A$$

is of finite type, and so is

$$\mathbb{P}_A^n \rightarrow \operatorname{Spec} A.$$

**Exercise 1.53.**

- (a) Show that every open embedding is locally of finite type, and hence that every quasi-compact open embedding is of finite type. Show that every open embedding into a locally Noetherian scheme is of finite type.
- (b) Show that the composition of two morphisms locally of finite type is locally of finite type. Deduce that the composition of two morphisms of finite type is of finite type.
- (c) Suppose  $\pi : X \rightarrow Y$  is locally of finite type, and  $Y$  is locally Noetherian. Show that  $X$  is also locally Noetherian. If  $\pi : X \rightarrow Y$  is of finite type and  $Y$  is Noetherian, show that  $X$  is Noetherian.

**Solution 1.54.**

(a) Let  $j : U \hookrightarrow X$  be an open immersion. This is local on  $X$ , so let  $\operatorname{Spec} A \subset X$  be an affine open subset. Then

$$j^{-1}(\operatorname{Spec} A) = U \cap \operatorname{Spec} A$$

is an open subset of  $\operatorname{Spec} A$ , hence can be written as a union of distinguished opens:

$$U \cap \operatorname{Spec} A = \bigcup_i D(f_i).$$

Each  $D(f_i) \cong \operatorname{Spec} A_{f_i}$ , and  $A_{f_i} = A[f_i^{-1}]$  is a finitely generated  $A$ -algebra. Thus  $j$  is locally of finite type.

If  $j$  is quasi-compact, then for each affine open  $\operatorname{Spec} A \subset X$ , the open subset  $U \cap \operatorname{Spec} A$  is quasi-compact, so from the cover

$$U \cap \operatorname{Spec} A = \bigcup_i D(f_i)$$

we may extract a finite subcover

$$U \cap \operatorname{Spec} A = \bigcup_{m=1}^r D(f_{i_m}).$$

Hence  $j^{-1}(\operatorname{Spec} A)$  is quasi-compact. Therefore  $j$  is quasi-compact and locally of finite type, so  $j$  is of finite type.

If  $X$  is locally Noetherian, then every affine open  $\operatorname{Spec} A \subset X$  has  $A$  Noetherian, and hence every open subset of  $\operatorname{Spec} A$  is quasi-compact. Therefore every open immersion into  $X$  is quasi-compact. Since every open immersion is locally of finite type, it follows that every open immersion into a locally Noetherian scheme is of finite type.

(b) Let  $X \xrightarrow{f} Y \xrightarrow{g} Z$  be morphisms locally of finite type. This is local on  $Z$ , so take an affine open  $\operatorname{Spec} C \subset Z$ , and let  $\operatorname{Spec} B \subset Y$  lie over it, and  $\operatorname{Spec} A \subset X$  lie over  $\operatorname{Spec} B$ .

Then  $B$  is a finitely generated  $C$ -algebra, and  $A$  is a finitely generated  $B$ -algebra. Hence

$$A \cong B[x_1, \dots, x_m]/I \cong C[y_1, \dots, y_n, x_1, \dots, x_m]/J,$$

so  $A$  is finitely generated over  $C$ . Thus  $g \circ f$  is locally of finite type.

If  $f$  and  $g$  are of finite type, then they are also quasi-compact, and compositions of quasi-compact morphisms are quasi-compact. Hence  $g \circ f$  is of finite type.

(c) Suppose  $\pi : X \rightarrow Y$  is locally of finite type and  $Y$  is locally Noetherian. This is local on  $Y$ , so assume  $Y = \operatorname{Spec} A$  with  $A$  Noetherian. Let  $\operatorname{Spec} B \subset X$  be an affine open mapping into  $\operatorname{Spec} A$ . Then  $B$  is a finitely generated  $A$ -algebra, hence Noetherian. Thus  $X$  is locally Noetherian.

If  $\pi$  is of finite type and  $Y$  is Noetherian, then  $X$  is quasi-compact and locally Noetherian. A quasi-compact locally Noetherian scheme is Noetherian. Hence  $X$  is Noetherian.

**Theorem 1.55** (Chevalley). *If  $f : X \rightarrow Y$  is a morphism of finite type between Noetherian schemes, then the image of any constructible subset of  $X$  is constructible in  $Y$ .*

**Remark 1.56.** *In particular,  $f(X)$  is a constructible subset of  $Y$ . Thus if  $f$  is dominant (i.e. the image of  $f$  is dense in  $Y$ ), its image contains a dense open subset of  $Y$ . This is saying something nontrivial because the target  $Y$  may have dense subsets which are not open (e.g.  $\mathbb{Q} \subset \mathbb{R}$  is dense but not open), but Chevalley's theorem says that the image of a morphism of finite type cannot be one of those.*

**Definition 1.57.** *A ring map  $A \rightarrow B$  is **finite** if  $B$  is finitely generated as an  $A$ -module. A morphism of schemes  $f : X \rightarrow Y$  is finite if it is affine and for every affine open  $\operatorname{Spec} A \subseteq Y$ , the inverse image  $f^{-1}(\operatorname{Spec} A)$  is affine, say  $\operatorname{Spec} B$ , with  $B$  finite as an  $A$ -module.*

**Example 1.58.** *Closed immersions are finite. Working locally, we are thinking about*

a morphism  $\text{Spec } A/I \rightarrow \text{Spec } A$  for some ideal  $I \subseteq A$ . The ring  $A/I$  is generated as an  $A$ -module by the single element 1, so the morphism is finite.

The morphism  $k[t] \rightarrow k[u]$  given by  $t \mapsto p(u)$  for some polynomial  $p$  of degree  $n$  is finite, since  $k[u]$  is generated as a  $k[t]$ -module by  $1, u, \dots, u^{n-1}$ .

Finite morphisms are precisely the morphisms which are affine and projective.

**Example 1.59.** The open embedding  $\mathbb{A}_k^2 \setminus \{(0,0)\} \rightarrow \mathbb{A}_k^2$  has finite fibers, but is not affine (as  $\mathbb{A}_k^2 \setminus \{(0,0)\}$  is not affine) and hence not finite. Since the complement has codimension 2 and  $\mathbb{A}_k^2$  is normal, every regular function on  $U$  extends uniquely to  $\mathbb{A}_k^2$ . Thus

$$\Gamma(U, \mathcal{O}_U) = k[x, y].$$

If  $U$  were affine, then

$$U \cong \text{Spec } \Gamma(U, \mathcal{O}_U) = \text{Spec } k[x, y] = \mathbb{A}_k^2,$$

which is impossible since  $U \neq \mathbb{A}_k^2$ .

In general, open immersions are not finite because they are not affine, but even when they are affine, they need not be finite. In fact, an open immersion is finite exactly when it is also a closed immersion onto an open-and-closed subset. So for connected schemes, the only finite open immersions are isomorphisms onto connected components.

**Exercise 1.60.** Show that the open embedding  $\mathbb{A}_{\mathbb{C}}^1 \setminus \{0\} \rightarrow \mathbb{A}_{\mathbb{C}}^1$  has finite fibers and is affine, but is not finite.

**Solution 1.61.** It is clear that the morphism is affine, since  $\mathbb{A}_{\mathbb{C}}^1 \setminus \{0\}$  is affine, given by the nonvanishing of  $x$ . It has finite fibers, since the fiber over 0 is empty and the fiber over any nonzero point is a single point. However, it is not finite, since the corresponding ring map  $\mathbb{C}[x] \rightarrow \mathbb{C}[x, x^{-1}]$  is not finite, as  $x^{-1}$  is not integral over  $\mathbb{C}[x]$ .

**Exercise 1.62.** Show that if  $f : X \rightarrow \text{Spec } k$  is a finite morphism, then  $X$  is a finite union of discrete points, each with residual field a finite extension of  $k$ .

**Solution 1.63.** If  $f : X \rightarrow \text{Spec } k$  is finite, then  $X$  is affine, say

$$X = \text{Spec } A$$

with  $A$  a finite  $k$ -algebra. Since  $A$  is finite-dimensional as a  $k$ -vector space, every prime quotient  $A/\mathfrak{p}$  is a finite-dimensional domain over  $k$ , hence a field. Thus every prime ideal of  $A$  is maximal.

Now  $A$  can only have finitely many maximal ideals, since if  $\mathfrak{m}_1, \dots, \mathfrak{m}_n$  are distinct maximal ideals, then they are pairwise coprime, so the Chinese remainder theorem gives

an surjection

$$A \rightarrow \prod_{i=1}^n A/\mathfrak{m}_i$$

so the dimension of  $A$  as a  $k$ -vector space is at least  $n$ .

Hence  $X$  is a finite set of closed points. For each point  $x \in X$ , the residue field is

$$\kappa(x) = A/\mathfrak{m}_x,$$

which is a finite extension of  $k$ . Thus  $X$  is a finite union of discrete points, each with residue field, finite over  $k$  again by the finite-dimensionality of  $A$  as a  $k$ -vector space.

More generally, a similar argument shows that any Artinian ring has only finitely many maximal ideals, and every prime ideal is maximal.

**Exercise 1.64.** Show that finite morphisms have finite fibers, i.e. for every point  $y \in Y$ , the fiber  $X_y$  is a finite scheme over  $\kappa(y)$ .

**Solution 1.65.** Let  $f : X \rightarrow Y$  be a finite morphism, and let  $y \in Y$ . Since the question is local on  $Y$ , we may assume  $Y = \operatorname{Spec} B$  and  $X = \operatorname{Spec} A$  where  $A$  is a finite  $B$ -module. Let  $y$  correspond to a prime ideal  $\mathfrak{q} \subset B$ .

The fiber over  $y$  is

$$X_y \cong \operatorname{Spec}(A \otimes_B \kappa(\mathfrak{q})).$$

Note that prime ideals of  $A \otimes_B \kappa(\mathfrak{q})$  correspond exactly to primes  $\mathfrak{p} \subset A$  such that  $\mathfrak{p} \cap B = \mathfrak{q}$ .

Since  $A$  is finite as a  $B$ -module, the ring  $A \otimes_B \kappa(\mathfrak{q})$  is a finite-dimensional  $\kappa(\mathfrak{q})$ -vector space. Thus  $X_y$  has only finitely many points. Thus finite morphisms have finite fibers.

**Exercise 1.66.** Show that the base change  $f' : X \times_Y Y' \rightarrow Y'$  of a finite morphism  $f : X \rightarrow Y$  is finite.

**Solution 1.67.** This is local on  $Y$  and  $Y'$ , so we may assume

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B, \quad Y' = \operatorname{Spec} A'$$

Since  $f$  is finite,  $B$  is a finite  $A$ -algebra. Now

$$X' = X \times_Y Y' \cong \operatorname{Spec}(B \otimes_A A').$$

So it is enough to show that  $B \otimes_A A'$  is a finite  $A'$ -module. But if  $b_1, \dots, b_n$  generate  $B$  as an  $A$ -module, then the elements  $b_i \otimes 1$  generate  $B \otimes_A A'$  as an  $A'$ -module, by  $A$ -linearity of the tensor product.

### 1.2.5 Integrality and some commutative algebra

**Definition 1.68.** A ring map  $B \rightarrow A$  is **integral** if every element  $a \in A$  satisfies a monic polynomial with coefficients in  $B$ . An integral ring morphism is an **integral extension** if it is injective.

A morphism of schemes is **integral** if it is affine and for every affine open  $\text{Spec } A \subseteq Y$ , the inverse image  $f^{-1}(\text{Spec } A)$  is affine, say  $\text{Spec } B$ , with  $B$  integral over  $A$ .

The key point is topological. In particular, if  $A \rightarrow B$  is integral, then  $\text{Spec } B \rightarrow \text{Spec } A$  is closed and surjective. This geometry is a consequence of the algebraic fact known as the **lying over** and **going up** theorems.

Note that being an integral morphism is preserved by quotienting in the source and in the target, and localizing in the source. It is not preserved by localizing in the target.

Also note that if  $b_1, b_2$  are integral over  $A$ , then so are  $b_1 + b_2$  and  $b_1 b_2$ . Thus the set of elements of  $B$  that are integral over  $A$  form a subalgebra of  $B$ , called the **integral closure** of  $A$  in  $B$ . If  $A$  is integrally closed in its field of fractions, then we say that  $A$  is **integrally closed**. Also note that integrality is transitive: if  $A \rightarrow B$  and  $B \rightarrow C$  are integral, then so is  $A \rightarrow C$ .

**Lemma 1.69.** *Suppose  $\varphi : B \rightarrow A$  is a ring morphism. Then  $a \in A$  is integral over  $B$  if and only if it is contained in a subalgebra of  $A$  that is a finitely generated  $B$ -module.*

*Proof.* If  $a$  satisfies a monic polynomial of degree  $n$ , then the  $B$ -submodule of  $A$  generated by  $1, a, \dots, a^{n-1}$  is closed under multiplication, and hence is a subalgebra of  $A$ .

Conversely, assume that  $a$  is contained in a subalgebra  $A' \subset A$  that is a finitely generated  $B$ -module. Choose a finite generating set  $m_1, \dots, m_n$  of  $A'$  (as a  $B$ -module). Then

$$am_i = \sum_j b_{ij} m_j \quad \text{for some } b_{ij} \in B.$$

Thus

$$(aI_n - [b_{ij}]_{ij}) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (1)$$

where  $I_n$  is the  $n \times n$  identity matrix. Recall that any  $n \times n$  matrix  $M$  has an adjugate matrix  $\text{adj}(M)$  such that

$$\text{adj}(M) M = \det(M) I_n.$$

Multiplying (1) by  $\text{adj}(aI_n - [b_{ij}])$ , we obtain

$$\det(aI_n - [b_{ij}]) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Thus  $\det(aI_n - [b_{ij}])$  annihilates each  $m_i$ , hence every element of  $A'$ . In particular, it annihilates 1 so

$$\det(aI_n - [b_{ij}]) = 0 \quad \text{in } A.$$

Expanding the determinant gives a monic polynomial in  $a$  with coefficients in  $B$ , so  $a$  is integral over  $B$ .  $\square$

**Proposition 1.70.** *If  $B$  is finite as an  $A$ -module, then  $B$  is integral over  $A$ .*

**Theorem 1.71** (Lying Over). *Suppose  $\varphi : B \rightarrow A$  is an integral extension. Then for any prime ideal  $\mathfrak{q} \subset B$ , there is a prime ideal  $\mathfrak{p} \subset A$  such that*

$$\mathfrak{p} \cap B = \mathfrak{q}.$$

Note that  $B$  need not be Noetherian, and  $A$  need not be finitely generated over  $B$ .

**Exercise 1.72.** *Show that the special case where  $A$  is a field translates to: if  $B \subset A$  and  $A$  is integral over  $B$ , then  $B$  is a field.*

**Solution 1.73.** *Let  $b \in B$ . Then  $b^{-1} \in A$  is integral over  $B$ , so there is a monic polynomial*

$$(b^{-1})^n + a_{n-1}(b^{-1})^{n-1} + \cdots + a_0 = 0$$

*with  $a_i \in B$ . Multiplying by  $b^n$  gives*

$$1 + a_{n-1}b + \cdots + a_0b^n = 0.$$

*Thus  $b^{-1} \in B$  so  $B$  is a field.*

*Proof.* We first reduce to the case where  $(B, \mathfrak{q})$  is a local ring. By localizing at  $\mathfrak{q}$  (which preserves integrality), we may assume this.

Let  $\mathfrak{p}$  be any maximal ideal of  $A$ . Consider the diagram

$$\begin{array}{ccc} A & \longrightarrow & A/\mathfrak{p} \\ \downarrow & & \downarrow \\ B & \longrightarrow & B/(\mathfrak{p} \cap B) \end{array}$$

The right vertical map is an integral extension. Since  $A/\mathfrak{p}$  is a field, it follows that  $B/(\mathfrak{p} \cap B)$  is also a field. Hence  $\mathfrak{p} \cap B$  is maximal, and therefore equals  $\mathfrak{q}$ . Using the correspondence between prime ideals of  $A$  and prime ideals of  $A_{\mathfrak{q}}$ , we can contract the prime ideal  $\mathfrak{p}$  back to get the desired prime ideal  $\mathfrak{p}$  of  $A$ .  $\square$

The same adjugate trick can be used to prove Nakayama's lemma, so we will do this now. They seem to have no connection conceptually.

**Lemma 1.74** (Nakayama's lemma). *Let  $I$  be an ideal of a ring  $A$ , and let  $M$  be a finitely generated  $A$ -module such that  $IM = M$ . Then there is an element  $a \in A$  with  $a \equiv 1 \pmod{I}$  such that  $aM = 0$ .*

*Proof.* Say  $M$  is generated by  $m_1, \dots, m_n$ . Then as  $M = IM$ , we have  $m_i = \sum_j a_{ij}m_j$  for some  $a_{ij} \in I$ . Thus

$$(I_n - Z) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$



where  $Z = (a_{ij})$ . Multiplying both sides on the left by  $\text{adj}(I_n - Z)$ , we obtain

$$\det(I_n - Z) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Expanding the determinant gives something that is  $1 \pmod{I}$ , since  $Z$  has entries in  $I$ .  $\square$

Suppose  $I$  is contained in all of the maximal ideals of  $A$ . Then  $a \equiv 1 \pmod{I}$  is a unit. Otherwise,  $(a)$  is not the whole ring and therefore is contained in some maximal ideal  $\mathfrak{m}$ . But then  $a \equiv 1 \pmod{\mathfrak{m}}$ , contradiction.

Thus an application of Nakayama's lemma says that  $aM = 0$  for some unit  $a$ , so  $M = 0$ .

**Lemma 1.75** (Nakayama's lemma). *Suppose  $A$  is a ring,  $I$  is an ideal of  $A$  contained in all maximal ideals, and  $M$  is a finitely generated  $A$ -module. Suppose  $M = IM$ . Then  $M = 0$ .*

The most interesting case is when  $A$  is a local ring, and  $I$  is the maximal ideal.

**Exercise 1.76** (Nakayama's lemma). *Suppose  $(A, \mathfrak{m})$  is a local ring, and let  $M$  be a finitely generated  $A$ -module. Suppose  $f_1, \dots, f_n \in M$  are such that their images generate  $M/\mathfrak{m}M$ . Then  $f_1, \dots, f_n$  generate  $M$ .*

**Solution 1.77.** *Let  $N \subseteq M$  be the submodule generated by  $f_1, \dots, f_n$ . Since the images of  $f_1, \dots, f_n$  generate  $M/\mathfrak{m}M$ , every class in  $M/\mathfrak{m}M$  is represented by an element of  $N$ . Equivalently,*

$$M = N + \mathfrak{m}M.$$

*Passing to the quotient by  $N$ , we obtain*

$$\begin{aligned} M/N &= (N + \mathfrak{m}M)/N \\ &= \mathfrak{m}M/N \end{aligned}$$

*Now  $M/N$  is a finitely generated  $A$ -module, because  $M$  is finitely generated. Since  $(A, \mathfrak{m})$  is local, the ideal  $\mathfrak{m}$  is contained in every maximal ideal of  $A$ ; in fact it is the unique maximal ideal. Therefore Nakayama's lemma implies that  $M/N = 0$ , so  $f_1, \dots, f_n$  generate  $M$ .*

**Example 1.78.** *Finite morphisms are integral, but integral morphisms need not have finite fibers. Let  $k$  be a field,*

$$B = k[x], \quad A = \bigcup_{n \geq 1} k[x^{1/n}] \subset k(x^{1/\infty}).$$

*Every element of  $A$  is integral over  $B$ , so  $\text{Spec } A \rightarrow \text{Spec } B$  is integral. But  $A$  is not finite over  $B$ . Look at the fiber over  $(x) \in \text{Spec } B$ . The primes of  $A$  lying over  $(x)$  correspond*

to primes of

$$A \otimes_B \kappa((x)) \cong A/(x).$$

where  $\kappa((x))$  is the residue field of  $B$  at  $(x)$ , which is  $k$ . Now

$$A/(x) \cong k[t_1, t_2, \dots]/(t_1, t_2^2, t_3^3, \dots)$$

where  $t_n$  is the image of  $x^{1/n}$ . This ring has infinitely many prime ideals, so the fiber is infinite.

Note that this counterexample is not of finite type. We can ask whether or not this is the only way for an integral morphism to fail to have finite fibers. The answer is indeed yes.

**Proposition 1.79.** *An integral morphism which is locally of finite type is finite.*

*Proof.* This is local on  $Y$ , so take

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B.$$

By assumption we can write

$$B = A[b_1, \dots, b_n].$$

with each  $b_i$  integral over  $A$ , so satisfies a monic polynomial:

$$b_i^{d_i} + a_{i,1}b_i^{d_i-1} + \dots + a_{i,d_i} = 0, \quad a_{i,j} \in A.$$

From these relations, every power  $b_i^m$  with  $m \geq d_i$  can be expressed as an  $A$ -linear combination of  $1, b_i, \dots, b_i^{d_i-1}$ .

Thus any monomial in the  $b_i$  reduces to an  $A$ -linear combination of finitely many monomials:

$$b_1^{e_1} \dots b_n^{e_n}, \quad 0 \leq e_i < d_i.$$

There are finitely many such monomials and they generate  $B$  as an  $A$ -module.  $\square$

It is also useful to know that integral and finite morphisms are preserved by base change and composition.

### 1.2.6 Quasicompact, quasiseparated, and proper morphisms

Let  $f : X \rightarrow Y$  be a morphism of schemes. The correct analogue of compactness is quasicompactness.

**Definition 1.80.** *The morphism  $f : X \rightarrow Y$  is **quasicompact** if the inverse image of every affine open in  $Y$  is quasi-compact, i.e. if it can be covered by finitely many affine opens. A scheme  $X$  is **quasicompact** if the morphism  $X \rightarrow \operatorname{Spec} \mathbb{Z}$  is quasicompact, i.e. if  $X$  can be covered by finitely many affine opens.*

**Example 1.81.** Every affine scheme  $\operatorname{Spec} A \rightarrow \operatorname{Spec} \mathbb{Z}$  is quasicompact, since  $\operatorname{Spec} A$  can be covered by the single affine open  $\operatorname{Spec} A$ .

**Example 1.82.** The morphism  $\bigsqcup_{i \in \mathbb{N}} \operatorname{Spec} k \rightarrow \operatorname{Spec} k$  is not quasi-compact, since the preimage of  $\operatorname{Spec} k$  is an infinite disjoint union of affine schemes, which is not quasi-compact.

**Example 1.83.** Let  $A = k[x_1, x_2, \dots]$ . Then  $\operatorname{Spec} A \rightarrow \operatorname{Spec} k$  is quasi-compact but not of finite type.

Let  $f : X \rightarrow Y$  be a morphism of schemes. The correct analogue of the Hausdorff condition is separatedness.

**Definition 1.84.** The **diagonal morphism** of  $f$  is

$$\Delta_f : X \rightarrow X \times_Y X, \quad x \mapsto (x, x).$$

The morphism  $f$  is called **quasi-separated** if  $\Delta_f$  is quasi-compact, i.e. if the inverse image of every affine open in  $X \times_Y X$  is quasi-compact. The morphism  $f$  is called **separated** if  $\Delta_f$  is a closed immersion.

**Remark 1.85.** A scheme  $X$  is quasi-separated iff the intersection of any two affine open subsets is quasi-compact. A scheme  $X$  is separated iff the intersection of any two affine open subsets is affine.

**Example 1.86.** Every affine morphism is separated. In particular the diagonal morphism of an affine morphism corresponds to the multiplication map of rings and being surjective is therefore a closed immersion. In particular every affine scheme is separated over  $\operatorname{Spec} \mathbb{Z}$ .

**Example 1.87.** Projective space  $\mathbb{P}_Y^n \rightarrow Y$  is separated. Cover  $\mathbb{P}_Y^n \times_Y \mathbb{P}_Y^n$  by the opens  $U_i \times_Y U_j$  where  $U_i$  is the standard affine open in  $\mathbb{P}_Y^n$  where the  $i$ -th homogeneous coordinate is nonzero. The intersection of  $U_i \times_Y U_j$  with the diagonal is the vanishing of  $x_i/x_j \subset U_i$ . A distinguished open in an affine scheme is affine, so the intersection of two affine opens in  $\mathbb{P}_Y^n$  is affine, thus  $\mathbb{P}_Y^n$  is separated over  $Y$ .

**Example 1.88** (Affine line with doubled origin). **The line with doubled origin is not separated.** Intuitively, the two origins cannot be distinguished by disjoint neighborhoods, and scheme-theoretically this failure is measured by the diagonal not being closed.

The line with doubled origin is obtained by gluing two copies of  $\mathbb{A}_k^1$  along the open  $D(t)$  via the identity map. The diagonal morphism is not closed. To test whether  $\Delta_X : X \rightarrow X \times X$  is a closed immersion, it is enough to check on an open cover of  $X \times X$ . In particular, look at the open subset  $U_1 \times U_2 \subset X \times X$  where  $U_1$  and  $U_2$  are the two copies of  $\mathbb{A}_k^1 \setminus \{0\}$ . The inverse image of  $U_1 \times U_2$  under the diagonal is exactly the set of points of  $X$  lying

in both  $U_1$  and  $U_2$ , namely the overlap  $\Delta_X^{-1}(U_1 \times U_2) = U_1 \cap U_2 = D(t)$

So on this chart the diagonal restricts to the morphism

$$D(t) \longrightarrow U_1 \times U_2 \cong \operatorname{Spec} k[t_1, t_2]$$

given by  $t_1 \mapsto t$  and  $t_2 \mapsto t$ . Its image is therefore  $\{(a, a) : a \neq 0\}$ , which is not closed in  $U_1 \times U_2$  because it does not contain the point  $(0, 0)$ .

**It is however quasi-separated.** Cover  $X$  by the two affine opens  $X = X_1 \cup X_2$ , with  $X_i \cong \operatorname{Spec} k[t]$ .

Then  $X \times X$  is covered by the four affine opens  $X_i \times X_j$  ( $i, j \in \{1, 2\}$ ). It suffices to show that for each  $i, j$ , the inverse image  $\Delta_X^{-1}(X_i \times X_j)$  is quasi-compact.

- If  $i = j$ , then  $\Delta_X^{-1}(X_i \times X_i) = X_i$ , which is affine, hence quasi-compact.
- If  $i \neq j$ , then  $\Delta_X^{-1}(X_i \times X_j) = X_i \cap X_j = D(t)$ , which is affine, hence quasi-compact.

Therefore  $\Delta_X$  is quasi-compact, so  $X$  is quasi-separated. Since each of these intersections is affine, it appears that we have just shown that  $X$  is separated, but this is not the case because there exist other affine opens  $U, V \subset X$  whose intersection is not affine. The subtle point is that it is not enough to just check the affine opens in the given cover; we must check all affine opens. This is a contrast to quasi-separatedness or quasicompactness, where it suffices to check the affine opens in a given cover.

**Example 1.89** (Affine line with infinitely many origins is not quasi-separated). Let  $X$  be obtained by gluing countably many copies

$$X_i \cong \operatorname{Spec} k[t]$$

along the common open subset  $D(t) = \operatorname{Spec} k[t, t^{-1}]$  via the identity.

This defines a scheme with infinitely many distinct origins  $\{0_i\}$ .

Let  $t$  denote the global function induced by the coordinate on each chart. For any subset  $I \subset \mathbb{N}$ , define

$$f_I := \prod_{i \in I} t.$$

Then  $f_I$  defines a global function on  $X$ , and

$$D(f_I)$$

is an affine open subset.

Choose a partition  $\mathbb{N} = I \sqcup J$  into two infinite subsets, and set

$$U := D(f_I), \quad V := D(f_J).$$

Then  $U$  and  $V$  are affine open subsets of  $X$ .

Their intersection is

$$U \cap V = D(f_I f_J) = D\left(\prod_{n \in \mathbb{N}} t\right),$$

which is the complement of all origins. Hence

$$U \cap V \cong \bigsqcup_{n \in \mathbb{N}} D(t).$$

This is an infinite disjoint union of affine schemes, hence not quasi-compact. Therefore  $X$  is not quasi-separated.

**Definition 1.90.** A morphism  $f : X \rightarrow Y$  is called **universally closed** if for every morphism  $Y' \rightarrow Y$ , the base change

$$f' : X \times_Y Y' \rightarrow Y'$$

is a closed map on underlying topological spaces.

**Exercise 1.91.** Show that  $\pi : \mathbb{A}_k^1 \rightarrow \operatorname{Spec} k$  is closed but not universally closed. Thus  $\pi$  is not proper.

Recall that we saw that if  $B \rightarrow A$  is integral, then  $\operatorname{Spec} A \rightarrow \operatorname{Spec} B$  is closed. Since being integral is preserved by base change, it follows that integral morphisms are universally closed.

**Proposition 1.92.** A morphism of schemes is integral if and only if it is affine and universally closed.

**Definition 1.93.** A morphism  $f : X \rightarrow Y$  is called **proper** if it is of finite type, separated, and universally closed.

This is the algebro-geometric replacement for compactness.

**Example 1.94.** A closed immersion is proper. On affines,  $A \rightarrow A/I$  is finitely generated so the morphism is of finite type. Closed immersions are affine morphisms and thus separated. A closed immersion is a homeomorphism onto a closed subset. Closed sets remain closed under any base change.

**Remark 1.95.** Proper morphisms are stable under composition and base change. This is because each of the three conditions in the definition of properness is stable under composition and base change.

Regarding composition. The condition of finite type is checked by thinking about finitely generated  $B$ -algebras where  $B$  is a finitely generated  $A$ -algebra. If  $X \rightarrow Y \rightarrow Z$  are separated, the diagonal morphism of  $X \rightarrow Z$  factors through the diagonal morphism of  $X \rightarrow Y$ . There is a natural map  $X \times_Y X \rightarrow X \times_Z X$  which is a closed immersion, being

the base change of the diagonal morphism of  $Y \rightarrow Z$ . A composition of closed immersions is clearly a closed immersion. Universally closed is obvious.

Regarding base change. The condition of finite type is checked by thinking about the finite generation of  $B \otimes_A A'$  as an  $A'$ -algebra where  $B$  is a finitely generated  $A$ -algebra. Closed immersions are stable under base change. Universally closed is obvious.

**Definition 1.96.** A morphism of schemes  $f : X \rightarrow Y$  is called **projective** if it factors as a closed immersion  $X \hookrightarrow \mathbb{P}_Y^n$  for some  $n$ , followed by the projection  $\mathbb{P}_Y^n \rightarrow Y$ . Equivalently, if  $f$  is isomorphic to the relative Proj of a finitely generated graded algebra over  $Y$ .

**Theorem 1.97.** Projective  $A$ -schemes are proper over  $A$ .

*Proof.* Let  $X \rightarrow \operatorname{Spec} A$  be a projective  $A$ -scheme. Then by definition the structure morphism factors as

$$X \hookrightarrow \mathbb{P}_A^n \rightarrow \operatorname{Spec} A$$

for some  $n$ , where the first map is a closed immersion. Since closed immersions are proper, and compositions of proper morphisms are proper, it suffices to show that

$$\mathbb{P}_A^n \rightarrow \operatorname{Spec} A$$

is proper.

We have already seen that this morphism is of finite type and separated. Therefore it remains to show that it is universally closed.

Now  $\mathbb{P}_A^n = \mathbb{P}_{\mathbb{Z}}^n \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} A$ . So to prove that  $\mathbb{P}_A^n \rightarrow \operatorname{Spec} A$  is universally closed, it suffices to show that for every scheme  $X$ , the projection

$$\mathbb{P}_X^n := \mathbb{P}_{\mathbb{Z}}^n \times_{\operatorname{Spec} \mathbb{Z}} X \rightarrow X$$

is a closed map.

Being a closed map is local on the target. Hence, after covering  $X$  by affine open subschemes, it suffices to prove that

$$\mathbb{P}_B^n \rightarrow \operatorname{Spec} B$$

is closed for every ring  $B$ .

But this is exactly the Fundamental Theorem of Elimination Theory, which says that given a system of homogeneous polynomial equations with coefficients in  $B$ , the set of  $b \in \operatorname{Spec} B$  such that the system has a solution over  $\kappa(b)$  is Zariski closed in  $\operatorname{Spec} B$ .  $\square$

**Proposition 1.98.** Let  $f : X \rightarrow S$  be a morphism of schemes with  $S$  locally Noetherian. Then  $f$  is finite if and only if  $f$  is proper and has finite fibers.

*Proof.* Finite morphisms are proper. Indeed, a finite morphism is affine and of finite type, hence separated. It is also universally closed since it is integral. We have also already seen that finite morphisms have finite fibers.

Conversely,  $\square$

**Proposition 1.99.** *If  $X$  is proper over  $S$  and  $Y$  is separated over  $S$ , then every  $S$ -morphism  $f : X \rightarrow Y$  is proper.*

*Proof.* The graph morphism

$$\Gamma_f : X \rightarrow X \times_S Y$$

is a closed immersion because  $Y$  is separated over  $S$ . The projection

$$X \times_S Y \rightarrow Y$$

is proper, since it is the base change of the proper morphism  $X \rightarrow S$ . Thus  $f$  factors as

$$X \xrightarrow{\Gamma_f} X \times_S Y \rightarrow Y,$$

a composition of proper morphisms, and is therefore proper.  $\square$

**Exercise 1.100.** *Show that a proper integral curve  $C$  over an algebraically closed field has  $h^0(C, \mathcal{O}_C) = 1$ .*

**Solution 1.101.** *A regular function on  $C$  gives a morphism to  $\mathbb{A}_k^1$ . Since  $C$  is proper and  $\mathbb{A}_k^1$  is separated, the morphism  $C \rightarrow \mathbb{A}_k^1$  is proper. The image of a proper morphism is closed, so the image of  $C$  in  $\mathbb{A}_k^1$  is a closed subset. Since  $k$  is algebraically closed, the only closed subsets of  $\mathbb{A}_k^1$  are finite sets of points and  $\mathbb{A}_k^1$  itself. The image of an irreducible space under a continuous map is irreducible. Moreover, the image of a proper morphism, equipped with the reduced induced subscheme structure, is proper. This rules out all the possibilities except for a single point. Thus every regular function on  $C$  is constant, so  $h^0(C, \mathcal{O}_C) = 1$ .*

**Remark 1.102.** *Note that properness is a scheme-theoretic property, in particular one has 2 one-point topological spaces  $X = \operatorname{Spec} k$  and  $Y = \operatorname{Spec} K$  where  $K$  is a field not finitely generated as a  $k$ -algebra, for example  $k(t)$  the field of rational functions. The map  $X \rightarrow X$  is proper but the map  $Y \rightarrow X$  is not proper, since it is not of finite type. Both maps are simply the constant map on the underlying topological spaces.*

### 1.2.7 Valuative criterion of properness

The most efficient way to test properness in practice is often via valuation rings.

**Definition 1.103.** *A valuation ring is an integral domain  $R$  with fraction field  $K$  such that for every  $x \in K^\times$ , either  $x \in R$  or  $x^{-1} \in R$ .*

**Theorem 1.104** (Valuative criterion for separatedness). *Let  $f : X \rightarrow Y$  be a morphism of finite type. Then  $f$  is separated if and only if for every valuation ring  $R$  with fraction*

field  $K$ , every commutative diagram

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec} R & \longrightarrow & Y \end{array}$$

admits at most one dotted lift

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & X \\ \downarrow & \nearrow \text{dotted} & \downarrow f \\ \operatorname{Spec} R & \longrightarrow & Y. \end{array}$$

**Theorem 1.105** (Valuative criterion for properness). *Let  $f : X \rightarrow Y$  be a morphism of finite type. Then  $f$  is proper if and only if for every valuation ring  $R$  with fraction field  $K$ , every commutative diagram*

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec} R & \longrightarrow & Y \end{array}$$

admits a unique dotted lift

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & X \\ \downarrow & \nearrow \text{dotted} & \downarrow f \\ \operatorname{Spec} R & \longrightarrow & Y. \end{array}$$

**Example 1.106.** *The valuative criterion shows that affine space  $\mathbb{A}_k^1$  is not proper over  $k$ . In particular, let  $R = k[t]_{(t)}$  be the local ring at 0 and  $K = k(t)$  its fraction field. Then  $K$ -points of  $\mathbb{A}_k^1$  are given by elements of  $K$ , and the morphism  $\operatorname{Spec} K \rightarrow \mathbb{A}_k^1$  given by  $t^{-1}$  does not extend to a morphism  $\operatorname{Spec} R \rightarrow \mathbb{A}_k^1$  because  $t^{-1}$  is not an element of  $R$ .*

More generally, if  $X$  is an affine scheme over  $k$  and the structure morphism  $X \rightarrow \operatorname{Spec} k$  is proper, then  $X$  must be a finite set of points. This is because the coherent pushforward theorem guarantees us that the global sections of  $\mathcal{O}_X$  is a finite-dimensional  $k$ -algebra, and moreover  $X \cong \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$  since  $X$  is affine, so  $X$  is a finite set of points.

**Example 1.107.** *The valuative criterion shows that  $\mathbb{P}_k^n$  is proper over  $k$ : a morphism*

$$\operatorname{Spec} K \rightarrow \mathbb{P}_k^n$$

*is given by homogeneous coordinates  $[x_0 : \cdots : x_n]$  with  $x_i \in K$ , and after scaling we may assume all  $x_i \in R$  and at least one is a unit, giving an extension*

$$\operatorname{Spec} R \rightarrow \mathbb{P}_k^n.$$



### 1.2.8 Flat, smooth, and étale morphisms

These three notions measure how a family varies over the base.

**Definition 1.108.** Let  $A \rightarrow B$  be a ring map. We say that  $B$  is **flat over**  $A$  if the functor

$$- \otimes_A B$$

is exact on the category of  $A$ -modules. Equivalently,  $B$  is flat over  $A$  if  $\mathrm{Tor}_i^A(B, M) = 0$  for all  $i > 0$  and all  $A$ -modules  $M$ .

**Definition 1.109.** A morphism of schemes  $f : X \rightarrow Y$  is flat if for every point  $x \in X$ , the local ring  $\mathcal{O}_{X,x}$  is flat over  $\mathcal{O}_{Y,f(x)}$ .

Flatness says that fibers vary without sudden jumps caused by torsion. It is a subtle but central notion.

**Theorem 1.110** (Local criterion for flatness). Let  $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$  be a local homomorphism of Noetherian local rings, and let  $M$  be a finitely generated  $B$ -module. Then  $M$  is flat over  $A$  if and only if

$$\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0.$$

**Corollary 1.111.** Let  $(R, \mathfrak{m})$  be a Noetherian local ring, and let  $M$  be a finitely generated  $R$ -module. Then  $M$  is flat over  $R$  if and only if  $M$  is free over  $R$ .

*Proof.* Suppose  $M$  is flat over  $R$ . Since  $M$  is finitely generated, choose elements  $x_1, \dots, x_r \in M$  whose images form a  $k = R/\mathfrak{m}$ -basis of  $M/\mathfrak{m}M$ . These define a surjection

$$\varphi : R^r \rightarrow M.$$

Let  $K = \ker \varphi$ , so we have an exact sequence

$$0 \rightarrow K \rightarrow R^r \rightarrow M \rightarrow 0.$$

Tensoring with  $k$  remains exact on the left because  $M$  is flat, giving

$$0 \rightarrow K \otimes_R k \rightarrow k^r \rightarrow M \otimes_R k \rightarrow 0.$$

But the map

$$k^r \rightarrow M \otimes_R k \cong M/\mathfrak{m}M$$

is an isomorphism by the choice of the generators  $x_1, \dots, x_r$ . Hence

$$K/\mathfrak{m}K = 0.$$

By Nakayama's lemma,  $K = 0$ . Therefore  $\varphi$  is an isomorphism, so  $M$  is free.  $\square$

We see that over Noetherian local rings, flatness is equivalent to freeness, which in particular implies torsion-freeness. If moreover, the ring is 1-dimensional, that is a discrete valuation ring, then flatness is equivalent to being torsion-free, and this is precisely detected by the vanishing of  $\text{Tor}_1$  with the residue field.

**Corollary 1.112.** *Let  $R$  be a discrete valuation ring with residue field  $k$ . Then a finitely generated  $R$ -module  $M$  is flat over  $R$  if and only if it is torsion-free.*

*Proof.* If  $M$  is a finitely generated  $R$ -module, then

$$M \cong R^r \oplus \bigoplus_i R/(\pi^{n_i}).$$

The free summand contributes no  $\text{Tor}_1$ . Using the exact sequence

$$0 \rightarrow R \xrightarrow{\pi} R \rightarrow k \rightarrow 0,$$

and tensoring with  $M$ , we get

$$\text{Tor}_1^R(M, k) \cong \ker(M \xrightarrow{\pi} M).$$

Thus

$$\text{Tor}_1^R(M, k) \cong M[\pi] = \{m \in M : \pi m = 0\}.$$

For a torsion summand  $R/(\pi^n)$ , the element  $\pi^{n-1}$  is killed by  $\pi$ , so

$$\text{Tor}_1^R(R/(\pi^n), k) \cong k.$$

Therefore

$$\text{Tor}_1^R(M, k) = 0$$

if and only if  $M$  has no  $\pi$ -torsion. For finitely generated modules over a DVR, this is equivalent to being torsion-free, hence equivalent to being free, and therefore equivalent to being flat.  $\square$

**Example 1.113.** *The localization map of any multiplicative set is flat. This is because localization is an exact functor on modules. Free modules are flat, and being direct summands of free modules, projective modules are flat. This is because given an exact sequence of  $A$ -modules*

$$0 \rightarrow M' \otimes (P \oplus Q) \rightarrow M \otimes (P \oplus Q) \rightarrow M'' \otimes (P \oplus Q) \rightarrow 0$$

*we can split it into two exact sequences because the maps are block diagonal.*

**Example 1.114.** *If  $A$  is a ring, then  $A[x_1, \dots, x_n]$  is flat over  $A$ . This is because they are free, albeit of infinite rank. Flatness is preserved by base change, composition, and localization in the source and target.*

**Exercise 1.115.** *Show that if  $\mathcal{F}$  is a locally free sheaf on a scheme  $X$ , then  $\mathbb{P}\mathcal{F} \rightarrow X$  is flat.*

**Solution 1.116.**  $\mathbb{P}(\mathcal{F})$  is locally on  $X$  just  $\mathbb{P}_X^{r-1}$ , and projective space over a base is flat because it is covered by affine spaces over that base. On each such chart, the morphism to  $U$  is induced by a polynomial ring over  $\mathcal{O}_U$ . Polynomial rings are free, hence flat, over the base ring. Therefore each  $D_+(x_i) \rightarrow U$  is flat.

**Example 1.117.** Open embeddings are flat. This is because they are localizations on the level of rings, and localization is an exact functor on modules.

**Exercise 1.118.** Show that a flat morphism induces an exact functor on quasi-coherent sheaves via pullback.

**Solution 1.119.** Recall that for any  $\mathcal{O}_Y$ -module  $\mathcal{F}$ ,

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

Now  $f^{-1}$  is exact on sheaves of abelian groups. This is because  $(f^{-1}\mathcal{F})_x \cong \mathcal{F}_{f(x)}$  and exactness can be checked on stalks. So if

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

is a short exact sequence of quasi-coherent sheaves on  $Y$ , then

$$0 \rightarrow f^{-1}\mathcal{F}' \rightarrow f^{-1}\mathcal{F} \rightarrow f^{-1}\mathcal{F}'' \rightarrow 0$$

is still exact. So it remains to check that tensoring with  $\mathcal{O}_X$  over  $f^{-1}\mathcal{O}_Y$  preserves exactness. This is exactly flatness, checked on stalks. For any  $x \in X$  with  $y = f(x)$ ,

$$(f^*\mathcal{F})_x \cong \mathcal{F}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}.$$

Since  $f$  is flat,  $\mathcal{O}_{X,x}$  is a flat  $\mathcal{O}_{Y,y}$ -module. Therefore tensoring with  $\mathcal{O}_{X,x}$  is exact, so

$$0 \rightarrow \mathcal{F}'_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \rightarrow \mathcal{F}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \rightarrow \mathcal{F}''_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \rightarrow 0$$

is exact for every  $x \in X$ .

Hence

$$0 \rightarrow f^*\mathcal{F}' \rightarrow f^*\mathcal{F} \rightarrow f^*\mathcal{F}'' \rightarrow 0$$

is exact on stalks, and therefore exact as a sequence of sheaves.

**Example 1.120.** A nonexample of a flat map is a quotient by a nonzerodivisor. For instance, if  $A = k[x]$  and  $B = A/(x)$ , then  $B$  is not flat over  $A$  because  $\text{Tor}_A^1(B, k) \neq 0$ .

To see this, use the free resolution of  $B$ :

$$0 \rightarrow F_1 = A \xrightarrow{x} F_0 = A \rightarrow B \rightarrow 0.$$

Tensoring with  $k = A/(x)$  over  $A$ :

$$0 \rightarrow k \xrightarrow{x} k \rightarrow B \otimes_A k \rightarrow 0.$$

Since  $x = 0$  in  $k$ , the first map is zero:

$$0 \rightarrow k \xrightarrow{0} k \rightarrow B \otimes_A k \rightarrow 0.$$

Taking homology at  $F_1 \otimes_A k$  gives

$$\mathrm{Tor}_A^1(B, k) = \ker(0)/\mathrm{im}(0) = k \neq 0.$$

Thus  $B$  is not flat over  $A$ . Geometrically, the map  $\mathrm{Spec} B \rightarrow \mathrm{Spec} A$  is the inclusion of the origin into the affine line, which has generic fiber  $B \otimes_A k(t) = (A/(t)) \otimes_A k(t) = 0$  but special fiber  $B \otimes_A k = k$ , so the fiber jumps in dimension.

Cohomology commutes with flat base change.

**Theorem 1.121.** *Suppose*

$$\begin{array}{ccc} X' & \xrightarrow{\rho'} & X \\ \pi' \downarrow & & \downarrow \pi \\ Y' & \xrightarrow{\rho} & Y \end{array}$$

*is a Cartesian diagram, and  $\pi$  (and thus  $\pi'$ ) is quasicompact and separated (so higher pushforwards of quasicoherent sheaves by  $\pi$  and  $\pi'$  exist). Suppose also that  $\rho$  is flat, and  $\mathcal{F}$  is a quasicoherent sheaf on  $X$ . Then the natural “push-pull” morphisms*

$$\rho^* R^i \pi_* \mathcal{F} \cong R^i \pi'_* (\rho'^* \mathcal{F})$$

*are isomorphisms. In particular, on the level of fibers, we have  $(R^i \pi_* \mathcal{F}) \otimes \kappa(y) \cong H^i(X_y, \mathcal{F}_y)$  for every  $y \in Y$ .*

*Proof.* The point is that flatness allows us to tensor the Čech complex computing  $R^i \pi_* \mathcal{F}$  with  $\mathcal{O}_{Y'}$  to get a Čech complex computing  $R^i \pi'_* (\rho'^* \mathcal{F})$ , and since flatness preserves exactness, the cohomology of the two complexes are isomorphic.

The result about the cohomology of the fiber would follow by pretending that the inclusion of  $y$  into  $Y$  is flat, which is not true in general. For example the inclusion of the closed point  $0$  into  $\mathbb{A}_k^1$  is not flat. However, the trick is to reduce to a situation where such an inclusion is flat. To do this, we replace  $Y$  by the reduced closure  $Y' = \overline{\{y\}}_{\mathrm{red}}$ . Then  $y$  is the generic point of  $Y'$ . If  $Y' = \mathrm{Spec} A$  is integral affine, then  $k(y) = \mathrm{Frac}(A)$  and  $\mathrm{Spec} k(y) \rightarrow Y'$  corresponds to the localization  $A \rightarrow \mathrm{Frac}(A)$  which is flat.

This does the trick because  $k(y)$  as a sheaf on  $Y$  is supported on  $Y' = \overline{\{y\}}_{\mathrm{red}}$  so after pulling back to  $X$ ,  $f^* k(y)$  is supported on  $X' = X \times_Y Y'$ . Therefore  $\mathcal{F} \otimes f^* k(y)$  is also supported on  $X'$ . Since a sheaf supported on a closed subscheme  $i : X' \hookrightarrow X$  is the same as the pushforward of a sheaf on  $X'$ , we can rewrite

$$\mathcal{F} \otimes f^* k(y) \cong i_* (\mathcal{F}|_{X'} \otimes f'^* k(y))$$

Then  $H^i(X, \mathcal{F} \otimes f^* k(y)) \cong H^i(X', \mathcal{F}|_{X'} \otimes f'^* k(y))$  since the pushforward along a closed immersion is exact. Meanwhile the fiber does not change:

$$X_y = X \times_Y \mathrm{Spec} k(y) \cong X' \times_{Y'} \mathrm{Spec} k(y).$$

So both sides of the claimed isomorphism can just be checked over  $Y'$ .  $\square$

**Definition 1.122.** *An embedded point of a scheme  $X$  is a point  $x \in X$  such that the maximal ideal  $\mathfrak{m}_x$  is an associated prime of 0 in the local ring  $\mathcal{O}_{X,x}$ . Equivalently, if every  $f \in \mathfrak{m}_x$  is a zero divisor in  $\mathcal{O}_{X,x}$ .*

In commutative algebra, an associated prime of an  $A$ -module  $M$  is a prime ideal  $\mathfrak{p}$  such that there exists an element  $m \in M$  with  $\text{Ann}(m) = \mathfrak{p}$ . The existence of such an  $m$  gives a submodule of  $M$  isomorphic to  $A/\mathfrak{p}$ , which can be interpreted geometrically as scheme theoretic data not visible to the underlying topological space. For an  $A$ -module  $M$ , and a submodule  $N \subset M$ , one defines the associated primes of  $N$  in  $M$  by  $\text{Ass}_A(M/N)$ .

In general, every Noetherian ring  $A$  has a finite number of minimal primes, which correspond to the generic points of irreducible components of the spectrum. If  $A$  is reduced, then the minimal primes are exactly the associated primes of 0. In general one only has

$$\text{Min}(A) \subseteq \text{Ass}_A(A).$$

Let us explain why. Let  $\mathfrak{p} \in \text{Min}(A)$  and localize at  $\mathfrak{p}$ . Set

$$R = A_{\mathfrak{p}}, \quad \mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}.$$

Since  $\mathfrak{p}$  is minimal, the local ring  $R$  has only one prime ideal, namely  $\mathfrak{m}$ . Hence

$$\sqrt{(0)} = \mathfrak{m}.$$

Because  $R$  is Noetherian, the nilradical is nilpotent. Therefore  $\mathfrak{m}^N = 0$  for some  $N \gg 0$ . Choose  $N$  minimal with this property. Then  $\mathfrak{m}^{N-1} \neq 0$  and choose a nonzero  $r \in \mathfrak{m}^{N-1}$ . Then  $\mathfrak{m}r \subseteq \mathfrak{m}^N = 0$ . Thus  $\mathfrak{m} \subseteq \text{Ann}_R(r)$ . Since  $r \neq 0$ , the annihilator  $\text{Ann}_R(r)$  is a proper ideal. But  $\mathfrak{m}$  is maximal, so

$$\text{Ann}_R(r) = \mathfrak{m}.$$

Hence

$$\mathfrak{m} \in \text{Ass}_R(R).$$

Using the localization behavior of associated primes,

$$\text{Ass}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}) = \{\mathfrak{q}A_{\mathfrak{p}} : \mathfrak{q} \in \text{Ass}_A(A), \mathfrak{q} \subseteq \mathfrak{p}\},$$

and since  $\mathfrak{p}$  is minimal, the only prime  $\mathfrak{q} \subseteq \mathfrak{p}$  is  $\mathfrak{p}$  itself. Therefore  $\mathfrak{p} \in \text{Ass}_A(A)$ . If  $A$  is Noetherian and reduced, then

$$\text{Ass}_A(A) = \text{Min}(A).$$

The inclusion  $\text{Min}(A) \subseteq \text{Ass}_A(A)$  holds for every Noetherian ring, as above. Reducedness is used to prove the reverse inclusion. Indeed, if  $A$  is reduced, then the zero divisors in  $A$  are exactly the union of the minimal primes:

$$\text{ZDiv}(A) = \bigcup_{\mathfrak{p} \in \text{Min}(A)} \mathfrak{p}.$$

Now suppose  $\mathfrak{q} \in \text{Ass}_A(A)$ . Then  $\mathfrak{q} = \text{Ann}_A(a)$  for some nonzero  $a \in A$ . Every element of  $\mathfrak{q}$  kills  $a$ , so every element of  $\mathfrak{q}$  is a zero divisor. Hence

$$\mathfrak{q} \subseteq \text{ZDiv}(A) = \bigcup_{\mathfrak{p} \in \text{Min}(A)} \mathfrak{p}.$$

By prime avoidance,

$$\mathfrak{q} \subseteq \mathfrak{p}$$

for some minimal prime  $\mathfrak{p}$ . But  $\mathfrak{q}$  is itself a prime containing  $(0)$ , and  $\mathfrak{p}$  is minimal over  $(0)$ , so this forces  $\mathfrak{q} = \mathfrak{p}$ . Therefore every associated prime is minimal, and hence

$$\text{Ass}_A(A) = \text{Min}(A).$$

**Example 1.123.** *Let  $A = k[x, y]/(x^2, xy)$ . The radical of the defining ideal is*

$$\sqrt{(x^2, xy)} = (x),$$

*so the underlying topological space is the line  $V(x)$ . Thus the only minimal prime is  $(x)$ .*

*However, there is also an embedded associated prime at the origin. Indeed, in  $A$ , the element  $x$  is nonzero, but*

$$x^2 = 0, \quad yx = 0.$$

*Therefore  $(x, y)x = 0$  and in fact  $\text{Ann}_A(x) = (x, y)$ . So we have*

$$\text{Ass}_A(A) = \{(x), (x, y)\}.$$

*The prime  $(x)$  corresponds to the visible line, while  $(x, y)$  corresponds to a hidden embedded point at the origin. Thus  $A$  contains an extra copy of the residue field at the origin. This is the embedded point.*

Primary decomposition gives another way to understand associated primes. Let  $A$  be Noetherian and let  $I \subset A$  be an ideal. A primary decomposition of  $I$  is an expression

$$I = Q_1 \cap \cdots \cap Q_r$$

where each  $Q_i$  is  $\mathfrak{p}_i$ -primary, meaning

$$\sqrt{Q_i} = \mathfrak{p}_i.$$

In a minimal primary decomposition, the radicals of the primary components are precisely the associated primes of  $A/I$ :

$$\text{Ass}_A(A/I) = \{\sqrt{Q_1}, \dots, \sqrt{Q_r}\}.$$

Thus primary decomposition decomposes the closed subscheme cut out by  $I$  into primary pieces supported at the associated primes.

The minimal primes over  $I$  correspond to the irreducible components of  $V(I)$ , while the associated primes of  $A/I$  that are not minimal over  $I$  correspond to embedded components.

For the zero ideal, a primary decomposition

$$0 = Q_1 \cap \cdots \cap Q_r$$

displays the associated primes of the whole scheme:

$$\text{Ass}_A(A) = \{\sqrt{Q_1}, \dots, \sqrt{Q_r}\}.$$

Recall that a morphism of schemes  $f : X \rightarrow Y$  is **dominant** if the image of  $f$  is dense in  $Y$ . Equivalently, if the image of  $f$  contains the generic point of  $Y$ . If  $X$  and  $Y$  are both affine schemes, this is the same as saying that the induced map on rings is injective. If  $X$  and  $Y$  are both locally of finite type over a field, then  $f$  sends closed points to closed points. This is because locally,  $f$  is modeled by  $B \rightarrow A \rightarrow A/\mathfrak{m}$  for  $A, B$  finite type  $k$ -algebras, then the image  $B \rightarrow A/\mathfrak{m}$  is an integral domain that is finite-dimensional as a  $k$ -vector space and so the kernel must necessarily be a maximal ideal. So if the generic point of  $Y$  is hit, then it must've necessarily been hit by a nonclosed point.

**Proposition 1.124.** *Let  $f : X \rightarrow Y$  be a morphism of schemes, with  $Y$  integral and regular of dimension 1. Then  $f$  is flat if and only if every associated point  $x \in X$  maps to the generic point of  $Y$ . In particular, if  $X$  is reduced, this says that every irreducible component of  $X$  dominates  $Y$ .*

*Proof.* First suppose that  $f$  is flat, and let  $x \in X$  be a point whose image  $y = f(x)$  is a closed point of  $Y$ . Then  $\mathcal{O}_{Y,y}$  is a discrete valuation ring. Let

$$t \in \mathfrak{m}_y - \mathfrak{m}_y^2$$

be a uniformizing parameter. Then  $t$  is not a zero divisor in  $\mathcal{O}_{Y,y}$ . Since  $f$  is flat,

$$f^\#t \in \mathfrak{m}_x$$

is not a zero divisor in  $\mathcal{O}_{X,x}$ , so  $x$  is not an associated point of  $X$ .

Conversely, suppose that every associated point of  $X$  maps to the generic point of  $Y$ . To show that  $f$  is flat, we must show that for any  $x \in X$ , letting  $y = f(x)$ , the local ring  $\mathcal{O}_{X,x}$  is flat over  $\mathcal{O}_{Y,y}$ . If  $y$  is the generic point, then  $\mathcal{O}_{Y,y}$  is a field, so there is nothing to prove.

If  $y$  is a closed point, then  $\mathcal{O}_{Y,y}$  is a discrete valuation ring, so by the local criterion for flatness over a DVR, we must show that  $\mathcal{O}_{X,x}$  is a torsion-free  $\mathcal{O}_{Y,y}$ -module. If it is not, then  $f^\#t$  must be a zero divisor in  $\mathcal{O}_{X,x}$ , where  $t$  is a uniformizing parameter of  $\mathcal{O}_{Y,y}$ . Therefore  $f^\#t$  is contained in some associated prime ideal  $\mathfrak{p}$  of  $(0)$  in  $\mathcal{O}_{X,x}$ . Then  $\mathfrak{p}$  determines a point  $x' \in X$ , which is an associated point of  $X$ , and whose image by  $f$  is  $y$ , which is a contradiction.

Finally, note that if  $X$  is reduced, its associated points are just the generic points of its irreducible components, so our condition says that each irreducible component of  $X$  dominates  $Y$ .  $\square$

**Example 1.125.** *Let  $Y$  be a curve with a node, and let*

$$f : X \rightarrow Y$$

be the normalization map. Then  $f$  is not flat. Indeed, if  $f$  were flat, then  $f_*\mathcal{O}_X$  would be a flat sheaf of  $\mathcal{O}_Y$ -modules. This is not immediately from the definition of flatness, and we are crucially using the fact that  $f$  is finite here. We elaborate further. Since  $f$  is affine, the sheaf  $f_*\mathcal{O}_X$  corresponds to the  $R$ -module  $S$ . Thus  $f_*\mathcal{O}_X$  is flat over  $Y$  if and only if  $S$  is flat over  $R$ .

Flatness of  $S$  over  $R$  can be checked after localizing at primes of  $R$ , so this is equivalent to requiring

$$S_{\mathfrak{p}} \text{ is flat over } R_{\mathfrak{p}} \quad \text{for all } \mathfrak{p} \in \operatorname{Spec} R.$$

On the other hand,  $f$  is flat if and only if, for every prime  $\mathfrak{q} \in \operatorname{Spec} S$  with  $\mathfrak{p} = \mathfrak{q} \cap R$ , the local ring  $S_{\mathfrak{q}}$  is flat over  $R_{\mathfrak{p}}$ .

Since  $S$  is finite over  $R$ , for each  $\mathfrak{p} \in \operatorname{Spec} R$  there is a finite product decomposition

$$S_{\mathfrak{p}} \cong \prod_{\mathfrak{q}|\mathfrak{p}} S_{\mathfrak{q}},$$

where  $\mathfrak{q}$  ranges over the finitely many primes of  $S$  lying over  $\mathfrak{p}$ . Therefore  $S_{\mathfrak{p}}$  is flat over  $R_{\mathfrak{p}}$  if and only if each factor  $S_{\mathfrak{q}}$  is flat over  $R_{\mathfrak{p}}$ .

Hence

$$S \text{ is flat over } R \iff S_{\mathfrak{q}} \text{ is flat over } R_{\mathfrak{q} \cap R} \text{ for all } \mathfrak{q} \in \operatorname{Spec} S.$$

This is precisely the equivalence between flatness of  $f_*\mathcal{O}_X$  and flatness of  $f$ .

Since  $f$  is finite,  $f_*\mathcal{O}_X$  is coherent. Hence, by the structure theorem for coherent flat sheaves on a regular curve,  $f_*\mathcal{O}_X$  would be locally free. This is a consequence of the fact that a coherent flat module over a regular local ring is free, which can be proved by Nakayama's lemma. Moreover, since  $f$  is generically an isomorphism,  $f_*\mathcal{O}_X$  has rank 1. Therefore  $f_*\mathcal{O}_X$  would be an invertible sheaf on  $Y$ .

But this is impossible at the node. If  $Q \in Y$  is the node, then there are two points

$$P_1, P_2 \in X$$

lying over  $Q$ . Thus the stalk

$$(f_*\mathcal{O}_X)_Q$$

has contributions from both branches:

$$(f_*\mathcal{O}_X)_Q \cong \mathcal{O}_{X,P_1} \oplus \mathcal{O}_{X,P_2}$$

as an  $\mathcal{O}_{Y,Q}$ -module. Consequently it needs two generators at  $Q$ , whereas an invertible sheaf is locally free of rank 1 and hence needs only one generator locally. Therefore  $f_*\mathcal{O}_X$  cannot be locally free of rank 1, and  $f$  cannot be flat.

**Proposition 1.126.** *Let  $Y$  be a regular, integral scheme of dimension 1 and let  $P$  be a closed point of  $Y$ . Let  $X \subset \mathbb{P}_{Y-P}^n$  be a closed subscheme that is flat over  $Y - P$ . Then there exists a unique closed subscheme  $\overline{X} \subset \mathbb{P}_Y^n$  that is flat over  $Y$  and restricts to  $X$  over  $Y - P$ .*



In particular, the Hilbert scheme  $\mathbb{P}_k^n$  whose  $T$ -points are closed subschemes of  $\mathbb{P}_T^n$  is separated and proper over the base  $\text{Spec } k$ .

*Proof.* Let  $U = Y - \{P\}$ . Take  $\overline{X}$  to be the scheme-theoretic closure of  $X \subset \mathbb{P}_U^n$  inside  $\mathbb{P}_Y^n$ . We first show that  $\overline{X}$  is flat over  $Y$ . The key point is that taking the scheme-theoretic closure introduces no new associated points along the boundary over  $P$ . Indeed, affine-locally on  $\mathbb{P}_Y^n$ , write

$$\mathbb{P}_Y^n = \text{Spec } A, \quad \mathbb{P}_U^n = D(t) \subset \text{Spec } A,$$

where  $t$  is a local equation for  $P$  on  $Y$ . If  $X \subset D(t)$  is cut out by an ideal

$$I \subset A_t,$$

then its scheme-theoretic closure is cut out by the contraction

$$J = I \cap A.$$

Thus

$$\overline{X} = \text{Spec}(A/J), \quad X = \text{Spec}(A_t/I).$$

By construction, the natural map

$$A/J \longrightarrow (A/J)_t \cong A_t/I$$

is injective. Equivalently, no nonzero element of  $A/J$  is killed by a power of  $t$ . Hence  $t$  is not contained in any associated prime of  $A/J$ . Therefore every associated point of  $\overline{X}$  lies in the open subset  $D(t) = \mathbb{P}_U^n$ .

Since the restriction of  $\overline{X}$  to  $\mathbb{P}_U^n$  is  $X$ , the associated points of  $\overline{X}$  are exactly the associated points of  $X$ . But  $X$  is flat over  $U$ , so by Proposition 1.124 every associated point of  $X$  maps to the generic point of  $U$ , equivalently to the generic point of  $Y$ . Hence every associated point of  $\overline{X}$  maps to the generic point of  $Y$ . Applying Proposition 1.124 again, we conclude that  $\overline{X} \rightarrow Y$  is flat.

It remains to prove uniqueness. Suppose  $Z \subset \mathbb{P}_Y^n$  is another closed subscheme, flat over  $Y$ , whose restriction to  $\mathbb{P}_U^n$  is  $X$ . Since  $\overline{X}$  is the scheme-theoretic closure of  $X$ , it is the smallest closed subscheme of  $\mathbb{P}_Y^n$  restricting to  $X$  over  $U$ . Therefore

$$\overline{X} \subseteq Z$$

scheme-theoretically.

Again affine-locally, let  $Z$  be cut out by an ideal

$$J' \subset A.$$

The condition that  $Z|_{D(t)} = X$  says

$$J'_t = I.$$

Since

$$J = I \cap A,$$

we have

$$J' \subseteq J.$$

If  $J' \neq J$ , then the quotient

$$J/J' \subset A/J'$$

is a nonzero submodule supported on  $V(t)$ , because

$$(J/J')_t = 0.$$

Thus  $A/J'$  has a nonzero submodule supported over the closed point  $P$ . Consequently  $Z$  has an associated point mapping to  $P$ .

But this contradicts Proposition 1.124, since  $Z$  is flat over  $Y$ , and therefore all associated points of  $Z$  must map to the generic point of  $Y$ . Hence  $J' = J$  locally, so

$$Z = \overline{X}.$$

Therefore  $\overline{X}$  is the unique flat closed subscheme of  $\mathbb{P}_Y^n$  extending  $X$ .  $\square$

**Example 1.127** (A flat degeneration of a twisted cubic). *We now calculate explicitly a flat family in which the general fiber is a twisted cubic in  $\mathbb{P}^3$ , while the special fiber is a nodal plane cubic together with an embedded point.*

*We work affine-locally near the double point. Use coordinates  $x, y$  on  $\mathbb{A}^2$  and coordinates  $x, y, z$  on  $\mathbb{A}^3$ . Let  $X_1 \subset \mathbb{A}^3$  be given parametrically by*

$$x = t^2 - 1, \quad y = t^3 - t, \quad z = t.$$

*Since*

$$t = z, \quad t^2 = x + 1, \quad t^3 = y + z,$$

*this is an affine piece of the twisted cubic.*

*For  $a \neq 0$ , define  $X_a \subset \mathbb{A}^3$  parametrically by*

$$x = t^2 - 1, \quad y = t^3 - t, \quad z = at.$$

*Each nonzero  $a$  gives an affine piece of a twisted cubic since the parameterization is just a reparameterization of the original one. As  $a \rightarrow 0$ , this family is obtained by projecting the twisted cubic onto the  $(x, y)$ -plane. The image is the nodal cubic*

$$y^2 = x^2(x + 1).$$

*To compute the flat limit, we consider the total family inside*

$$\mathbb{A}_a^1 \times \mathbb{A}_{x,y,z}^3.$$

*Eliminating  $t$  from*

$$x = t^2 - 1, \quad y = t^3 - t, \quad z = at,$$

*and taking the scheme-theoretic closure over all of  $\mathbb{A}_a^1$ , one obtains the ideal*

$$I = \left( a^2(x + 1) - z^2, ax(x + 1) - yz, xz - ay, y^2 - x^2(x + 1) \right) \subset k[a, x, y, z].$$

Thus the total space  $\overline{X} \subset \mathbb{A}_a^1 \times \mathbb{A}_{x,y,z}^3$  is defined by

$$\overline{X} = V(I).$$

This is the scheme-theoretic closure of the family over  $\mathbb{A}^1 - \{0\}$ , and it is flat over  $\mathbb{A}^1$ .

Setting  $a = 0$ , we obtain the special fiber  $X_0$ . Its ideal is

$$I_0 = (z^2, yz, xz, y^2 - x^2(x+1)) \subset k[x, y, z].$$

Therefore

$$X_0 = V(z^2, yz, xz, y^2 - x^2(x+1)) \subset \mathbb{A}^3.$$

The support of  $X_0$  is the nodal cubic curve in the plane  $z = 0$ :

$$C = V(z, y^2 - x^2(x+1)).$$

Indeed, taking radicals gives

$$\sqrt{I_0} = (z, y^2 - x^2(x+1)).$$

Thus set-theoretically,  $X_0$  is just the nodal plane cubic.

However,  $X_0$  has extra scheme structure at the node. Away from the node, if  $x \neq 0$ , then the equation

$$xz = 0$$

implies

$$z = 0.$$

Thus  $X_0$  is reduced away from the node. At the node  $(0,0,0)$ , however, the element  $z$  satisfies

$$z^2 = 0, \quad xz = 0, \quad yz = 0.$$

So  $z$  is a nonzero nilpotent element supported at the node. Thus  $X_0$  has an embedded point at the node.

In particular, although the support of  $X_0$  lies inside the plane  $\mathbb{A}^2 = \{z = 0\}$ , the scheme  $X_0$  is not a closed subscheme of that plane. Indeed, a closed subscheme of the plane  $z = 0$  would have ideal containing  $z$ . Equivalently,  $z$  would vanish as a function on its structure sheaf. But in  $\mathcal{O}_{X_0}$ , the element  $z$  is not zero; it is only nilpotent. Thus

$$|X_0| \subset \mathbb{A}^2, \quad \text{but} \quad X_0 \not\subset \mathbb{A}^2$$

as a closed subscheme.

This also explains the Hilbert polynomial. A twisted cubic in  $\mathbb{P}^3$  has Hilbert polynomial

$$P_{\text{twisted}}(m) = 3m + 1.$$

On the other hand, a plane cubic has Hilbert polynomial

$$P_{\text{plane cubic}}(m) = 3m.$$

Since the family is flat, the Hilbert polynomial of the fibers must be constant. Therefore the special fiber cannot be merely the nodal plane cubic. It must contain an additional length-1 contribution. This contribution appears as the embedded point at the node.

Recall that the Hilbert polynomial of a closed subscheme  $X \subset \mathbb{P}_k^n$  is the unique polynomial  $P_X(m) \in \mathbb{Q}[m]$  such that  $P_X(m) = \dim_k H^0(X, \mathcal{O}_X(m))$  for all  $m \gg 0$ . By the vanishing of higher cohomology for large twists, the Hilbert polynomial can also be computed as

$$P_X(m) = \chi(X, \mathcal{O}_X(m))$$

If  $X$  is a curve, then an application of Riemann-Roch gives that the Hilbert polynomial has the form

$$P_X(m) = \deg(X)m + 1 - g_a(X),$$

where  $\deg(X)$  is the degree of  $X$  and  $g_a(X)$  is the arithmetic genus of  $X$ , defined as the quantity  $1 - \chi(X, \mathcal{O}_X)$ .

**Theorem 1.128.** *Let  $T$  be an integral Noetherian scheme, and let  $X \subset \mathbb{P}_T^n$  be a closed subscheme. For each point  $t \in T$ , let  $P_t \in \mathbb{Q}[z]$  be the Hilbert polynomial of the fiber*

$$X_t \subset \mathbb{P}_{\kappa(t)}^n.$$

*Then  $X$  is flat over  $T$  if and only if the Hilbert polynomial  $P_t$  is independent of  $t$ .*

In particular, dimension, degree, and arithmetic genus of the fibers are constant in a flat family.

**Definition 1.129.** *A morphism  $f : X \rightarrow Y$  is **smooth of relative dimension  $n$**  if it is locally of finite type, flat, and for every geometric point  $\bar{y} \rightarrow Y$ , the fiber  $X_{\bar{y}}$  is regular of dimension  $n$ .*

**Definition 1.130.** *A morphism  $f : X \rightarrow Y$  is **formally smooth** if for every affine  $Y$ -scheme  $T$  and every closed subscheme  $T_0 \subset T$  defined by a nilpotent ideal, the natural map*

$$\mathrm{Hom}_Y(T, X) \longrightarrow \mathrm{Hom}_Y(T_0, X)$$

*is surjective.*

**Theorem 1.131** (Smoothness Criterion). *Let  $f : X \rightarrow Y$  be locally of finite presentation. The following are equivalent:*

1.  *$f$  is smooth of relative dimension  $n$ .*
2.  *$f$  is formally smooth of relative dimension  $n$ .*
3.  *$f$  is flat and  $\Omega_{X/Y}$  is locally free of rank  $n$ .*

**Theorem 1.132** (Affine Jacobian Criterion). *Suppose*

$$X = \mathrm{Spec} B = \mathrm{Spec} A[x_1, \dots, x_m]/(f_1, \dots, f_r), \quad Y = \mathrm{Spec} A.$$

*Then  $f$  is smooth of relative dimension  $m - r$  at a point if and only if:*

- the matrix

$$\begin{pmatrix} \frac{\partial f_i}{\partial x_j} \end{pmatrix}$$

has rank  $r$  at that point, and

- $B$  is flat over  $A$ .

Equivalently, some  $r \times r$  minor of the Jacobian matrix is invertible in the local ring.

**Proposition 1.133.** *If  $f : X \rightarrow Y$  is smooth, then all geometric fibers are smooth. In particular, this means that for every geometric point  $\bar{y} \rightarrow Y$ , the fiber  $X_{\bar{y}} = X \times_Y \text{Spec } \kappa(\bar{y})$  is a smooth variety over  $\kappa(\bar{y})$ .*

**Example 1.134.** *The projection*

$$\mathbb{A}_Y^n \rightarrow Y$$

*is smooth of relative dimension  $n$ .*

**Example 1.135.** *The morphism*

$$\text{Spec } k[x]/(x^2) \rightarrow \text{Spec } k$$

*is not smooth because it has a nonreduced fiber, and smooth morphisms have regular fibers, in particular reduced. However, it is flat because every module over a field is flat.*

**Example 1.136.** *Let  $A = k[t]$  and  $B = A[x]/(x^2 - t)$ . As an  $A$ -module, every element of  $B$  can be written uniquely as  $a + bx$  for  $a, b \in A$ . Therefore  $B \cong A \oplus A$  as an  $A$ -module, so  $B$  is flat over  $A$ . However, the fibers are not all isomorphic:*

- *For  $t \neq 0$ : the fiber is  $\text{Spec } k[x]/(x^2 - t)$ , which is reduced (two distinct points if  $k$  is algebraically closed).*
- *For  $t = 0$ : the fiber is  $\text{Spec } k[x]/(x^2)$ , which is nonreduced.*

*Moreover, the morphism  $\text{Spec } B \rightarrow \text{Spec } A$  is not smooth, because the fiber over  $t = 0$  is singular. At the fiber over  $t = 0$  which looks like  $\text{Spec } k[x]/(x^2)$ , the scheme is 0-dimensional but the cotangent space  $\mathfrak{m}/\mathfrak{m}^2$  is 1-dimensional. Thus flatness does not imply reduced fibers or that the fibers are isomorphic, and flatness does not imply smoothness.*

**Example 1.137.** *A smooth morphism need not be étale. For example, the projection  $\mathbb{A}_Y^n \rightarrow Y$  is smooth of relative dimension  $n$ , but it is not étale unless  $n = 0$ .*

**Exercise 1.138.** *Give an example of a morphism of schemes that is unramified morphism but not étale.*

**Solution 1.139.** *We consider a closed immersion which is not flat. The inclusion of the origin into the affine line  $k[t] \rightarrow k[t]/t$  is not flat because tensoring the short exact*

sequence

$$0 \rightarrow k[t] \rightarrow k[t] \rightarrow k[t]/t \rightarrow 0$$

by  $k[t]/t$  gives

$$0 \rightarrow k[t]/t \rightarrow k[t]/t \rightarrow k[t]/t \rightarrow 0$$

which is not exact. The module of relative differentials is zero because if  $D : A/I \rightarrow M$  is an  $A$ -linear derivation, then  $D(\bar{a}) = 0$  must be zero for every  $a \in A$  and so  $D$  is zero.

**Exercise 1.140.** Give an example of a morphism of schemes that is finite but not étale.

**Solution 1.141.** The inclusion of the origin into the affine line  $k[t] \rightarrow k[t]/t$  is finite because  $k[t]/t$  is a finite  $k[t]$ -module, but it is not étale because it is not flat.

**Definition 1.142.** A morphism  $f : X \rightarrow Y$  is **étale** if it is smooth of relative dimension 0.

Equivalently, étale means unramified (i.e.  $\Omega_{X/Y} = 0$ ) and flat of finite presentation. Étale morphisms are the algebro-geometric analogue of local isomorphisms.

**Example 1.143.** If  $k$  is a field, then

$$\mathrm{Spec} k[t, t^{-1}] \rightarrow \mathrm{Spec} k[s, s^{-1}], \quad s \mapsto t^n$$

is étale whenever  $n$  is invertible in  $k$ . It is flat because we can write  $A = k[s, s^{-1}]$  and  $B = A[t]/(t^n - s)$ , so  $B$  is free of rank  $n$  as an  $A$ -module. It is unramified because  $\Omega_{B/A} = Bdt/(nt^{n-1}dt) = 0$  since  $nt^{n-1}$  is invertible in  $B$ .

**Example 1.144.** Recall that a field extension  $L/k$  is separable if every element of  $L$  has minimal polynomial over  $k$  with distinct roots, i.e. separable. This is a purely characteristic  $p$  phenomenon, since in characteristic zero every field extension is separable.  $\mathrm{Spec} L \rightarrow \mathrm{Spec} k$  is étale if and only if  $L$  is a finite separable extension of  $k$ . If  $\mathrm{Spec} L \rightarrow \mathrm{Spec} k$  is étale, then  $L$  is a finite extension of  $k$  because the morphism is of finite type, and  $L$  is separable over  $k$  because  $\Omega_{L/k} = 0$ . Conversely, if  $L$  is a finite separable extension of  $k$ , then  $\mathrm{Spec} L \rightarrow \mathrm{Spec} k$  is étale. This is because by the primitive element theorem, we can write  $L = k[x]/(f)$  where  $f$  is a separable polynomial, and the Jacobian criterion for étaleness says that this morphism is étale if and only if  $f' \in L$  is invertible, which is equivalent to  $f$  being separable. Any repeated root of  $f$  would be a common root of  $f$  and  $f'$ , so  $f'$  is invertible in  $L$  if and only if  $f$  has no repeated roots, i.e.  $f$  is separable.

**Example 1.145.** The map  $\mathrm{Spec} A[x]/(f) \rightarrow \mathrm{Spec} A$  with  $f'$  invertible in  $A$  is étale. There is a standard exact sequence:

$$(f)/(f)^2 \longrightarrow \Omega_{A[x]/A} \otimes B \longrightarrow \Omega_{B/A} \rightarrow 0.$$

Now  $\Omega_{A[x]/A} \cong A[x] \cdot dx$  so tensoring with  $B$  and using the relation  $f = 0$  gives the relation  $df = 0$ . Since  $df = f'(x) dx$ , we obtain

$$\Omega_{B/A} \cong B dx / (f'(x) dx) \cong B / (f'(x)) \cdot dx.$$

Étale requires  $\Omega_{B/A} = 0$  which is equivalent to  $f'(x)$  being invertible in  $B$ .

**Example 1.146.** *Open immersions are étale.*

**Exercise 1.147.** *Show that  $\text{Spec } k \rightarrow \text{Spec } k[t]/(t^2)$  is not an étale morphism. In general, classify the étale morphisms to  $\text{Spec } k$  and to  $\text{Spec } \bar{k}[t]/(t^2)$ .*

**Solution 1.148.** *The morphism*

$$\text{Spec } k \rightarrow \text{Spec } k[t]/(t^2)$$

*corresponds to the quotient map*

$$A := k[t]/(t^2) \rightarrow k = A/(t).$$

*This is not étale because it is not flat: over the local ring  $A$ , the module  $k$  is not free, hence not flat.*

*More generally, an étale morphism to  $\text{Spec } k$  is the same as a finite étale  $k$ -scheme. Being étale, we can conclude that it is dimension zero, so it is a finite disjoint union of discrete points, each with residue field which must be a finite separable extension of  $k$  by étaleness. Hence the morphism is of the form*

$$\text{Spec} \left( \prod_{i=1}^r L_i \right) \rightarrow \text{Spec } k$$

*with each  $L_i/k$  a finite separable field extension.*

*If  $A = \bar{k}[t]/(t^2)$ , then every étale  $A$ -algebra  $B$  satisfies  $B/tB$  is finite étale over  $\bar{k} = A/(t)$  because étaleness is preserved by base change. Therefore*

$$B/tB \cong \bar{k}^r.$$

*Write*

$$\bar{k}^r = \bar{k}e_1 \times \cdots \times \bar{k}e_r,$$

*where the  $e_i$  are the standard orthogonal idempotents:*

$$e_i^2 = e_i, \quad e_i e_j = 0 \ (i \neq j), \quad \sum_i e_i = 1.$$

*Since  $(t) \subset A$  is nilpotent, the ideal  $tB \subset B$  is also nilpotent. Idempotents lift uniquely across nilpotent ideals, so each  $e_i \in B/tB$  lifts to an idempotent  $\tilde{e}_i \in B$ . These lifts remain orthogonal and sum to 1. Therefore*

$$B \cong \prod_{i=1}^r B_i, \quad B_i := \tilde{e}_i B.$$

So we reduce to the case where  $B$  is connected, meaning  $B/tB \cong \bar{k}$ . Assume now  $B$  is connected étale over  $A$ . Because  $B$  is finite étale over  $A$ , it is finite projective over  $A$ . Since  $A$  is local, finite projective  $A$ -modules are free. Thus

$$B \cong A^n$$

as an  $A$ -module for some rank  $n$ .

Reduce modulo  $t$  implies that  $n = 1$ .

$$B/tB \cong (A^n)/t(A^n) \cong (A/t)^n \cong \bar{k}^n$$

So  $B$  is free of rank 1 as an  $A$ -module, hence  $B \cong A$  as an  $A$ -algebra. Therefore the only étale morphisms to  $\text{Spec } \bar{k}[t]/(t^2)$  are finite disjoint unions of copies of  $\text{Spec } A$ , i.e. of the form

$$\bigsqcup_{i=1}^r \text{Spec } A \rightarrow \text{Spec } A.$$

Thus the étale morphisms to  $\text{Spec } \bar{k}[t]/(t^2)$  are exactly the finite disjoint unions

$$\bigsqcup_{i=1}^r \text{Spec } A \rightarrow \text{Spec } A.$$

Note that the same argument shows the same conclusion holds for any local ring whose maximal ideal is nilpotent and whose residue field is algebraically closed. The general result relevant here is for Henselian local rings. Let  $(R, \mathfrak{m}, \kappa)$  be a henselian local ring. The category of finite étale ring extensions  $R \rightarrow S$  is equivalent to the category of finite étale  $\kappa$ -algebras via the functor  $S \mapsto S/\mathfrak{m}S$ .

The connection is that nilpotent is about finite order liftings whereas Henselian is about infinite order liftings. Note that local rings with nilpotent maximal ideal are  $\mathfrak{m}$ -adically complete, and thus Henselian.

This exercise shows that Henselian local rings are “as simply connected as their residue field.” In general we have seen that a field is simply connected in the étale topology if and only if it is separably closed. This is because finite étale covers of  $\text{Spec } k$  are disjoint unions  $\text{Spec } L$  with  $L/k$  a finite separable extension.

**Remark 1.149.** Smooth morphisms are stable under base change and composition, and the same is true for étale morphisms.

**Example 1.150.** An étale cover of a scheme  $Y$  is a finite surjective étale morphism  $X \rightarrow Y$ . An étale cover is said to be trivial if it is a disjoint union of copies of  $Y$ . We will show that  $\mathbb{P}^1$  has no nontrivial étale covers, that is it is simply connected in the étale topology. Indeed, if  $X \rightarrow \mathbb{P}^1$  is a finite surjective étale morphism, then  $X \rightarrow \text{Spec } k$ , being the composition of two maps which are both smooth and proper, is also smooth and proper. Therefore  $X$  is also a smooth proper curve, and the morphism  $X \rightarrow \mathbb{P}^1$  is a nonconstant morphism of smooth proper curves. We may further assume that  $X$  is connected. Since  $f$  is separable (by the étale assumption), an application of Riemann-Hurwitz shows that

$$2g(X) - 2 = \deg(f)(2g(\mathbb{P}^1) - 2)$$



where  $R = 0$  since the morphism is étale and in particular unramified. We see that the left hand side is negative, so  $g(X) = 0$  and  $\deg f = 1$ . Therefore  $X \cong \mathbb{P}^1$  and  $f$  is an isomorphism, so the étale cover is trivial.

Thus finite étale morphisms are simultaneously the algebraic analogue of local diffeomorphisms and of covering spaces.

**Proposition 1.151** (Two interpretations of finite étale covers). *Let  $f : X \rightarrow Y$  be a finite étale morphism of constant degree  $n$ . Then  $f$  is locally a trivial  $n$ -sheeted cover in the étale topology: there exists an étale cover  $U \rightarrow Y$  such that*

$$X \times_Y U \cong \bigsqcup_{i=1}^n U.$$

*Proof.* Consider the projection

$$X \times_Y X \rightarrow X.$$

It has a tautological section, namely the diagonal

$$\Delta : X \rightarrow X \times_Y X.$$

Because  $f$  is unramified, the diagonal is an open immersion. Because  $f$  is finite, hence separated, the diagonal is also a closed immersion. Therefore  $\Delta(X)$  is an open-and-closed subscheme of  $X \times_Y X$ . Thus

$$X \times_Y X \cong X \sqcup X',$$

where  $X' \rightarrow X$  is again finite étale, now of degree  $n - 1$ .  $\square$

**Remark 1.152** (Affine form of the splitting step). *Let  $A \rightarrow B$  be a finite étale  $A$ -algebra of rank  $n$ . After base change along  $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$ , the algebra of the pulled-back cover is  $B \otimes_A B$ . The diagonal section corresponds to the multiplication map  $\mu : B \otimes_A B \rightarrow B$ . Let  $J = \ker(\mu)$ . Since  $A \rightarrow B$  is unramified,*

$$J/J^2 \cong \Omega_{B/A} = 0.$$

*As  $J$  is finitely generated,  $J/J^2 = 0$  implies that  $J$  is generated by an idempotent, i.e.  $J = (1 - e)(B \otimes_A B)$ . This follows from Nakayama's lemma which says that if  $M = IM$  for some ideal  $I$ , then there exists  $x \in I$  such that  $(1 - x)M = 0$ . Equivalently, there is an idempotent  $e \in B \otimes_A B$  such that  $B \cong (B \otimes_A B)_e$  because inverting  $e$  is the same as killing  $1 - e$ , which is the same as killing  $J$ . Thus*

$$\operatorname{Spec} B \cong D(e) \subset \operatorname{Spec}(B \otimes_A B)$$

*is an open-and-closed component.*

## 1.3 Quasicoherent sheaves

### 1.3.1 Definitions and basic properties

**Definition 1.153.** *Let  $X$  be a scheme. An  $\mathcal{O}_X$ -module  $\mathcal{F}$  is **quasi-coherent** if for every affine open subset  $U = \operatorname{Spec} A \subset X$ , there exists an  $A$ -module  $M$  such that*

$$\mathcal{F}|_U \cong \widetilde{M},$$

where  $\widetilde{M}$  is the sheaf associated to  $M$ .

**Definition 1.154.** If  $M$  is an  $A$ -module, the associated sheaf  $\widetilde{M}$  on  $\operatorname{Spec} A$  is defined by

$$\widetilde{M}(D(f)) = M_f.$$

**Proposition 1.155.** Quasi-coherence is a local property: an  $\mathcal{O}_X$ -module  $\mathcal{F}$  is quasi-coherent if and only if there exists an affine open cover  $\{U_i\}$  of  $X$  such that  $\mathcal{F}|_{U_i}$  is quasi-coherent for all  $i$ .

**Proposition 1.156.** The category of quasi-coherent sheaves on a scheme  $X$  is an abelian subcategory of the category of all  $\mathcal{O}_X$ -modules. In particular, it is closed under kernels, cokernels, and extensions.

**Definition 1.157.** A quasi-coherent sheaf  $\mathcal{F}$  on a scheme  $X$  is **coherent** if:

1.  $\mathcal{F}$  is of finite type (i.e. locally finitely generated), and
2. for every open  $U \subset X$  and every morphism  $\mathcal{O}_X^{\oplus n}|_U \rightarrow \mathcal{F}|_U$ , the kernel is of finite type.

**Proposition 1.158.** If  $X = \operatorname{Spec} A$  is affine, then:

- quasi-coherent sheaves correspond to  $A$ -modules,
- coherent sheaves correspond to finitely generated  $A$ -modules (when  $A$  is Noetherian).

**Remark 1.159.** On a Noetherian scheme, a quasi-coherent sheaf is coherent if and only if it is locally finitely generated.

**Example 1.160.** On any scheme  $X$ , the structure sheaf  $\mathcal{O}_X$  is quasi-coherent (and in fact coherent).

**Example 1.161.** If  $X = \operatorname{Spec} A$  is an affine scheme, if  $Y \subseteq X$  is the closed subscheme defined by an ideal  $\mathfrak{a} \subseteq A$  and if  $i : Y \rightarrow X$  is the inclusion morphism, then  $i_*(\mathcal{O}_Y)$  is a quasi-coherent (in fact coherent)  $\mathcal{O}_X$ -module. Indeed, it is isomorphic to  $(A/\mathfrak{a})^\sim$ .

**Example 1.162.** If  $U$  is an open subscheme of a scheme  $X$ , with inclusion map  $j : U \rightarrow X$ , then the sheaf  $j_!(\mathcal{O}_U)$  obtained by extending  $\mathcal{O}_U$  by zero outside of  $U$ , i.e. the sheafification of the presheaf

$$j_!(\mathcal{O}_U)(V) = \begin{cases} \mathcal{O}_U(V) & V \subseteq U, \\ 0 & \text{otherwise} \end{cases}$$

is an  $\mathcal{O}_X$ -module, but it is not in general quasi-coherent.

For example, suppose  $X$  is integral, and  $V = \operatorname{Spec} A$  is any open affine subset of  $X$  not contained in  $U$ . Then  $j_!(\mathcal{O}_U)|_V$  has no global sections over  $V$  because a section in  $j_!(\mathcal{O}_U)(V)$  must live in  $\mathcal{O}_U(V \cap U)$  and vanish near  $V \setminus U$ , meaning that there exists an open  $W \subset V$  with  $V \setminus U \subset W$  so that  $s|_{W \cap U} = 0$ . But  $W \cap U$  is an open subset of  $V \cap U$ , and so  $s$  vanishes on a nonempty open subset of an integral scheme, meaning  $s = 0$ . However,  $j_!(\mathcal{O}_U)|_V$  is not the zero sheaf, since it has nonzero stalks. Note that the stalks of  $j_!(\mathcal{O}_U)$  are the same as the stalks of  $\mathcal{O}_U$ . Hence it cannot be of the form  $\widetilde{M}$  for any  $A$ -module  $M$ .

**Example 1.163.** Building off of the previous example, recall that the closed subschemes of  $X$  are in bijection with the quasi-coherent ideal sheaves of  $\mathcal{O}_X$ . Let

$$X = \mathbb{A}_k^1 = \operatorname{Spec} k[t], \quad U = D(t) = X \setminus \{0\},$$

and let  $j : U \hookrightarrow X$  be the open immersion. Define the sheaf  $\mathcal{I} := j_!\mathcal{O}_U \subseteq \mathcal{O}_X$ . This is an ideal sheaf of  $\mathcal{O}_X$ , but it is not quasi-coherent. Now consider  $\mathcal{O}_X/\mathcal{I}$ . The stalks of  $\mathcal{O}_X/\mathcal{I}$  are 0 if  $x \in U$  and  $\mathcal{O}_{X,0}$  if  $x = 0$ .

Therefore the support is  $Z = \{0\}$ . The resulting closed ringed subspace is  $(\{0\}, \mathcal{O}_{X,0})$ . So the underlying topological space is a single point, but its ring of sections is the local ring  $\mathcal{O}_{X,0} = k[t]_{(t)}$ . A one-point scheme must be the spectrum of a local ring with exactly one prime ideal, hence the spectrum of a field. But our example has a local ring with two prime ideals, the zero ideal and the maximal ideal  $(t)$ . Hence the one-point ringed space  $(\{0\}, k[t]_{(t)})$  is not a scheme.

**Example 1.164.** If  $Y$  is a closed subscheme of a scheme  $X$ , then the sheaf  $\mathcal{O}_X|_Y$  is not in general quasi-coherent on  $Y$ . In fact, it is not even an  $\mathcal{O}_Y$ -module in general.

**Example 1.165.** Let  $X$  be an integral noetherian scheme, and let  $\mathcal{K}$  be the constant sheaf with group  $K$  equal to the function field of  $X$ . Then  $\mathcal{K}$  is a quasi-coherent  $\mathcal{O}_X$ -module, but it is not coherent unless  $X$  is reduced to a point.

**Proposition 1.166.** Let  $X$  be a scheme. The kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent. Any extension of quasi-coherent sheaves is quasi-coherent. If  $X$  is noetherian, the same is true for coherent sheaves.

### 1.3.2 Modules on affine and projective schemes

**Theorem 1.167.** Let  $X = \operatorname{Spec} A$  be an affine scheme. The functor

$$M \longmapsto \widetilde{M}$$

defines an equivalence of categories between the category of  $A$ -modules and the category of quasi-coherent sheaves on  $X$ . Under this equivalence:

- $\Gamma(X, \widetilde{M}) \cong M$ ,
- morphisms of sheaves correspond to module homomorphisms.

- For  $f \in A$ ,

$$\widetilde{M}(D(f)) \cong M_f.$$

**Definition 1.168.** Let  $S$  be a graded ring and let  $M$  be a graded  $S$ -module. We define the sheaf associated to  $M$  on  $\text{Proj } S$ , denoted by  $\widetilde{M}$ , as follows. For each  $\mathfrak{p} \in \text{Proj } S$ , let  $M_{(\mathfrak{p})}$  be the group of elements of degree 0 in the localization  $T^{-1}M$ , where  $T$  is the multiplicative system of homogeneous elements of  $S$  not in  $\mathfrak{p}$ . For any open subset  $U \subseteq \text{Proj } S$  we define  $\widetilde{M}(U)$  to be the set of functions  $s$  from  $U$  to  $\coprod_{\mathfrak{p} \in U} M_{(\mathfrak{p})}$  which are locally fractions. This means that for every  $\mathfrak{p} \in U$ , there is a neighborhood  $V$  of  $\mathfrak{p}$  in  $U$ , and homogeneous elements  $m \in M$  and  $f \in S$  of the same degree, such that for every  $\mathfrak{q} \in V$ , we have  $f \notin \mathfrak{q}$ , and  $s(\mathfrak{q}) = m/f$  in  $M_{(\mathfrak{q})}$ . We make  $\widetilde{M}$  into a sheaf with the obvious restriction maps.

**Proposition 1.169.** Let  $S$  be a graded ring, and  $M$  a graded  $S$ -module. Let  $X = \text{Proj } S$ .

(a) For any  $\mathfrak{p} \in X$ , the stalk  $(\widetilde{M})_{\mathfrak{p}} = M_{(\mathfrak{p})}$ .

(b) For any homogeneous  $f \in S_+$ , we have

$$\widetilde{M}|_{D_+(f)} \cong (M_{(f)})^\sim$$

via the isomorphism of  $D_+(f)$  with  $\text{Spec } S_{(f)}$  where  $M_{(f)}$  denotes the group of elements of degree 0 in the localized module  $M_f$ .

(c)  $\widetilde{M}$  is a quasi-coherent  $\mathcal{O}_X$ -module. If  $S$  is noetherian and  $M$  is finitely generated, then  $\widetilde{M}$  is coherent.

This construction gives a functor from the category of graded  $S$ -modules to the category of quasi-coherent  $\mathcal{O}_X$ -modules. It is not an equivalence of categories, but it becomes one once we restrict to the full subcategory of graded  $S$ -modules which are **saturated**.

**Definition 1.170.** Let  $\mathcal{F}$  be a quasi-coherent sheaf on  $\text{Proj } S$ . Then we define the graded  $S$ -module  $\Gamma_*(\mathcal{F})$  by

$$\Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n)).$$

A module  $M$  is **saturated** if the natural map  $M \rightarrow \Gamma_*(\widetilde{M})$ , which takes  $m \in M_n$  to the global section  $s_m \in \Gamma(X, \widetilde{M}(n))$  which on a standard open  $D_+(f)$ ,  $s_m|_{D_+(f)} = \frac{m}{1} \in (M_f)_0$ , is an isomorphism.

**Theorem 1.171.** The functor  $M \mapsto \widetilde{M}$  defines an equivalence of categories between the category of saturated graded  $S$ -modules and the category of quasi-coherent  $\mathcal{O}_X$ -modules on  $X = \text{Proj } S$ .

It is important to remember that two graded  $S$ -modules can give rise to the same sheaf on  $\text{Proj } S$ . However there is a best representative of the isomorphism class of a sheaf, which is the saturation of any module giving rise to that sheaf. Also note that two ideal

sheaves on  $\text{Proj } S$  cut out the same closed subscheme if and only if they have the same saturation.

**Example 1.172.** *Let*

$$S = k[x_0, x_1], \quad S_+ = (x_0, x_1), \quad M = S/(x_0, x_1).$$

*Then every element of  $M$  is killed by  $S_+$ , so  $M$  is entirely  $S_+$ -torsion. On any standard open  $D_+(f)$  with  $f \in S_+$  homogeneous, we have*

$$\widetilde{M}(D_+(f)) = (M_f)_0 = 0,$$

*because  $f$  acts by 0 on  $M$ , so localizing at  $f$  kills the module. Hence*

$$\widetilde{M} = 0.$$

*Consequently,*

$$\Gamma_*(\widetilde{M}) = 0,$$

*so the saturation of  $M$  is zero. Geometrically,  $M$  is supported at the irrelevant ideal  $S_+$ , which is invisible on  $\text{Proj } S$ .*

**Definition 1.173.** *Let  $S$  be a graded ring, and let  $X = \text{Proj } S$ . For any  $n \in \mathbb{Z}$ , we define the sheaf  $\mathcal{O}_X(n)$  to be  $S(n)^\sim$ . We call  $\mathcal{O}_X(1)$  **the twisting sheaf of Serre**. For any sheaf of  $\mathcal{O}_X$ -modules,  $\mathcal{F}$ , we denote by  $\mathcal{F}(n)$  the twisted sheaf  $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$ .*

**Proposition 1.174.** *Let  $S$  be a graded ring and let  $X = \text{Proj } S$ . Assume that  $S$  is generated by  $S_1$  as an  $S_0$ -algebra. The sheaf  $\mathcal{O}_X(n)$  is locally free of rank 1 for every  $n \in \mathbb{Z}$ .*

We end this section with some special results about sheaves on a projective scheme over a noetherian ring.

**Definition 1.175.** *Let  $X$  be a scheme and let  $\mathcal{F}$  be a sheaf of  $\mathcal{O}_X$ -modules. We say that  $\mathcal{F}$  is **generated by global sections** if there is a family of global sections  $\{s_i\}_{i \in I}$ ,  $s_i \in \Gamma(X, \mathcal{F})$ , such that for each  $x \in X$ , the images of  $s_i$  in the stalk  $\mathcal{F}_x$  generate that stalk as an  $\mathcal{O}_{X,x}$ -module.*

*Note that  $\mathcal{F}$  is generated by global sections if and only if  $\mathcal{F}$  can be written as a quotient of a free sheaf. Indeed, the generating sections  $\{s_i\}_{i \in I}$  define a surjective morphism of sheaves*

$$\bigoplus_{i \in I} \mathcal{O}_X \longrightarrow \mathcal{F},$$

*and conversely.*

**Example 1.176.** *Any quasi-coherent sheaf on an affine scheme is generated by global sections. Indeed, if  $\mathcal{F} = \widetilde{M}$  on  $\text{Spec } A$ , any set of generators for  $M$  as an  $A$ -module will do.*

**Example 1.177.** Let  $X = \text{Proj } S$ , where  $S$  is a graded ring which is generated by  $S_1$  as an  $S_0$ -algebra. Then the elements of  $S_1$  give global sections of  $\mathcal{O}_X(1)$  which generate it.

**Theorem 1.178 (Serre).** Let  $X$  be a projective scheme over a noetherian ring  $A$ , let  $\mathcal{O}_X(1)$  be a very ample invertible sheaf on  $X$ , and let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. Then there is an integer  $n_0$  such that for all  $n \geq n_0$ , the sheaf  $\mathcal{F}(n)$  can be generated by a finite number of global sections.

**Theorem 1.179.** Let  $k$  be a field, let  $A$  be a finitely generated  $k$ -algebra, let  $X$  be a projective scheme over  $A$ , and let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. Then  $\Gamma(X, \mathcal{F})$  is a finitely generated  $A$ -module. In particular, if  $A = k$ ,  $\Gamma(X, \mathcal{F})$  is a finite-dimensional  $k$ -vector space.

### 1.3.3 Operations: tensor product, Hom, pullback, pushforward

**Definition 1.180.** If  $\mathcal{F}, \mathcal{G}$  are  $\mathcal{O}_X$ -modules, their tensor product is

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}.$$

If both are quasi-coherent, then so is the tensor product. The sheaf  $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$  assigns to each open set  $U$  the group

$$\text{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

**Definition 1.181.** Let  $f : X \rightarrow Y$  be a morphism of schemes. The **pullback** of  $\mathcal{F}$  is

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

The **pushforward** is

$$f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U)).$$

**Proposition 1.182.** If  $f : X \rightarrow Y$  is a morphism of schemes, then  $f^*$  preserves quasi-coherence. If  $f$  is quasi-compact and separated, then  $f_*$  preserves quasi-coherence. It is not true in general that  $f_*$  preserves coherence, even if  $X$  and  $Y$  are noetherian. However if  $f$  is proper and  $X$  and  $Y$  are noetherian, then  $f_*$  does preserve coherence.

**Example 1.183.** Quasicoherent pushforward can fail without the quasi-compact and separated hypothesis. Let  $X_i \cong \text{Spec } \mathbb{Z}$  for  $i \in \mathbb{N}$ , and set

$$X = \bigsqcup_{i \in \mathbb{N}} X_i, \quad S = \text{Spec } \mathbb{Z}.$$

Let  $f : X \rightarrow S$  be the morphism induced by the identities on each component. We show that  $f_*\mathcal{O}_X$  is not quasi-coherent.

Recall that for a quasi-coherent sheaf  $\mathcal{F}$  on  $S$ , the natural map

$$m_U : \mathcal{F}(S) \otimes_{\mathcal{O}_S(S)} \mathcal{O}_S(U) \rightarrow \mathcal{F}(U)$$

is an isomorphism for every affine open  $U \subset S$ .

Take  $U = D(2) \subset S$ . Then

$$f_*\mathcal{O}_X(S) = \prod_{i \in \mathbb{N}} \mathbb{Z}, \quad f_*\mathcal{O}_X(U) = \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right].$$

Thus the map becomes

$$\left( \prod_{i \in \mathbb{N}} \mathbb{Z} \right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{1}{2}\right] \longrightarrow \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right].$$

This map is not surjective: elements in the image have a uniform bound on the power of 2 in denominators. This is because  $\mathbb{Z}[1/2]$  is a filtered colimit of free  $\mathbb{Z}$ -modules  $\mathbb{Z}[1/2] = \varinjlim_n 2^{-n}\mathbb{Z}$ , so any element comes from some fixed  $n$ . However elements in the codomain like

$$\left( \frac{1}{2^i} \right)_{i \in \mathbb{N}} \in \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right]$$

do not. Hence  $m_U$  is not surjective, so  $f_*\mathcal{O}_X$  is not quasi-coherent.

**Proposition 1.184.** *The functor on sheaves of abelian groups  $f^{-1}$  is exact. The functor on  $\mathcal{O}_X$ -modules  $f^*$  is exact if  $f$  is flat. The functor  $f_*$  is left exact, but not in general right exact.*

The failure of  $f_*$  to be right exact leads to the need for sheaf cohomology, which measures the failure of exactness of  $f_*$ .

Pushforward and pullback are adjoint in the following sense.

**Proposition 1.185.** *For any  $\mathcal{O}_X$ -module  $\mathcal{F}$  and any  $\mathcal{O}_Y$ -module  $\mathcal{G}$ , there is a natural isomorphism*

$$\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

*Proof.* On stalks we have

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}((f^*\mathcal{G})_x, \mathcal{F}_x) \cong \mathrm{Hom}_{\mathcal{O}_{Y,y}}(\mathcal{G}_y, \mathcal{F}_x).$$

Note that the formula for the stalk of the pushforward is

$$(f_*\mathcal{F})_y \cong \varinjlim_{y \in U} \mathcal{F}(f^{-1}(U))$$

which in general only maps to  $\mathcal{F}_x$  for  $x \in f^{-1}(y)$ , but since the statement is local on  $X$  we can just pass to  $\mathcal{F}_x$ .s Since

$$(f^*\mathcal{G})_x \cong \mathcal{G}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x},$$

this becomes the familiar extension–restriction of scalars adjunction

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}, \mathcal{F}_x) \cong \mathrm{Hom}_{\mathcal{O}_{Y,y}}(\mathcal{G}_y, \mathcal{F}_x)$$

Then the statement becomes the familiar identification

$$\mathrm{Hom}_B(M \otimes_A B, N) \cong \mathrm{Hom}_A(M, N)$$

which is obviously true.  $\square$

### 1.3.4 Locally free sheaves and vector bundles

**Definition 1.186.** An  $\mathcal{O}_X$ -module  $\mathcal{F}$  is **locally free of rank  $r$**  if there exists an open cover  $\{U_i\}$  such that

$$\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus r}.$$

A **vector bundle** on  $X$  is a locally free sheaf of finite rank.

If  $\mathcal{F}$  and  $\mathcal{G}$  are locally free of finite rank, then so is  $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ ,  $\mathcal{F} \otimes \mathcal{G}$ , and the dual  $\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$ .

**Proposition 1.187.** If  $X = \operatorname{Spec} A$ , then locally free sheaves correspond to projective  $A$ -modules.

**Proposition 1.188.** If  $A$  is a local ring, then every projective  $A$ -module is free. Hence locally free sheaves over  $\operatorname{Spec} A$  are free.

**Exercise 1.189.** Let  $f : X \rightarrow Y$  be a morphism of schemes.

- (a) Show by example that if  $\mathcal{F}$  is coherent on  $X$ , then  $f_*\mathcal{F}$  need not be coherent on  $Y$ , even if  $X$  and  $Y$  are varieties over a field  $k$ .
- (b) If  $f$  is a finite morphism of noetherian schemes, and if  $\mathcal{F}$  is coherent on  $X$ , then  $f_*\mathcal{F}$  is coherent on  $Y$ .

**Solution 1.190.**

- (a) Consider  $\operatorname{Spec} k[t] \rightarrow \operatorname{Spec} k$  and the coherent sheaf  $\mathcal{O}_{\operatorname{Spec} k[t]}$  on  $\operatorname{Spec} k[t]$ . The pushforward of  $\mathcal{O}_{\operatorname{Spec} k[t]}$  to  $\operatorname{Spec} k$  is the quasi-coherent sheaf corresponding to the  $k$ -module  $k[t]$ . This is not a finite-dimensional  $k$ -vector space, hence not a coherent sheaf on  $\operatorname{Spec} k$ .
- (b) This is local on  $Y$ , so assume

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B,$$

where  $B$  is a finite  $A$ -algebra. Let  $\mathcal{F}$  be coherent on  $X$ . Since  $X$  is noetherian affine,  $\mathcal{F}$  is of the form

$$\mathcal{F} \cong \widetilde{M}$$

for some finitely generated  $B$ -module  $M$ .

Now for any  $A$ -module  $N$ , the associated quasi-coherent sheaf on  $Y$  is  $\widetilde{N}$ . In the affine case, pushforward along  $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$  sends  $\widetilde{M}$  to  $\widetilde{M}$ , where  $M$  is viewed as an  $A$ -module via  $A \rightarrow B$ . More precisely, for any distinguished open  $D(f) \subseteq \operatorname{Spec} A$ ,

$$f_*\widetilde{M}(D(f)) = \widetilde{M}(\operatorname{Spec} B \times_{\operatorname{Spec} A} D(f)) = \widetilde{M}(\operatorname{Spec} B_f) = M_f.$$



Thus

$$f_*\mathcal{F} \cong \widetilde{M}$$

as a quasi-coherent sheaf on  $Y = \operatorname{Spec} A$ .

It remains to show that  $M$  is finitely generated as an  $A$ -module. Since  $B$  is finite over  $A$ , choose  $b_1, \dots, b_r \in B$  generating  $B$  as an  $A$ -module. Since  $M$  is finitely generated as a  $B$ -module, choose  $m_1, \dots, m_s \in M$  generating  $M$  over  $B$ . Then every element of  $M$  is a  $B$ -linear combination of the  $m_j$ , hence an  $A$ -linear combination of the finitely many elements

$$b_i m_j \quad (1 \leq i \leq r, 1 \leq j \leq s).$$

So  $M$  is finitely generated as an  $A$ -module. Therefore  $f_*\mathcal{F} \cong \widetilde{M}$  is a coherent sheaf on  $Y$ .

One may ask when the pushforward of a vector bundle is a vector bundle. Knowing the morphism is proper is not enough, as the following example shows. The answer is Grauert's theorem.

**Example 1.191.** *Properness alone does not guarantee that the pushforward of a vector bundle is locally free. Let*

$$i : \operatorname{Spec} k \hookrightarrow \mathbb{A}_k^1$$

*be the inclusion of the origin. This is a proper morphism, since every closed immersion is proper. The structure sheaf  $\mathcal{O}_{\operatorname{Spec} k}$  is a vector bundle of rank 1 on  $\operatorname{Spec} k$ . However,  $i_*\mathcal{O}_{\operatorname{Spec} k}$  is the skyscraper sheaf at the origin on  $\mathbb{A}_k^1$ , which is not locally free.*

**Theorem 1.192** (Grauert). *Let  $\pi : X \rightarrow S$  be a proper flat morphism, with  $S$  reduced and locally Noetherian, and let  $\mathcal{F}$  be a coherent sheaf on  $X$  flat over  $S$ , meaning that for every  $x$ , the stalk  $\mathcal{F}_x$  is a flat  $\mathcal{O}_{S, \pi(x)}$ -module via the structure morphism  $\mathcal{O}_{S, \pi(x)} \rightarrow \mathcal{O}_{X, x}$ .*

*Suppose that for some  $p \geq 0$  the function*

$$s \longmapsto h^p(X_s, \mathcal{F}_s)$$

*is locally constant on  $S$ .*

*Then:*

1.  $R^p\pi_*\mathcal{F}$  is locally free on  $S$ ;
2. for every morphism  $f : T \rightarrow S$ , the natural base change map

$$f^*(R^p\pi_*\mathcal{F}) \longrightarrow R^p\pi'_*(\mathcal{F}_T)$$

*is an isomorphism, where  $\pi' : X_T := X \times_S T \rightarrow T$ .*

In particular, every morphism of smooth projective curves over a field is proper and flat, so the pushforward of a vector bundle is always a vector bundle. This could also be seen

by a direct argument. Since the local rings of a smooth curve are DVRs, it suffices to show that  $f_*\mathcal{E}$  is torsion-free. This follows from the constancy of the function

$$y \mapsto h^0(X_y, \mathcal{E}_y),$$

since any torsion would cause the fiber dimension to jump. Hence  $f_*\mathcal{E}$  is locally free. This function is constant because we are in the curve case, so the fibers are zero-dimensional, and the Euler characteristic is constant in flat families. Also we have the vanishing of  $R^1f_*$  because  $f$  is affine.

**Theorem 1.193** (Projection formula). *Let  $f : X \rightarrow Y$  be a morphism of schemes, let  $\mathcal{F}$  be a quasi-coherent sheaf on  $X$ , and let  $\mathcal{E}$  be a locally free sheaf on  $Y$ . Then there is a natural isomorphism*

$$f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*\mathcal{F} \otimes \mathcal{E}.$$

**Example 1.194** (Projection formula on an affine chart). *Let  $A \rightarrow B$  be a ring map, corresponding to a morphism*

$$f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A.$$

*If  $M$  is a  $B$ -module and  $N$  an  $A$ -module, then the projection formula says*

$$f_*(\widetilde{M} \otimes f^*\widetilde{N}) \cong f_*\widetilde{M} \otimes \widetilde{N}.$$

*On global sections this becomes an isomorphism of  $A$ -modules*

$$\operatorname{Res}_A^B(M \otimes_B (B \otimes_A N)) \cong \operatorname{Res}_A^B(M) \otimes_A N$$

*where the left hand side obviously simplifies to  $\operatorname{Res}_A^B(M \otimes_A N)$ . Hence the projection formula is just the statement that restriction of scalars commutes with tensor product.*

The adjunction  $\operatorname{Hom}_B(B \otimes_A N, M) \cong \operatorname{Hom}_A(N, \operatorname{Res}_A^B M)$  is exactly the sheaf adjunction  $\operatorname{Hom}_X(f^*\mathcal{G}, \mathcal{F}) \cong \operatorname{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$ .

## 1.4 Divisors

### 1.4.1 Weil divisors

Let  $X$  be an Noetherian integral separated scheme which is regular in codimension one, meaning that the local ring at every codimension one point is a regular local ring. We need the regularity assumption to ensure that the local ring at every codimension one point is a discrete valuation ring. Recall that the notion of a 1-dimensional regular local ring is precisely a discrete valuation ring.

The most important examples of such schemes are nonsingular varieties over a field and Noetherian normal schemes. Recall that a variety over a field is a separated integral (reduced and irreducible) scheme of finite type. On a nonsingular variety, the local ring at every closed point is regular, hence all local rings are regular, since they are localizations of local rings at closed points. In particular, every  $x \in X$  lies in the closure of some closed point  $y \in X$ . Then  $\mathcal{O}_{X,x} = (\mathcal{O}_{X,y})_{\mathfrak{p}_x}$  and now use the fact that regularity

is preserved by localization. A Noetherian ring is **regular** if every localization at a prime ideal is a regular local ring. A Noetherian local ring  $(R, \mathfrak{m})$  is **regular** if and only if  $\dim R = \dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2$  where  $\dim R$  is the Krull dimension of  $R$ . It is a theorem that a regular local ring is a UFD, so in particular integrally closed.

On a Noetherian normal scheme, any local ring of dimension one is an integrally closed domain, hence a discrete valuation ring, and therefore regular. Recall that a scheme is **normal** if all of its local rings are normal, i.e. integrally closed domains. A fact is that an affine variety  $X$  is normal if and only if its coordinate ring is a normal domain, i.e. a domain integrally closed in its field of fractions. For a Noetherian scheme, if  $x$  has codimension one, then  $\dim \mathcal{O}_{X,x} = 1$  by a standard argument about chains of prime ideals. The key commutative algebra fact is that if  $A$  is a Noetherian local domain of dimension 1 and  $A$  is integrally closed, then  $A$  is a discrete valuation ring. A DVR has principal maximal ideal, so  $\dim_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$  is regular local ring.

Recall that an integral scheme has the very important property that if two rational functions agree on a nonempty open subset, then they are equal. This is the same thing as saying that  $\mathcal{K}_X$  is constant or  $K(U) = K(X)$  for all nonempty  $U$ . Take  $f, g \in K(X)$ . Assume  $f|_U = g|_U$ . Then  $h := f - g$  restricts to 0 on  $U$ . Since  $X$  is integral, every rational function is represented on some nonempty affine open  $\text{Spec } A$  by an element of  $\text{Frac}(A)$ , and if it vanishes on a nonempty open subset of  $\text{Spec } A$ , then it is zero in  $\text{Frac}(A)$ . Indeed, write

$$h = \frac{a}{b} \in \text{Frac}(A), \quad b \neq 0.$$

If  $h$  vanishes on a nonempty distinguished open  $D(s) \subseteq U \cap \text{Spec } A$ , then  $a/1 = 0$  in  $A_s$ , so for some  $n$ ,

$$s^n a = 0.$$

Since  $A$  is a domain and  $s \neq 0$ , this implies  $a = 0$ , hence  $h = 0$ . Thus  $f = g$ .

**Exercise 1.195.** *Can normality be checked on an affine open cover? If  $X$  is a normal scheme and  $\text{Spec } A \subset X$  is an affine open, does it follow that  $A$  is a normal domain?*

**Solution 1.196.** *A scheme  $X$  is normal if and only if all local rings  $\mathcal{O}_{X,x}$  are normal domains. Since this is a local condition on stalks, it can be checked on any open cover, in particular on an affine open cover.*

*If  $\text{Spec } A \subseteq X$  is an open affine subscheme, then  $A$  is a normal ring. However,  $A$  need not be a domain unless  $U$  is irreducible, or unless  $X$  itself is integral.*

*Normality for a scheme usually means all local rings are normal domains, but the scheme need not be irreducible. A disconnected union of normal varieties is normal, and an affine open in it can have a ring that is normal but not a domain.*

**Definition 1.197.** *A **Weil divisor** on  $X$  is a formal finite sum*

$$D = \sum_{Z \subset X} n_Z [Z],$$

where the sum runs over irreducible closed subsets  $Z$  of codimension 1, and  $n_Z \in \mathbb{Z}$ . A Weil divisor  $D$  is **effective** if all  $n_Z \geq 0$ .

**Definition 1.198.** Let  $K(X)$  be the function field of  $X$ . For  $f \in K(X)^\times$ , the **principal divisor** associated to  $f$  is

$$\operatorname{div}(f) = \sum_Z \operatorname{ord}_Z(f) [Z],$$

where  $\operatorname{ord}_Z(f)$  is the valuation at the codimension 1 point corresponding to  $Z$ . Two Weil divisors  $D_1, D_2$  are **linearly equivalent** if

$$D_1 - D_2 = \operatorname{div}(f)$$

for some  $f \in K(X)^\times$ . The **class group**  $\operatorname{Cl}(X)$  is the group of Weil divisors modulo linear equivalence.

**Proposition 1.199.** Let  $A$  be a Noetherian domain. Then  $A$  is a unique factorization domain if and only if  $X = \operatorname{Spec} A$  is normal and  $\operatorname{Cl}(X) = 0$ .

*Proof.* In the forward direction, first we check that if  $A$  is a UFD, then  $A$  is integrally closed. Let  $A$  be a UFD, and let  $K = \operatorname{Frac}(A)$ . Suppose  $x \in K$  is integral over  $A$ . Write

$$x = \frac{a}{b}$$

with  $a, b \in A$  and  $\gcd(a, b) = 1$ . Since  $x$  is integral over  $A$ , it satisfies a monic polynomial

$$x^n + c_{n-1}x^{n-1} + \cdots + c_0 = 0 \quad \text{with } c_i \in A.$$

Multiplying by  $b^n$  gives

$$a^n + c_{n-1}a^{n-1}b + \cdots + c_0b^n = 0,$$

so

$$a^n = -b(c_{n-1}a^{n-1} + \cdots + c_0b^{n-1}).$$

Thus  $b \mid a^n$ . Since  $A$  is a UFD and  $\gcd(a, b) = 1$ , this forces  $b$  to be a unit. Hence  $x \in A$ . Therefore  $A$  is integrally closed.

Assume  $R$  is a UFD and let  $\mathfrak{p}$  be a height 1 prime ideal. Take  $x \in \mathfrak{p}$  nonzero and let  $x = a_1 \cdots a_n$  be a factorization into irreducibles. Since  $\mathfrak{p}$  is prime we see that  $a_i \in \mathfrak{p}$  for some  $i$ . The ideal  $(a_i)$  is prime. Since  $\mathfrak{p}$  has height 1 we conclude that  $(a_i) = \mathfrak{p}$ .

Conversely, assume  $R$  is a domain so that every height 1 prime ideal is principal. Since  $R$  is Noetherian every nonzero nonunit element  $x$  has a factorization into irreducibles. Now recall that if  $R$  is a domain and every nonzero, nonunit factors into irreducibles, then  $R$  is a UFD if and only if every irreducible element is prime. So it suffices to show that every irreducible element is prime. Let  $\mathfrak{p}$  be a prime minimal over  $(x)$ . Then  $\mathfrak{p}$  has height 1. This is because if  $\mathfrak{p}$  is minimal over  $(x)$ , then the only prime ideal of  $R_{\mathfrak{p}}$  containing  $x$  is the maximal ideal  $\mathfrak{p}R_{\mathfrak{p}}$ . This is true because the primes of  $R_{\mathfrak{p}}$  correspond 1-to-1 with the primes of  $R$  contained in  $\mathfrak{p}$ . Then  $\dim(R_{\mathfrak{p}}) = 1$  if  $x$  is not nilpotent in  $R_{\mathfrak{p}}$ .

This is because if  $R$  is a local Noetherian ring of dimension 1, then  $\dim R = 1$  if and only if there exists  $x \in \mathfrak{m}$  such that  $x$  is not nilpotent and  $V(x) = \{\mathfrak{q} \in \operatorname{Spec} R : x \in \mathfrak{q}\}$  is

precisely the closed point  $\{\mathfrak{m}\}$ . Now apply this fact and the minimality of  $\mathfrak{p}$ , along with the fact that  $\dim R_{\mathfrak{p}} = \text{ht}(\mathfrak{p})$ .

So we see that  $\mathfrak{p}$  is height one and therefore by assumption  $\mathfrak{p} = (y)$ . Hence  $x = yz$  and  $z$  is a unit as  $x$  is irreducible. Thus  $(x) = (y)$  and we see that  $x$  is prime.

So what we must show is that if  $A$  is an integrally closed domain, then every prime ideal of height 1 is principal if and only if  $\text{Cl}(\text{Spec } A) = 0$ .

One direction is easy: if every prime ideal of height 1 is principal, consider a prime divisor  $Y \subseteq X = \text{Spec } A$ . Then  $Y$  corresponds to a prime ideal  $\mathfrak{p}$  of height 1. If  $\mathfrak{p}$  is generated by an element  $f \in A$ , then clearly the divisor of  $f$  is

$$(f) = Y.$$

Thus every prime divisor is principal, so  $\text{Cl}(X) = 0$ .

For the converse, suppose  $\text{Cl}(X) = 0$ . Let  $\mathfrak{p}$  be a prime ideal of height 1, and let  $Y$  be the corresponding prime divisor. Then there is an  $f \in K$ , the quotient field of  $A$ , with

$$(f) = Y.$$

We will show that in fact  $f \in A$  and that  $f$  generates  $\mathfrak{p}$ . Since  $v_Y(f) = 1$ , we have  $f \in A_{\mathfrak{p}}$ , and  $f$  generates  $\mathfrak{p}A_{\mathfrak{p}}$ . If  $\mathfrak{p}' \subset A$  is any other prime ideal of height 1, then  $\mathfrak{p}'$  corresponds to a prime divisor  $Y' \subset X$ , and  $v_{Y'}(f) = 0$ , so  $f \in A_{\mathfrak{p}'}$ .

Now the algebraic result below implies that  $f \in A$ . In fact,

$$f \in A \cap \mathfrak{p}A_{\mathfrak{p}} = \mathfrak{p}$$

Now to show that  $f$  generates  $\mathfrak{p}$ , let  $g$  be any other element of  $\mathfrak{p}$ . Then  $v_Y(g) \geq 1$  and  $v_{Y'}(g) \geq 0$  for all  $Y' \neq Y$ . Hence

$$v_{Y'}(g/f) \geq 0$$

for all prime divisors  $Y'$  (including  $Y$ ). Thus

$$g/f \in A_{\mathfrak{p}'}$$

for all  $\mathfrak{p}'$  of height 1, so by the algebraic result again,  $g/f \in A$ . In other words,  $g \in Af$ , which shows that  $\mathfrak{p}$  is a principal ideal, generated by  $f$ .  $\square$

We have the following algebraic result which is often referred to as "algebraic Hartogs' lemma". It is saying that regularity can be checked at the generic points of all prime divisors.

**Theorem 1.200.** *Let  $A$  be an integrally closed Noetherian domain. Then*

$$A = \bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}},$$

*where the intersection is taken over all prime ideals of height 1. Moreover, if  $A$  is normal, then each  $A_{\mathfrak{p}}$  is a discrete valuation ring.*

**Remark 1.201.** *I was originally confused about what this lemma is really saying. If  $A$  is dimension 1, then the statement is completely obvious because every prime ideal is either zero or maximal. This is because if  $\mathfrak{p}$  is a nonzero prime ideal, then  $\mathfrak{p}$  is properly contained in a maximal ideal  $\mathfrak{m}$ , and the chain  $0 \subset \mathfrak{p} \subseteq \mathfrak{m}$  would witness that  $\dim A \geq 2$ , a contradiction. Then in dimension 1, the theorem says that*

$$A = \bigcap_{\mathfrak{p} \text{ of height } 1} A_{\mathfrak{p}} = \bigcap_{\mathfrak{m} \text{ maximal}} A_{\mathfrak{m}}$$

*which is true for all Noetherian domains, not just integrally closed ones. The reason is as follows. Let  $x \in \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$ . Define*

$$I = \{a \in A : ax \in A\}.$$

*This is an ideal of  $A$ , and  $I \neq 0$  because  $x \in K$ , so if  $x = b/c$  with  $b, c \in A$ ,  $c \neq 0$ , then  $c \in I$ .*

*Now suppose  $x \notin A$ . Then  $1 \notin I$ , so  $I$  is a proper ideal. Hence  $I \subseteq \mathfrak{m}$  for some maximal ideal  $\mathfrak{m}$ . But  $x \in A_{\mathfrak{m}}$ , so we can write*

$$x = \frac{u}{s} \quad \text{with } u \in A, \ s \notin \mathfrak{m}.$$

*Then*

$$sx = u \in A,$$

*so by definition of  $I$ ,*

$$s \in I \subseteq \mathfrak{m}.$$

*This contradicts  $s \notin \mathfrak{m}$ . Therefore  $I$  cannot be proper, so  $I = A$ . Hence  $1 \in I$ , which means that  $x \in A$ .*

**Remark 1.202.** *In general, recall that we have the notion of a Dedekind domain, which is a Noetherian integrally closed domain of dimension 1. Let  $C$  be a nonsingular geometrically integral affine algebraic curve over a field  $k$ . Then the coordinate ring  $k[C]$  of regular functions on  $C$  is a Dedekind domain, regardless if  $C$  is smooth.*

*Another example of a Dedekind domain is the ring of integers  $\mathcal{O}_K$  of a number field  $K$ . The prime ideals of  $\mathcal{O}_K$  correspond to the closed points of  $\text{Spec } \mathcal{O}_K$ , because every nonzero prime ideal is maximal. An interesting example here to study is  $\mathbb{Z}[\sqrt{5}]$  which is not normal because  $5 \cong 1 \pmod{4}$  and the element  $(1 + \sqrt{5})/2$  is integral over  $\mathbb{Z}[\sqrt{5}]$  but not in  $\mathbb{Z}[\sqrt{5}]$ . Even though it is not normal, the conclusion of the theorem still holds, because it is a Noetherian domain of dimension 1. Another funny example to study here is  $\mathbb{Z}[\sqrt{-5}]$  which is normal but not a UFD, because the ideal  $(2, 1 + \sqrt{-5})$  is not principal. Here, we have a chain of ideals  $(0) \subset (2) \subset (2, 1 + \sqrt{-5}) \subset \mathcal{O}_K$ , but the point is that  $(2) = (2, 1 + \sqrt{-5})^2$ , is not prime, so the chain does not witness that  $\dim \mathcal{O}_K \geq 2$ .*

**Example 1.203.** *If  $X$  is affine  $n$ -space  $\mathbb{A}_k^n$  over a field  $k$ , then*

$$\text{Cl}(X) = 0.$$

*Indeed,*

$$X = \text{Spec } k[x_1, \dots, x_n],$$

*and the polynomial ring is a UFD.*

**Example 1.204.** If  $A$  is a Dedekind domain, then  $\text{Cl}(\text{Spec } A)$  is just the ideal class group of  $A$ , as defined in algebraic number theory. Thus the first proposition generalizes the fact that  $A$  is a UFD if and only if its ideal class group is 0.

**Proposition 1.205.** Let  $X$  be the projective space  $\mathbb{P}_k^n$  over a field  $k$ . For any divisor

$$D = \sum n_i Y_i,$$

define the degree of  $D$  by

$$\deg D = \sum n_i \deg Y_i,$$

where  $\deg Y_i$  is the degree of the hypersurface  $Y_i$ . Let  $H$  be the hyperplane  $x_0 = 0$ . Then:

- (a) if  $D$  is any divisor of degree  $d$ , then  $D \sim dH$ ;
- (b) for any  $f \in K^\times$ ,  $\deg(f) = 0$ ;
- (c) the degree function gives an isomorphism

$$\deg : \text{Cl}(X) \xrightarrow{\sim} \mathbb{Z}.$$

*Proof.* Let  $S = k[x_0, \dots, x_n]$  be the homogeneous coordinate ring of  $X$ . If  $g$  is a homogeneous element of degree  $d$ , we can factor it into irreducible polynomials  $g = g_1^{n_1} \cdots g_r^{n_r}$ . Then  $g_i$  defines a hypersurface  $Y_i$  of degree  $d_i = \deg g_i$ , and we can define the divisor of  $g$  to be  $(g) = \sum n_i Y_i$ . Then  $\deg(g) = d$ .

Now a rational function  $f$  on  $X$  is a quotient  $g/h$  of homogeneous polynomials of the same degree. Clearly  $(f) = (g) - (h)$ , so we see that  $\deg(f) = 0$ , which proves (b).

If  $D$  is any divisor of degree  $d$ , we can write it as a difference  $D_1 - D_2$  of effective divisors of degrees  $d_1, d_2$  with  $d_1 - d_2 = d$ . Let  $D_1 = (g_1)$  and  $D_2 = (g_2)$ . This is possible because an irreducible hypersurface in  $\mathbb{P}^n$  corresponds to a homogeneous prime ideal of height 1 in  $S$ , which is principal. Taking power products we can get any effective divisor as  $(g)$  for some homogeneous  $g$ . Now  $D - dH = (f)$  where  $f = g_1/x_0^d g_2$  is a rational function on  $X$ . This proves (a). Statement (c) follows from (a), (b), and the fact that  $\deg H = 1$ .  $\square$

**Proposition 1.206.** Let  $X$  satisfy the hypotheses at the beginning of this section, and let  $Z$  be a proper closed subset of  $X$ , and let  $U = X - Z$  be the complement. Then:

- (a) there is a surjective homomorphism

$$\text{Cl } X \rightarrow \text{Cl } U$$

defined by

$$D = \sum n_i Y_i \mapsto \sum n_i (Y_i \cap U),$$

where we ignore those  $Y_i \cap U$  which are empty;

- (b) if  $\text{codim}(Z, X) \geq 2$ , then

$$\text{Cl } X \rightarrow \text{Cl } U$$

is an isomorphism;

(c) if  $Z$  is an irreducible subset of codimension 1, then there is an exact sequence

$$\mathbb{Z} \longrightarrow \mathrm{Cl} X \longrightarrow \mathrm{Cl} U \longrightarrow 0,$$

where the first map is defined by

$$1 \longmapsto 1 \cdot Z.$$

*Proof.* (a) If  $Y$  is a prime divisor on  $X$ , then  $Y \cap U$  is either empty or a prime divisor on  $U$ . If  $f \in K^\times$ , and

$$(f) = \sum n_i Y_i,$$

then considering  $f$  as a rational function on  $U$ , we have

$$(f)_U = \sum n_i (Y_i \cap U),$$

so indeed we have a homomorphism  $\mathrm{Cl} X \rightarrow \mathrm{Cl} U$ . It is surjective because every prime divisor of  $U$  is the restriction of its closure in  $X$ .

(b) The groups  $\mathrm{Div} X$  and  $\mathrm{Cl} X$  depend only on subsets of codimension 1, so removing a closed subset  $Z$  of codimension  $\geq 2$  does not change anything.

(c) The kernel of

$$\mathrm{Cl} X \rightarrow \mathrm{Cl} U$$

consists of divisors whose support is contained in  $Z$ . If  $Z$  is irreducible, the kernel is just the subgroup of  $\mathrm{Cl} X$  generated by  $1 \cdot Z$   $\square$

**Example 1.207.** Let  $Y$  be an irreducible curve of degree  $d$  in  $\mathbb{P}_k^2$ . Then

$$\mathrm{Cl}(\mathbb{P}^2 - Y) = \mathbb{Z}/d\mathbb{Z}.$$

**Example 1.208.** We show that the quadric cone  $X$  in  $\mathbb{A}_k^3$  is normal. In equations we have  $X = \mathrm{Spec} A$  where

$$A = k[x, y, z]/(xy - z^2)$$

and we want to show that  $A$  is integrally closed. Write  $A = k[s^2, t^2, st]$ . The semigroup of exponents of monomials in  $s^2, t^2, st$  is saturated and the integral closure of a monomial subring is again a monomial subring corresponding to the saturation of the semigroup of exponents, so  $A$  is integrally closed.

By the same reasoning, we can show that the cuspidal cubic curve  $Y$  in  $\mathbb{A}_k^2$  defined by  $y^2 - x^3 = 0$  is not normal, because  $k[x, y]/(y^2 - x^3) = k[s^2, s^3]$  is not integrally closed. The integral closure is  $k[s]$ , which corresponds to the normalization map  $\mathbb{A}_k^1 \rightarrow Y$  given by  $s \mapsto (s^2, s^3)$ .

**Example 1.209.** We compute the normalization of the nodal cubic curve  $Y$  in  $\mathbb{A}_k^2$  defined by  $y^2 - x^3 - x^2 = 0$ . This equation plays a key role in the flat degeneration of a family of smooth elliptic curves to a nodal cubic curve, which is a standard example of a flat family with nonreduced special fiber. Set  $t = y/x$  then the equation becomes  $x = t^2 - 1$  and  $y = xt = t^3 - t$ . Thus the normalization of  $Y$  is  $\mathbb{A}_k^1$  with coordinate  $t$ , and the normalization map  $\mathbb{A}_k^1 \rightarrow Y$  is given by  $t \mapsto (t^2 - 1, t^3 - t)$ . This map replaces the node  $(x, y) = (0, 0)$  with two distinct points  $t = 1$  and  $t = -1$ .



**Example 1.210.** Let  $k$  be a field, let  $X$  be the affine quadric cone in  $\mathbb{A}_k^3$ . We will show that

$$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z},$$

and that it is generated by a ruling of the cone, say

$$Y : y = z = 0.$$

Note that  $Y$  is a prime divisor. To see this we check that  $X$  is integral so codimension-1 irreducible implies prime divisor, and that  $Y$  is irreducible closed of codimension 1. First,  $A$  is a domain since  $xy - z^2$  is irreducible in  $k[x, y, z]$ . To see that  $Y$  is irreducible, we compute  $A/\mathfrak{p} \cong k[x]$ , which is a domain, so  $\mathfrak{p}$  is prime and  $Y$  is irreducible. For codimension  $\dim A = 2$  (hypersurface in  $\mathbb{A}^3$ ) and  $\dim A/\mathfrak{p} = 1$ , so  $\mathrm{ht}(\mathfrak{p}) = 1$  and  $Y$  has codimension 1.

We have an exact sequence

$$\mathbb{Z} \longrightarrow \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(X - Y) \longrightarrow 0,$$

where the first map sends  $1 \mapsto 1 \cdot Y$ . Now  $Y$  can be cut out set-theoretically by the function  $y$ . In fact, the divisor of  $y$  is

$$\mathrm{div}(y) = 2 \cdot Y,$$

because  $z$  generates the maximal ideal of the local ring at the generic point of  $Y$ . This is because  $x \notin \mathfrak{p}$  so  $x$  becomes invertible in  $A_{\mathfrak{p}}$  and so  $y = z^2/x$  implies  $\mathfrak{p}A_{\mathfrak{p}} = (y, z)A_{\mathfrak{p}} = zA_{\mathfrak{p}}$ , so  $z$  is a uniformizer at the generic point, and  $y$  has valuation 2 at the generic point. Since  $Y$  is set theoretically cut out by  $y$ , we have

$$X - Y = D(y) = \mathrm{Spec} A_y.$$

Now

$$A_y = k[x, y, y^{-1}, z]/(xy - z^2).$$

In this ring,

$$x = y^{-1}z^2,$$

so we can eliminate  $x$ , and find

$$A_y \cong k[y, y^{-1}, z].$$

This is a UFD and so  $\mathrm{Cl}(X - Y) = 0$ . Thus we see that  $\mathrm{Cl}(X)$  is generated by  $Y$ , and that

$$2 \cdot Y = 0.$$

It remains to show that  $Y$  itself is not a principal divisor. Since  $A$  is integrally closed, it is equivalent to show that the prime ideal of  $Y$ , namely

$$\mathfrak{p} = (y, z),$$

is not principal. Here we are using the following fact to establish this equivalence. Let  $A$  be a Noetherian normal domain, and let  $\mathfrak{p}$  be a height-1 prime. Then

$$Y = V(\mathfrak{p})$$

is a prime divisor. The claim is  $Y$  is a principal Weil divisor if and only if  $\mathfrak{p}$  is a principal ideal. If  $\mathfrak{p} = (f)$  for some  $f \in A$ , then  $\text{ord}_{\mathfrak{p}}(f) = 1$  and  $\text{ord}_{\mathfrak{p}'}(f) = 0$  for all other height-1 primes  $\mathfrak{p}'$ , so  $\text{div} f = Y$ . Conversely, if  $Y$  is a principal Weil divisor, then  $Y = (f)$  for some  $f \in K(X)^\times$  and  $\text{ord}_{\mathfrak{p}}(f) = 1$  and  $\text{ord}_{\mathfrak{p}'}(f) = 0$  for all other height-1 primes  $\mathfrak{p}'$ , so  $f \in A_{\mathfrak{p}}$  and  $f$  generates  $\mathfrak{p}A_{\mathfrak{p}}$ . Also  $f$  is a unit in  $A_{\mathfrak{p}'}$  for all other height-1 primes  $\mathfrak{p}'$ .

Let  $I = (f) \subset A$ . Then for every height-1 prime  $\mathfrak{q}$ :

$$IA_{\mathfrak{q}} = \begin{cases} \mathfrak{p}A_{\mathfrak{p}} & \text{if } \mathfrak{q} = \mathfrak{p}, \\ A_{\mathfrak{q}} & \text{if } \mathfrak{q} \neq \mathfrak{p}. \end{cases}$$

But  $\mathfrak{p}$  has the same property:

$$\mathfrak{p}A_{\mathfrak{q}} = \begin{cases} \mathfrak{p}A_{\mathfrak{p}} & \text{if } \mathfrak{q} = \mathfrak{p}, \\ A_{\mathfrak{q}} & \text{if } \mathfrak{q} \neq \mathfrak{p}. \end{cases}$$

In a normal domain,

$$I = \bigcap_{\text{ht}(\mathfrak{q})=1} IA_{\mathfrak{q}}.$$

So:

$$(f) = \bigcap IA_{\mathfrak{q}} = \bigcap \mathfrak{p}A_{\mathfrak{q}} = \mathfrak{p}.$$

Let

$$\mathfrak{m} = (x, y, z),$$

and note that  $\mathfrak{m}/\mathfrak{m}^2$  is a 3-dimensional vector space over  $k$  generated by  $\bar{x}, \bar{y}, \bar{z}$ , the images of  $x, y, z$ . Now  $\mathfrak{p} \subseteq \mathfrak{m}$ , and the image of  $\mathfrak{p}$  in  $\mathfrak{m}/\mathfrak{m}^2$  contains  $\bar{y}$  and  $\bar{z}$ . Hence  $\mathfrak{p}$  cannot be a principal ideal.

**Example 1.211.** We compute  $\text{Cl}(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z} \oplus \mathbb{Z}$  where the isomorphism is given by pulling back the hyperplane class from each factor. Let  $X = \mathbb{P}_k^1 \times \mathbb{P}_k^1$ . Write  $H_1 = \{\infty\} \times \mathbb{P}^1$  and  $H_2 = \mathbb{P}^1 \times \{\infty\}$ . Then  $U := X \setminus (H_1 \cup H_2) \cong \mathbb{A}^1 \times \mathbb{A}^1$ . Since  $k[x, y]$  is a UFD, we have  $\text{Cl}(U) = 0$ .

Now use the standard exact sequence for removing codimension-1 components: if  $Z \subset X$  is a union of irreducible codimension-1 closed subsets, then

$$\bigoplus_i \mathbb{Z}[Z_i] \longrightarrow \text{Cl}(X) \longrightarrow \text{Cl}(X \setminus Z) \longrightarrow 0$$

Applying this to  $Z = H_1 \cup H_2$  gives  $\mathbb{Z}[H_1] \oplus \mathbb{Z}[H_2] \longrightarrow \text{Cl}(X) \longrightarrow 0$ , so  $\text{Cl}(X)$  is generated by  $H_1$  and  $H_2$ . As for the injectivity, suppose that  $aH_1 + bH_2$  is principal. Then since  $X$  is smooth, we can identify  $\text{Cl}(X)$  with the Picard group  $\text{Pic}(X)$ , and the line bundle associated to  $aH_1 + bH_2$  is  $\mathcal{O}_X(aH_1 + bH_2)$ . Then  $\mathcal{O}_X(aH_1 + bH_2)$  is trivial, and pulling back along the inclusion  $\mathbb{P}^1 \times * \hookrightarrow X$  gives a trivial line bundle on  $\mathbb{P}^1$ . Now recall that if  $D$  is a Cartier divisor on any scheme  $X$  and  $C \subset X$  is a closed subscheme not contained in  $\text{Supp}(D)$ , then

$$\mathcal{O}_X(D)|_C \cong \mathcal{O}_C(D|_C),$$

where  $D|_C$  is the induced Cartier divisor on  $C$ . The defining equation of  $aH_1$ , when restricted to  $\mathbb{P}^1 \times *$ , vanishes to order  $a$  at the point  $\infty$ . Therefore,  $\mathcal{O}_{\mathbb{P}^1}(aH_1|_{\mathbb{P}^1})$  is trivial which implies  $a = 0$ . Similarly, pulling back along  $* \times \mathbb{P}^1$  gives  $b = 0$ . Thus  $H_1$  and  $H_2$  are linearly independent in  $\text{Cl}(X)$ , so  $\text{Cl}(X) \cong \mathbb{Z} \oplus \mathbb{Z}$ .

**Example 1.212.** Consider a quadric surface  $X$  in  $\mathbb{P}^3$  defined by a homogeneous degree 2 polynomial. We can identify this as the locus of singular matrices in  $\mathbb{P}^3$  where we think of  $\mathbb{P}^3$  as the projectivization of the space of  $2 \times 2$  matrices. Then  $X$  is isomorphic to  $\mathbb{P}^1 \times \mathbb{P}^1$  since every singular matrix has a unique 1-dimensional kernel and a unique 1-dimensional cokernel. The hyperplane class  $\mathcal{O}_{\mathbb{P}^3}(1)$  restricts to  $\mathcal{O}_X(1, 1)$  under this isomorphism, because the restriction to  $\mathbb{P}^1 \times *$  is  $\mathcal{O}_{\mathbb{P}^1}(1)$ . This is saying that the composition  $\mathbb{P}^1 \hookrightarrow X \hookrightarrow \mathbb{P}^3$  is a linear embedding. Equivalently, we have the Segre embedding  $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$  given by  $([x_0 : x_1], [y_0 : y_1]) \mapsto [x_0y_0 : x_0y_1 : x_1y_0 : x_1y_1]$  which corresponds to the choice of line bundle  $\mathcal{O}_{\mathbb{P}^1}(1) \boxtimes \mathcal{O}_{\mathbb{P}^1}(1)$  on  $\mathbb{P}^1 \times \mathbb{P}^1$ . This by definition is the pullback of  $\mathcal{O}_{\mathbb{P}^3}(1)$  under the Segre embedding.

### 1.4.2 Cartier divisors

**Definition 1.213.** Let  $X$  be a scheme. A **Cartier divisor** is given by an open cover  $\{U_i\}$  together with functions  $f_i \in \mathcal{O}_X(U_i)^\times$  such that on overlaps

$$f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Equivalently, a Cartier divisor is a global section of the sheaf of groups  $K_X^\times/\mathcal{O}_X^\times$ , where  $K_X$  is the sheaf of total quotient rings of  $\mathcal{O}_X$ . A Cartier divisor is **principal** if it is given globally by a single rational function  $f \in K(X)^\times$ .

**Definition 1.214.** A Cartier divisor is **effective** if it can be represented by functions  $f_i$  that are regular and nonvanishing on  $U_i$ . In that case we define the associated subscheme of codimension 1,  $Y$ , to be the closed subscheme defined by the ideal sheaf  $\mathcal{I}$  which is locally generated by  $f_i$ .

This gives a correspondence between effective Cartier divisors and locally principal closed subschemes. The ideal sheaf  $\mathcal{I}$  of the corresponding closed subscheme is  $\mathcal{O}_X(-D)$  because we are thinking about functions which vanish along  $D$ .

### 1.4.3 Relation between Weil and Cartier divisors

**Proposition 1.215.** Let  $X$  be an integral, separated noetherian scheme, all of whose local rings are unique factorization domains (in which case we say  $X$  is **locally factorial**). Then the group  $\text{Div } X$  of Weil divisors on  $X$  is isomorphic to the group of Cartier divisors  $\Gamma(X, \mathcal{K}^*/\mathcal{O}^*)$ , and furthermore, the principal Weil divisors correspond to the principal Cartier divisors under this isomorphism.

*Proof.* First note that  $X$  is normal, hence satisfies the conditions at the beginning of this section, since a UFD is integrally closed. Thus it makes sense to talk about Weil divisors. Since  $X$  is integral, the sheaf  $\mathcal{K}$  is just the constant sheaf corresponding to the function

field  $K$  of  $X$ . This is because the sheaf  $\mathcal{K}$  is the sheaf which on an open set  $U$  invert the nonzerodivisors in  $\mathcal{O}_X(U)$ , and since  $X$  is integral, the nonzerodivisors are precisely the nonzero elements of  $K$ , so  $\mathcal{K}(U) = K$  for every nonempty open set  $U$ . Note that on an integral scheme, if two rational functions agree on a nonempty open set, then they are equal.

Now let a Cartier divisor be given by  $\{(U_i, f_i)\}$  where  $\{U_i\}$  is an open cover of  $X$ , and  $f_i \in \Gamma(U_i, \mathcal{K}^*) = K^*$ . We define the associated Weil divisor as follows. For each prime divisor  $Y$ , take the coefficient of  $Y$  to be  $\text{ord}_Y(f_i)$ , where  $i$  is any index for which  $Y \cap U_i \neq \emptyset$ . If  $j$  is another such index, then  $f_i/f_j$  is invertible on  $U_i \cap U_j$ , so  $\text{ord}_Y(f_i/f_j) = 0$  and  $\text{ord}_Y(f_i) = \text{ord}_Y(f_j)$ . Thus we obtain a well-defined Weil divisor

$$D = \sum \text{ord}_Y(f_i) Y$$

on  $X$ . The sum is finite because  $X$  is noetherian.

Conversely, if  $D$  is a Weil divisor on  $X$ , let  $x \in X$  be any point. Then  $D$  induces a Weil divisor  $D_x$  on the local scheme  $\text{Spec } \mathcal{O}_{X,x}$ . Since  $\mathcal{O}_{X,x}$  is a UFD,  $D_x$  is a principal divisor, so let  $D_x = (f_x)$  for some  $f_x \in K$ . Now the principal divisor  $(f_x)$  on  $X$  has the same restriction to  $\text{Spec } \mathcal{O}_{X,x}$  as  $D$ , hence they differ only at prime divisors which do not pass through  $x$ . There are only finitely many of these which have a nonzero coefficient in  $D$  or  $(f_x)$ , so there is an open neighborhood  $U_x$  of  $x$  such that  $D$  and  $(f_x)$  have the same restriction to  $U_x$ . Covering  $X$  with such open sets  $U_x$ , the functions  $f_x$  give a Cartier divisor on  $X$ .

Note that if  $f, f'$  give the same Weil divisor on an open set  $U$ , then  $f/f'$  has order zero at every prime divisor intersecting  $U$ , so  $f/f'$  is a unit in  $\mathcal{O}_{X,\mathfrak{p}}$  for every  $\mathfrak{p}$  of height 1 with  $\mathfrak{p} \cap U \neq \emptyset$ . Since  $X$  is normal, we have  $\mathcal{O}_X(U) = \bigcap_{\substack{\mathfrak{p} \in U \\ \text{ht}(\mathfrak{p})=1}} \mathcal{O}_{X,\mathfrak{p}} \subset K(X)$  and so we can conclude that  $f/f' \in \mathcal{O}_X(U)^\times$ . Thus the Cartier divisor we constructed from  $D$  is well-defined. These two constructions are inverse to each other, so we see that the groups of Weil divisors and Cartier divisors are isomorphic. Furthermore it is clear that the principal divisors correspond to each other.  $\square$

**Remark 1.216.** *Since a regular local ring is a UFD, this proposition applies in particular to any regular integral separated noetherian scheme. A scheme is regular if all of its local rings are regular local rings.*

**Remark 1.217.** *If  $X$  is a normal scheme, which is not necessarily locally factorial, we can define a subgroup of  $\text{Div } X$  consisting of the locally principal Weil divisors:  $D$  is locally principal if  $X$  can be covered by open sets  $U$  such that  $D|_U$  is principal for each  $U$ . Then the above proof shows that the Cartier divisors are the same as the locally principal Weil divisors.*

**Example 1.218.** *Let  $X$  be the affine quadric cone*

$$\text{Spec } k[x, y, z]/(xy - z^2)$$

*treated above. The ruling  $Y$  is a Weil divisor which is not locally principal in the neighborhood of the vertex of the cone. Indeed, our earlier proof shows that its prime ideal*

$\mathfrak{p}A_{\mathfrak{m}}$  is not a principal ideal even in the local ring  $A_{\mathfrak{m}}$ . Thus  $Y$  does not correspond to a Cartier divisor.

On the other hand  $2Y$  is locally principal, and in fact principal. So in this case the group of Cartier divisors modulo principal divisors is 0, whereas

$$\mathrm{Cl} X \cong \mathbb{Z}/2\mathbb{Z}.$$

#### 1.4.4 Divisors and line bundles

**Definition 1.219.** To a Cartier divisor  $D$ , one associates an invertible sheaf  $\mathcal{O}_X(D)$ , defined by gluing the trivial line bundles on  $U_i$  via the transition functions  $f_i/f_j$  on overlaps. Explicitly, the sections of  $\mathcal{O}_X(D)$  over an open set  $U$  are given by

$$\mathcal{O}_X(D)(U) = \{s \in K(X) : \mathrm{div}(s|_{U_i \cap U}) + D|_{U_i \cap U} \geq 0 \text{ for all } i\}.$$

Note that if  $D$  is a principal Cartier divisor given by a global rational function  $f$ , then  $\mathcal{O}_X(D) \cong \mathcal{O}_X$  via the isomorphism sending a section  $s$  to  $fs$ .

**Proposition 1.220.** On any scheme  $X$ , the association  $D \mapsto \mathcal{O}_X(D)$  gives an injective homomorphism from the group of Cartier divisors modulo principal divisors to the Picard group  $\mathrm{Pic}(X)$  of isomorphism classes of line bundles on  $X$ . If  $X$  is integral, then this map is an isomorphism.

*Proof.* The map  $D \mapsto \mathcal{O}_X(D)$  maps into the invertible subsheaves of  $\mathcal{K}$ . Thus we have only to show that every invertible sheaf is isomorphic to a subsheaf of  $\mathcal{K}$ , which in this case is the constant sheaf  $K$ , where  $K$  is the function field of  $X$ .

So let  $\mathcal{L}$  be any invertible sheaf, and consider the sheaf

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{K}.$$

On any open set  $U$  where  $\mathcal{L} \cong \mathcal{O}_X$ , we have

$$\mathcal{L} \otimes \mathcal{K} \cong \mathcal{K},$$

so it is a constant sheaf on  $U$ . Now because  $X$  is irreducible, it follows that any sheaf whose restriction to each open set of a covering of  $X$  is constant is in fact a constant sheaf. Thus  $\mathcal{L} \otimes \mathcal{K}$  is isomorphic to the constant sheaf  $\mathcal{K}$ , and the natural map

$$\mathcal{L} \rightarrow \mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

expresses  $\mathcal{L}$  as a subsheaf of  $\mathcal{K}$ .  $\square$

More precisely, from  $D$  one gets an invertible sheaf  $\mathcal{O}(D)$  along with a meromorphic section  $s_D$  such that  $(s_D) = D$ . Effective Cartier divisors  $D$  are precisely those for which the meromorphic section  $s_D$  is regular.

One can think of  $s_D$  as the constant function 1. In particular, recall that  $\mathcal{O}(D)$  is locally isomorphic to  $\mathcal{O}_R$  by picking local defining equations  $\eta_\alpha$  for  $D$  on an open cover  $U_\alpha$ .

Recall that on a smooth variety there is an equivalence between Cartier divisors and Weil divisors. Then the isomorphism

$$\mathcal{O}(D)|_{U_\alpha} \rightarrow \mathcal{O}_R|_{U_\alpha}$$

is given by multiplication by  $\eta_\alpha$ . Then the canonical meromorphic section  $s_D$ , when restricted to  $U_\alpha$ , is given by  $\eta_\alpha$  which has divisor  $D|_{U_\alpha}$ .

Therefore, there is an isomorphism of abelian groups

$\text{Cl}(R) \rightarrow$  subgroup of  $\text{Pic}(R)$  those invertible sheaves admitting meromorphic sections

$$D \mapsto (\mathcal{O}(D), 1)$$

and the RHS is all of  $\text{Pic}(R)$  whenever every invertible sheaf admits a meromorphic section, which is the case for example when  $R$  is integral.

**Remark 1.221.** *Note that in general, we have flat pullback and proper pushforward for prime Weil divisors, and they are compatible with linear equivalence and intersection products, so we get flat pullback and proper pushforward for Weil divisor classes. We also have a pullback for Cartier divisors for any morphism, defined by precomposing with the morphism on the local equations. This is compatible with linear equivalence, for the reason that if  $D - D' = (f)$  and  $g$  is a morphism, then  $g^*D - g^*D' = (f \circ g)$ .*

*However, in general we do not have a pushforward for Cartier divisors, because the pushforward of a locally principal subscheme need not be locally principal.*

**Example 1.222.** *An effective Cartier divisor  $D$  on an affine scheme  $\text{Spec } A$  need not be principal. Let  $A$  be a Dedekind domain with nontrivial class group, e.g.,*

$$A = \mathbb{Z}[\sqrt{-5}].$$

*Let  $\mathfrak{p} = (2, 1 + \sqrt{-5})$ . Then*

- $\mathfrak{p}$  is a non-principal prime ideal.
- But  $\mathfrak{p}$  is invertible (all nonzero ideals in a Dedekind domain are invertible).

*Then  $D$  is an effective Cartier divisor because  $\mathfrak{p}$  is invertible and hence locally principal. If  $I$  is an invertible ideal then,  $I \cdot J = A$  for some ideal  $J$ , so at any prime  $\mathfrak{p}$  of  $A$ ,  $I_{\mathfrak{p}} \cdot J_{\mathfrak{p}} = A_{\mathfrak{p}}$ , so  $I_{\mathfrak{p}}$  is finitely generated projective of rank 1 over the local ring  $A_{\mathfrak{p}}$ . Projective modules over local rings are free and so therefore  $I_{\mathfrak{p}}$  is generated by a single element, so  $I$  is locally principal.*

*However  $D$  is not principal because  $\mathfrak{p}$  is not principal (note that on an affine scheme, principal Cartier divisors correspond precisely to principal ideals).*

**Example 1.223.** *Let  $X$  be the cuspidal cubic curve  $y^2z = x^3$  in  $\mathbb{P}_k^2$ , with  $\text{char } k \neq 2$ . In this case  $X$  is not regular in codimension 1, namely it has a singular point at  $Z = (0, 0, 1)$  which is not a regular point of  $X$ . On the affine chart  $z = 1$ , the cuspidal cubic is*

$$X = \text{Spec } A, \quad A = k[x, y]/(y^2 - x^3),$$

and the cusp is the maximal ideal  $\mathfrak{m} = (x, y)$ . The local ring  $A_{\mathfrak{m}}$  is not regular because  $\dim A_{\mathfrak{m}} = 1$  but  $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$  since  $\mathfrak{m} = (x, y)$  and  $\mathfrak{m}^2 = (x^2, xy, y^2)$ , so the classes of  $x$  and  $y$  are linearly independent modulo  $\mathfrak{m}^2$ .

Hence  $A_{\mathfrak{m}}$  is not regular and therefore not a DVR, since DVRs are precisely the regular local rings of dimension 1. Note that  $A$  is also not integrally closed: the element  $t = y/x \in \text{Frac}(A)$  satisfies  $t^2 = x \in A$  so it is integral over  $A$ , but  $t \notin A$ . Indeed,  $A \cong k[t^2, t^3] \subset k(t)$  and  $t \notin k[t^2, t^3]$ .

Thus we cannot talk about Weil divisors on  $X$ . However, we can talk about the group  $\text{CaCl } X$  of Cartier divisor classes modulo principal divisors. We will show:

(a) there is a surjective degree homomorphism

$$\deg : \text{CaCl } X \rightarrow \mathbb{Z};$$

(b) there is a one-to-one correspondence between the set of nonsingular closed points of  $X$  and the kernel of the degree map, which makes it into a group variety; and in fact

(c) there is a natural isomorphism of group varieties between  $\text{CaCl}^0 X$  and the additive group  $G_a$  of the field  $k$ .

To define the degree of a Cartier divisor on  $X$ , note that any Cartier divisor is linearly equivalent to one whose local function is invertible in some neighborhood of the singular point  $Z = (0, 0, 1)$ . To see this, consider an open neighborhood  $V$  of  $Z$  on which the divisor is represented by a single function  $g \in K(X)^\times$ . Now subtract the principal divisor  $(g)$ . On  $V$ , the new local equation becomes 1, which is invertible. Then this Cartier divisor corresponds to a Weil divisor

$$D = \sum n_i P_i$$

on  $X - Z$ , and we define the degree of the original divisor to be

$$\deg D = \sum n_i.$$

Suppose  $D$  and  $D'$  are linearly equivalent Cartier divisors, and both are represented by local equations invertible near  $Z$ . Then

$$D' - D = (f)$$

for some  $f \in K(X)^\times$  and since  $D$  and  $D'$  are both represented by local equations invertible near  $Z$ , we have that  $f$  is invertible at  $Z$ . If  $f \in K$  is invertible at  $Z$ , then the principal divisor  $(f)$  on  $X - Z$  has degree 0. In particular,  $f$  has neither zero nor pole at  $Z$  and so the difference  $D' - D$  is in fact supported on  $U$ , so it is enough to consider  $D'|_U - D|_U = \text{div}(f|_U)$ , which we argue has degree 0.

Let  $\nu : \widetilde{X} \rightarrow X$  be the normalization. For the cuspidal cubic,

$$\widetilde{X} \cong \mathbb{P}^1.$$

Also,  $\nu$  is an isomorphism over  $U = X - \{Z\}$ . So the closed points of  $U$  correspond exactly to the points of  $\widetilde{X}$  away from the point lying over  $Z$ .

Now pull  $f$  back to  $\widetilde{X}$ . Since  $\widetilde{X}$  is a complete nonsingular curve,

$$\deg \operatorname{div}_{\widetilde{X}}(\nu^* f) = 0.$$

Because  $f$  is invertible at  $Z$ , the pullback  $\nu^* f$  has no zero or pole at the point(s) of  $\widetilde{X}$  over  $Z$ . So all zeros and poles of  $\nu^* f$  occur over  $U$ . Hence

$$\deg(f)|_U = 0.$$

Thus the degree of a Cartier divisor on  $X$  is well-defined, and it passes to linear equivalence classes to give a surjective homomorphism

$$\deg : \operatorname{CaCl} X \rightarrow \mathbb{Z}.$$

Now let  $P_0$  be the point  $(0, 1, 0)$  as in the case of the nonsingular cubic curve. To each closed point  $P \in X - Z$ , we associate the Cartier divisor  $D_P$  which is 1 in a neighborhood of  $Z$ , and which corresponds to the Weil divisor  $P - P_0$  on  $X - Z$ .

First note this map is injective: if  $P \neq Q$  are two points in  $X - Z$ , and if  $D_P \sim D_Q$ , then there is an  $f \in K^*$ , which is invertible at  $Z$ , and such that

$$(f) = P - Q \quad \text{on } X - Z.$$

Then  $f$  gives a morphism of  $X$  to  $\mathbb{P}^1$ , which must be birational because it is degree 1. But then we would have a local map of local rings  $\mathcal{O}_{\mathbb{P}^1, 0} \rightarrow \mathcal{O}_{X, Z}$  which is dominant. Recall that we say a map of local rings  $A \rightarrow B$  is dominant if it is local, injective, and  $A$  and  $B$  have the same fraction field. The injectivity and fraction field condition follow immediately from the birationality. The following proposition lemma shows that the local map is in fact an isomorphism, which is a contradiction since  $\mathcal{O}_{\mathbb{P}^1, 0}$  is regular and  $\mathcal{O}_{X, Z}$  is not.

**Proposition 1.224.** *Let  $A$  be a DVR with fraction field  $K$ . If  $B$  is a local domain with fraction field  $K$  that dominates  $A$ , then  $B = A$ .*

*Proof.* Let  $v : K^\times \rightarrow \mathbb{Z}$  be the valuation of  $A$ . Then

$$A = \{x \in K : v(x) \geq 0\}.$$

If  $B$  is local and dominates  $A$ , then for any  $x \in B$ , if  $v(x) < 0$  then  $x^{-1} \in A \subset B$ , so  $x$  would be a unit of  $B$ , contradicting  $v(x) < 0$ . Hence  $v(x) \geq 0$  for all  $x \in B$ , so  $B \subset A$ . Since  $A \subset B$ , we get  $A = B$ .  $\square$

For the moment suppose that  $X$  is a nonsingular cubic curve. Then the kernel of the degree map is isomorphic to the group of closed points of  $X$  under the group law defined by the chord-tangent construction. Let  $D \in \operatorname{Cl} X$  be in the kernel of the degree map.



Then we may write

$$D = \sum_i n_i (P_i - P_0)$$

for a fixed base point  $P_0 \in X$ . For any point  $R \in X$ , let the line  $P_0R$  meet  $X$  in a third point  $T$ , counted with multiplicity (if  $R = P_0$ , we take the tangent line at  $P_0$ , so  $T = P_0$ ).

Then we have

$$P_0 + R + T \sim 3P_0,$$

so

$$R - P_0 \sim -(T - P_0).$$

If  $n_i < 0$ , take  $P_i = R$ . Replacing  $P_i$  by  $T$ , we obtain a linearly equivalent divisor in which the coefficient becomes  $-n_i > 0$ . Repeating this process, we may assume

$$D = \sum_i n_i (P_i - P_0), \quad n_i > 0 \text{ for all } i.$$

We prove by induction on  $\sum_i n_i$  that  $D \sim P - P_0$  for some point  $P \in X$ .

If  $\sum_i n_i = 1$ , there is nothing to prove.

Now suppose  $\sum_i n_i \geq 2$ . Choose two points  $P, Q$  among the  $P_i$  (possibly equal). Let the line  $PQ$  meet  $X$  in a third point  $R$ , and let the line  $P_0R$  meet  $X$  in a third point  $T$ .

Then

$$P + Q + R \sim 3P_0, \quad P_0 + R + T \sim 3P_0,$$

so subtracting,

$$(P - P_0) + (Q - P_0) \sim (T - P_0).$$

Thus we may replace  $P$  and  $Q$  by  $T$ , obtaining a divisor linearly equivalent to  $D$  with total coefficient  $\sum_i n_i - 1$ . By induction,

$$D \sim P - P_0$$

for some point  $P$ .

The same is true for the cuspidal cubic curve, except that the group law is now given by the additive group structure on  $\mathbb{A}^1$ .

To show that every divisor in  $\text{CaCl}^0 X$  is linearly equivalent to  $D_P$  for some closed point  $P \in X - Z$ , we proceed exactly as in the case of the nonsingular cubic curve. The only difference is to note that the geometric constructions described there remain inside  $X - Z$ . Thus the group  $\text{CaCl}^0 X$  is in one-to-one correspondence with the set of closed points of  $X - Z$ , making it into a group variety.

In this case we are able to identify the group variety as  $G_a$ . Indeed,  $X$  is a rational curve, and

$$X - Z \cong \mathbb{A}_k^1.$$

More precisely, define a morphism

$$G_a = \text{Spec } k[t] \rightarrow X - Z$$

by

$$t \mapsto (t, 1, t^3).$$

This is clearly an isomorphism of varieties and one can show that it is compatible with the group structures. Thus we obtain an isomorphism of group varieties

$$\mathrm{CaCl}^0 X \cong G_a.$$

**Example 1.225.** Let  $X$  be the nonsingular cubic curve

$$y^2z = x^3 - xz^2 \subset \mathbb{P}_k^2,$$

Let  $\mathcal{L} = \mathcal{O}(P_0)$  be the invertible sheaf associated to a point  $P_0 \in X$ .

Then  $\mathcal{L}$  is ample, because

$$\mathcal{L}(3P_0) \cong \mathcal{O}_X(1),$$

and  $\mathcal{O}_X(1)$  is very ample. On the other hand,  $\mathcal{L}$  is not very ample, because  $\mathcal{L}(P_0)$  is not generated by global sections. Indeed, if it were generated by global sections, then there is a section  $s$  of  $\mathcal{L}(P_0)$  which does not vanish at  $P_0$ . Then the divisor of  $s$  must equal  $Q - P_0$  for some point  $Q \in X$ , so  $P_0$  would be linearly equivalent to some other point  $Q \in X$ , which is impossible since  $X$  is not rational. Thus an ample line bundle need not be very ample.

**Example 1.226.** If  $D$  is a divisor on a complete nonsingular curve  $X$ , then  $\mathcal{O}(D)$  is ample if and only if  $\deg D > 0$ . This is a consequence of the Riemann–Roch theorem.

**Proposition 1.227.** Let  $Y$  be a scheme and let  $E$  be a vector bundle on  $Y$ . Let  $\mathbf{P}(E) = \mathrm{Proj}(\mathrm{Sym}^\bullet E^\vee)$  be the projectivization of a vector bundle  $E$  on a scheme  $Y$  parameterizing lines in the fibers of  $E$ . Then there is a natural isomorphism of abelian groups

$$\mathrm{Pic}(\mathbf{P}(E)) \cong \mathrm{Pic}(Y) \oplus \mathbf{Z}[\mathcal{O}_{\mathbf{P}(E)}(1)]$$

where  $\mathcal{O}_{\mathbf{P}(E)}(1)$  is the quotient line bundle on  $\mathbf{P}(E)$ , the dual of the tautological line bundle.

*Proof.* Define

$$\Phi : \mathrm{Pic}(Y) \oplus \mathbf{Z} \rightarrow \mathrm{Pic}(\mathbf{P}(E)), \quad (\mathcal{L}, n) \mapsto \pi^* \mathcal{L} \otimes \mathcal{O}_{\mathbf{P}(E)}(n).$$

To prove surjectivity, let  $\mathcal{M} \in \mathrm{Pic}(\mathbf{P}(E))$ . For each  $y \in Y$ , the restriction

$$\mathcal{M}|_{\mathbf{P}(E_y)}$$

is a line bundle on  $\mathbf{P}^r$ , hence isomorphic to  $\mathcal{O}_{\mathbf{P}^r}(n)$  for a unique integer  $n$ . This integer is locally constant in  $y$ , so after restricting to a connected component of  $Y$ , it is constant. Replacing  $\mathcal{M}$  by  $\mathcal{M}(-n)$ , we may assume

$$\mathcal{M}|_{\mathbf{P}(E_y)} \cong \mathcal{O}_{\mathbf{P}(E_y)}$$

for all  $y$ . Then the seesaw theorem implies that  $\mathcal{M}$  is pulled back from  $Y$ .

For injectivity, if

$$\pi^* \mathcal{L} \otimes \mathcal{O}(n) \cong \pi^* \mathcal{L}' \otimes \mathcal{O}(n'),$$

then restricting to any fiber  $\mathbf{P}^r$  gives  $\mathcal{O}_{\mathbf{P}^r}(n) \cong \mathcal{O}_{\mathbf{P}^r}(n')$ , hence  $n = n'$ . Cancelling  $\mathcal{O}(n)$  and pushing forward then gives  $\mathcal{L} \cong \mathcal{L}'$ . Therefore  $\Phi$  is an isomorphism.  $\square$

**Exercise 1.228.** Consider the compactification of the line bundle  $\mathcal{O}(n)$  over  $\mathbb{P}^1$ . What is its Picard group? What are its generators? What is the difference between the zero section and the "section at infinity" of this bundle?

**Solution 1.229.** Let

$$X = \mathbf{P}(\mathcal{O}_{\mathbf{P}^1} \oplus \mathcal{O}_{\mathbf{P}^1}(n))$$

with projection

$$\pi : X \rightarrow \mathbf{P}^1.$$

This is the Hirzebruch surface  $F_n$ , i.e. the compactification of the total space of  $\mathcal{O}_{\mathbf{P}^1}(n)$  by adding the section at infinity. We will show that

$$\text{Pic}(X) \cong \mathbf{Z} \oplus \mathbf{Z},$$

generated by a fiber  $f$  of  $\pi$  and by either the infinity section  $s_\infty$  or the zero section  $s_0$ . Note that all fibers are linearly equivalent because on  $\mathbf{P}^1$  all points are linearly equivalent, and pulli back by  $\pi$  gives  $\pi^*([p]) \sim \pi^*([q])$ .

The two distinguished sections differ by

$$s_0 \sim s_\infty + nf$$

and can be distinguished by their self-intersection numbers

$$s_0^2 = n, \quad s_\infty^2 = -n.$$

We are thinking about sections as morphisms

$$s : \mathbf{P}^1 \rightarrow F_n$$

such that  $\pi \circ s = \text{id}_{\mathbf{P}^1}$ . Equivalently, in the projective bundle

$$F_n = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(n)),$$

sections correspond to line subbundles of  $\mathcal{O} \oplus \mathcal{O}(n)$ .

The two distinguished sections  $s_0$  and  $s_\infty$  come from the two natural inclusions of line bundles into  $\mathcal{O} \oplus \mathcal{O}(n)$ . These are the zero section and the section at infinity. There are, in general, many other sections. On the open subset identified with the total space of  $\mathcal{O}(n)$ , any global section

$$\sigma \in H^0(\mathbf{P}^1, \mathcal{O}(n))$$

defines a section of  $\pi$  whose image is disjoint from  $s_\infty$ ; geometrically, this is the graph of  $\sigma$ . Thus the "finite sections" are precisely the sections landing in the total-space chart, and for  $n \geq 0$  they are parametrized by the vector space  $H^0(\mathbf{P}^1, \mathcal{O}(n))$ .

All such finite sections are linearly equivalent to the zero section  $s_0$ . By contrast,  $s_\infty$  is the unique section of self-intersection  $-n$  when  $n > 0$ , which is why it is distinguished. A general section of  $F_n \rightarrow \mathbf{P}^1$  need not stay in the finite part and may meet the section at infinity. In fact, the intersection product tells you how many times it meets the section at infinity:

$$s \cdot s_\infty = (s_\infty + af) \cdot s_\infty = -n + a$$

Finally we check that the self intersections of the two distinguished sections are as claimed, by taking the degree of the normal bundle. Given a line subbundle  $L \hookrightarrow E$  corresponding to a section  $s$  and its normal bundle is

$$N_{s/X} \cong \text{Hom}(L, E/L)$$

Thus

$$N_{s_0/X} \cong \text{Hom}(\mathcal{O}, \mathcal{O}(n)) \cong \mathcal{O}(n), \quad N_{s_\infty/X} \cong \text{Hom}(\mathcal{O}(n), \mathcal{O}) \cong \mathcal{O}(-n),$$

so

$$s_0^2 = n, \quad s_\infty^2 = -n.$$

Consequently, if  $f$  denotes the class of a fiber, then

$$s_0 \sim s_\infty + nf.$$

## 1.5 Line bundles and maps to projective space

Let  $A$  be a ring and consider projective space

$$\mathbb{P}_A^n = \text{Proj } A[x_0, \dots, x_n].$$

On  $\mathbb{P}_A^n$  we have the invertible sheaf  $\mathcal{O}(1)$ , and the homogeneous coordinates

$$x_0, \dots, x_n \in \Gamma(\mathbb{P}_A^n, \mathcal{O}(1))$$

are global sections. These sections generate  $\mathcal{O}(1)$ , i.e. for each point  $P \in \mathbb{P}_A^n$ , their images generate the stalk  $\mathcal{O}(1)_P$  as a module over  $\mathcal{O}_P$ .

Now let  $X$  be a scheme over  $A$ , and let  $\varphi : X \rightarrow \mathbb{P}_A^n$  be an  $A$ -morphism. Then  $\mathcal{L} := \varphi^*\mathcal{O}(1)$  is an invertible sheaf on  $X$ , and the global sections

$$s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L}), \quad i = 0, \dots, n,$$

generate  $\mathcal{L}$ . Generation of a line bundle by sections is preserved under pullback because it is a stalkwise statement and tensoring preserves generators. Conversely,  $\mathcal{L}$  together with the sections  $s_i$  determine  $\varphi$ .

**Theorem 1.230.** *Let  $A$  be a ring and let  $X$  be a scheme over  $A$ .*

- (a) *If  $\varphi : X \rightarrow \mathbb{P}_A^n$  is an  $A$ -morphism, then  $\varphi^*\mathcal{O}(1)$  is an invertible sheaf on  $X$ , generated by the global sections  $s_i = \varphi^*(x_i)$ ,  $i = 0, \dots, n$*

(b) Conversely, let  $\mathcal{L}$  be an invertible sheaf on  $X$ , and suppose  $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$  are global sections which generate  $\mathcal{L}$ . Then there exists a unique  $A$ -morphism  $\varphi : X \rightarrow \mathbb{P}_A^n$  so that  $\mathcal{L} \cong \varphi^* \mathcal{O}(1)$  and  $s_i = \varphi^*(x_i)$  for all  $i$ .

*Proof.* Part (a) is immediate from the discussion above. For (b), suppose given  $\mathcal{L}$  and generating sections  $s_0, \dots, s_n$ . For each  $i$ , define

$$X_i := \{ P \in X \mid (s_i)_P \notin \mathfrak{m}_P \mathcal{L}_P \}.$$

Then each  $X_i$  is open, and since the  $s_i$  generate  $\mathcal{L}$ , the sets  $X_i$  cover  $X$ .

Recall that the standard open subset

$$U_i = \{x_i \neq 0\} \subset \mathbb{P}_A^n$$

is isomorphic to

$$U_i \cong \operatorname{Spec} A[y_0, \dots, \hat{y}_i, \dots, y_n], \quad \text{where } y_j = x_j/x_i.$$

We define a morphism  $X_i \rightarrow U_i$  by giving a ring homomorphism

$$A[y_0, \dots, \hat{y}_i, \dots, y_n] \longrightarrow \Gamma(X_i, \mathcal{O}_X),$$

sending  $y_j \mapsto s_j/s_i$ . This is well-defined because for  $P \in X_i$ , the section  $s_i$  is a unit in  $\mathcal{L}_P$ , so  $s_j/s_i$  is a well-defined section of  $\mathcal{O}_X$  on  $X_i$ . These morphisms  $X_i \rightarrow U_i$  agree on overlaps and hence glue to a morphism

$$\varphi : X \rightarrow \mathbb{P}_A^n.$$

By construction,  $\mathcal{L} \cong \varphi^* \mathcal{O}(1)$  and  $s_i = \varphi^*(x_i)$ . Uniqueness is clear from the construction.  $\square$

In general, one can test very ampleness of a line bundle by checking whether or not the line bundle separates points and tangent vectors.

**Proposition 1.231.** *Let  $k$  be an algebraically closed field, let  $X$  be a projective scheme over  $k$ , and let  $\varphi : X \rightarrow \mathbb{P}_k^n$  be a morphism corresponding to an invertible sheaf  $\mathcal{L}$  and sections*

$$s_0, \dots, s_n \in \Gamma(X, \mathcal{L}).$$

*Let*

$$V := \operatorname{span}_k \{s_0, \dots, s_n\} \subseteq \Gamma(X, \mathcal{L}).$$

*Then  $\varphi$  is a closed immersion if and only if:*

(1) *For any two distinct closed points  $P, Q \in X$ , there exists  $s \in V$  such that*

$$s_P \in \mathfrak{m}_P \mathcal{L}_P \quad \text{and} \quad s_Q \notin \mathfrak{m}_Q \mathcal{L}_Q,$$

*or vice versa.*

(2) *For each closed point  $P \in X$ , the set  $\{s \in V \mid s_P \in \mathfrak{m}_P \mathcal{L}_P\}$  spans the  $k$ -vector space  $\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P$ .*

**Example 1.232.** Let  $X = \mathbb{P}_k^1$  and  $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1)$ . Then  $\Gamma(X, \mathcal{L}) = \text{span}\{x_0, x_1\}$ . This gives the identity map  $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ . Let  $P \neq Q \in \mathbb{P}^1$ . Then at least one of  $x_0, x_1$  vanishes at  $P$  but not at  $Q$ , since homogeneous coordinates distinguish points. Hence  $V$  separates points. Fix  $P \in \mathbb{P}^1$ . Choose affine coordinate  $t = x_1/x_0$  near  $P$  (assuming  $x_0(P) \neq 0$ ). Then  $\mathcal{O}(1)|_{D_+(x_0)} \cong \mathcal{O}_{\mathbb{A}^1}$ , and sections correspond to linear polynomials in  $t$ .

The sections vanishing at  $P$  are multiples of  $(t - t(P))$ , and these generate  $\mathfrak{m}_P/\mathfrak{m}_P^2$ , which is 1-dimensional. Hence tangent vectors are separated. Therefore  $\mathcal{O}_{\mathbb{P}^1}(1)$  is very ample.

**Definition 1.233.** Let  $X$  be a scheme and  $\mathcal{L}$  an invertible sheaf. Let  $V \subseteq \Gamma(X, \mathcal{L})$  be a finite-dimensional  $A$ -submodule. The **base locus** of  $V$  is

$$\text{Bs}(V) := \{x \in X \mid s_x \in \mathfrak{m}_x \mathcal{L}_x \text{ for all } s \in V\}.$$

Equivalently, it is the set of points where all sections in  $V$  vanish.

**Definition 1.234.** If  $D$  is a Cartier divisor on  $X$ , the **complete linear system** associated to  $D$  is

$$|D| := \mathbb{P}(\Gamma(X, \mathcal{O}_X(D))).$$

Points of  $|D|$  correspond to effective divisors linearly equivalent to  $D$ . A nonzero section  $s \in \Gamma(X, \mathcal{O}_X(D))$  defines an effective divisor  $(s) + D$ .

**Exercise 1.235.** Give some examples of nonconstant maps  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^2$ . Do there exist any embeddings  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^2$ ? Do there exist nonconstant maps  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^n$ ? Do there exist any embeddings  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$ ?

**Solution 1.236.** A nonconstant map  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^2$  is given by three global sections of an invertible sheaf on  $\mathbb{P}^1 \times \mathbb{P}^1$ . For example, we can take the sections  $x_0y_1, x_1y_0, x_0y_0 + x_1y_1$  of the invertible sheaf  $\mathcal{O}(1, 1)$ . One can quickly check that these three sections have no common zero.

There cannot be any embeddings  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^2$ , because  $\mathbb{P}^1 \times \mathbb{P}^1$  is a surface and since  $\mathbb{P}^2$  is an irreducible surface, the image of  $\mathbb{P}^1 \times \mathbb{P}^1$  would have to be all of  $\mathbb{P}^2$ , which is impossible since they have different Picard groups, for instance.

A nonconstant map  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^n$  can be given by projecting onto one of the factors and then embedding that factor into  $\mathbb{P}^n$ . For example, we can take the sections  $x_0, x_1$  of  $\mathcal{O}(1, 0)$  to get a map  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$ , and then compose with the standard embedding  $\mathbb{P}^1 \rightarrow \mathbb{P}^n$  given by the  $n - 1$ -th Veronese embedding.

There cannot be any embeddings  $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$  for dimension reasons. There are many ways to see that a closed subscheme  $X$  of a scheme  $Y$  must have  $\dim X \leq \dim Y$ . For instance, recall that the dimension of a scheme is the supremum of the lengths of chains of irreducible closed subsets. If  $X$  is a closed subscheme of  $Y$ , then any chain of irreducible closed subsets in  $X$  is also a chain of irreducible closed subsets in  $Y$ , so  $\dim X \leq \dim Y$ .

### 1.5.1 Very ampleness and embeddings

**Definition 1.237.** An invertible sheaf  $\mathcal{L}$  on  $X$  is **very ample** if there exists a closed immersion

$$X \hookrightarrow \mathbb{P}_A^n$$

such that

$$\mathcal{L} \cong \mathcal{O}_X(1).$$

**Definition 1.238.** An invertible sheaf  $\mathcal{L}$  is **ample** if for every coherent sheaf  $\mathcal{F}$ , there exists  $n_0$  such that for all  $n \geq n_0$ , the sheaf  $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$  is generated by global sections.

**Proposition 1.239.** If  $\mathcal{L}$  is very ample, then it is basepoint-free.

*Proof.* The embedding  $X \hookrightarrow \mathbb{P}^n$  is given by global sections of  $\mathcal{L}$ , and at every point at least one coordinate is nonzero.  $\square$

**Proposition 1.240.** Let  $\mathcal{L}$  be very ample. Then the associated morphism

$$\varphi : X \rightarrow \mathbb{P}_A^n$$

is a closed immersion.

**Theorem 1.241** (Serre). Let  $X$  be a projective scheme over a Noetherian ring  $A$ , and let  $\mathcal{L}$  be ample. Then for  $n \gg 0$ ,  $\mathcal{L}^{\otimes n}$  is very ample.

**Example 1.242.** Let  $Q$  be the nonsingular quadric surface  $xy = zw$  in  $\mathbb{P}_k^3$  over a field  $k$ . We have seen that  $\text{Pic}(Q) \cong \mathbb{Z} \oplus \mathbb{Z}$  and so we speak of invertible sheaves of type  $(a, b)$ , with  $a, b \in \mathbb{Z}$ .

Now  $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ . If  $a, b > 0$ , then we have Veronese embeddings  $\mathbb{P}^1 \rightarrow \mathbb{P}^a$  and  $\mathbb{P}^1 \rightarrow \mathbb{P}^b$  whose product and composition with the Segre embedding gives a closed immersion

$$Q = \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^a \times \mathbb{P}^b \rightarrow \mathbb{P}^n,$$

which corresponds to an invertible sheaf of type  $(a, b)$  on  $Q$ . Thus for any  $a, b > 0$ , the corresponding invertible sheaf is very ample, and hence ample.

On the other hand, if  $\mathcal{L}$  is of type  $(a, b)$  with either  $a < 0$  or  $b < 0$ , then by restricting to a fiber of the product  $\mathbb{P}^1 \times \mathbb{P}^1$ , one sees that  $\mathcal{L}$  is not generated by global sections. Hence if  $a \leq 0$  or  $b \leq 0$ ,  $\mathcal{L}$  cannot be ample.

Therefore, on  $Q$ , an invertible sheaf  $\mathcal{L}$  of type  $(a, b)$  is ample if and only if it is very ample, which holds if and only if  $a, b > 0$ .

### 1.5.2 The tautological bundles on projective space

Let  $V$  be a finite-dimensional vector space over a field  $k$ . We write  $\mathbf{P}(V)$  for the projective space of lines in  $V$ , i.e. 1-dimensional linear subspaces  $\ell \subset V$ .

There are two natural line bundles on  $\mathbf{P}(V)$ , dual to one another.

**Definition 1.243.** *The **tautological line bundle** on  $\mathbf{P}(V)$  is the line subbundle*

$$\mathcal{O}_{\mathbf{P}(V)}(-1) \subset V \otimes_k \mathcal{O}_{\mathbf{P}(V)}$$

*whose fiber at a point  $[\ell] \in \mathbf{P}(V)$  is the line  $\ell$  itself:*

$$\mathcal{O}_{\mathbf{P}(V)}(-1)_{[\ell]} = \ell.$$

*Its dual line bundle is denoted*

$$\mathcal{O}_{\mathbf{P}(V)}(1) := \mathcal{O}_{\mathbf{P}(V)}(-1)^\vee.$$

Thus for each point  $[\ell] \in \mathbf{P}(V)$ , we have

$$\mathcal{O}_{\mathbf{P}(V)}(1)_{[\ell]} = \ell^\vee.$$

Equivalently, if one thinks of  $\mathbf{P}(V)$  as parametrizing 1-dimensional quotient spaces of  $V$ , then  $\mathcal{O}_{\mathbf{P}(V)}(1)$  is the universal quotient line bundle.

**Proposition 1.244.** *There is a natural isomorphism*

$$H^0(\mathbf{P}(V), \mathcal{O}_{\mathbf{P}(V)}(1)) \cong V^\vee.$$

*Proof.* Let  $\varphi \in V^\vee$  be a linear functional. For each point  $[\ell] \in \mathbf{P}(V)$ , we may restrict  $\varphi$  to the line  $\ell$ , obtaining

$$\varphi|_\ell \in \ell^\vee = \mathcal{O}_{\mathbf{P}(V)}(1)_{[\ell]}.$$

Thus  $\varphi$  defines a section  $s_\varphi$  of  $\mathcal{O}_{\mathbf{P}(V)}(1)$ , given fiberwise by

$$s_\varphi([\ell]) = \varphi|_\ell.$$

This construction is linear in  $\varphi$ , so it gives a linear map

$$V^\vee \longrightarrow H^0(\mathbf{P}(V), \mathcal{O}_{\mathbf{P}(V)}(1)).$$

To see that this is an isomorphism, choose a basis of  $V$ , so that  $\mathbf{P}(V) \cong \mathbf{P}^n$ . Then  $\mathcal{O}_{\mathbf{P}(V)}(1)$  identifies with the usual hyperplane bundle  $\mathcal{O}_{\mathbf{P}^n}(1)$ , whose global sections are precisely the linear homogeneous polynomials. These are naturally identified with  $V^\vee$ .  $\square$

The zero locus of the section  $s_\varphi$  is easy to describe. Indeed,

$$s_\varphi([\ell]) = 0 \iff \varphi|_\ell = 0 \iff \ell \subset \ker(\varphi).$$

Therefore

$$Z(s_\varphi) = \mathbf{P}(\ker \varphi),$$

which is a hyperplane in  $\mathbf{P}(V)$ . This is why  $\mathcal{O}_{\mathbf{P}(V)}(1)$  is called the hyperplane bundle.



**Remark 1.245.** *The punctured affine cone*

$$V \setminus \{0\} \longrightarrow \mathbf{P}(V), \quad v \longmapsto [kv],$$

is naturally the total space of  $\mathcal{O}_{\mathbf{P}(V)}(-1)$  with the zero section removed. Indeed, over a point  $[\ell]$ , the fiber of  $\mathcal{O}_{\mathbf{P}(V)}(-1)$  is the line  $\ell$ , and removing the zero vector gives precisely the set of nonzero vectors in that line.

In general, if  $\mathcal{E}$  is a vector bundle on  $X$ , then its total space is

$$\underline{\mathrm{Spec}}_X(\mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}^\vee)).$$

The reason is that if  $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$ , then

$$\mathrm{Sym}(\mathcal{E}^\vee)|_U \cong \mathcal{O}_U[t_1, \dots, t_r],$$

so

$$\underline{\mathrm{Spec}}_U(\mathrm{Sym}(\mathcal{E}^\vee)) \cong U \times \mathbf{A}^r.$$

These local affine spaces glue exactly as the vector bundle glues.

**Example 1.246** (The Euler sequence on projective space). *Let  $V$  be a vector space of dimension  $n + 1$  over a field  $k$ . Recall that  $\mathcal{O}_{\mathbf{P}(V)}(-1)$  is the tautological line bundle on  $\mathbf{P}(V)$ , which identifies the line  $\ell$  at a point  $[\ell]$  with the fiber of  $\mathcal{O}_{\mathbf{P}(V)}(-1)$  at that point. In particular, the inclusion  $\ell \subset V$  globalizes to a short exact sequence of vector bundles on  $\mathbf{P}(V)$ :*

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}(V)}(-1) \longrightarrow V \otimes_k \mathcal{O}_{\mathbf{P}(V)} \longrightarrow Q \longrightarrow 0$$

where  $Q$  is the quotient bundle. The graph construction identifies the tangent bundle  $T_{\mathbf{P}(V)}$  with  $\mathrm{Hom}(\mathcal{O}_{\mathbf{P}(V)}(-1), Q)$ . In particular, tensoring the above sequence by  $\mathcal{O}_{\mathbf{P}(V)}(1)$  gives the **Euler sequence**:

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}(V)} \longrightarrow V \otimes_k \mathcal{O}_{\mathbf{P}(V)}(1) \longrightarrow T_{\mathbf{P}(V)} \longrightarrow 0.$$

Taking determinants identifies the canonical bundle of  $\mathbf{P}(V)$  as  $\mathcal{O}_{\mathbf{P}(V)}(-n - 1)$ .

**Example 1.247.** *One can also directly compute the canonical bundle of  $\mathbf{P}(V)$  using transition functions. Let  $t_1 = x_1/x_0, \dots, t_n = x_n/x_0$  be affine coordinates on the standard open subset  $U_0 = \{x_0 \neq 0\}$  and let  $s_1 = x_0/x_1, \dots, s_n = x_n/x_1$  be affine coordinates on the standard open subset  $U_1 = \{x_1 \neq 0\}$ . Then  $dt_1 \wedge \dots \wedge dt_n$  is a local trivialization of the canonical bundle on  $U_0$ , and  $ds_1 \wedge \dots \wedge ds_n$  is a local trivialization on  $U_1$ . On the overlap  $U_0 \cap U_1$ , we have  $s_1 = 1/t_1$  and  $s_i = t_i/t_1$  for  $i > 1$ , so*

$$\begin{aligned} ds_1 \wedge \dots \wedge ds_n &= d(1/t_1) \wedge d(t_2/t_1) \wedge \dots \wedge d(t_n/t_1) \\ &= (-1/t_1^2)dt_1 \wedge \frac{t_1 dt_2 - t_2 dt_1}{t_1^2} \wedge \dots \\ &= (-1/t_1^{n+1})dt_1 \wedge dt_2 \wedge \dots \wedge dt_n \end{aligned}$$

since all of the terms involving  $dt_1$  vanish, and each of the  $n$  terms involving  $dt_i$  for  $i > 1$  contributes a factor of  $t_1/t_1^2$ . Thus the transition function for the canonical bundle on  $U_0 \cap U_1$  is  $-1/t_1^{n+1}$ , which identifies the canonical bundle as  $\mathcal{O}_{\mathbf{P}(V)}(-n - 1)$ .

### 1.5.3 Relative Spec and affine morphisms

Let  $X$  be a scheme, and let  $\mathcal{A}$  be a quasi-coherent sheaf of  $\mathcal{O}_X$ -algebras. We define the **relative spectrum** of  $\mathcal{A}$  over  $X$ , denoted

$$\underline{\mathrm{Spec}}_X(\mathcal{A}),$$

to be the scheme obtained by gluing the affine schemes

$$\mathrm{Spec} \mathcal{A}(U)$$

over affine open subsets  $U = \mathrm{Spec} R \subset X$ .

**Theorem 1.248.** *The construction*

$$\mathcal{A} \longmapsto \underline{\mathrm{Spec}}_X(\mathcal{A})$$

*defines an anti-equivalence between the category of quasi-coherent  $\mathcal{O}_X$ -algebras and the category of affine morphisms  $f : Y \rightarrow X$ . The inverse functor sends an affine morphism  $f : Y \rightarrow X$  to the quasi-coherent  $\mathcal{O}_X$ -algebra*

$$f_*\mathcal{O}_Y.$$

**Remark 1.249.** *This is the relative version of the affine correspondence  $A \leftrightarrow \mathrm{Spec} A$ . In particular, a morphism  $f : Y \rightarrow X$  is affine if and only if*

$$Y \cong \underline{\mathrm{Spec}}_X(f_*\mathcal{O}_Y).$$

**Example 1.250.** *If  $\mathcal{E}$  is a vector bundle on  $X$ , its total space is*

$$\mathrm{Tot}(\mathcal{E}) := \underline{\mathrm{Spec}}_X(\mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}^\vee)).$$

*Indeed, on any open set  $U \subset X$  where  $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$ , we have*

$$\mathrm{Sym}(\mathcal{E}^\vee)|_U \cong \mathcal{O}_U[t_1, \dots, t_r],$$

*so*

$$\mathrm{Tot}(\mathcal{E})|_U \cong U \times \mathbb{A}^r.$$

*Thus  $\mathrm{Tot}(\mathcal{E}) \rightarrow X$  is an affine morphism which is locally the projection  $U \times \mathbb{A}^r \rightarrow U$ .*

## 1.6 Differentials

### 1.6.1 Kähler differentials $\Omega_{X/Y}$

**Definition 1.251.** *Let  $A \rightarrow B$  be a ring homomorphism. The **module of Kähler differentials**  $\Omega_{B/A}$  is the  $B$ -module generated by symbols  $db$  for  $b \in B$ , subject to:*

$$\begin{aligned} d(b + b') &= db + db', \\ d(bb') &= b db' + b' db, \\ da &= 0 \quad \text{for all } a \in A. \end{aligned}$$

There is a universal derivation

$$d : B \rightarrow \Omega_{B/A}.$$

**Definition 1.252.** For a morphism of schemes  $f : X \rightarrow Y$ , the **sheaf of relative differentials**  $\Omega_{X/Y}$  is the quasi-coherent sheaf obtained by gluing  $\Omega_{B/A}$  on affine opens. If  $X$  is a scheme over  $\text{Spec } k$ , define  $\Omega_X := \Omega_{X/k}$ .

**Example 1.253.** If  $B = A[x_1, \dots, x_n]$ , then

$$\Omega_{B/A} \cong \bigoplus_{i=1}^n B dx_i.$$

**Definition 1.254.** An  $A$ -derivation from  $B$  to a  $B$ -module  $M$  is an  $A$ -linear map  $D : B \rightarrow M$  satisfying the Leibniz rule  $D(bb') = bD(b') + b'D(b)$ .

The sheaf of relative differentials is the universal source of derivations in the sense that for any  $B$ -module  $M$ , there is a natural isomorphism

$$\text{Der}_A(B, M) \cong \text{Hom}_B(\Omega_{B/A}, M),$$

sending  $D$  to the unique  $B$ -linear map  $\Omega_{B/A} \rightarrow M$  with  $db \mapsto D(b)$ .

**Proposition 1.255.**  $\Omega_{X/Y}$  is quasi-coherent, and coherent if  $X \rightarrow Y$  is of finite type.

**Definition 1.256.** A morphism  $f : X \rightarrow Y$  is **smooth of relative dimension**  $n$  if  $f$  is locally of finite presentation,  $f$  is flat, and  $\Omega_{X/Y}$  is locally free of rank  $n$ .

**Example 1.257.** If  $X$  is smooth over  $k$ , then  $\Omega_X$  is locally free and  $\omega_X := \wedge^{\dim X} \Omega_X$  is the canonical bundle.

### 1.6.2 Exact sequences for differentials

**Proposition 1.258.** Let  $f : X \rightarrow Y$  be a morphism of schemes over  $S$ . Then there is a natural right exact sequence of sheaves which that absolute differentials decompose into "directions coming from the base" and "new directions along the fibers"

$$f^* \Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

where working on an open  $V \subseteq Y$ , this is  $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V)) \rightarrow \Omega_{X/S}(f^{-1}(V))$ , the structure map followed by the universal derivation  $d : \mathcal{O}_X(f^{-1}(V)) \rightarrow \Omega_{X/S}(f^{-1}(V))$ . The composition is  $S$ -linear and satisfies the Leibniz rule so it factors through  $f^* \Omega_{Y/S}$ , giving  $\Omega_{Y/S} \rightarrow f_* \Omega_{X/S}$  and by adjunction  $f^* \Omega_{Y/S} \rightarrow \Omega_{X/S}$ .

For a sequence of ring maps  $A \rightarrow B \rightarrow C$ , this corresponds to

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0$$

where the maps are  $b db' \otimes 1 \mapsto b db$  the image of  $b$  in  $C$ , and  $dc \otimes 1 \mapsto dc$ .

If  $f$  is smooth, or more generally flat and locally of finite presentation, then the first map is injective and we get a short exact sequence. This is because flatness implies that no nonzero element  $\sum db_i \otimes c_i$  maps to zero.

**Proposition 1.259** (Conormal sequence). *Let  $i : Z \hookrightarrow X$  be a closed immersion with ideal sheaf  $\mathcal{I}$ . Then there is an exact sequence*

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_X|_Z \longrightarrow \Omega_Z \longrightarrow 0.$$

where  $\mathcal{I}/\mathcal{I}^2 = N_{Z/X}^\vee$  is the conormal bundle of  $Z$  in  $X$ .

If  $B = A/I$ , then this corresponds to:

$$I/I^2 \longrightarrow \Omega_A \otimes_A B \longrightarrow \Omega_B \longrightarrow 0,$$

where the first map sends  $f \mapsto df$ . If

$$B = A[x_1, \dots, x_n]/(f_1, \dots, f_r),$$

then

$$\Omega_{B/A} \cong \frac{\bigoplus B dx_i}{\langle df_j \rangle}, \quad \text{where} \quad df_j = \sum_i \frac{\partial f_j}{\partial x_i} dx_i.$$

**Remark 1.260.** *Let  $Z \hookrightarrow X$  be a closed immersion, locally given by  $A \rightarrow A/I$ . The conormal bundle at a point  $z \in Z$  has fiber  $(T_z X/T_z Z)^*$ , and we will see why it should be modeled by  $I/I^2$ .*

*The thing to remember is that  $T_z X = \text{Der}_k(A, k)$  and  $T_z Z = \text{Der}_k(A/I, k)$ . In particular, a tangent vector  $v \in T_z X$  comes from a tangent vector on  $Z$  exactly when it kills  $I$  so  $v \in T_z Z$  iff  $v(f) = 0$  for all  $f \in I$ . Define  $\phi_v : I \rightarrow k$  by  $\phi_v(f) = v(f)$ .*

*If  $f, g \in I$ , then  $v(fg) = f(z)v(g) + g(z)v(f) = 0$ , so  $\phi_v$  factors through  $I/I^2 \rightarrow k$ . Therefore each tangent vector  $v \in T_z X$  gives a linear functional  $\bar{\phi}_v \in \text{Hom}_k((I/I^2) \otimes_{A/I} k, k)$ . Also it is clear that if the linear functional  $\bar{\phi}_v$  is zero, then  $v$  as a derivation on  $A$  kills any  $f \in I$ .*

*So the induced map  $T_z X/T_z Z \rightarrow \text{Hom}_k((I/I^2) \otimes_{A/I} k, k)$  is injective, and in fact it is an isomorphism.*

$$T_z X/T_z Z \cong \text{Hom}_k((I/I^2) \otimes_{A/I} k, k)$$

and globally we have

$$(I/I^2) \otimes_{A/I} k \cong (T_z X/T_z Z)^\vee$$

**Proposition 1.261.** *Let  $D \hookrightarrow X$  be an effective Cartier divisor. Then  $\mathcal{I}_D$  is locally principal and*

$$\mathcal{I}_D/\mathcal{I}_D^2 \cong \mathcal{O}_D(-D).$$

*In this case the conormal sequence becomes short exact:*

$$0 \longrightarrow \mathcal{O}_D(-D) \longrightarrow \Omega_X|_D \longrightarrow \Omega_D \longrightarrow 0.$$

The normal bundle is a line bundle, reflecting that a Cartier divisor has codimension 1. Dualizing gives

$$0 \longrightarrow T_D \longrightarrow T_X|_D \longrightarrow \mathcal{O}_D(D) \longrightarrow 0.$$

### 1.6.3 Tangent space via $\text{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$

**Definition 1.262.** *Let  $X$  be a scheme,  $x \in X$  with residue field  $k(x)$ . The **Zariski tangent space** is*

$$T_x X := \text{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x)).$$

We can also give an interpretation of the tangent space in terms of the dual numbers. Let  $k[\epsilon]/(\epsilon^2)$  be the ring of dual numbers, and let  $\text{Spec } k[\epsilon]/(\epsilon^2)$  be the corresponding scheme, which has a single point with residue field  $k$ . Then we can think about the  $k[\epsilon]/(\epsilon^2)$ -points of  $X$  via the functor of points interpretation

$$X(k[\epsilon]/(\epsilon^2)) = \text{Hom}(\text{Spec } k[\epsilon]/(\epsilon^2), X)$$

If we base the map  $\text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow X$  at  $x$ , then the tangent space  $T_x X$  can be identified with the set of maps  $\text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow X$  that send the unique point of  $\text{Spec } k[\epsilon]/(\epsilon^2)$  to  $x$ . In other words,  $T_x X$  is the set of first-order infinitesimal deformations of the point  $x$ . A morphism  $f : \text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow X$  with image at  $x$  induces a local ring map

$$f^\# : \mathcal{O}_{X,x} \rightarrow k[\epsilon]/(\epsilon^2).$$

Because it is local, the maximal ideal  $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$  must map into the maximal ideal of  $k[\epsilon]/(\epsilon^2)$ , namely  $(\epsilon) = \epsilon k$  so

$$f^\#(\mathfrak{m}_x) \subseteq \epsilon k.$$

Now  $\epsilon k$  is naturally identified with  $k$  via multiplication by  $\epsilon$ . Also, because  $\epsilon^2 = 0$ , if  $u, v \in \mathfrak{m}_x$ , then

$$f^\#(uv) = f^\#(u)f^\#(v) \in (\epsilon)^2 = 0.$$

So  $\mathfrak{m}_x^2$  maps to zero. Therefore the map factors through

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \epsilon k \cong k.$$

Therefore, we see that the Zariski tangent space at a point  $x \in X$  identifies with the space of maps from the dual numbers based at  $x$ .

**Proposition 1.263.** *There is a natural identification*

$$T_x X \cong \text{Der}_{k(x)}(\mathcal{O}_{X,x}, k(x)).$$

**Proposition 1.264.** *If  $X$  is locally of finite type over  $k$ , then*

$$T_x X \cong \text{Hom}_{k(x)}(\Omega_{X/k} \otimes k(x), k(x)).$$

**Corollary 1.265.**  $\dim T_x X \geq \dim X$ , with equality iff  $X$  is smooth at  $x$ .

### 1.6.4 Relative differentials and base change

**Proposition 1.266** (Base change). *Given a cartesian square*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

*there is a natural isomorphism*

$$\Omega_{X'/Y'} \cong \Omega_{X/Y} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}.$$

*Proof.* Let  $X \rightarrow Y$  be modeled by a ring map  $A \rightarrow B$ , and consider the base change along  $A \rightarrow A'$  so that  $B' := B \otimes_A A'$ . Then the local statement is that  $\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$ . The point is relative differentiation kills base scalars, so base change only enlarges coefficients and does not create new differential directions.  $\square$

The stalkwise version states that  $f : X \rightarrow Y$  and  $x \in X$  mapping to  $y \in Y$ ,

$$\Omega_{X/Y} \otimes k(x) \cong \Omega_{X_y/k(y)} \otimes_{k(y)} k(x).$$

In particular, the fiber of  $\Omega_{X/Y}$  at  $x$  computes the tangent space of the fiber  $X_y$ .

**Theorem 1.267** (Smoothness criterion). *Let  $f : X \rightarrow Y$  be locally of finite type. Then  $f$  is smooth at  $x$  iff:*

- $\Omega_{X/Y}$  is locally free near  $x$ ,
- $\dim_{k(x)}(\Omega_{X/Y} \otimes k(x)) = \dim_x(X_y)$ .

### 1.6.5 More examples

**Example 1.268.** *Let  $B = k[x, y]/(xy)$ . Then*

$$d(xy) = x dy + y dx,$$

*so*

$$\Omega_{B/k} = \frac{B dx \oplus B dy}{\langle x dy + y dx \rangle}.$$

*At the origin  $(x, y) = (0, 0)$ , the relation  $x dy + y dx$  vanishes, so the dimension of the tangent space is 2. However, the curve has dimension 1, so the origin is a singular point.*

**Example 1.269.** *Another example is the dual numbers  $B = k[x]/(x^2)$ . We compute  $d(x^2) = 2x dx = 0 \Rightarrow x dx = 0$  so*

$$\Omega_{B/k} \cong B dx / (x dx)$$

*At the closed point,  $\dim T = 1$  but the scheme is 0-dimensional.*

**Example 1.270.** A field extension  $L/k$  is separable if and only if  $\Omega_{L/k} = 0$ . A basic nonexample in characteristic  $p > 0$  is

$$k(t^p) \subset k(t).$$

This extension is not separable, hence not étale. Indeed, if we set  $u = t^p$ , then

$$k(t) \cong k(u)[t]/(t^p - u).$$

Therefore  $\Omega_{k(t)/k(u)}$  is generated by  $dt$  subject to the relation

$$d(t^p - u) = 0.$$

But in characteristic  $p$ ,

$$d(t^p) = pt^{p-1}dt = 0,$$

and also  $du = 0$  in relative differentials over  $k(u)$ . So this imposes no condition on  $dt$ . Thus

$$\Omega_{k(t)/k(t^p)} \cong k(t) dt \neq 0.$$

## 1.7 Cohomological Techniques

Recall that we say an object  $I$  in an abelian category  $\mathcal{A}$  is **injective** if for every monomorphism  $A \rightarrow B$  and every morphism  $A \rightarrow I$ , there exists a morphism  $B \rightarrow I$  making the diagram commute:

$$\begin{array}{ccc} A & \longrightarrow & I \\ \downarrow & \nearrow \text{dashed} & \\ B & & \end{array}$$

**Theorem 1.271.** The category of sheaves of abelian groups on a topological space  $X$  has enough injectives, i.e. for any sheaf  $\mathcal{F}$  on  $X$ , there exists a monomorphism  $\mathcal{F} \rightarrow \mathcal{I}$  into an injective sheaf  $\mathcal{I}$ .

This allows us to define sheaf cohomology as the right derived functors of the global sections functor  $\Gamma(X, -)$ . We will give the definition now. Take an injective resolution of a sheaf  $\mathcal{F}$  on  $X$ :

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1 \rightarrow \mathcal{I}^2 \rightarrow \dots$$

Applying the global sections functor  $\Gamma(X, -)$  gives a complex of abelian groups

$$0 \rightarrow \Gamma(X, \mathcal{I}^0) \rightarrow \Gamma(X, \mathcal{I}^1) \rightarrow \Gamma(X, \mathcal{I}^2) \rightarrow \dots$$

**Definition 1.272.** The **sheaf cohomology** groups  $H^i(X, \mathcal{F})$  are defined as the cohomology groups of the complex of global sections of an injective resolution of  $\mathcal{F}$ :

$$H^i(X, \mathcal{F}) = R^i\Gamma(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{I}^\bullet))$$

However, in practice it is quite difficult to work with injective resolutions of sheaves. Instead, we use other types of resolutions which are easier to construct.

**Definition 1.273.** Let  $F$  be a left exact functor on  $\text{AbSh}(X)$ . A sheaf  $\mathcal{I}$  is  **$F$ -acyclic** if  $R^i F(\mathcal{I}) = 0$  for all  $i > 0$ .

The point of  $F$ -acyclic sheaves is that they can be used to compute the derived functors of  $F$ . More precisely, if  $\mathcal{F}$  is a sheaf on  $X$  and  $\mathcal{I}^\bullet$  is a resolution of  $\mathcal{F}$  by  $F$ -acyclic sheaves, then

$$R^i F(\mathcal{F}) = H^i(F(\mathcal{I}^\bullet))$$

**Definition 1.274.** A sheaf  $\mathcal{F}$  on a topological space  $X$  is **flabby** if for every inclusion of open sets  $V \subseteq U \subseteq X$ , the restriction map  $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$  is surjective.

**Definition 1.275.** A sheaf  $\mathcal{F}$  on a topological space  $X$  is **soft** if for every closed set  $K \subseteq X$ , the restriction map is surjective.

$$\mathcal{F}(X) \rightarrow \mathcal{F}(K)$$

**Proposition 1.276.** Flabby sheaves are soft on any topological space.

**Theorem 1.277.** On a paracompact Hausdorff space  $X$ , soft sheaves are  $\Gamma(X, -)$ -acyclic.

**Definition 1.278.** A sheaf  $\mathcal{F}$  on a topological space  $X$  is **fine** if for every locally finite open cover  $\{U_i\}_{i \in I}$  of  $X$ , there exists a partition of unity subordinate to the cover, i.e. a collection of sheaf endomorphisms  $\{\varphi_i : \mathcal{F} \rightarrow \mathcal{F}\}_{i \in I}$  such that

- For each  $i \in I$ , the support of  $\varphi_i$  is contained in  $U_i$ ,
- The sum  $\sum_{i \in I} \varphi_i$  is the identity endomorphism of  $\mathcal{F}$ .

**Theorem 1.279.** Fine sheaves are soft. Sheaves of modules over fine sheaves of rings are soft.

**Example 1.280.** Let  $X$  be a paracompact Hausdorff space, and let  $\mathcal{C}_X$  be the sheaf of continuous real-valued functions on  $X$ . Then  $\mathcal{C}_X$  is a fine sheaf because we can construct partitions of unity using continuous bump functions subordinate to any locally finite open cover of  $X$ . In particular,  $\mathcal{C}_X$  has no higher cohomology.

**Example 1.281.** Let  $\mathcal{E}_X^k$  be the sheaf of smooth  $k$ -forms on a smooth manifold  $X$ . Then  $\mathcal{E}_X^k$  is a fine sheaf because we can construct partitions of unity using smooth bump functions subordinate to any locally finite open cover of  $X$ .

**Example 1.282** (Complex of sheaves of  $\mathbb{Z}$  singular cochains). Consider the presheaf  $\mathcal{C}^k$  on a paracompact Hausdorff space  $X$  defined by

$$\mathcal{C}^k(U) = C_{\text{sing}}^k(U, \mathbb{Z})$$



where  $C_{\text{sing}}^k(U, \mathbb{Z})$  is the group of singular  $k$ -cochains on  $U$  with integer coefficients. The restriction maps are given by restricting cochains to smaller open sets. The sheafification  $\mathcal{S}^k = \mathcal{C}^{k+}$  is called the sheaf of singular  $k$ -cochains. The complex of sheaves

$$\cdots \rightarrow \mathcal{S}^{k-1} \rightarrow \mathcal{S}^k \rightarrow \mathcal{S}^{k+1} \rightarrow \cdots$$

with the usual coboundary maps is a fine resolution of the constant sheaf  $\mathbb{Z}_X$  on  $X$ . This is because partitions of unity can be constructed for singular cochains by using barycentric subdivisions of simplices.

The complex  $(\mathcal{C}^\bullet(X), d)$  is exact on stalks except for when  $k = 0$ , where the cohomology is  $\mathbb{Z}$ . Thus this complex is a fine resolution of the constant sheaf  $\mathbb{Z}_X$ . This shows that sheaf cohomology with coefficients in  $\mathbb{Z}_X$  can be computed using the complex of singular cochains, and hence agrees with singular cohomology.

**Exercise 1.283.** Let  $S^1$  be the circle with its usual topology, and let  $\mathbb{Z}$  be the constant sheaf  $\mathbb{Z}$ .

1. Show that  $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$ .
2. Let  $\mathcal{R}$  be the sheaf of germs of continuous real-valued functions on  $S^1$ . Show that  $H^1(S^1, \mathcal{R}) = 0$ .

**Solution 1.284.** We first prove that  $H^1(S^1, \mathcal{R}) = 0$ . The sheaf  $\mathcal{R}$  is a fine sheaf. Indeed, since  $S^1$  is paracompact Hausdorff, every open cover admits a partition of unity subordinate to it. If  $\{U_i\}$  is an open cover of  $S^1$  and  $\{\rho_i\}$  is a partition of unity subordinate to this cover, then multiplication by  $\rho_i$  defines an endomorphism of  $\mathcal{R}$  whose support is contained in  $U_i$ , and the sum of these endomorphisms is the identity. Hence  $\mathcal{R}$  is fine.

Fine sheaves on paracompact spaces are acyclic for the global sections functor. Therefore  $H^i(S^1, \mathcal{R}) = 0$  for all  $i > 0$ . In particular,  $H^1(S^1, \mathcal{R}) = 0$ .

Now we compute  $H^1(S^1, \mathbb{Z})$ . There is a short exact sequence of sheaves

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{R} \rightarrow \mathcal{R}/\mathbb{Z} \rightarrow 0.$$

The map  $\mathbb{Z} \rightarrow \mathcal{R}$  sends a locally constant integer-valued function to the corresponding continuous real-valued function. The quotient  $\mathcal{R}/\mathbb{Z}$  is the sheaf of germs of continuous  $\mathbb{R}/\mathbb{Z}$ -valued functions.

Taking the long exact sequence in cohomology gives

$$H^0(S^1, \mathcal{R}) \rightarrow H^0(S^1, \mathcal{R}/\mathbb{Z}) \rightarrow H^1(S^1, \mathbb{Z}) \rightarrow H^1(S^1, \mathcal{R}).$$

Since  $H^1(S^1, \mathcal{R}) = 0$ , we get an isomorphism

$$H^1(S^1, \mathbb{Z}) \cong H^0(S^1, \mathcal{R}/\mathbb{Z}) / \text{im}(H^0(S^1, \mathcal{R}) \rightarrow H^0(S^1, \mathcal{R}/\mathbb{Z})).$$

Now  $H^0(S^1, \mathcal{R}) = C(S^1, \mathbb{R})$  and  $H^0(S^1, \mathcal{R}/\mathbb{Z}) = C(S^1, \mathbb{R}/\mathbb{Z})$ . Therefore

$$H^1(S^1, \mathbb{Z}) \cong C(S^1, \mathbb{R}/\mathbb{Z}) / \{\text{continuous maps } S^1 \rightarrow \mathbb{R}/\mathbb{Z} \text{ lifting to } S^1 \rightarrow \mathbb{R}\}.$$

A continuous map  $f : S^1 \rightarrow \mathbb{R}/\mathbb{Z}$  lifts to a continuous map  $\tilde{f} : S^1 \rightarrow \mathbb{R}$  if and only if  $f$  has winding number 0. Therefore  $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$ .

**Theorem 1.285.** *Let  $X$  be a Noetherian topological space of dimension  $n$ . Then  $H^i(X, \mathcal{F}) = 0$  for all  $i > n$  and all sheaves of abelian groups on  $X$ .*

**Theorem 1.286.** *Let  $\mathcal{F}_i$  be a direct system of sheaves of abelian groups on a topological space  $X$ . Let  $\mathcal{F} = \varinjlim \mathcal{F}_i$  be the colimit sheaf. Then there is a natural isomorphism*

$$H^p(X, \mathcal{F}) \cong \varinjlim H^p(X, \mathcal{F}_i)$$

*In particular, colimits are exact in the category of sheaves of abelian groups.*

A remark that this theorem does not hold for limits. In particular, if  $\mathcal{F}_i$  is an inverse system of sheaves of abelian groups on a topological space  $X$ , then the natural map

$$H^p(X, \varprojlim \mathcal{F}_i) \rightarrow \varprojlim H^p(X, \mathcal{F}_i)$$

is not an isomorphism in general. The failure of this map to be an isomorphism is measured by the derived functors of the inverse limit, denoted  $R^1 \varprojlim$ . In particular, there is a natural exact sequence

$$0 \rightarrow R^1 \varprojlim H^{p-1}(X, \mathcal{F}_i) \rightarrow H^p(X, \varprojlim \mathcal{F}_i) \rightarrow \varprojlim H^p(X, \mathcal{F}_i) \rightarrow 0$$

Limits commute with cohomology if the inverse system  $\{H^p(X, \mathcal{F}_i)\}$  satisfies the Mittag-Leffler condition, which means that for each  $p$  and each  $i$ , there exists  $j \geq i$  such that the image of the map  $H^p(X, \mathcal{F}_j) \rightarrow H^p(X, \mathcal{F}_i)$  is equal to the image of the map  $H^p(X, \mathcal{F}_k) \rightarrow H^p(X, \mathcal{F}_i)$  for all  $k \geq j$ . In particular, if the maps in the inverse system  $\{H^p(X, \mathcal{F}_i)\}$  are surjective for all  $p$  and all  $i$ , then limits commute with cohomology.

**Example 1.287.** *On an irreducible topological space  $X$ , any constant sheaf  $\underline{A}$  is flasque. This is because if any open subset  $U \subseteq X$  is also irreducible, in particular connected, then the restriction maps of  $\underline{A}$  are all identity maps. Thus every restriction map is surjective. In particular*

$$H^i(X, \underline{A}) = 0 \quad \text{for all } i > 0$$

**Exercise 1.288.** *Let  $X = \mathbb{A}_k^1$  be the affine line over an infinite field  $k$ . Let  $P, Q$  be distinct closed points of  $X$ , and let  $U = X - \{P, Q\}$ . Show that  $H^1(X, \mathbb{Z}_U) \neq 0$ .*

*More generally, let  $Y \subset X = \mathbb{A}_k^n$  be the union of  $n+1$  hyperplanes in suitably general position, and let  $U = X - Y$ . Show that  $H^n(X, \mathbb{Z}_U) \neq 0$ . Thus the bound in the cohomological dimension theorem for noetherian topological spaces is best possible.*

**Solution 1.289.** *Let  $j : U \hookrightarrow X$  be the open immersion. We write  $\mathbb{Z}_U := j_* \mathbb{Z}$  for the constant sheaf on  $U$ , extended by zero to  $X$ .*

*First consider the case*

$$X = \mathbb{A}_k^1, \quad Y = \{P, Q\}, \quad U = X - Y.$$

There is a short exact sequence of sheaves on  $X$

$$0 \rightarrow \mathbb{Z}_U \rightarrow \mathbb{Z}_X \rightarrow \mathbb{Z}_Y \rightarrow 0.$$

Taking cohomology gives an exact sequence

$$0 \rightarrow H^0(X, \mathbb{Z}_U) \rightarrow H^0(X, \mathbb{Z}_X) \rightarrow H^0(X, \mathbb{Z}_Y) \rightarrow H^1(X, \mathbb{Z}_U) \rightarrow \cdots.$$

Since  $X = \mathbb{A}_k^1$  is irreducible, we have  $H^0(X, \mathbb{Z}_X) = \mathbb{Z}$ . On the other hand,  $Y = \{P, Q\}$  has two connected components, so  $H^0(X, \mathbb{Z}_Y) = H^0(Y, \mathbb{Z}_Y) = \mathbb{Z} \oplus \mathbb{Z}$ . The map on cohomology is the diagonal map  $\mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}$  sending 1 to  $(1, 1)$ . Its cokernel is  $\mathbb{Z}$  and by exactness, this cokernel injects into  $H^1(X, \mathbb{Z}_U)$ . Therefore  $H^1(X, \mathbb{Z}_U) \neq 0$ . Now consider the general case. Let  $X = \mathbb{A}_k^n$  and let  $Y = H_0 \cup H_1 \cup \cdots \cup H_n$  be a union of  $n + 1$  hyperplanes in suitably general position. Again we have a short exact sequence

$$0 \rightarrow \mathbb{Z}_U \rightarrow \mathbb{Z}_X \rightarrow \mathbb{Z}_Y \rightarrow 0.$$

Since  $X = \mathbb{A}_k^n$  is irreducible noetherian, the constant sheaf  $\mathbb{Z}_X$  is flasque. Hence

$$H^i(X, \mathbb{Z}_X) = 0 \quad \text{for all } i > 0.$$

The long exact sequence in cohomology therefore gives an isomorphism

$$H^n(X, \mathbb{Z}_U) \cong H^{n-1}(X, i_! \mathbb{Z}_Y) = H^{n-1}(X, i_* \mathbb{Z}_Y) = H^{n-1}(Y, \mathbb{Z}_Y)$$

Note that  $i_! = i_*$  agree for closed immersions, and the second equality follows from the fact that being a closed immersion,

$$\Gamma(X, i_* \mathcal{F}) = \Gamma(Y, \mathcal{F})$$

which follows directly from the definition of  $i_*$ , and also  $i_*$  is exact. Thus it remains to show that

$$H^{n-1}(Y, \mathbb{Z}_Y) \neq 0.$$

The hyperplanes are in general position, so every intersection of at most  $n$  of them is nonempty, while the intersection of all  $n + 1$  hyperplanes is empty:

$$H_{i_0} \cap \cdots \cap H_{i_r} \neq \emptyset \quad \text{for } r \leq n - 1,$$

but

$$H_0 \cap H_1 \cap \cdots \cap H_n = \emptyset.$$

Moreover, every nonempty intersection

$$H_{i_0} \cap \cdots \cap H_{i_r}$$

is an affine space, hence irreducible. Therefore the constant sheaf on each nonempty intersection has no higher cohomology.

The closed cover

$$Y = H_0 \cup H_1 \cup \cdots \cup H_n$$

has nerve equal to the boundary of an  $n$ -simplex. Indeed, all proper intersections are nonempty, but the full intersection is empty. Hence the nerve is

$$\partial\Delta^n \simeq S^{n-1}.$$

Since the intersections are acyclic for the constant sheaf  $\mathbb{Z}$ , the cohomology of  $Y$  with coefficients in  $\mathbb{Z}_Y$  is computed by the Čech cohomology of this cover. Therefore

$$H^{n-1}(Y, \mathbb{Z}_Y) \cong H^{n-1}(\partial\Delta^n, \mathbb{Z}) \cong \mathbb{Z}.$$

It follows that

$$H^n(X, \mathbb{Z}_U) \cong H^{n-1}(Y, \mathbb{Z}_Y) \cong \mathbb{Z}.$$

In particular,

$$H^n(X, \mathbb{Z}_U) \neq 0.$$

Thus for the noetherian topological space  $X = \mathbb{A}_k^n$  of dimension  $n$ , there exists a sheaf of abelian groups  $\mathbb{Z}_U$  such that

$$H^n(X, \mathbb{Z}_U) \neq 0.$$

Therefore the vanishing bound

$$H^i(X, \mathcal{F}) = 0 \quad \text{for } i > \dim X$$

is best possible.

**Theorem 1.290.** *Let  $X$  be a Noetherian scheme. Then the following conditions are equivalent:*

- $X$  is affine.
- $H^i(X, \mathcal{F}) = 0$  for all  $i > 0$  and all quasi-coherent sheaves  $\mathcal{F}$  on  $X$ .
- $H^i(X, \mathcal{I}) = 0$  for all coherent ideal sheaves  $\mathcal{I}$  on  $X$ .

A relative version of the vanishing theorem holds for affine morphisms.

**Theorem 1.291.** *If  $f : X \rightarrow Y$  is an affine morphism and  $\mathcal{F}$  is a quasi-coherent sheaf on  $X$ , then*

$$R^i f_* \mathcal{F} = 0 \quad \text{for all } i > 0.$$

In particular,

$$H^i(X, \mathcal{F}) = H^i(Y, f_* \mathcal{F}) \quad \text{for all } i \geq 0$$

### 1.7.1 Čech cohomology

Let  $X$  be a scheme and  $\mathcal{F}$  a sheaf of abelian groups.

**Definition 1.292.** Let  $\mathfrak{U} = \{U_i\}$  be an open cover of  $X$ . The **Čech complex** is a complex of abelian groups defined by

$$\check{C}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_p}),$$

with the alternating differential

$$\begin{aligned} d : \check{C}^p(\mathfrak{U}, \mathcal{F}) &\rightarrow \check{C}^{p+1}(\mathfrak{U}, \mathcal{F}) \\ (d\alpha)_{i_0, \dots, i_{p+1}} &= \sum_{j=0}^{p+1} (-1)^j \alpha_{i_0, \dots, \widehat{i_j}, \dots, i_{p+1}}|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}} \end{aligned}$$

The cohomology is denoted  $\check{H}^p(\mathfrak{U}, \mathcal{F})$ . We can also write down the Čech complex of sheaves. Let

$$\mathfrak{C}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} j_{i_0, \dots, i_p, *}(\mathcal{F}|_{U_{i_0} \cap \dots \cap U_{i_p}})$$

with the same alternating differential. Then the complex

$$0 \rightarrow \mathcal{F} \rightarrow \mathfrak{C}^0(\mathfrak{U}, \mathcal{F}) \rightarrow \mathfrak{C}^1(\mathfrak{U}, \mathcal{F}) \rightarrow \dots$$

is a resolution of  $\mathcal{F}$ , flasque if  $\mathcal{F}$  is flasque. Picking an injective resolution of  $\mathcal{F}$  and comparing the two resolutions gives a morphism from the Čech complex to the injective resolution, and hence a morphism on cohomology. In other words, there is a natural map

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$$

where the RHS is derived functor cohomology.

**Theorem 1.293.** If  $X$  is quasi-compact and separated, and  $\mathcal{F}$  is quasi-coherent, then Čech cohomology with respect to any affine open cover computes sheaf cohomology.

*Proof.* For  $p = 0$ , the desired map is an isomorphism by the preceding result. For the general case, embed  $\mathcal{F}$  into a flasque quasicoherent sheaf  $\mathcal{G}$ , and let  $\mathcal{R}$  be the quotient:

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{R} \rightarrow 0.$$

For each  $i_0 < \dots < i_p$ , the open set

$$U_{i_0, \dots, i_p} := U_{i_0} \cap \dots \cap U_{i_p}$$

is affine, since it is an intersection of affine open subsets of a separated scheme. Since  $\mathcal{F}, \mathcal{G}, \mathcal{R}$  are quasicoherent, we have an exact sequence

$$0 \rightarrow \mathcal{F}(U_{i_0, \dots, i_p}) \rightarrow \mathcal{G}(U_{i_0, \dots, i_p}) \rightarrow \mathcal{R}(U_{i_0, \dots, i_p}) \rightarrow 0.$$

Taking products over all  $i_0 < \dots < i_p$ , we obtain a short exact sequence of Čech complexes

$$0 \rightarrow \check{C}^\bullet(\mathfrak{U}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathfrak{U}, \mathcal{G}) \rightarrow \check{C}^\bullet(\mathfrak{U}, \mathcal{R}) \rightarrow 0.$$

Hence there is a long exact sequence in Čech cohomology. Since  $\mathcal{G}$  is flasque, its higher Čech cohomology vanishes, so for  $p > 0$  we have

$$\check{H}^p(\mathfrak{U}, \mathcal{G}) = 0.$$

Therefore the long exact sequence gives an exact sequence

$$0 \rightarrow \check{H}^0(\mathfrak{U}, \mathcal{F}) \rightarrow \check{H}^0(\mathfrak{U}, \mathcal{G}) \rightarrow \check{H}^0(\mathfrak{U}, \mathcal{R}) \rightarrow \check{H}^1(\mathfrak{U}, \mathcal{F}) \rightarrow 0$$

and isomorphisms

$$\check{H}^p(\mathfrak{U}, \mathcal{R}) \xrightarrow{\sim} \check{H}^{p+1}(\mathfrak{U}, \mathcal{F}) \quad \text{for all } p \geq 1.$$

Now compare this with the long exact sequence in ordinary sheaf cohomology associated to

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{R} \rightarrow 0.$$

Since  $\mathcal{G}$  is flasque, it is acyclic, so  $H^q(X, \mathcal{G}) = 0$  for  $q > 0$ . Using the case  $p = 0$ , we know that

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) \cong H^0(X, \mathcal{F}), \quad \check{H}^0(\mathfrak{U}, \mathcal{G}) \cong H^0(X, \mathcal{G}), \quad \check{H}^0(\mathfrak{U}, \mathcal{R}) \cong H^0(X, \mathcal{R}).$$

Comparing the two long exact sequences, we conclude that the natural map

$$\check{H}^1(\mathfrak{U}, \mathcal{F}) \rightarrow H^1(X, \mathcal{F})$$

is an isomorphism.

Finally,  $\mathcal{R}$  is again quasicoherent, so the same argument applies to  $\mathcal{R}$ . Since

$$\check{H}^p(\mathfrak{U}, \mathcal{R}) \cong \check{H}^{p+1}(\mathfrak{U}, \mathcal{F})$$

and

$$H^p(X, \mathcal{R}) \cong H^{p+1}(X, \mathcal{F})$$

for  $p \geq 1$ , induction on  $p$  gives

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$$

for all  $p \geq 0$ .  $\square$

More generally, any open cover  $\mathfrak{U}$  such that all finite intersections of opens in  $\mathfrak{U}$  are acyclic for  $\mathcal{F}$  will compute the cohomology of  $\mathcal{F}$ . The separatedness condition guarantees that finite intersections of affine opens are affine, hence acyclic for quasi-coherent sheaves.

**Exercise 1.294.** Let  $U = \mathbb{A}_k^2 \setminus \{(0, 0)\}$  be the punctured affine plane over a field  $k$ . Show that  $H^1(U, \mathcal{O}_U)$  is infinite-dimensional over  $k$ . Conclude that  $U$  is not affine.

**Solution 1.295.** Let

$$U = \mathbb{A}_k^2 \setminus \{(0, 0)\} = D(x) \cup D(y).$$

Since  $D(x)$ ,  $D(y)$ , and  $D(x) \cap D(y) = D(xy)$  are affine, Čech cohomology with respect to this cover computes  $H^1(U, \mathcal{O}_U)$ . Since  $C^2 = 0$  for this cover,  $\mathcal{O}(D(xy)) = \ker(C^1 \rightarrow C^2)$ ,

so  $H^1(U, \mathcal{O}_U)$  is the cokernel of the map  $C^0 \rightarrow C^1$ . In other words, we divide out by differences  $f|_{D(xy)} - g|_{D(xy)}$  where  $f \in \mathcal{O}(D(x))$  and  $g \in \mathcal{O}(D(y))$ . But allowing each of  $f$  and  $g$  to be zero, we can divide out by all differences of the form  $f|_{D(xy)}$  with  $f \in \mathcal{O}(D(x))$  or  $f \in \mathcal{O}(D(y))$ . Hence

$$H^1(U, \mathcal{O}_U) \cong \frac{\mathcal{O}(D(xy))}{\mathcal{O}(D(x)) + \mathcal{O}(D(y))}$$

Now

$$\mathcal{O}(D(x)) = k[x, x^{-1}, y], \quad \mathcal{O}(D(y)) = k[x, y, y^{-1}],$$

and

$$\mathcal{O}(D(xy)) = k[x, x^{-1}, y, y^{-1}].$$

Therefore

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, x^{-1}, y, y^{-1}]}{k[x, x^{-1}, y] + k[x, y, y^{-1}]}.$$

A monomial  $x^i y^j$  lies in  $k[x, x^{-1}, y]$  if and only if  $j \geq 0$ , and it lies in  $k[x, y, y^{-1}]$  if and only if  $i \geq 0$ . Hence the only monomials which survive in the quotient are those with

$$i < 0, \quad j < 0.$$

It follows that

$$H^1(U, \mathcal{O}_U) \cong \bigoplus_{i, j < 0} k \cdot x^i y^j,$$

so  $H^1(U, \mathcal{O}_U)$  is isomorphic to the  $k$ -vector space spanned by

$$\{x^i y^j \mid i, j < 0\}.$$

In particular, it is infinite-dimensional over  $k$ .

Since  $\mathcal{O}_U$  is quasi-coherent, if  $U$  were affine then we would have

$$H^1(U, \mathcal{O}_U) = 0$$

Thus  $U$  is not affine. This also shows that it is not proper since it has infinite-dimensional cohomology.

**Exercise 1.296.** Let  $X = \mathbb{A}_k^4$  be the affine 4-space over a field  $k$ . Let

$$Y = V(x_1, x_2) \cup V(x_3, x_4) \subset X$$

be the union of two planes meeting at the origin, and let

$$U = X - Y.$$

Show that  $H^2(U, \mathcal{O}_U) \neq 0$ . Conclude that  $Y$  is not a set-theoretic complete intersection, and that the projective closure of  $Y$  in  $\mathbb{P}_k^4$  is also not a set-theoretic complete intersection.

**Solution 1.297.** Let  $R = k[x_1, x_2, x_3, x_4]$  and  $X = \operatorname{Spec} R$ . Set

$$Y_1 = V(x_1, x_2), \quad Y_2 = V(x_3, x_4), \quad Y = Y_1 \cup Y_2.$$

Then

$$U = X - Y = (X - Y_1) \cap (X - Y_2).$$

Now

$$X - Y_1 = D(x_1) \cup D(x_2), \quad X - Y_2 = D(x_3) \cup D(x_4).$$

Thus  $U$  is covered by the four affine opens

$$U_{13} = D(x_1 x_3), \quad U_{14} = D(x_1 x_4), \quad U_{23} = D(x_2 x_3), \quad U_{24} = D(x_2 x_4).$$

Since  $X$  is separated, all finite intersections of these opens are affine. Hence Čech cohomology with respect to this cover computes  $H^i(U, \mathcal{O}_U)$ .

We now isolate one monomial subcomplex of the Čech complex. Order the four opens as

$$U_1 = U_{13}, \quad U_2 = U_{14}, \quad U_3 = U_{23}, \quad U_4 = U_{24}.$$

Consider the Laurent monomial

$$m := \frac{1}{x_1 x_2 x_3 x_4}.$$

This monomial appears in the coordinate ring of an intersection precisely when that intersection inverts all four variables  $x_1, x_2, x_3, x_4$ .

It does not appear on any single  $U_i$ . Among pairwise intersections, it appears only on

$$U_1 \cap U_4 = D(x_1 x_2 x_3 x_4) \quad \text{and} \quad U_2 \cap U_3 = D(x_1 x_2 x_3 x_4).$$

It appears on every triple intersection and on the quadruple intersection. Therefore the part of the Čech complex spanned by this monomial is

$$0 \rightarrow k^2 \xrightarrow{d^1} k^4 \xrightarrow{d^2} k \rightarrow 0,$$

where the  $k^2$  has coordinates corresponding to the pairs  $(1, 4)$  and  $(2, 3)$ , the  $k^4$  has coordinates corresponding to the triples  $(1, 2, 3)$ ,  $(1, 2, 4)$ ,  $(1, 3, 4)$ , and  $(2, 3, 4)$ , and the last  $k$  corresponds to the quadruple intersection.

If a 1-cochain has values  $a$  on  $U_1 \cap U_4$  and  $b$  on  $U_2 \cap U_3$ , then its coboundary has triple components

$$d^1(a, b) = (b, -a, -a, b).$$

Indeed, this follows directly from the alternating Čech differential.

On the other hand, for a 2-cochain with triple components  $(u, v, w, z)$ , the next differential is

$$d^2(u, v, w, z) = z - w + v - u.$$

Hence the 2-cochain

$$\alpha = (0, m, 0, -m)$$



is a cocycle, since

$$d^2\alpha = (-m) - 0 + m - 0 = 0.$$

But  $\alpha$  is not a coboundary. If it were equal to  $d^1(a, b) = (b, -a, -a, b)$ , then comparing components would give simultaneously

$$b = 0, \quad -a = m, \quad -a = 0, \quad b = -m,$$

which is impossible. Thus  $\alpha$  defines a nonzero class in  $H^2(U, \mathcal{O}_U)$ . Therefore

$$H^2(U, \mathcal{O}_U) \neq 0.$$

Now suppose  $Y$  were a set-theoretic complete intersection. Since  $\text{codim}_X(Y) = 2$ , this would mean that  $Y$  is cut out set-theoretically by two equations. Hartshorne's cohomological criterion says that if  $Y \subset X$  is cut out set-theoretically by  $r$  equations, then

$$H^i(X - Y, \mathcal{F}) = 0 \quad \text{for all } i \geq r$$

for all quasicoherent sheaves  $\mathcal{F}$  on  $X$ . Taking  $\mathcal{F} = \mathcal{O}_X$  and  $r = 2$ , we would get

$$H^2(X - Y, \mathcal{O}_X) = 0,$$

contradicting the computation above. Therefore  $Y$  is not a set-theoretic complete intersection.

Finally, the projective closure  $\bar{Y} \subset \mathbb{P}_k^4$  is also not a set-theoretic complete intersection. Indeed, if  $\bar{Y}$  were cut out set-theoretically by two homogeneous equations in  $\mathbb{P}_k^4$ , then restricting to the affine chart  $x_0 \neq 0$  would cut out  $Y \subset \mathbb{A}_k^4$  set-theoretically by two equations, contradicting the affine case.

**Exercise 1.298.** Let  $X$  be a subscheme of  $\mathbb{P}_k^2$  defined by a single homogeneous equation

$$f(x_0, x_1, x_2) = 0$$

of degree  $d$ . Do not assume that  $f$  is irreducible. Assume that  $(1, 0, 0)$  is not on  $X$ . Show that  $X$  can be covered by the two open affine subsets

$$U = X \cap \{x_1 \neq 0\}, \quad V = X \cap \{x_2 \neq 0\}.$$

Compute the Čech complex

$$\Gamma(U, \mathcal{O}_X) \oplus \Gamma(V, \mathcal{O}_X) \rightarrow \Gamma(U \cap V, \mathcal{O}_X)$$

explicitly, and show that

$$\dim H^0(X, \mathcal{O}_X) = 1, \quad \dim H^1(X, \mathcal{O}_X) = \frac{1}{2}(d-1)(d-2).$$

**Solution 1.299.** Since  $(1, 0, 0) \notin X$ , the coefficient of  $x_0^d$  in  $f$  is nonzero. After rescaling  $f$ , we may assume that this coefficient is 1.

Let

$$U = X \cap D_+(x_1), \quad V = X \cap D_+(x_2).$$

These two opens cover  $X$ , because the only point of  $\mathbb{P}_k^2$  not contained in  $D_+(x_1) \cup D_+(x_2)$  is  $(1, 0, 0)$ , which is not on  $X$ .

On  $U$ , set

$$x = \frac{x_0}{x_1}, \quad y = \frac{x_2}{x_1}.$$

Then

$$\Gamma(U, \mathcal{O}_X) = A := k[x, y]/(F), \quad F = f(x, 1, y).$$

Since  $f$  has coefficient 1 in front of  $x_0^d$ , the polynomial  $F$  is monic of degree  $d$  in  $x$ . Therefore  $A$  is a free  $k[y]$ -module with basis

$$1, x, \dots, x^{d-1}.$$

On the intersection  $U \cap V$ , we invert  $y$ , so

$$\Gamma(U \cap V, \mathcal{O}_X) = A_y.$$

Thus  $A_y$  has a  $k$ -basis given by monomials

$$x^i y^n, \quad 0 \leq i \leq d-1, \quad n \in \mathbb{Z}.$$

Now on  $V$ , set

$$u = \frac{x_0}{x_2}, \quad v = \frac{x_1}{x_2}.$$

In terms of  $x, y$ , we have

$$u = \frac{x}{y}, \quad v = \frac{1}{y}.$$

Therefore

$$\Gamma(V, \mathcal{O}_X) = k[u, v]/(f(u, v, 1)).$$

Again  $f(u, v, 1)$  is monic of degree  $d$  in  $u$ , so  $\Gamma(V, \mathcal{O}_X)$  is a free  $k[v]$ -module with basis

$$1, u, \dots, u^{d-1}.$$

Inside  $A_y$ , the monomial  $u^i v^m$  becomes

$$u^i v^m = \left(\frac{x}{y}\right)^i \left(\frac{1}{y}\right)^m = x^i y^{-i-m}.$$

Thus  $\Gamma(V, \mathcal{O}_X)$  is spanned inside  $A_y$  by monomials  $x^i y^n$  with

$$0 \leq i \leq d-1, \quad n \leq -i.$$

On the other hand,  $\Gamma(U, \mathcal{O}_X) = A$  is spanned by monomials  $x^i y^n$  with

$$0 \leq i \leq d-1, \quad n \geq 0.$$

The Čech complex for the cover  $X = U \cup V$  is

$$0 \rightarrow \Gamma(U, \mathcal{O}_X) \oplus \Gamma(V, \mathcal{O}_X) \xrightarrow{\delta} \Gamma(U \cap V, \mathcal{O}_X) \rightarrow 0,$$

where

$$\delta(a, b) = b|_{U \cap V} - a|_{U \cap V}.$$

Therefore

$$H^0(X, \mathcal{O}_X) = \ker(\delta) = \Gamma(U, \mathcal{O}_X) \cap \Gamma(V, \mathcal{O}_X) \subset A_y.$$

From the monomial descriptions above, the only monomial lying in both  $\Gamma(U, \mathcal{O}_X)$  and  $\Gamma(V, \mathcal{O}_X)$  is 1. Hence

$$H^0(X, \mathcal{O}_X) \cong k,$$

so

$$\dim H^0(X, \mathcal{O}_X) = 1.$$

Similarly,

$$H^1(X, \mathcal{O}_X) \cong \frac{\Gamma(U \cap V, \mathcal{O}_X)}{\Gamma(U, \mathcal{O}_X) + \Gamma(V, \mathcal{O}_X)} = \frac{A_y}{A + \Gamma(V, \mathcal{O}_X)}.$$

The quotient has basis given by monomials  $x^i y^n$  with

$$0 \leq i \leq d-1, \quad -i < n < 0.$$

For fixed  $i$ , the possible values are

$$n = -1, -2, \dots, -(i-1).$$

Thus there are  $i-1$  such monomials for  $i \geq 2$ , and none for  $i = 0, 1$ . Therefore

$$\dim H^1(X, \mathcal{O}_X) = \sum_{i=2}^{d-1} (i-1) = 1 + 2 + \dots + (d-2) = \frac{(d-1)(d-2)}{2}.$$

Hence

$$\dim H^0(X, \mathcal{O}_X) = 1, \quad \dim H^1(X, \mathcal{O}_X) = \frac{1}{2}(d-1)(d-2).$$

**Example 1.300.** We compute the cohomology of the line with the double origin. Let  $X = U \cup V$  where  $U = \operatorname{Spec} k[x]$  and  $V = \operatorname{Spec} k[y]$  are two copies of the affine line, and  $U \cap V = \operatorname{Spec} k[x, x^{-1}]$  is the punctured affine line, with the restriction maps given by  $x \mapsto x$  and  $y \mapsto x$ . Then the Čech complex for the cover  $X = U \cup V$  is

$$0 \rightarrow k[x] \oplus k[y] \xrightarrow{\delta} k[x, x^{-1}] \rightarrow 0,$$

where  $\delta(f(x), g(y)) = g(x) - f(x)$ . The kernel of  $\delta$  consists of pairs  $(f(x), f(y))$  where  $f$  is a polynomial, so  $H^0(X, \mathcal{O}_X) \cong k[x]$ . The cokernel of  $\delta$  is spanned by monomials  $x^n$  with  $n < 0$ , so  $H^1(X, \mathcal{O}_X)$  is infinite-dimensional over  $k$ .

### 1.7.2 Cohomology of $\mathbb{P}^n$

Let  $k$  be a field and  $X = \mathbb{P}_k^n$ .

**Theorem 1.301.** *For all  $m \in \mathbb{Z}$ :*

1.  $H^0(\mathbb{P}^n, \mathcal{O}(m))$  is the space of homogeneous polynomials of degree  $m$ :

$$H^0(\mathbb{P}^n, \mathcal{O}(m)) = \begin{cases} S_m & m \geq 0, \\ 0 & m < 0. \end{cases}$$

2.  $H^i(\mathbb{P}^n, \mathcal{O}(m)) = 0$  for  $0 < i < n$ .

3. Top cohomology comes exactly from monomials  $x^a$  with all exponents negative and with total degree  $m$ :

$$H^n(\mathbb{P}^n, \mathcal{O}(m)) \cong \begin{cases} 0 & m > -n - 1, \\ k^{\binom{-m-1}{n}} & m \leq -n - 1. \end{cases}$$

*Sketch of the computation.* Let

$$X = \mathbb{P}_k^n = \text{Proj } S, \quad S = k[x_0, \dots, x_n],$$

and cover  $X$  by the standard affine opens

$$\mathfrak{U} = \{U_i = D_+(x_i)\}_{i=0}^n.$$

For a subset  $I = \{i_0, \dots, i_p\}$ , write  $U_I = U_{i_0} \cap \dots \cap U_{i_p}$ . Then

$$\mathcal{O}_X(m)(U_I) = (S_{x_I})_m, \quad x_I := x_{i_0} \cdots x_{i_p}.$$

Thus

$$\check{C}^p(\mathfrak{U}, \mathcal{O}_X(m)) = \bigoplus_{|I|=p+1} (S_{x_I})_m.$$

A basis for  $(S_{x_I})_m$  is given by Laurent monomials

$$x^a = x_0^{a_0} \cdots x_n^{a_n}, \quad a_i \in \mathbb{Z}, \quad \sum_{i=0}^n a_i = m,$$

such that  $a_j \geq 0$  for every  $j \notin I$ . Equivalently, the variables with negative exponent must all belong to  $I$ .

For such an exponent vector  $a = (a_0, \dots, a_n)$ , set

$$N(a) := \{i \mid a_i < 0\}.$$

Then the monomial  $x^a$  appears in the summand indexed by  $I$  if and only if  $N(a) \subseteq I$ . Hence the Čech complex decomposes as a direct sum of monomial subcomplexes:

$$\check{C}^\bullet(\mathfrak{U}, \mathcal{O}_X(m)) = \bigoplus_{\substack{a \in \mathbb{Z}^{n+1} \\ a_0 + \cdots + a_n = m}} C^\bullet(a),$$

where  $C^\bullet(a)$  is spanned by the single monomial  $x^a$ .

Fix such an  $a$ , and let  $r = |N(a)|$ . Then

$$C^p(a) = 0 \quad \text{if } p + 1 < r,$$

and for  $p + 1 \geq r$  we have

$$C^p(a) = \bigoplus_{\substack{I \supseteq N(a) \\ |I|=p+1}} k \cdot x^a.$$

Thus  $C^\bullet(a)$  is the simplicial cochain complex of a simplex on the remaining vertices  $\{0, \dots, n\} \setminus N(a)$ , shifted by  $r - 1$ .

There are three cases.

**Case 1:**  $N(a) = \emptyset$ . Then all exponents are nonnegative, so  $x^a$  is an honest homogeneous polynomial of degree  $m$ . In this case  $C^\bullet(a)$  is the ordinary Čech complex of the full simplex, which has the cohomology of a point:

$$H^q(C^\bullet(a)) = \begin{cases} k & q = 0, \\ 0 & q > 0. \end{cases}$$

Summing over all such monomials gives

$$H^0(\mathbb{P}^n, \mathcal{O}(m)) = \begin{cases} S_m & m \geq 0, \\ 0 & m < 0. \end{cases}$$

**Case 2:**  $\emptyset \neq N(a) \neq \{0, \dots, n\}$ . Let  $M = \{0, \dots, n\} \setminus N(a)$  and write each index set as

$$I = N(a) \sqcup J, \quad J \subseteq M.$$

Then

$$C^p(a) \cong \bigoplus_{\substack{J \subseteq M \\ |J|=p+1-|N(a)|}} k.$$

Thus  $C^\bullet(a)$  is the augmented simplicial cochain complex of the simplex on the vertex set  $M$ , shifted by  $|N(a)| - 1$ . It is augmented because  $J = \emptyset$  occurs when  $p = r - 1$ . So the complex starts with  $C^{r-1}(a) \cong k$  then comes  $C^r(a) \cong k^{|M|}$  and higher terms.

Since a simplex is contractible, its augmented cochain complex is exact. Therefore

$$H^q(C^\bullet(a)) = 0 \quad \text{for all } q.$$

**Case 3:**  $N(a) = \{0, \dots, n\}$ . Then every exponent is negative. In this case the only subset  $I$  containing  $N(a)$  is the full set  $\{0, \dots, n\}$ , so the complex is concentrated in degree  $n$ :

$$C^p(a) = 0 \quad (p < n), \quad C^n(a) = k \cdot x^a.$$

Hence

$$H^q(C^\bullet(a)) = \begin{cases} k & q = n, \\ 0 & q \neq n. \end{cases}$$

Thus top cohomology is spanned by monomials  $x^a$  with all  $a_i < 0$  and  $\sum a_i = m$ .

Writing

$$a_i = -b_i - 1, \quad b_i \geq 0,$$

the condition  $\sum a_i = m$  becomes

$$\sum_{i=0}^n b_i = -m - n - 1.$$

Therefore such monomials exist if and only if

$$m \leq -n - 1.$$

When this holds, a basis for  $H^n(\mathbb{P}^n, \mathcal{O}(m))$  is given by the classes

$$\left[ \frac{1}{x_0^{b_0+1} \cdots x_n^{b_n+1}} \right], \quad b_i \geq 0, \quad \sum_{i=0}^n b_i = -m - n - 1.$$

In fact this basis is more naturally interpreted in terms of rational top differential forms on  $\mathbb{P}^n$ . Let

$$\Omega = \sum_{i=0}^n (-1)^i x_i dx_0 \wedge \cdots \widehat{dx_i} \cdots \wedge dx_n$$

be the standard homogeneous  $(n+1)$ -form on  $\mathbb{A}^{n+1} \setminus \{0\}$ . Then one can check that  $\Omega$  has weight  $n+1$  with respect to the standard scaling action. Then a monomial  $x^a$  with all exponents negative corresponds to the rational differential form

$$\frac{\Omega}{x_0^{b_0+1} \cdots x_n^{b_n+1}},$$

which descends to a well-defined rational section of the canonical bundle  $\omega_{\mathbb{P}^n}$ .  $\square$

**Theorem 1.302** (Serre duality and vanishing). *Let  $X$  be a smooth projective scheme of dimension  $n$  over a field  $k$ , and let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. Then there is a natural perfect pairing*

$$H^i(X, \mathcal{F}) \times \text{Ext}_{\mathcal{O}_X}^{n-i}(\mathcal{F}, \omega_X) \longrightarrow k,$$

*functorial in  $\mathcal{F}$ , which induces canonical isomorphisms*

$$H^i(X, \mathcal{F})^\vee \cong \text{Ext}_{\mathcal{O}_X}^{n-i}(\mathcal{F}, \omega_X).$$

*Here  $\omega_X := \Omega_{X/k}^n$  is the canonical (dualizing) sheaf.*

**Corollary 1.303.** *If  $\mathcal{F}$  is locally free, then*

$$H^i(X, \mathcal{F})^\vee \cong H^{n-i}(X, \mathcal{F}^\vee \otimes \omega_X).$$

**Theorem 1.304** (Serre vanishing). *Let  $X$  be a projective scheme over a Noetherian ring  $A$ , and let  $\mathcal{L}$  be an ample line bundle. Then for every coherent sheaf  $\mathcal{F}$ , there exists  $n_0$  such that for all  $n \geq n_0$ :*

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) = 0 \quad \text{for all } i > 0.$$

**Corollary 1.305.** *For  $n \gg 0$ , the sheaf  $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$  is generated by global sections.*

**Exercise 1.306.** *Let  $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$  be a short exact sequence of coherent sheaves on a Noetherian scheme  $X$ . Show that  $\chi(X, F) = \chi(X, F') + \chi(X, F'')$ , where  $\chi(X, F) = \sum_{i=0}^{\infty} (-1)^i \dim_k H^i(X, F)$  is the Euler characteristic of  $F$ .*

**Solution 1.307.** *This follows immediately from the long exact sequence in cohomology associated to the short exact sequence of sheaves and the alternating sum of dimensions of a long exact sequence of vector spaces is zero.*

**Exercise 1.308.** *Let  $X$  be a projective scheme over a field  $k$ , let  $\mathcal{O}_X(1)$  be a very ample invertible sheaf on  $X$  over  $k$ , and let  $\mathcal{F}$  be a coherent sheaf on  $X$ .*

1. *Show that there is a polynomial  $P(z) \in \mathbb{Q}[z]$  such that*

$$\chi(\mathcal{F}(n)) = P(n)$$

*for all  $n \in \mathbb{Z}$ .*

2. *If  $X = \mathbb{P}_k^r$  and  $M = \Gamma_*(\mathcal{F})$ , show that this agrees with the usual Hilbert polynomial of the graded  $S = k[x_0, \dots, x_r]$ -module  $M$ .*

**Solution 1.309.** *We prove the first statement by induction on  $d = \dim \text{Supp } \mathcal{F}$ .*

*If  $d = 0$ , then  $\mathcal{F}$  is supported on finitely many closed points. Twisting by an invertible sheaf does not change the length of a zero-dimensional coherent sheaf, so  $\chi(\mathcal{F}(n))$  is independent of  $n$ . Hence it is given by a constant polynomial.*

*Now suppose  $d > 0$ . Choose a section  $s \in H^0(X, \mathcal{O}_X(1))$  whose zero locus does not contain any generic point of  $\text{Supp } \mathcal{F}$ . Multiplication by  $s$  gives a map*

$$\mathcal{F}(-1) \xrightarrow{s} \mathcal{F}.$$

*Let  $\mathcal{R}$  be its kernel and  $\mathcal{Q}$  its cokernel. Then*

$$0 \rightarrow \mathcal{R} \rightarrow \mathcal{F}(-1) \rightarrow \mathcal{F} \rightarrow \mathcal{Q} \rightarrow 0.$$

*Since  $s$  is generically injective on  $\mathcal{F}$ , both  $\mathcal{R}$  and  $\mathcal{Q}$  have support of dimension strictly smaller than  $d$ .*

*Twisting by  $\mathcal{O}_X(n)$  gives*

$$0 \rightarrow \mathcal{R}(n) \rightarrow \mathcal{F}(n-1) \rightarrow \mathcal{F}(n) \rightarrow \mathcal{Q}(n) \rightarrow 0.$$

*By additivity of Euler characteristic,*

$$\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = \chi(\mathcal{Q}(n)) - \chi(\mathcal{R}(n)).$$

*By induction, the right-hand side is a polynomial function of  $n$ . Hence the finite difference of  $\chi(\mathcal{F}(n))$  is polynomial in  $n$ .*

A standard property of numerical polynomials says that if  $a(n) - a(n-1)$  is polynomial in  $n$ , then  $a(n)$  is given by a polynomial in  $\mathbb{Q}[n]$ . Therefore there exists  $P_{\mathcal{F}}(z) \in \mathbb{Q}[z]$  such that

$$\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$$

for all  $n \in \mathbb{Z}$ .

Now let  $X = \mathbb{P}_k^r$ , and let

$$M = \Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} H^0(\mathbb{P}_k^r, \mathcal{F}(n)).$$

Then

$$M_n = H^0(\mathbb{P}_k^r, \mathcal{F}(n)).$$

By Serre vanishing, for  $n \gg 0$ ,

$$H^i(\mathbb{P}_k^r, \mathcal{F}(n)) = 0 \quad \text{for all } i > 0.$$

Hence for  $n \gg 0$ ,

$$\chi(\mathcal{F}(n)) = \dim_k H^0(\mathbb{P}_k^r, \mathcal{F}(n)) = \dim_k M_n.$$

Thus the Hilbert polynomial of  $\mathcal{F}$  agrees for large  $n$  with the Hilbert function of  $M$ . Therefore it is the usual Hilbert polynomial of the graded  $S$ -module  $M$ .

### 1.7.3 Semicontinuity of cohomology, base change, and seesaw principle

We begin with a foundational result about the coherent pushforward of cohomology.

**Theorem 1.310.** *Let  $f : X \rightarrow Y$  be a proper morphism of Noetherian schemes, and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . Then  $R^i f_* \mathcal{F}$  is coherent on  $Y$  for all  $i \geq 0$ .*

**Theorem 1.311** (Semicontinuity). *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, and let  $\mathcal{F}$  be a coherent sheaf on  $X$ , flat over  $Y$ . Then for each  $i \geq 0$ , the function*

$$y \mapsto h^i(y, \mathcal{F}) := \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$$

*is upper semicontinuous on  $Y$ . Moreover, the function*

$$y \mapsto \sum_{i=0}^{\infty} (-1)^i h^i(y, \mathcal{F})$$

*is locally constant on  $Y$ .*

Equivalently, for each integer  $N$ , the subset  $\{y \in Y : h^i(y, \mathcal{F}) \geq N\}$  is closed in  $Y$  for each  $i$ . In particular the dimension of sheaf cohomology can only jump up upon specialization, as is demonstrated by the following example.

**Example 1.312.** *Let  $X = E \times E \rightarrow E$  be the trivial family of an elliptic curve  $E$ , and let  $\mathcal{L}$  be a line bundle on  $X$  whose restriction  $\mathcal{L}_y$  varies in  $\text{Pic}^0(E)$ , with  $\mathcal{L}_0 \cong \mathcal{O}_E$*



and  $\mathcal{L}_y \not\cong \mathcal{O}_E$  for  $y \neq 0$ . For example, we can take  $\mathcal{L}$  to be the line bundle of the divisor  $\Delta - (E \times \{0\})$ , where  $\Delta$  is the diagonal in  $E \times E$ .

Then  $\mathcal{L}$  is flat over  $Y$ . Note that because  $\mathcal{L}$  is locally free,  $\mathcal{L}_x \cong \mathcal{O}_{X_x}$  for every  $x \in X$ , and so it is enough to check that  $X \rightarrow E$  is flat. The latter fact holds because the map  $X \rightarrow E$  is just the projection from a product, which is flat. Locally on  $E$ , if  $E$  is covered by affine opens  $\text{Spec } A$ , and  $Y$  by affine opens  $\text{Spec } R$ , then

$$X|_{\text{Spec } A \times \text{Spec } R} = \text{Spec}(A \otimes_k R),$$

and  $A \otimes_k R$  is flat over  $R$ , because it is actually a free  $R$ -module once you choose a  $k$ -basis of  $A$ . Another way to see this is that  $E \times_{\text{Spec } k} E \rightarrow E$  is the base change of the flat morphism  $E \rightarrow \text{Spec } k$  and the structure map of a variety to  $\text{Spec } k$  is always flat, and flatness is preserved under base change.

Then we see that

$$h^0(E, \mathcal{L}_y) = \begin{cases} 1, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

This is because a nontrivial degree 0 line bundle on an elliptic curve has no nonzero sections, while  $\mathcal{O}_E$  has one. This is true on any smooth projective curve  $C$  for example. If a line bundle  $\mathcal{L}$  is degree zero and has nonzero section  $s$  then  $s$  has an effective divisor of zeros  $D$  and  $\mathcal{L} \cong \mathcal{O}_C(D)$  and  $\deg D = 0$  and  $D \geq 0$  implies that  $D = 0$  and  $\mathcal{L} \cong \mathcal{O}_C$ .

This also shows that one cannot naively think of  $\pi_*\mathcal{L}$  as obtained by "gluing together" the spaces  $H^0(E, \mathcal{L}_p)$ . Indeed, in this example one has

$$\pi_*\mathcal{L} = 0,$$

To see this, it is enough to check that for every open  $U \subseteq X$  the stalk of  $\pi_*\mathcal{L}$  at the generic point of  $U$  is zero (because  $X = E \times E$  is integral). But the restriction of  $\mathcal{L}$  to the generic point of  $U$  is a nontrivial degree 0 line bundle since  $\eta_U$  is not equal to the closed point 0 of  $E$ , so it has no nonzero sections, and hence the stalk of  $\pi_*\mathcal{L}$  at  $\eta_U$  is zero.

Thus this example shows cohomology does not commute with taking fibers everywhere. Let's closely examine the failure of cohomology and base change here.

The relevant base change map in degree 0 is

$$\varphi^0(p) : (\pi_*\mathcal{L}) \otimes k(p) \longrightarrow H^0(E, \mathcal{L}_p).$$

Since  $\pi_*\mathcal{L} = 0$ , the source is always 0. So

$$\varphi^0(p) : 0 \longrightarrow H^0(E, \mathcal{L}_p).$$

Hence:

- if  $p \neq p_0$ , then  $H^0(E, \mathcal{L}_p) = 0$ , so  $\varphi^0(p)$  is  $0 \rightarrow 0$ , an isomorphism;
- if  $p = p_0$ , then  $H^0(E, \mathcal{L}_{p_0}) \cong k$ , so  $\varphi^0(p_0)$  is  $0 \rightarrow k$ , not surjective, hence not an isomorphism.

*So cohomology and base change fails exactly at the special point  $p_0$ .*

The semicontinuity theorem also implies that the locus where line bundles are trivial is Zariski closed, which is made precise by the following result.

**Proposition 1.313.** *Suppose  $X \rightarrow Y$  is a flat projective morphism with integral fibers. Let  $\mathcal{L}$  be a line bundle on  $X$ , which is automatically flat over  $Y$  since it is locally free. Then the locus of  $y \in Y$  where  $\mathcal{L}_y$  is trivial on  $X_y$  is a closed set.*

*Proof.* The locus where

$$h^0(X_y, \mathcal{L}_y) \geq 1$$

is closed by the Semicontinuity Theorem, and the same holds for the locus where

$$h^0(X_y, \mathcal{L}_y^\vee) \geq 1.$$

Now invoke the fact that if  $\mathcal{L}$  is an invertible sheaf on an integral projective scheme  $X$  such that both  $\mathcal{L}$  and  $\mathcal{L}^\vee$  have non-zero sections, then  $\mathcal{L}$  is trivial. In the proof of this fact, we can remove projective if we know that the only global functions on  $X$  are the constants.

This is because the multiplication map if  $s$  and  $t$  are nonzero sections of  $\mathcal{L}$  and  $\mathcal{L}^\vee$ , respectively, they are both nonzero in the stalk at the generic point. Otherwise they would be the zero section; this is where we use that  $X$  is integral.

Under the natural map

$$\mathcal{L} \otimes \mathcal{L}^\vee \longrightarrow \mathcal{O}_X,$$

the section  $s \otimes t$  maps to  $st$ , which is also nonzero. But the only global functions, i.e. global sections of  $\mathcal{O}_X$ , are the constants, so  $st$  is a nonzero constant. It follows that  $s$  is nowhere vanishing, for otherwise  $st$  would vanish somewhere. Hence  $\mathcal{L}$  has a nowhere vanishing section, and therefore  $\mathcal{L} \cong \mathcal{O}_X$ .  $\square$

**Theorem 1.314** (Cohomology and Base Change). *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, and let  $\mathcal{F}$  be a coherent sheaf on  $X$ , flat over  $Y$ . Let  $y \in Y$ . Then:*

1. *If the natural map*

$$\varphi^i(y) : (R^i f_* \mathcal{F}) \otimes_{\mathcal{O}_Y} k(y) \longrightarrow H^i(X_y, \mathcal{F}_y)$$

*is surjective, then it is an isomorphism, and the same holds for all  $y'$  in a suitable neighborhood of  $y$ .*

2. *Assume that  $\varphi^i(y)$  is surjective. Then the following are equivalent:*

(a)  $\varphi^{i-1}(y)$  is surjective;

(b)  $R^i f_* \mathcal{F}$  is locally free in a neighborhood of  $y$ .

**Remark 1.315.** *One naive thing one might try to do to understand the failure of cohomology and base change is to take a Čech complex  $C^\bullet$  computing  $H^i(X, \mathcal{F})$  and tensor it with  $k(s)$ . The problem is that tensoring with  $k(s)$  is not exact in general, since  $k(s)$  is usually not flat over  $A$ . Thus cohomology need not commute with tensor product:*

$$H^i(C^\bullet) \otimes_A k(s) \not\cong H^i(C^\bullet \otimes_A k(s)).$$

*Moreover, for a general family and a general sheaf, the terms of the Čech complex need not be free or even flat over  $A$ , and a given Čech cover need not compute the actual sheaf cohomology.*

With this result in mind, we are ready to state the Seesaw principle, which is a powerful criterion for recognizing when a line bundle on a family of integral projective schemes is pulled back from the base.

**Theorem 1.316** (Seesaw principle: recognition of pullbacks). *Let  $\pi : X \rightarrow Y$  be a flat projective morphism of Noetherian schemes with geometrically integral fibers, and let  $\mathcal{L}$  be an invertible sheaf on  $X$ . Assume that for every  $y \in Y$ ,*

$$\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}.$$

*Then there exists an invertible sheaf  $\mathcal{M}$  on  $Y$  such that*

$$\mathcal{L} \cong \pi^* \mathcal{M}.$$

*Moreover,  $\mathcal{M}$  is unique up to unique isomorphism.*

*Discussion of the mechanism.* The key input is cohomology and base change applied to  $\mathcal{L}$ . Since each fiber  $X_y$  is proper and geometrically integral over  $k(y)$ , we have

$$H^0(X_y, \mathcal{O}_{X_y}) \cong k(y).$$

To see this, fix the fiber

$$X_y = X \times_Y \operatorname{Spec} k(y),$$

and suppose it is proper and geometrically integral over  $k(y)$ . Set  $K := H^0(X_y, \mathcal{O}_{X_y})$ . Because  $X_y$  is proper over  $k(y)$ ,  $K$  is a finite  $k(y)$ -algebra. Because  $X_y$  is integral,  $K$  is a domain, hence a field finite over  $k(y)$ . We need to see that this extension is trivial. Extend scalars from  $k(y)$  to an algebraic closure  $\overline{k(y)}$ . Then

$$X_{\overline{k(y)}} := X_y \times_{k(y)} \operatorname{Spec} \overline{k(y)}.$$

Geometric integrality means exactly that this new scheme is still integral.

But global functions also base change in degree 0 here (the map  $\operatorname{Spec} \overline{k(y)} \rightarrow \operatorname{Spec} k(y)$  is flat!), so

$$H^0(X_{\overline{k(y)}}, \mathcal{O}_{X_{\overline{k(y)}}}) \cong K \otimes_{k(y)} \overline{k(y)}.$$

So if  $K \otimes_{k(y)} \overline{k(y)}$  is not a domain, then  $X_{\overline{k(y)}}$  cannot be integral, because an integral scheme has its ring of global sections a domain.

If  $K/k(y)$  separable, then over the algebraic closure,  $K$  splits:

$$K \otimes_{k(y)} \overline{k(y)} \cong \overline{k(y)}^{[K:k(y)]}.$$

So it is a product of fields and in particular not a domain. A product ring corresponds geometrically to several components, so the base change becomes reducible.

If  $K/k(y)$  is purely inseparable, then after base change, nilpotents appear. Consider the simplest case where in characteristic  $p$  we have a degree  $p$  extension

$$k \subset k(\alpha), \quad \alpha^p = a \in k, \quad \alpha \notin k.$$

Over an algebraic closure  $\bar{k}$ ,

$$K \otimes_k \bar{k} \cong \bar{k}[t]/(t - \beta)^p$$

for some  $\beta \in \bar{k}$ . This ring has nilpotents, so it is not reduced, hence not integral. So in this case the base change becomes nonreduced. Thus proper and geometrically integral implies that  $H^0(X_y, \mathcal{O}_{X_y}) \cong k(y)$ . By the hypothesis  $\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}$ , it follows that

$$H^0(X_y, \mathcal{L}_y) \cong k(y),$$

so in particular

$$h^0(X_y, \mathcal{L}_y) = 1 \quad \text{for all } y \in Y.$$

Now consider the base change map

$$\varphi^0(y) : (\pi_* \mathcal{L}) \otimes_{\mathcal{O}_Y} k(y) \longrightarrow H^0(X_y, \mathcal{L}_y).$$

Because the target has constant dimension 1, cohomology and base change implies that  $\pi_* \mathcal{L}$  is locally free of rank 1 on  $Y$ , hence an invertible sheaf. Set

$$\mathcal{M} := \pi_* \mathcal{L}.$$

There is a natural adjunction morphism

$$\pi^* \pi_* \mathcal{L} \longrightarrow \mathcal{L},$$

that is,

$$\pi^* \mathcal{M} \longrightarrow \mathcal{L}.$$

To show that this is an isomorphism, it suffices to check on fibers. Restricting to  $X_y$ , the induced map is

$$H^0(X_y, \mathcal{L}_y) \otimes_{k(y)} \mathcal{O}_{X_y} \longrightarrow \mathcal{L}_y.$$

Since  $\mathcal{L}_y \cong \mathcal{O}_{X_y}$  and  $H^0(X_y, \mathcal{L}_y) \cong k(y)$ , this is the canonical map

$$k(y) \otimes_{k(y)} \mathcal{O}_{X_y} \longrightarrow \mathcal{O}_{X_y},$$

hence an isomorphism for every  $y \in Y$ . Therefore

$$\pi^* \mathcal{M} \xrightarrow{\sim} \mathcal{L},$$

as claimed. Uniqueness is immediate: if  $\pi^* \mathcal{M}_1 \cong \pi^* \mathcal{M}_2$ , then pushing forward and using  $\pi_* \mathcal{O}_X \cong \mathcal{O}_Y$  gives  $\mathcal{M}_1 \cong \mathcal{M}_2$ .  $\square$

**Exercise 1.317.** Let  $X$  be an integral projective scheme over an algebraically closed field  $k$ , and assume that

$$H^1(X, \mathcal{O}_X) = 0.$$

Let  $T$  be a connected scheme of finite type over  $k$ .

1. If  $\mathcal{L}$  is an invertible sheaf on  $X \times T$ , show that the invertible sheaves  $\mathcal{L}_t := \mathcal{L}|_{X \times \{t\}}$  on  $X_t := X \times \{t\} \cong X$  are isomorphic for all closed points  $t \in T$ .
2. Show that

$$\text{Pic}(X \times T) \cong \text{Pic}(X) \times \text{Pic}(T).$$

Do not assume that  $T$  is reduced.

**Exercise 1.318.** Let  $X$  be an integral projective scheme over an algebraically closed field  $k$ , and assume that  $H^1(X, \mathcal{O}_X) = 0$ . Let  $T$  be a connected scheme of finite type over  $k$ .

1. If  $\mathcal{L}$  is an invertible sheaf on  $X \times T$ , show that the invertible sheaves

$$\mathcal{L}_t := \mathcal{L}|_{X \times \{t\}}$$

on

$$X_t := X \times \{t\} \cong X$$

are isomorphic for all closed points  $t \in T$ .

2. Show that

$$\text{Pic}(X \times T) \cong \text{Pic}(X) \times \text{Pic}(T).$$

Do not assume that  $T$  is reduced.

**Solution 1.319.** Let  $p : X \times T \rightarrow T$  and  $q : X \times T \rightarrow X$  be the projections. Since  $X$  is integral and projective over the algebraically closed field  $k$ , we have  $H^0(X, \mathcal{O}_X) = k$ .

1. Let  $\mathcal{L} \in \text{Pic}(X \times T)$ . We show that the class of  $\mathcal{L}_t$  is locally constant in  $t$ . Fix a closed point  $t_0 \in T$  and set  $M = \mathcal{L}_{t_0}$ . Put  $\mathcal{F} = \mathcal{L} \otimes q^* M^{-1}$ . Then  $\mathcal{F}_{t_0} \cong \mathcal{O}_X$ , so  $h^0(X, \mathcal{F}_{t_0}) = 1$  and  $h^1(X, \mathcal{F}_{t_0}) = h^1(X, \mathcal{O}_X) = 0$ . Since  $h^1(X, \mathcal{F}_{t_0}) = 0$ , upper semicontinuity gives an open neighborhood  $U \subset T$  of  $t_0$  such that  $h^1(X, \mathcal{F}_t) = 0$  for all  $t \in U$ . In particular,  $R^1 p_* \mathcal{F}$  vanishes on  $U$ . The base change map

$$\varphi^1(t) : (R^1 p_* \mathcal{F}) \otimes k(t) \longrightarrow H^1(X, \mathcal{F}_t)$$

has target zero for all  $t \in U$ , so it is surjective. Since  $R^1 p_* \mathcal{F}$  vanishes on  $U$ , it is locally free of rank 0 on  $U$ . This implies that  $\varphi^0(t) : (p_* \mathcal{F}) \otimes k(t) \longrightarrow H^0(X, \mathcal{F}_t)$  is an isomorphism for all  $t \in U$ . In particular,  $h^0(X, \mathcal{F}_t) = 1$  for all  $t \in U$ , since

$$h^0(X, \mathcal{F}_t) = \chi(\mathcal{F}_t) = \chi(\mathcal{F}_{t_0}) = \chi(\mathcal{O}_X) = 1.$$

A local generator of  $p_* \mathcal{F}$  gives a section  $s : \mathcal{O}_{X \times U} \rightarrow \mathcal{F}|_{X \times U}$  which is nonvanishing on the fiber over  $t_0$ . Let  $Z = V(s)$  be the zero locus of  $s$ . Then  $Z$  is a closed subset of  $X \times U$  which does not meet the fiber over  $t_0$ .

Since  $p$  is proper, the image  $p(Z)$  is a closed subset of  $U$ . Since  $Z$  does not meet the fiber over  $t_0$ ,  $p(Z)$  does not contain  $t_0$ . Hence there is an open neighborhood of  $t_0$  in  $U$  which does not meet  $p(Z)$ . Shrinking  $U$  to this neighborhood, we can assume that  $p(Z) = \emptyset$ , so  $s$  is nowhere vanishing on  $X \times U$ . Hence  $\mathcal{F}|_{X \times U} \cong \mathcal{O}_{X \times U}$ , so  $\mathcal{L}_t \cong M$  for all closed  $t \in U$ . Thus  $t \mapsto [\mathcal{L}_t]$  is locally constant, and since  $T$  is connected, all  $\mathcal{L}_t$  are isomorphic.

2. Consider the natural map

$$\mathrm{Pic}(X) \times \mathrm{Pic}(T) \rightarrow \mathrm{Pic}(X \times T), \quad (M, N) \mapsto q^*M \otimes p^*N.$$

It is injective: if  $q^*M \otimes p^*N \cong \mathcal{O}_{X \times T}$ , then restricting to a closed fiber  $X_t$  gives  $M \cong \mathcal{O}_X$ ; then pulling back along any closed point  $x \in X(k)$  gives  $N \cong \mathcal{O}_T$ .

For surjectivity, let  $\mathcal{L} \in \mathrm{Pic}(X \times T)$ . By (a), all  $\mathcal{L}_t$  are isomorphic. Choose a closed point  $t_0 \in T$  and set  $M = \mathcal{L}_{t_0}$ . Put  $\mathcal{F} = \mathcal{L} \otimes q^*M^{-1}$ . Then  $\mathcal{F}_t \cong \mathcal{O}_X$  for every closed  $t \in T$ , so  $h^0(X, \mathcal{F}_t) = 1$  and  $h^1(X, \mathcal{F}_t) = 0$ . By cohomology and base change,  $N := p_*\mathcal{F}$  is an invertible sheaf on  $T$ ,

We need to check that the evaluation map  $p^*N \rightarrow \mathcal{F}$  is an isomorphism on every closed fiber. Since base change holds, we have  $N \otimes k(t) \cong H^0(X_t, \mathcal{F}_t)$ . Therefore the restriction of the evaluation map to  $X_t$  is

$$H^0(X_t, \mathcal{F}_t) \otimes_k \mathcal{O}_{X_t} \rightarrow \mathcal{F}_t$$

Since  $\mathcal{F}_t \cong \mathcal{O}_X$ , this identifies with the canonical map

$$H^0(X, \mathcal{O}_X) \otimes_k \mathcal{O}_X \rightarrow \mathcal{O}_X$$

which is an isomorphism. Therefore  $\mathcal{F} \cong p^*N$ , and thus  $\mathcal{L} \cong q^*M \otimes p^*N$ . Hence

$$\mathrm{Pic}(X \times T) \cong \mathrm{Pic}(X) \times \mathrm{Pic}(T).$$

Note that in general, in order to identify the fiber of the evaluation map  $\varepsilon : p^*N \rightarrow \mathcal{F}$  with the map  $H^0(X_t, \mathcal{F}_t) \otimes_k \mathcal{O}_{X_t} \rightarrow \mathcal{F}_t$ , we need to use the fact that the base change map  $\varphi^0(t) : N \otimes k(t) \rightarrow H^0(X_t, \mathcal{F}_t)$  is an isomorphism. Without base change, one can only identify  $p^*N \otimes k(t) = p_*\mathcal{F} \otimes k(t) \otimes \mathcal{O}_{X_t}$  and this may not equal  $H^0(X_t, \mathcal{F}_t) \otimes_k \mathcal{O}_{X_t}$ .

#### 1.7.4 Higher direct image sheaves

If  $f : X \rightarrow Y$  is a morphism of schemes and  $\mathcal{F}$  is a sheaf on  $X$ , then the pushforward  $f_*\mathcal{F}$  is a sheaf on  $Y$ . Therefore  $f_*$  is a functor from the category of sheaves on  $X$  to the category of sheaves on  $Y$ . This functor is left exact, but not necessarily right exact. The right derived functors of  $f_*$  are denoted by  $R^if_*$  and are called the **higher direct image functors**.

We explicitly construct  $R^if_*$  as we do  $R\Gamma$ . Let  $\mathcal{I}^\bullet$  be an injective resolution of  $\mathcal{F}$ . Then  $f_*\mathcal{I}^\bullet$  is a complex of sheaves on  $Y$ , and we define

$$R^if_*\mathcal{F} := \mathcal{H}^i(f_*\mathcal{I}^\bullet).$$

to be the  $i$ -th cohomology sheaf of this complex. The resulting  $R^i f_*$  is independent of the choice of injective resolution, and it is a functor from the category of sheaves on  $X$  to the category of sheaves on  $Y$ .

It is true that  $R^i f_* \mathcal{F}$  is the sheafification of the presheaf

$$U \mapsto H^i(f^{-1}(U), \mathcal{F})$$

In particular, the stalk of  $R^i f_* \mathcal{F}$  at a point  $y \in Y$  is given by

$$(R^i f_* \mathcal{F})_y \cong \varinjlim_{y \in U} H^i(f^{-1}(U), \mathcal{F}).$$

Moreover it is also true that for affine open subsets  $U \subseteq Y$ , we have

$$H^0(U, R^q \pi_* \mathcal{F}) \cong H^q(\pi^{-1}(U), \mathcal{F})$$

because of the degeneration of the Leray spectral sequence for  $\pi_U : \pi^{-1}(U) \rightarrow U$

$$E_2^{p,q} = H^p(U, R^q \pi_* \mathcal{F}|_U) \Rightarrow H^{p+q}(\pi^{-1}(U), \mathcal{F})$$

Now use the fact that  $U$  is affine to conclude that  $H^p(U, -) = 0$  for  $p > 0$ .

**Theorem 1.320.** *Suppose  $\pi : X \rightarrow Y$  is a quasi-compact separated morphism of schemes. Then:*

(a) *For each  $i \geq 0$ , the higher direct image*

$$R^i \pi_* : \text{QCoh}_X \rightarrow \text{QCoh}_Y$$

*is a covariant functor.*

(b) *There is a natural isomorphism of functors*

$$R^0 \pi_* \cong \pi_*.$$

(c) *If*

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$$

*is a short exact sequence of sheaves on  $X$ , then there is an induced long exact sequence of higher direct image sheaves on  $Y$ :*

$$0 \rightarrow R^0 \pi_* \mathcal{F} \rightarrow R^0 \pi_* \mathcal{G} \rightarrow R^0 \pi_* \mathcal{H} \rightarrow R^1 \pi_* \mathcal{F} \rightarrow R^1 \pi_* \mathcal{G} \rightarrow R^1 \pi_* \mathcal{H} \rightarrow \cdots$$

(d) *If  $\pi$  is projective,  $Y$  is locally Noetherian, and  $\mathcal{F}$  is a coherent sheaf on  $X$ , then for all  $i$ ,*

$$R^i \pi_* \mathcal{F}$$

*is a coherent sheaf on  $Y$ .*

**Example 1.321** (Computation for the projection  $\mathbf{P}^1 \times Y \rightarrow Y$ ). *Let  $Y$  be a scheme, and let*

$$p : \mathbf{P}^1 \times Y \rightarrow Y, \quad q : \mathbf{P}^1 \times Y \rightarrow \mathbf{P}^1$$

*be the two projections. We compute the higher direct images of  $q^*\mathcal{O}_{\mathbf{P}^1}(n)$ .*

*For every affine open  $U = \operatorname{Spec} A \subseteq Y$ , we have*

$$p^{-1}(U) \cong \mathbf{P}_A^1, \quad q^*\mathcal{O}_{\mathbf{P}^1}(n)|_{p^{-1}(U)} \cong \mathcal{O}_{\mathbf{P}_A^1}(n).$$

*Hence*

$$(R^i p_* q^* \mathcal{O}_{\mathbf{P}^1}(n))(U) \cong H^i(\mathbf{P}_A^1, \mathcal{O}_{\mathbf{P}_A^1}(n)).$$

*Since cohomology of  $\mathcal{O}_{\mathbf{P}^1}(n)$  commutes with extension of scalars in this case (because all of the terms in the Čech complex for the standard affine cover are free  $A$ -modules), we have*

$$H^i(\mathbf{P}_A^1, \mathcal{O}_{\mathbf{P}_A^1}(n)) \cong H^i(\mathbf{P}_k^1, \mathcal{O}_{\mathbf{P}^1}(n)) \otimes_k A.$$

*Therefore*

$$R^i p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong H^i(\mathbf{P}^1, \mathcal{O}_{\mathbf{P}^1}(n)) \otimes_k \mathcal{O}_Y.$$

*Now recall the standard cohomology of line bundles on  $\mathbf{P}^1$ :*

$$h^0(\mathbf{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0, \end{cases} \quad h^1(\mathbf{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n \leq -2. \end{cases}$$

*Thus*

$$p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong \begin{cases} \mathcal{O}_Y^{\oplus(n+1)}, & n \geq 0, \\ 0, & n < 0, \end{cases}$$

*and*

$$R^1 p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong \begin{cases} 0, & n \geq -1, \\ \mathcal{O}_Y^{\oplus(-n-1)}, & n \leq -2. \end{cases}$$

*Moreover,*

$$R^i p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) = 0 \quad \text{for all } i \geq 2,$$

*since the fibers of  $p$  are 1-dimensional.*

### 1.7.5 Formal functions

Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, let  $\mathcal{F}$  be a coherent sheaf on  $X$ , and let  $y \in Y$ . For each  $n \geq 1$ , define

$$X_n = X \times_Y \operatorname{Spec}(\mathcal{O}_{Y,y}/\mathfrak{m}_y^n).$$

Thus  $X_1 = X_y$  is the fiber over  $y$ , and for  $n > 1$ ,  $X_n$  is an infinitesimal thickening of  $X_y$ . Let

$$v : X_n \rightarrow X$$

be the natural map, and set

$$\mathcal{F}_n = v^* \mathcal{F}.$$



For each  $n$ , there are natural base-change maps

$$R^i f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y,y}/\mathfrak{m}_y^n \longrightarrow R^i f'_*(\mathcal{F}_n),$$

where  $f' : X_n \rightarrow \operatorname{Spec}(\mathcal{O}_{Y,y}/\mathfrak{m}_y^n)$  is the induced morphism. Since  $\operatorname{Spec}(\mathcal{O}_{Y,y}/\mathfrak{m}_y^n)$  is affine and supported at one point, the right-hand side is

$$H^i(X_n, \mathcal{F}_n).$$

As  $n$  varies, these groups form an inverse system. Passing to inverse limits gives a natural map

$$(R^i f_* \mathcal{F})_y^\wedge \longrightarrow \varprojlim_n H^i(X_n, \mathcal{F}_n).$$

**Theorem 1.322** (Theorem on Formal Functions). *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, let  $\mathcal{F}$  be a coherent sheaf on  $X$ , and let  $y \in Y$ . Then the natural map*

$$(R^i f_* \mathcal{F})_y^\wedge \longrightarrow \varprojlim_n H^i(X_n, \mathcal{F}_n)$$

*is an isomorphism for all  $i \geq 0$ .*

**Remark 1.323.** *The theorem says that the completed stalk of the higher direct image  $R^i f_* \mathcal{F}$  at  $y$  is determined by the cohomology of the infinitesimal thickenings of the fiber  $X_y$ .*

**Corollary 1.324.** *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, and set*

$$r = \max\{\dim X_y : y \in Y\}.$$

*Then*

$$R^i f_*(\mathcal{F}) = 0$$

*for all  $i > r$  and all coherent sheaves  $\mathcal{F}$  on  $X$ .*

*Proof.* For any  $y \in Y$ , the schemes  $X_n$  have the same underlying topological space as  $X_y$ . Hence  $\dim X_n = \dim X_y \leq r$ . Therefore

$$H^i(X_n, \mathcal{F}_n) = 0$$

for all  $i > r$ . By the theorem on formal functions,

$$(R^i f_* \mathcal{F})_y^\wedge = 0$$

for all  $y \in Y$  and all  $i > r$ . Since  $R^i f_* \mathcal{F}$  is coherent, this implies  $R^i f_* \mathcal{F} = 0$  for all  $i > r$ .  $\square$

In the last step, we are invoking a Nakayama/faithful flatness of completion argument for Noetherian local rings. In particular, if for a finitely generated module  $M$  over a Noetherian local ring  $(A, \mathfrak{m})$ , if  $\widehat{M} = 0$ , then  $M/\mathfrak{m}M = 0$ , so Nakayama gives  $M = 0$ .

**Corollary 1.325.** *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, and assume*

$$f_*\mathcal{O}_X = \mathcal{O}_Y.$$

*Then every fiber  $f^{-1}(y)$  is connected.*

*Proof.* Suppose, for contradiction, that

$$f^{-1}(y) = X' \sqcup X''$$

where  $X'$  and  $X''$  are nonempty disjoint closed subsets. Then for each  $n$ , the thickened fiber  $X_n$  decomposes as

$$X_n = X'_n \sqcup X''_n,$$

and hence

$$H^0(X_n, \mathcal{O}_{X_n}) = H^0(X'_n, \mathcal{O}_{X_n}) \oplus H^0(X''_n, \mathcal{O}_{X_n}).$$

Taking inverse limits and applying formal functions with  $\mathcal{F} = \mathcal{O}_X$  gives

$$\widehat{\mathcal{O}}_{Y,y} = (f_*\mathcal{O}_X)_y^\wedge \cong \varprojlim_n H^0(X_n, \mathcal{O}_{X_n}) = A' \oplus A'',$$

where

$$A' = \varprojlim_n H^0(X'_n, \mathcal{O}_{X_n}), \quad A'' = \varprojlim_n H^0(X''_n, \mathcal{O}_{X_n}).$$

But  $\widehat{\mathcal{O}}_{Y,y}$  is a local ring, and a local ring cannot be written as a nontrivial direct product. Indeed, a nontrivial direct product would give nontrivial idempotents, while a local ring has only the idempotents 0 and 1. This contradiction shows that  $f^{-1}(y)$  is connected.  $\square$

**Corollary 1.326** (Zariski's Main Theorem). *Let  $f : X \rightarrow Y$  be a birational projective morphism of Noetherian integral schemes, and assume that  $Y$  is normal. Then every fiber  $f^{-1}(y)$  is connected.*

*Proof.* By the previous corollary, it suffices to prove

$$f_*\mathcal{O}_X = \mathcal{O}_Y.$$

This question is local on  $Y$ , so assume  $Y = \operatorname{Spec} A$ . Since  $f$  is projective,  $f_*\mathcal{O}_X$  is coherent, so

$$B = \Gamma(Y, f_*\mathcal{O}_X) = \Gamma(X, \mathcal{O}_X)$$

is a finite  $A$ -module. Since  $f$  is birational and  $X, Y$  are integral,  $A$  and  $B$  have the same fraction field. Moreover,  $B$  is integral over  $A$ , because it is finite as an  $A$ -module. But  $A$  is integrally closed, since  $Y$  is normal. Hence

$$A = B.$$

Therefore

$$f_*\mathcal{O}_X = \mathcal{O}_Y,$$

and the fibers of  $f$  are connected.  $\square$

**Corollary 1.327** (Stein factorization). *Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes. Then  $f$  factors as*

$$X \xrightarrow{f'} Y' \xrightarrow{g} Y,$$

where  $f'$  is projective with connected fibers and  $g$  is finite.

*Proof.* Let  $Y' = \operatorname{Spec}_Y(f_*\mathcal{O}_X)$ . Since  $f_*\mathcal{O}_X$  is a coherent sheaf of  $\mathcal{O}_Y$ -algebras, the natural morphism  $g : Y' \rightarrow Y$  is finite. By construction, the morphism  $f : X \rightarrow Y$  factors through  $g$ , so we obtain

$$f' : X \rightarrow Y'.$$

Since  $g$  is separated and  $f$  is projective, the induced morphism  $f'$  is projective. Moreover, by construction,

$$f'_*\mathcal{O}_X = \mathcal{O}_{Y'}.$$

Therefore, by the connected-fibers corollary, the fibers of  $f'$  are connected. Hence

$$X \xrightarrow{f'} Y' \xrightarrow{g} Y$$

is the desired factorization.  $\square$

## 1.8 Curves

### 1.8.1 Valuations and local rings

**Definition 1.328.** A **curve** is a one-dimensional variety, i.e. an integral separated scheme of finite type over a field  $k$  such that every irreducible component has dimension 1.

**Definition 1.329.** Let  $K$  be a field. A **valuation** on  $K$  is a function

$$v : K^\times \rightarrow \mathbb{Z}$$

satisfying the following properties:

- For all  $x, y \in K^\times$ ,  $v(xy) = v(x) + v(y)$ .
- For all  $x, y \in K^\times$  with  $x + y \neq 0$ ,  $v(x + y) \geq \min\{v(x), v(y)\}$ .

The set  $R = \{x \in K : v(x) \geq 0\}$  is a subring of  $K$  called the **valuation ring** of  $v$ . It is a local ring with maximal ideal  $\mathfrak{m} = \{x \in K : v(x) > 0\} \cup \{0\}$ . If  $v(k^\times) = 0$ , then we say that  $v$  is a **discrete valuation** on  $K/k$ .

**Exercise 1.330.** Let  $k = \bar{k}$  and let  $K = k(t)$ . Show that every nontrivial discrete valuation

$$v : K^\times \rightarrow \mathbb{Z}$$

which is trivial on  $k^\times$  is either the valuation at a finite point  $t = a$ , for some  $a \in k$ , or the valuation at infinity. Thus the nontrivial discrete valuations of  $k(t)$  over  $k$  naturally correspond to the points of  $\mathbb{P}_k^1$ .

**Solution 1.331.** Let  $v : k(t)^\times \rightarrow \mathbb{Z}$  be a nontrivial discrete valuation such that  $v(c) = 0$  for all  $c \in k^\times$ . We study the values  $v(t - a)$  for  $a \in k$ . First, at most one of these values can be positive. Indeed, if  $a \neq b$  and  $v(t - a) > 0, v(t - b) > 0$  then the difference  $0 = v(b - a) \geq \min\{v(t - a), v(t - b)\} > 0$  is a contradiction.

Suppose first that  $v(t - a) > 0$  for some  $a \in k$ . Then for every  $b \neq a$ ,  $v(t - b) = 0$  for otherwise we would have  $0 = v(b - a) \geq \min\{v(t - a), v(t - b)\} > 0$ . Since  $k$  is algebraically closed, every polynomial factors as  $f(t) = c \prod_i (t - a_i)$ . Therefore only the factors with  $a_i = a$  contribute to  $v(f)$ , and  $v(f) = v(t - a) \operatorname{ord}_{t=a}(f)$ . Thus for every rational function  $f/g$ ,  $v(f/g) = v(t - a) \operatorname{ord}_{t=a}(f/g)$ . If  $v$  is normalized, this gives  $v = \operatorname{ord}_{t=a}$ .

Now suppose no  $v(t - a)$  is positive. Then  $v(t - a) \leq 0$  for all  $a \in k$ . If all these values were zero, then every polynomial, and hence every rational function, would have valuation zero, contradicting that  $v$  is nontrivial. Hence for some  $a \in k$ ,  $v(t - a) < 0$ . For any  $b \in k$ ,  $t - b = (t - a) + (a - b)$ , and since  $v(t - a) < 0$ ,  $v(a - b) = 0$ , we get  $v(t - b) = v(t - a)$ . Write  $v(t - a) = -m$  with  $m > 0$ . Then  $v(t - b) = -m$  for all  $b \in k$ . Therefore, if  $f(t)$  is a polynomial of degree  $d$ , then  $v(f) = -md$ . For a rational function  $f/g$ , we get  $v(f/g) = -m \deg(f) + m \deg(g) = m(\deg(g) - \deg(f))$ . But  $\operatorname{ord}_\infty(f/g) = \deg(g) - \deg(f)$ . Hence  $v = m \operatorname{ord}_\infty$ . Normalizing gives  $v = \operatorname{ord}_\infty$ . Thus every nontrivial discrete valuation of  $k(t)$  over  $k$  is, up to normalization, either  $\operatorname{ord}_{t=a}$  for some  $a \in k$ , or  $\operatorname{ord}_\infty$ . These are precisely the valuations corresponding to the points of  $\mathbb{P}_k^1$ .

Note that if  $k$  is not algebraically closed, for example if  $k = \mathbb{R}$ , then there would be additional valuations corresponding to the irreducible quadratics. However these valuations are still coming from the closed points of  $\mathbb{P}_k^1$ .

**Theorem 1.332.** Let  $(R, \mathfrak{m})$  be a Noetherian local domain of dimension 1. Then the following are equivalent:

1.  $R$  is a discrete valuation ring;
2.  $R$  is integrally closed;
3.  $R$  is a regular local ring;
4.  $\mathfrak{m}$  is a principal ideal.

A **Dedekind domain** is a Noetherian integrally closed domain of dimension 1. Being dimension 1 means that all the prime ideals of  $R$  are either  $(0)$  or maximal. The local rings of a Dedekind domain are discrete valuation rings.

**Example 1.333.** Let  $X$  be a nonsingular affine curve over  $\bar{k}$ , and let

$$R = \Gamma(X, \mathcal{O}_X)$$

be its ring of regular functions. Then  $R$  is a Noetherian domain of dimension 1.

Moreover,  $R$  is integrally closed. Indeed, integral closedness can be checked locally, it suffices to show that for every maximal ideal  $\mathfrak{m} \subset R$ , the local ring  $R_{\mathfrak{m}}$  is integrally closed.

But  $R_{\mathfrak{m}} \cong \mathcal{O}_{X,x}$  where  $x \in X$  is the point corresponding to  $\mathfrak{m}$ . Since  $X$  is nonsingular,  $\mathcal{O}_{X,x}$  is a regular local ring. Because  $X$  is a curve, this is a Noetherian local domain of dimension 1. Hence, by the equivalence between regular one-dimensional local domains and DVRs,  $\mathcal{O}_{X,x}$  is a DVR, and in particular is integrally closed.

**Theorem 1.334.** *Let  $R$  be a Dedekind domain that is a finitely generated  $k$ -algebra, and let  $K$  be its field of fractions. Let  $L$  be a finite extension of  $K$ , and let  $S$  be the integral closure of  $R$  in  $L$ . Then  $S$  is also a Dedekind domain, and is also a finitely generated  $k$ -algebra.*

**Lemma 1.335.** *Suppose  $(S, \mathfrak{n})$ ,  $(R, \mathfrak{m})$  are two discrete valuation rings of  $K/k$ , with  $S \subset R$  and  $\mathfrak{n} = S \cap \mathfrak{m}$ . Then  $S = R$  and  $\mathfrak{n} = \mathfrak{m}$ : they are the same valuation ring.*

**Theorem 1.336.** *Let  $K/\bar{k}$  be a finitely generated field extension of transcendence degree 1, and let  $x \in K^\times$ . Then there are only finitely many discrete valuations  $v$  of  $K/\bar{k}$  such that  $v(x) < 0$ .*

*Proof.* It is enough to prove the corresponding statement for zeros. Set  $y = x^{-1}$ . Then  $v(x) < 0$  if and only if  $v(y) > 0$ . Equivalently, if  $R_v$  is the valuation ring of  $v$  with maximal ideal  $\mathfrak{m}_v$ , then  $v(y) > 0$  if and only if  $y \in \mathfrak{m}_v$ . Thus we must show that there are only finitely many discrete valuation rings  $(R_v, \mathfrak{m}_v)$  of  $K/\bar{k}$  such that  $y \in \mathfrak{m}_v$ .

If  $y \in \bar{k}$ , then  $v(y) = 0$  for every valuation of  $K/\bar{k}$ , so there is nothing to prove. Hence assume  $y \notin \bar{k}$ . Since  $\text{trdeg}_{\bar{k}} K = 1$ , the element  $y$  is transcendental over  $\bar{k}$ , and  $K$  is a finite extension of  $\bar{k}(y)$ . Let  $B$  be the integral closure of  $\bar{k}[y]$  inside  $K$ . By finiteness of integral closure,  $B$  is a finitely generated  $\bar{k}$ -algebra. It is also a Dedekind domain with fraction field  $K$ .

Let  $(R_v, \mathfrak{m}_v)$  be a discrete valuation ring of  $K/\bar{k}$  such that  $y \in \mathfrak{m}_v$ . Then  $\bar{k}[y] \subset R_v$ . Since  $R_v$  is integrally closed in  $K$  (being a DVR), and  $B$  is the integral closure of  $\bar{k}[y]$  in  $K$ , we get  $B \subset R_v$ . Set  $\mathfrak{n} = \mathfrak{m}_v \cap B$ . Then  $\mathfrak{n}$  is a maximal ideal of  $B$ , and since  $y \in \mathfrak{m}_v$ , we have  $y \in \mathfrak{n}$ . Thus every such valuation gives a point  $\mathfrak{n} \in V(y) \subset \text{Spec } B$ .

Conversely, the valuation  $R_v$  is determined by this point. Indeed, localizing gives  $B_{\mathfrak{n}} \subset R_v$ . Since  $B$  is a Dedekind domain,  $B_{\mathfrak{n}}$  is a DVR. Both  $B_{\mathfrak{n}}$  and  $R_v$  are DVRs inside the same field  $K$ , and their maximal ideals are compatible:  $\mathfrak{n}B_{\mathfrak{n}} \subset \mathfrak{m}_v$ . By the elementary lemma that two DVRs  $S \subset R$  of the same field with compatible maximal ideals must be equal, we get  $B_{\mathfrak{n}} = R_v$ . Hence the valuations with  $y \in \mathfrak{m}_v$  inject into the set  $V(y) = \{\mathfrak{n} \in \text{Spec } B : y \in \mathfrak{n}\}$ .

It remains to note that  $V(y)$  is finite. Since  $B$  is a finitely generated  $\bar{k}$ -domain of dimension 1 and  $0 \neq y \in B$ , the quotient  $B/(y)$  is a zero-dimensional finitely generated  $\bar{k}$ -algebra. Hence it is Artinian, and therefore has only finitely many maximal ideals. Thus  $V(y)$  is finite.

Consequently, there are only finitely many valuations with  $v(y) > 0$ , and therefore only finitely many valuations with  $v(x) < 0$ .  $\square$

The construction of  $\text{Spec } B$  along with a maximal ideal  $\mathfrak{n} \subset V(y) \subset \text{Spec } B$  implies the following.

**Corollary 1.337.** *Let  $v$  be a discrete valuation of the field  $K/\bar{k}$ , a finitely generated function field of transcendence degree 1. Then there is a nonsingular curve  $C$  with function field  $K$ , and a point  $p \in C$ , such that  $v$  is the valuation induced by  $p$ .*

Note that the affine model for such a valuation may change depending on the choice of  $y$ . However, there is a common projective curve behind all of them. There is a unique smooth projective curve  $C$  with function field  $K$  over  $\bar{k}$ . Each choice of  $y \in K \setminus \bar{k}$  gives an affine open  $Y \subset C$  obtained by removing the finitely many poles of  $y$ .

**Corollary 1.338.** *Let  $y \in K \setminus \bar{k}$ . Then all discrete valuations of  $K/\bar{k}$  such that  $v(y) \geq 0$  are accounted for by points on the curve  $Y$ .*

*Proof.*  $v(y) \geq 0$  means  $y \in R_v$ . In particular  $y$  has a residue class in  $R_v/m_v$ . If  $y \equiv 0 \in R_v/m_v$ , then  $y \in m_v$ , so  $v(y) > 0$  and apply the proof of the previous theorem. If  $y \not\equiv 0 \in R_v/m_v$ , then  $y - a$  has valuation 0 for some  $a \in \bar{k}$ , so  $v(y - a) = 0$ . Here we are using the algebraic closedness of  $\bar{k}$  to identify the residue field of  $R_v$  with  $\bar{k}$ , so that  $\bar{k}$  sits inside  $K$  with valuation 0.  $\square$

**Theorem 1.339.** *Let  $X$  be a nonsingular curve, let  $p \in X$ , let  $Y$  be a projective variety, and let  $\phi: X - \{p\} \rightarrow Y$  be a morphism. Then there exists a unique morphism  $\bar{\phi}: X \rightarrow Y$  extending  $\phi$ .*

*Proof.* Since  $Y$  is a variety, it is separated. Hence uniqueness follows: two morphisms  $X \rightarrow Y$  which agree on the dense open subset  $X - \{p\}$  must agree on all of  $X$ .

It remains to prove existence. Since  $Y$  is projective, choose a closed immersion  $Y \hookrightarrow \mathbb{P}^n$ . It suffices to show that the composite  $X - \{p\} \rightarrow Y \hookrightarrow \mathbb{P}^n$  extends to a morphism  $X \rightarrow \mathbb{P}^n$ . Indeed, once such an extension exists, its image lands in  $Y$ , because  $Y \subset \mathbb{P}^n$  is closed and the image of the dense open subset  $X - \{p\}$  lies in  $Y$ .

Thus assume  $Y = \mathbb{P}^n$ , with homogeneous coordinates  $x_0, \dots, x_n$ . Let  $f: X - \{p\} \rightarrow \mathbb{P}^n$  be the given morphism. After changing homogeneous coordinates if necessary, we may assume that the image of  $f$  meets the open set where all  $x_i$  are nonzero. Then, for each  $i, j$ , the rational function  $x_i/x_j$  on  $\mathbb{P}^n$  pulls back to a rational function on  $X$ , which we denote by  $f_{ij} \in k(X)$ .

Let  $v = v_p$  be the valuation on  $k(X)$  associated to the nonsingular point  $p$ . Fix the index 0, and set  $r_i = v(f_{i0})$  for  $i = 0, \dots, n$ . Since  $f_{ij} = f_{i0}/f_{j0}$ , we have  $v(f_{ij}) = r_i - r_j$ . Choose  $k$  such that  $r_k$  is minimal among the  $r_i$ . Then for every  $i$ , we have  $v(f_{ik}) = r_i - r_k \geq 0$ . Hence each  $f_{ik}$  is regular at  $p$ .

Define a set-theoretic extension  $\bar{f}: X \rightarrow \mathbb{P}^n$  by setting  $\bar{f} = f$  on  $X - \{p\}$  and

$$\bar{f}(p) = [f_{0k}(p) : \dots : f_{nk}(p)].$$

This is a well-defined point of  $\mathbb{P}^n$ , since  $f_{kk} = 1$ , so the  $k$ th coordinate is 1.

It remains to check that  $\bar{f}$  is a morphism near  $p$ . Let  $U_k \subset \mathbb{P}^n$  be the affine chart where  $x_k \neq 0$ . Since  $\bar{f}(p) \in U_k$ , it is enough to show that  $\bar{f}$  is a morphism from some affine neighborhood of  $p$  into  $U_k$ . The coordinate ring of  $U_k$  is

$$k[x_0/x_k, \dots, \widehat{x_k/x_k}, \dots, x_n/x_k].$$

Under  $\bar{f}$ , the coordinate function  $x_i/x_k$  pulls back to  $f_{ik}$ . But each  $f_{ik}$  is regular at  $p$ , hence regular on some affine neighborhood of  $p$ . Therefore the functions  $f_{0k}, \dots, f_{nk}$  define a morphism from a neighborhood of  $p$  to  $U_k$ . This agrees with  $f$  on the punctured neighborhood, so  $\bar{f}$  is a morphism near  $p$ .

Thus  $f$  extends to a morphism  $X \rightarrow \mathbb{P}^n$ , and consequently  $\phi$  extends to a morphism  $X \rightarrow Y$ . This proves existence, and uniqueness was already shown.  $\square$

**Remark 1.340.** *There are two basic ways the extension theorem can fail.*

*First, the target must be proper. For example, the rational function  $t$  defines a rational map*

$$\mathbb{P}^1 \dashrightarrow \mathbb{A}^1$$

*which is regular on  $\mathbb{A}^1 \subset \mathbb{P}^1$  but has a pole at  $\infty$ . It cannot extend to a morphism  $\mathbb{P}^1 \rightarrow \mathbb{A}^1$ , since every global regular function on  $\mathbb{P}^1$  is constant.*

*Second, the source must be a nonsingular curve. For example,*

$$\mathbb{A}^2 \dashrightarrow \mathbb{P}^1, \quad (x, y) \mapsto [x : y],$$

*is defined away from the origin but cannot be extended over  $(0, 0)$ . Along the line  $y = mx$ , the limiting value is  $[1 : m]$ , which depends on the tangent direction. Algebraically, an extension with value in the chart  $X \neq 0$  would require  $y/x$  to be regular at the origin, while an extension with value in the chart  $Y \neq 0$  would require  $x/y$  to be regular.*

**Exercise 1.341.** *Show that if  $U$  and  $V$  are two open subsets of  $\mathbb{A}^1$ , then  $U \simeq V$  only if  $\mathbb{A}^1 - U$  and  $\mathbb{A}^1 - V$  have the same number of points. Is the converse true?*

**Solution 1.342.** *Let  $U = \mathbb{A}^1 - \{P_1, \dots, P_r\}$  and  $V = \mathbb{A}^1 - \{Q_1, \dots, Q_s\}$ . View both as open subsets of  $\mathbb{P}^1$ . Then  $\mathbb{P}^1 - U = \{\infty, P_1, \dots, P_r\}$  and  $\mathbb{P}^1 - V = \{\infty, Q_1, \dots, Q_s\}$ .*

*Suppose  $U \simeq V$ . Since  $U$  and  $V$  are dense open subsets of  $\mathbb{P}^1$ , this isomorphism gives a birational map  $\mathbb{P}^1 \dashrightarrow \mathbb{P}^1$ . Because  $\mathbb{P}^1$  is nonsingular and projective, the map and its inverse extend to morphisms  $\mathbb{P}^1 \rightarrow \mathbb{P}^1$ . These extensions are inverse to each other, since their compositions agree with the identity on dense open subsets. Hence the original isomorphism extends to an automorphism of  $\mathbb{P}^1$ .*

*Therefore the automorphism sends the complement of  $U$  in  $\mathbb{P}^1$  bijectively onto the complement of  $V$  in  $\mathbb{P}^1$ . Thus  $|\mathbb{P}^1 - U| = |\mathbb{P}^1 - V|$ , so  $r + 1 = s + 1$ , and hence  $r = s$ .*

*The converse is false in general. For example, over  $k = \mathbb{C}$ , consider  $U = \mathbb{A}^1 - \{0, 1, 2\}$  and  $V = \mathbb{A}^1 - \{0, 1, 3\}$ . If  $U \simeq V$ , then the isomorphism would extend to an automorphism*

of  $\mathbb{P}^1$  carrying  $\{\infty, 0, 1, 2\}$  onto  $\{\infty, 0, 1, 3\}$ . Checking the cross ratio of these four points shows that no such automorphism exists.

**Theorem 1.343.** *Every nonsingular separated curve  $C$  is quasiprojective.*

*Proof.* Since  $C$  is Noetherian, we may cover it by finitely many nonsingular affine open subsets

$$C = \bigcup_{i=1}^r U_i.$$

For each  $i$ , choose a closed immersion  $U_i \hookrightarrow \mathbb{P}^{n_i}$ . Let  $Y_i \subset \mathbb{P}^{n_i}$  be the projective closure of  $U_i$ .

The open immersion  $U_i \hookrightarrow Y_i$  defines a morphism from the open subset  $U_i \subset C$  to the projective variety  $Y_i$ . Since  $C$  is nonsingular and  $Y_i$  is projective, the extension theorem for morphisms from punctured nonsingular curves implies that this map extends uniquely to a morphism

$$\phi_i : C \rightarrow Y_i.$$

Taking the product of these morphisms gives

$$\phi = (\phi_1, \dots, \phi_r) : C \rightarrow \prod_{i=1}^r Y_i.$$

Since each  $Y_i$  is projective, their product is projective. By the Segre embedding, we may view  $\prod_i Y_i$  as a closed subvariety of some projective space  $\mathbb{P}^N$ . Let

$$Y = \overline{\phi(C)} \subset \mathbb{P}^N$$

be the closure of the image. Then  $Y$  is projective, and  $\phi(C)$  is a locally closed subset of  $Y$ . Since  $C$  is a curve and  $\phi$  is nonconstant on each irreducible component,  $Y$  has dimension 1.

We claim that  $\phi$  is an isomorphism from  $C$  onto its image  $\phi(C)$ . It is enough to construct a morphism

$$\psi : \phi(C) \rightarrow C$$

which is inverse to  $\phi$  on points.

Fix  $i$ . On  $U_i$ , the morphism  $\phi$  factors as

$$U_i \longrightarrow C \xrightarrow{\phi} \prod_j Y_j \longrightarrow Y_i,$$

and the composite is precisely the original open immersion

$$U_i \hookrightarrow Y_i.$$

Therefore, on the subset  $\phi(U_i) \subset \phi(C)$ , we may recover the point of  $U_i$  by projecting to  $Y_i$  and then using the inverse of the open immersion onto its image. Thus we obtain a morphism

$$\psi_i : \phi(U_i) \rightarrow U_i$$



which is inverse to  $\phi|_{U_i}$ .

On overlaps  $\phi(U_i) \cap \phi(U_j)$ , the morphisms  $\psi_i$  and  $\psi_j$  agree on the dense open subset corresponding to  $U_i \cap U_j$ . Since  $C$  is separated, morphisms into  $C$  which agree on a dense open subset agree wherever both are defined. Hence the  $\psi_i$  glue to a morphism

$$\psi : \phi(C) \rightarrow C.$$

By construction,  $\psi$  is inverse to  $\phi$  on points. Therefore  $\phi : C \rightarrow \phi(C)$  is an isomorphism. Thus  $C$  is isomorphic to the locally closed subset  $\phi(C)$  of the projective variety  $Y \subset \mathbb{P}^N$ . Hence  $C$  is quasiprojective.  $\square$

**Corollary 1.344.** *Every nonsingular separated curve is birational to a projective curve.*

**Remark 1.345.** *The proof invokes two important facts that I was not familiar.*

*Let  $f : X \rightarrow Y$  be a morphism of varieties over a field  $k$ . Then  $\dim \overline{f(X)} \leq \dim X$ . In particular, if  $X$  is a curve and  $f$  is nonconstant, then  $\overline{f(X)}$  is a curve.*

*Proof.* First assume  $X$  is irreducible, and let  $Z = \overline{f(X)}$  with the reduced induced structure. Then  $f$  factors through a dominant morphism  $X \rightarrow Z$ , hence induces an inclusion of function fields  $k(Z) \hookrightarrow k(X)$ . Therefore  $\text{trdeg}_k k(Z) \leq \text{trdeg}_k k(X)$ . Since dimension equals transcendence degree for irreducible varieties,  $\dim Z \leq \dim X$ .

In general, if  $X_1, \dots, X_r$  are the irreducible components of  $X$ , then  $\overline{f(X)} = \bigcup_i \overline{f(X_i)}$ , so  $\dim \overline{f(X)} = \max_i \dim \overline{f(X_i)} \leq \max_i \dim X_i = \dim X$ .  $\square$

*Let  $X$  and  $Y$  be reduced separated schemes. Suppose there are morphisms  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$  which are inverse to one another on the set of closed points. Then  $f$  and  $g$  are inverse isomorphisms.*

*Proof.* We show  $g \circ f = \text{id}_X$ ; the proof of  $f \circ g = \text{id}_Y$  is identical. Consider the two morphisms  $g \circ f, \text{id}_X : X \rightarrow X$ . Pull the diagonal  $\Delta_X \subset X \times X$  back along the morphism  $(g \circ f, \text{id}_X) : X \rightarrow X \times X$ . This gives a closed subscheme  $E \rightarrow X$  whose set of closed points  $E$  is the set of points  $x \in X$  such that the two maps to  $X$  agree as morphisms after passing to the residue field  $\kappa(x)$ . Being the pullback of a closed immersion,  $E \rightarrow X$  is a closed immersion.

By assumption,  $g \circ f$  and  $\text{id}_X$  agree on the underlying topological space, so  $E$  has underlying closed subset all of  $X$ . If  $\mathcal{I} \subset \mathcal{O}_X$  is the ideal sheaf defining  $E \hookrightarrow X$ , then  $\sqrt{\mathcal{I}} = 0$ . Since  $X$  is reduced,  $\mathcal{I} = 0$ . Hence  $E = X$ , so  $g \circ f = \text{id}_X$  as morphisms.

Similarly, using that  $Y$  is reduced and separated, one gets  $f \circ g = \text{id}_Y$ . Therefore  $f$  and  $g$  are inverse isomorphisms.  $\square$

**Proposition 1.346.** *Let  $C_1$  and  $C_2$  be separated nonsingular curves, and let*

$$\alpha : C_1 \dashrightarrow C_2$$

be a birational map. Then  $C_1$  and  $C_2$  can be glued along  $\alpha$ : there exists a nonsingular curve

$$C = U_1 \cup U_2$$

such that  $U_i \simeq C_i$ , and the birational map  $U_1 \dashrightarrow U_2$  induced by this gluing is  $\alpha$ .

*Proof.* Since  $C_1$  and  $C_2$  are birational, there are dense open subsets which are identified by  $\alpha$ . Thus we may choose a common dense open curve

$$V \subset C_1, \quad V \subset C_2.$$

By quasiprojectivity of nonsingular separated curves, we may regard each  $C_i$  as an open subvariety of a projective curve

$$C_i \subset Y_i \subset \mathbb{P}^{n_i}.$$

Consider the morphism

$$\phi : V \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$$

given by the two embeddings of  $V$  into  $C_1$  and  $C_2$ .

The rational map

$$\phi_1 : C_1 \dashrightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$$

extends to a morphism, because  $C_1$  is nonsingular and the target is projective. Similarly, we get a morphism

$$\phi_2 : C_2 \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}.$$

Moreover, each  $\phi_i$  is an isomorphism onto its image: for example, projecting

$$\phi_1(C_1) \subset \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$$

to the first factor recovers the original embedding of  $C_1$ .

Now define

$$C = \phi_1(C_1) \cup \phi_2(C_2) \subset \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}.$$

The two images contain the common open subset  $\phi(V)$ , and their union is precisely the gluing of  $C_1$  and  $C_2$  along the birational identification given by  $\alpha$ . Since each  $\phi_i(C_i)$  is nonsingular and the two curves agree on the overlap,  $C$  is nonsingular. Also, since  $C$  is a subvariety of a projective variety, it is separated.  $\square$

The above proposition suggests how to construct a projective curve birational to a given nonsingular separated curve  $C$ . Let  $K/\bar{k}$  be a finitely generated field extension of transcendence degree 1. Choose  $y \in K \setminus \bar{k}$  and let  $B$  be the integral closure of  $\bar{k}[y]$  in  $K$ . Then  $Y_0 = \text{Spec } B$  is a nonsingular affine curve with function field  $K$ . Its closed points correspond to the discrete valuations  $v$  of  $K/\bar{k}$  for which  $v(y) \geq 0$ .

The missing valuations are precisely those for which  $v(y) < 0$ . By the finiteness theorem, there are only finitely many such valuations, say  $v_1, \dots, v_r$ . For each  $v_i$ , choose a nonsingular affine curve  $Y_i$  with function field  $K$  such that  $v_i$  is represented by a closed point of  $Y_i$ . Since all the  $Y_i$  have the same function field  $K$ , they are birational. Gluing the finitely many curves  $Y_0, Y_1, \dots, Y_r$  along their common birational open subsets gives a nonsingular separated curve  $C$  with function field  $K$ .

By construction, every discrete valuation of  $K/\bar{k}$  occurs as the local ring at some point of  $C$ . Thus  $C$  has no missing valuation points. This is the precise sense in which one has glued finitely many birational affine curves together until the result is complete. Since  $C$  is nonsingular and separated, it is quasiprojective; and the valuative criterion then shows that  $C$  is proper. Hence  $C$  is projective.

**Lemma 1.347.** *Let  $X$  be a quasiprojective scheme over  $k$ . If  $X$  is proper over  $k$ , then  $X$  is projective over  $k$ .*

*Proof.* Since  $X$  is quasiprojective, there is an open immersion

$$j : X \hookrightarrow \bar{X}$$

with  $\bar{X}$  projective over  $k$ . Since  $X$  is proper over  $k$  and  $\bar{X}$  is separated over  $k$ , the morphism  $j : X \rightarrow \bar{X}$  is proper. But an open immersion which is proper is a closed immersion. Hence  $X$  is isomorphic to a closed subscheme of the projective  $k$ -scheme  $\bar{X}$ . Therefore  $X$  is projective over  $k$ .  $\square$

**Proposition 1.348.** *If two nonsingular projective curves  $C_1$  and  $C_2$  are birational, then they are isomorphic.*

*Proof.* Let

$$\phi_1 : C_1 \dashrightarrow C_2, \quad \phi_2 : C_2 \dashrightarrow C_1$$

be inverse birational maps. Since  $C_1$  and  $C_2$  are nonsingular and projective, the extension theorem for morphisms from nonsingular curves to projective varieties implies that both rational maps extend to morphisms

$$\phi_1 : C_1 \rightarrow C_2, \quad \phi_2 : C_2 \rightarrow C_1.$$

On the dense open subsets where the original birational maps are defined, the compositions satisfy

$$\phi_2 \circ \phi_1 = \text{id}_{C_1}, \quad \phi_1 \circ \phi_2 = \text{id}_{C_2}.$$

Since  $C_1$  and  $C_2$  are separated, two morphisms into them which agree on a dense open subset agree everywhere. Hence the above equalities hold globally. Therefore  $\phi_1$  and  $\phi_2$  are inverse isomorphisms.  $\square$

**Corollary 1.349.** *The following categories are equivalent, up to the usual contravariance of function fields:*

1. *nonsingular projective curves over  $\bar{k}$ , with dominant morphisms;*
2. *curves over  $\bar{k}$ , with dominant rational maps;*
3. *function fields of transcendence degree 1 over  $\bar{k}$ , with  $\bar{k}$ -homomorphisms.*

We state the algebraic version of Riemann–Hurwitz, which is valid for morphisms of smooth projective curves over an algebraically closed field  $k$ , subject to the condition that

the morphism is generically separable, meaning that the induced extension of function fields is separable.

For a finite separable morphism  $f : X \rightarrow Y$  of curves, we define the **ramification divisor** of  $f$  to be

$$R = \sum_{P \in X} \text{length}_{\mathcal{O}_{X,P}}((\Omega_{X/Y})_P) \cdot P.$$

**Proposition 1.350.** *Let  $f : X \rightarrow Y$  be a finite separable morphism of curves. Let  $K_X$  and  $K_Y$  be canonical divisors of  $X$  and  $Y$ , respectively. Then*

$$K_X \sim f^*K_Y + R.$$

*Proof.* By the exact sequence of differentials,

$$0 \rightarrow f^*\Omega_Y \rightarrow \Omega_X \rightarrow \Omega_{X/Y} \rightarrow 0.$$

Viewing  $R$  as a closed subscheme of  $X$ , the structure sheaf  $\mathcal{O}_R$  is isomorphic to  $\Omega_{X/Y}$ . Tensoring the exact sequence above with  $\Omega_X^{-1}$ , we get an exact sequence

$$0 \rightarrow f^*\Omega_Y \otimes \Omega_X^{-1} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_R \rightarrow 0.$$

Thus  $f^*\Omega_Y \otimes \Omega_X^{-1}$  is the ideal sheaf of  $R$ . Since the ideal sheaf of an effective divisor  $R$  is  $\mathcal{O}_X(-R)$ , we have

$$f^*\Omega_Y \otimes \Omega_X^{-1} \cong \mathcal{O}_X(-R) \quad \square$$

Taking degrees in the above relation gives the algebraic version of Riemann–Hurwitz.

$$2g(X) - 2 = (\deg f)(2g(Y) - 2) + \deg R.$$

Note that in positive characteristic, the ramification divisor  $R$  can have degree larger than  $\sum_{P \in X} (e_P - 1)$ , due to the presence of wild ramification. In particular, if  $p = \text{char}(k)$  divides  $e$  in the expression  $dt = (ew^{e-1} + \dots)dw$ , then the contribution to  $R$  from this point can be larger than  $e - 1$ .

**Definition 1.351.** *Let  $X$  be a scheme whose local rings are all of characteristic  $p$ . The **Frobenius morphism** of  $X$  is the morphism whose underlying map of topological spaces is the identity, and whose pullback on structure sheaves is given by the  $p$ th power map  $\mathcal{O}_X \rightarrow \mathcal{O}_X$ ,  $f \mapsto f^p$ .*

If  $\pi : X \rightarrow \text{Spec } k$  is the structure map, then the Frobenius morphism of  $X$  fails to be  $k$ -linear. Therefore, we define a new scheme  $X_{(p)}$  over  $k$  to be the same underlying scheme as  $X$ , but with the structure map given by  $F \circ \pi : X \rightarrow \text{Spec } k$ . Then the Frobenius morphism  $F : X \rightarrow X$  is a morphism of schemes over  $k$  from  $X_{(p)}$  to  $X$ . We call  $X_{(p)}$  the **Frobenius twist** of  $X$ , and there  $F : X_{(p)} \rightarrow X$  is the **relative Frobenius morphism**.

**Example 1.352.** *The map  $F : X_p \rightarrow X$  is ramified everywhere, with ramification index  $p$ . This is because in a local coordinate  $t$  at a point of  $X$ , the pullback of  $t$  under  $F$  is  $t^p$ . Since  $d(t^p) = pt^{p-1}dt = 0$ , the map  $f^*\Omega_X \rightarrow \Omega_{X_p}$  is the zero map, so  $\Omega_{X_p/X} \cong \Omega_{X_p}$  which is supported at every point of  $X_p$ . In particular, there is no ramification divisor.*

Over nonalgebraically closed fields, field extensions of transcendence degree 1 factor into separable and purely inseparable extensions. The separable extensions correspond to geometric morphisms of curves, while the purely inseparable extensions correspond to Frobenius twists.

**Proposition 1.353.** *Let  $f : X \rightarrow Y$  be a purely inseparable morphism of curves. Then  $X$  and  $Y$  are abstractly isomorphic, and  $f$  factors as a composition of  $k$ -linear Frobenius morphisms. In particular  $g(X) = g(Y)$ .*

**Theorem 1.354.** *Let  $f : X \rightarrow Y$  be a finite morphism of smooth projective curves. Then*

$$g(X) \geq g(Y).$$

*Proof.* The extension  $K(Y) \subseteq K(X)$  factors as a separable extension followed by a purely inseparable extension. Since purely inseparable extensions do not change genus, we reduce to the case where  $f$  is separable.

If  $g(Y) = 0$ , there is nothing to prove. Otherwise  $g(Y) \geq 1$ . By Riemann–Hurwitz,

$$2g(X) - 2 = n(2g(Y) - 2) + \deg R.$$

Equivalently,

$$g(X) = g(Y) + (n - 1)(g(Y) - 1) + \frac{1}{2} \deg R.$$

Since  $n - 1 \geq 0$ ,  $g(Y) - 1 \geq 0$ , and  $\deg R \geq 0$ , we get

$$g(X) \geq g(Y).$$

Equality in the separable case occurs only if  $n = 1$ , or  $g(Y) = 1$  and  $f$  is unramified.  $\square$

**Theorem 1.355** (Lüroth). *Let  $k \subseteq L \subseteq k(t)$ . If  $L \neq k$ , then*

$$L \cong k(u)$$

*for some  $u$ . In other words,  $L$  is purely transcendental over  $k$ .*

*Proof.* Since  $L \neq k$ , it has transcendence degree 1 over  $k$ , so  $L = K(Y)$  for some smooth projective curve  $Y$ . The inclusion

$$K(Y) = L \subseteq k(t) = K(\mathbb{P}^1)$$

corresponds to a finite morphism

$$\mathbb{P}^1 \rightarrow Y.$$

By the previous theorem,

$$0 = g(\mathbb{P}^1) \geq g(Y).$$

Hence  $g(Y) = 0$ . Therefore  $Y \cong \mathbb{P}^1$ , so

$$L = K(Y) \cong k(u)$$

for some parameter  $u$ .  $\square$

**Exercise 1.356.** Let  $Y$  be a curve over a field  $k$  with  $\text{char } k \neq 2$ . Show that finite étale morphisms  $f : X \rightarrow Y$  of degree 2 are classified by 2-torsion line bundles  $\mathcal{L} \in \text{Pic}(Y)$ , i.e. line bundles satisfying

$$\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y.$$

**Solution 1.357.** Let  $f : X \rightarrow Y$  be a finite étale morphism of degree 2. Then  $f_*\mathcal{O}_X$  is a locally free  $\mathcal{O}_Y$ -algebra of rank 2 because finite flat modules over local rings are finite free. There is a natural inclusion  $\mathcal{O}_Y \hookrightarrow f_*\mathcal{O}_X$  coming from pullback of functions. Let  $\mathcal{L}$  be the cokernel:

$$0 \rightarrow \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X \rightarrow \mathcal{L} \rightarrow 0.$$

Since  $f_*\mathcal{O}_X$  is locally free of rank 2, the quotient  $\mathcal{L}$  is locally free of rank 1, hence an invertible sheaf. Moreover taking determinants gives an isomorphism  $\mathcal{L} \cong \det(f_*\mathcal{O}_X)$

Because  $f$  is finite étale of degree 2, locally on  $Y$  it is isomorphic to

$$\text{Spec}(A \times A) \rightarrow \text{Spec } A.$$

Thus locally

$$f_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{O}_Y.$$

In particular,  $\det(f_*\mathcal{O}_X)$  is locally trivial, and the algebra structure implies

$$\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y.$$

Equivalently, the quotient line bundle  $\mathcal{L}$  is 2-torsion in  $\text{Pic}(Y)$ .

Conversely, suppose  $\mathcal{L}$  is an invertible sheaf with an isomorphism

$$\varphi : \mathcal{L}^{\otimes 2} \xrightarrow{\sim} \mathcal{O}_Y.$$

Define an  $\mathcal{O}_Y$ -algebra structure on

$$\mathcal{A} := \mathcal{O}_Y \oplus \mathcal{L}$$

by

$$(a, b)(a', b') = (aa' + \varphi(b \otimes b'), ab' + a'b).$$

Then set

$$X := \text{Spec}_Y(\mathcal{A}).$$

Since  $\mathcal{A}$  is locally free of rank 2 as an  $\mathcal{O}_Y$ -module, the morphism

$$f : X \rightarrow Y$$

is finite flat of degree 2.

It remains to check that it is étale. This is local on  $Y$ . Trivialize  $\mathcal{L}$  on an open set  $U$ , choosing a generator  $s$ . Under the trivialization, the isomorphism

$$\varphi : \mathcal{L}^{\otimes 2} \rightarrow \mathcal{O}_Y$$

sends

$$s \otimes s \mapsto u$$

for some unit  $u \in \mathcal{O}_Y(U)^\times$ . Then over  $U$ ,

$$\mathcal{A}|_U \cong \mathcal{O}_U[T]/(T^2 - u).$$

Because  $\text{char } k \neq 2$  and  $u$  is a unit, the derivative

$$2T$$

is invertible on the algebra  $\mathcal{O}_U[T]/(T^2 - u)$ , since  $T^2 = u$  is a unit. Therefore

$$\mathcal{O}_U[T]/(T^2 - u)$$

is étale over  $\mathcal{O}_U$ . Hence  $X \rightarrow Y$  is finite étale of degree 2.

Finally, we check that the two constructions are inverse. Starting from  $\mathcal{L}$ , the algebra

$$\mathcal{O}_Y \oplus \mathcal{L}$$

has an involution

$$(a, b) \mapsto (a, -b).$$

Thus  $\mathcal{O}_Y$  is the invariant part and  $\mathcal{L}$  is the anti-invariant part.

Conversely, starting from a degree 2 étale cover  $f : X \rightarrow Y$ , there is an involution

$$\tau : X \rightarrow X$$

which interchanges the two points in each geometric fiber. It induces an involution on  $f_*\mathcal{O}_X$ . Since  $\text{char } k \neq 2$ , we can decompose

$$f_*\mathcal{O}_X = (f_*\mathcal{O}_X)^+ \oplus (f_*\mathcal{O}_X)^-,$$

where

$$(f_*\mathcal{O}_X)^+ = \{a : \tau(a) = a\}, \quad (f_*\mathcal{O}_X)^- = \{a : \tau(a) = -a\}.$$

The invariant part is

$$(f_*\mathcal{O}_X)^+ \cong \mathcal{O}_Y,$$

and the anti-invariant part is a line bundle  $\mathcal{L}$ . Therefore

$$f_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{L}.$$

Multiplication in  $f_*\mathcal{O}_X$  sends

$$\mathcal{L} \otimes \mathcal{L} \rightarrow \mathcal{O}_Y,$$

because the product of two anti-invariant sections is invariant. This gives an isomorphism

$$\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y.$$

Thus the cover is recovered from the corresponding 2-torsion line bundle, and the two constructions are inverse.

### 1.8.2 Structure theory

**Theorem 1.358** (Local normal form and ramification index). *Let  $f : R \rightarrow S$  be a nonconstant holomorphic map between Riemann surfaces, and let  $p \in R$  with  $q := f(p)$ . Then there exist local holomorphic coordinates*

$$z : (R, p) \xrightarrow{\sim} (\mathbb{C}, 0), \quad w : (S, q) \xrightarrow{\sim} (\mathbb{C}, 0),$$

*and an integer  $k \geq 1$  such that*

$$w \circ f \circ z^{-1}(\zeta) = \zeta^k$$

*for all  $\zeta$  in a neighbourhood of 0. The integer  $k$  is uniquely determined and is called the **ramification index** of  $f$  at  $p$ .*

**Corollary 1.359.** *Let  $f : R \rightarrow S$  be a nonconstant holomorphic map between Riemann surfaces. Then  $f$  is an open map.*

**Corollary 1.360.** *Every holomorphic function defined everywhere on a compact connected Riemann surface is constant.*

**Exercise 1.361.** *Show that if  $R$  is a compact Riemann surface,  $R - \{p\}$  is affine for every  $p \in R$ .*

*Proof.* By Riemann-Roch for sufficiently large  $n$ , there exists

$$f \in H^0(C, \mathcal{O}_C(np)) \setminus H^0(C, \mathcal{O}_C((n-1)p))$$

This means  $f$  has pole of order exactly  $n$  at  $p$ , and no poles elsewhere.

So  $f$  defines a nonconstant morphism  $f : C \rightarrow \mathbf{P}^1$  whose only pole is at  $p$ . Therefore  $f^{-1}(\infty) = \{p\}$ . Restricting gives  $f|_U : U = C \setminus \{p\} \rightarrow \mathbf{A}^1$ .

Since any morphism from a projective curve to  $\mathbf{P}^1$  is finite when nonconstant,  $f$  is finite, hence so is  $f|_U$ . Therefore  $U$  is finite over  $\mathbf{A}^1$ , so  $U$  is affine.  $\square$

### 1.8.3 Genus and degree

**Theorem 1.362.** *Let  $f : R \rightarrow S$  be a nonconstant holomorphic map between compact Riemann surfaces. Pick  $s \in S$  and define the quantity*

$$d := \sum_{p \in f^{-1}(s)} e_p$$

*Then  $d$  is independent of the choice of  $s \in S$  and is called the **degree** of the map  $f$ .*

**Theorem 1.363** (Riemann–Hurwitz). *Let  $f : R \rightarrow S$  be a nonconstant holomorphic map of compact Riemann surfaces of degree  $d$ . Write*

$$B := \sum_{p \in R} (e_p - 1)p$$



for the ramification divisor of  $f$ . Then

$$2g_R - 2 = d(2g_S - 2) + \deg(B),$$

or equivalently,

$$\chi(R) = d\chi(S) - \sum_{p \in R} (e_p - 1).$$

*Proof via Euler characteristic.* If  $f$  were an unramified covering map, then  $R$  would be a  $d$ -fold cover of  $S$  and by the description of Euler characteristic as the alternating sum of cells in a cell decomposition, we would have

$$\chi(R) = d\chi(S).$$

However, ramification changes this count. We have to delete  $e_p - 1$  points of  $R$  for each ramification point  $p$ , and each deletion drops the Euler characteristic by 1. Thus we obtain the desired formula.  $\square$

*Proof via canonical bundles.* Let  $K_R$  and  $K_S$  denote the canonical bundles of  $R$  and  $S$ . Choose local coordinates  $w$  near  $p \in R$  and  $z$  near  $f(p)$  so that the map is written  $z = w^{e_p}$ . Then

$$f^*(dz) = d(w^{e_p}) = e_p w^{e_p-1} dw,$$

so  $f^*(dz)$  has a zero of order  $e_p - 1$  at  $p$ . Globalizing, one obtains an isomorphism of line bundles

$$K_R \cong f^*K_S \otimes \mathcal{O}_R(B),$$

where  $B = \sum (e_p - 1)p$  is the ramification divisor.

Taking degrees on both sides, and using  $\deg K_X = 2g_X - 2$ , yields

$$2g_R - 2 = d(2g_S - 2) + \sum_p (e_p - 1),$$

the desired identity.  $\square$

**Remark 1.364.** These two proofs are related by the fact a holomorphic 1-form  $\omega$  on a compact Riemann surface satisfies  $\deg(\omega) = 2g - 2 = -\chi(X)$ . In particular, there is the isomorphism of line bundles  $K_R = f^*K_S + B$ , which refines the Riemann–Hurwitz formula.

**Exercise 1.365.** Show that there are no nonconstant maps  $f : C \rightarrow D$  between smooth projective curves of genera  $g_C$  and  $g_D$  if  $g_C < g_D$  and  $g_D \geq 2$ .

**Solution 1.366.** If  $f : C \rightarrow D$  is a nonconstant map, then by Riemann–Hurwitz,

$$2g_C - 2 = \deg(f)(2g_D - 2) + \deg(B) \geq \deg(f)(2g_D - 2).$$

Since  $\deg(f) \geq 1$  and  $g_D \geq 2$ , we have  $2g_D - 2 > 0$ , so

$$2g_C - 2 \geq 2g_D - 2,$$

which implies  $g_C \geq g_D$ .

**Exercise 1.367.** Show that all maps  $\mathbb{P}^1 \rightarrow E$  are constant.

**Solution 1.368.** If  $f : \mathbb{P}^1 \rightarrow E$  is a nonconstant map, then pulling back a nonzero holomorphic 1-form on  $E$  gives a nonzero holomorphic 1-form on  $\mathbb{P}^1$ , which is impossible since  $K_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$  has no nonzero global sections.

**Theorem 1.369** (Adjunction formula). If  $X$  is a smooth variety and  $D \subset X$  is a smooth effective Cartier divisor, then

$$\omega_D \cong (\omega_X \otimes \mathcal{O}_X(D))|_D.$$

Equivalently, at the divisor-class level,

$$K_D = (K_X + D)|_D.$$

*Proof.* The conormal exact sequence

$$0 \rightarrow \mathcal{I}_D/\mathcal{I}_D^2 \rightarrow \Omega_X^1|_D \rightarrow \Omega_D^1 \rightarrow 0$$

becomes, since  $D$  is Cartier,

$$0 \rightarrow \mathcal{O}_X(-D)|_D \rightarrow \Omega_X^1|_D \rightarrow \Omega_D^1 \rightarrow 0.$$

Taking determinants gives the desired isomorphism.  $\square$

**Remark 1.370.** Geometrically, adjunction can be understood through residues. Suppose  $\dim X = n$ , and locally  $D \subset X$  is cut out by  $z = 0$ . Choose local coordinates  $z, x_1, \dots, x_{n-1}$  on  $X$ , so that  $x_1, \dots, x_{n-1}$  are coordinates along  $D$ . Then a local generator of  $\omega_X$  is  $dz \wedge dx_1 \wedge \dots \wedge dx_{n-1}$ , while a local generator of  $\omega_D$  is  $dx_1 \wedge \dots \wedge dx_{n-1}$ .

Any local section of  $\omega_X(D)$  can be written locally as

$$\omega = \frac{dz}{z} \wedge \alpha + \beta$$

where  $\alpha$  is a regular  $(n-1)$ -form and  $\beta$  is a regular  $n$ -form. Then define

$$\text{Res}_D(\omega) = \alpha|_D$$

If  $z' = uz$  for a unit  $u$ , then

$$\frac{dz'}{z'} = \frac{d(uz)}{uz} = \frac{dz}{z} + \frac{du}{u}$$

so the residue map gives a well-defined isomorphism  $\omega_D \cong \omega_X(D)|_D$ . Thus canonical forms on  $D$  are residues of ambient canonical forms with a simple pole along  $D$ . Allowing such a simple pole is exactly twisting by  $\mathcal{O}_X(D)$ , so

$$\omega_D \cong \omega_X(D)|_D = (\omega_X \otimes \mathcal{O}_X(D))|_D$$

**Exercise 1.371** (POAC 2.2). If  $C \subset \mathbb{P}^r$  is a 1-dimensional variety with normalization  $\varphi : \tilde{C} \rightarrow C$ , and  $X \subset \mathbb{P}^r$  is any subscheme that does not contain  $C$ , we define the multiplicity of intersection  $\text{mult}_p(C, X)$  at the point  $p \in X \cap C$  to be the sum of the lengths of the finite scheme  $\varphi^{-1}(X)$  at the points of  $\varphi^{-1}(p)$ . Show that  $C$  is singular at  $p$  if and only if  $\text{mult}_p(C, X) \geq 2$  for all  $X$  containing  $p$ . Show further that if  $H \subset \mathbb{P}^r$  is a hyperplane then

$$\deg C = \sum_{p \in C} \text{mult}_p(C, H).$$

**Solution 1.372.**

1. Let  $I_X$  be the ideal sheaf of  $X$  in  $C$ , that is it is a subsheaf of  $\mathcal{O}_C$  such that for any open set  $U \subseteq C$ ,  $I_X(U)$  is the ideal of  $\mathcal{O}_C(U)$  consisting of functions that vanish on  $X \cap U$ . By definition, we have the formula

$$\text{mult}_p(C, X) = \sum_{q \in \varphi^{-1}(p)} \ell(\mathcal{O}_{\varphi^{-1}(X), q}) = \sum_{q \in \varphi^{-1}(p)} \ell(\mathcal{O}_{\tilde{C}, q} / I_X \mathcal{O}_{\tilde{C}, q}).$$

where the last equality follows from the definition of the scheme theoretic pullback.

Recall that if  $f : X \rightarrow Y$  is a morphism of schemes and let  $Z \hookrightarrow Y$  be a closed subscheme defined by an ideal sheaf  $\mathcal{I}_Z \subset \mathcal{O}_Y$ . Then the scheme-theoretic preimage is  $f^{-1}(Z) := X \times_Y Z = \text{Spec}_X(\mathcal{O}_X / \mathcal{I}_Z \mathcal{O}_X)$ . So at the sheaf level,  $\mathcal{O}_{f^{-1}(Z)} = \mathcal{O}_X / \mathcal{I}_Z \mathcal{O}_X$ .

Recall that the local ring  $\mathcal{O}_{\tilde{C}, q}$  is a discrete valuation ring and every nonzero ideal in a DVR is principal, generated by a power of a uniformizer  $t$ .

Let  $q_1, \dots, q_s$  be the points of  $\tilde{C}$  over  $p$  (the branches). If  $s \geq 2$  (multibranch singularity, e.g. a node), then for any  $X \ni p$ ,  $\phi^* I_X$  vanishes at each branch, so  $\text{ord}_{q_i}(\phi^* I_X) \geq 1$  for all  $i$ . Hence  $\text{mult}_p(C, X) = \sum_i \text{ord}_{q_i}(\phi^* I_X) \geq s \geq 2$ . If  $s = 1$  (unibranch singularity, e.g. a cusp), let  $q$  be the unique point over  $p$  with parameter  $t$ . Locally we have  $A = \mathcal{O}_{C, p}$  is a 1-dimensional Noetherian local ring that is not regular (since  $p$  is singular).  $R = \mathcal{O}_{\tilde{C}, q}$  is a DVR, with uniformizer  $t$ . So  $A \subset R \subset \text{Frac}(A)$ . If  $A$  has an element of order 1 in  $R$ , then  $A = R$  is regular, contradicting the singularity of  $p$ . So  $\text{ord}_q(\phi^* I_X) \geq 2$  for all  $X \ni p$ , hence  $\text{mult}_p(C, X) = \text{ord}_q(\phi^* I_X) \geq 2$ .

Conversely, if  $p$  is nonsingular, then we can take  $X = H$  the tangent hyperplane to  $C$  at  $p$ . Then  $\phi^* I_H$  has order 1 at the unique point  $q$  over  $p$ , so  $\text{mult}_p(C, H) = 1$ .

Let  $L := \mathcal{O}_{\mathbb{P}^r}(1)|_C$ . A hyperplane  $H$  corresponds to a section  $s_H \in H^0(C, L)$ . Pulling back,  $\phi^* s_H \in H^0(\tilde{C}, \phi^* L)$  and its divisor is exactly  $\phi^{-1}(H)$ . Therefore

$$\deg C = \deg L = \deg \phi^* L = \deg \text{div}(\phi^* s_H) = \sum_{p \in C} \text{mult}_p(C, H)$$

This holds for any hyperplane  $H$  not containing  $C$ .

**Remark 1.373.** Note that if you take a nodal curve  $C \subset \mathbb{P}^2 \subset \mathbb{P}^3$  and pick a  $\mathbb{P}^1$  which passes through the node transversely, then the resulting intersection is in fact reduced

and smooth. For example if you take a nodal curve  $(w, xy)$  and  $(x - \beta w, y - \alpha w)$  and  $\alpha, \beta \in k$ . Then  $(w, xy) + (x - \beta w, y - \alpha w) = (w, x, y)$ , I was just suprised that you can start with something singular, intersect with something else and get something smooth. This is the basis of Bertini-type theorems: a general enough hyperplane cut of a singular variety often avoids carrying the singularity forward, unless you cut right along the tangent cone.

**Exercise 1.374.** Let  $Q = \mathbb{P}^1 \times \mathbb{P}^1$ , and let  $C \subset Q$  be a smooth curve of bidegree  $(4, 3)$ .

1. Compute the genus of  $C$ .
2. Let  $\pi := p_1|_C : C \rightarrow \mathbb{P}^1$  be the restriction to  $C$  of the first projection. Compute the degree of  $\pi$ , and determine the total ramification of  $\pi$ .
3. Let  $L = \mathcal{O}_Q(4, 3)$ . Compute  $p_{1*}L$ .
4. Let  $L_C = \mathcal{O}_C(C) = \mathcal{O}_Q(4, 3)|_C$ , the normal bundle of  $C$  in  $Q$ . Describe  $\pi_*L_C$ .

Note that something I learned in this exercise is that  $\mathcal{O}_Q(-C)$  is the ideal sheaf of  $C$  in  $Q$ , and  $\mathcal{O}_C(C)$  is the normal bundle of  $C$  in  $Q$ . The first makes sense because if one asks for functions vanishing along  $C$ , then these are of course the functions that cut out  $C$  as a subscheme. If  $f$  is a local defining equation for  $C$  in  $Q$ , then we are saying that  $1/f$  is a local generator of the normal bundle  $\mathcal{O}_Q(C)$ , which makes sense because a local section looks like  $g/f$  with  $g$  a regular function on  $Q$ , and restricting to  $C$  corresponds to modding out by  $\mathcal{O}_Q$  (recall the following exact sequence):

$$0 \rightarrow \mathcal{O}_Q \rightarrow \mathcal{O}_Q(C) \rightarrow \mathcal{O}_C(C) \rightarrow 0$$

so we remember only the first order data of  $g \bmod f$ .

**Solution 1.375.** Write  $F_1$  and  $F_2$  for the divisor classes of the two rulings on  $Q = \mathbb{P}^1 \times \mathbb{P}^1$ . Then  $C \sim 4F_1 + 3F_2$ . For smooth varieties  $K_{X \times Y} \cong p_X^*K_X \otimes p_Y^*K_Y$ . Therefore, the canonical class on  $Q$  is  $K_Q = -2F_1 - 2F_2$ . By adjunction,

$$\deg K_C = 2g(C) - 2 = C \cdot (C + K_Q) = (4F_1 + 3F_2) \cdot (2F_1 + F_2) = 4 \cdot 1 + 3 \cdot 2 = 10$$

It follows that  $g(C) = 6$ . In general for a smooth curve of type  $(a, b)$  on  $\mathbb{P}^1 \times \mathbb{P}^1$ , one has  $g(C) = (a - 1)(b - 1)$ .

**(2) Degree and ramification of  $\pi$ .** The degree of  $\pi = p_1|_C$  is the number of intersection points of  $C$  with a fiber of  $p_1$ , counted with multiplicity. A fiber of  $p_1$  has class  $F_1$ , so

$$\deg(\pi) = C \cdot F_1 = (4F_1 + 3F_2) \cdot F_1 = 3.$$

Now apply the Riemann–Hurwitz formula:

$$2g(C) - 2 = \deg(\pi)(2g(\mathbb{P}^1) - 2) + \deg(R),$$

where  $R$  is the ramification divisor. Since  $g(C) = 6$ ,  $g(\mathbb{P}^1) = 0$ , and  $\deg(\pi) = 3$ , we obtain

$$2 \cdot 6 - 2 = 3(-2) + \deg(R).$$

so  $\deg(R) = 16$ .

**(3) The pushforward  $p_{1*}L$ .** Since

$$L = \mathcal{O}_Q(4, 3) \cong p_1^* \mathcal{O}_{\mathbb{P}^1}(4) \otimes p_2^* \mathcal{O}_{\mathbb{P}^1}(3),$$

the projection formula gives

$$p_{1*}L \cong \mathcal{O}_{\mathbb{P}^1}(4) \otimes p_{1*}(p_2^* \mathcal{O}_{\mathbb{P}^1}(3)).$$

But

$$p_{1*}(p_2^* \mathcal{O}_{\mathbb{P}^1}(3)) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(3)) \otimes \mathcal{O}_{\mathbb{P}^1},$$

is the trivial vector bundle of rank 4 on  $\mathbb{P}^1$ . This is because  $p_1 : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$  is the projection from a product, and the sheaf  $p_2^* \mathcal{O}_{\mathbb{P}^1}(3)$  is constant in the first factor and equals  $\mathcal{O}_{\mathbb{P}^1}(3)$  along each fiber of  $p_1$ . Therefore

$$p_{1*} \mathcal{O}_Q(4, 3) \cong \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4}.$$

**(4) The pushforward  $\pi_* L_C$ .** Consider the exact sequence on  $Q$ :

$$0 \rightarrow \mathcal{O}_Q \rightarrow \mathcal{O}_Q(4, 3) \rightarrow \mathcal{O}_C(C) \rightarrow 0.$$

Pushing forward by  $p_1$ , and using

$$p_{1*} \mathcal{O}_Q \cong \mathcal{O}_{\mathbb{P}^1}, \quad R^1 p_{1*} \mathcal{O}_Q = 0, \quad R^1 p_{1*} \mathcal{O}_Q(4, 3) = 0,$$

we obtain an exact sequence

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^1} \rightarrow \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4} \rightarrow \pi_* \mathcal{O}_C(C) \rightarrow 0.$$

Hence  $\pi_* \mathcal{O}_C(C)$  is a rank 3 vector bundle on  $\mathbb{P}^1$ , namely the cokernel of

$$\mathcal{O}_{\mathbb{P}^1} \longrightarrow \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4}.$$

Its degree is

$$\deg(\pi_* \mathcal{O}_C(C)) = 4 \cdot 4 = 16.$$

By the Birkhoff–Grothendieck theorem,  $\pi_* \mathcal{O}_C(C)$  splits as a direct sum of three line bundles on  $\mathbb{P}^1$  whose degrees add up to 16, but the exact splitting type depends on the particular curve  $C$ .

**Exercise 1.376.** Reason about the following facts about a smooth plane curve  $C \subset \mathbb{P}^2$  of degree  $d$ :

$$1. \ g(C) = \frac{(d-1)(d-2)}{2}.$$

$$2. \ \omega_C \cong \mathcal{O}_C(d-3).$$

$$3. \ \deg \mathcal{O}_C(1) = d.$$

**Solution 1.377.** A smooth plane curve  $C \subset \mathbb{P}^2$  of degree  $d$  fits into

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \rightarrow \mathcal{O}_{\mathbb{P}^2} \rightarrow \mathcal{O}_C \rightarrow 0.$$

Taking Euler characteristics gives

$$1 - g(C) = \chi(\mathcal{O}_C) = \chi(\mathcal{O}_{\mathbb{P}^2}) - \chi(\mathcal{O}_{\mathbb{P}^2}(-d)) = 1 - \frac{(d-1)(d-2)}{2}$$

where we know that  $h^0(\mathcal{O}_{\mathbb{P}^2}) = 1$  and  $h^1(\mathcal{O}_{\mathbb{P}^2}) = h^2(\mathcal{O}_{\mathbb{P}^2}) = 0$ , and  $h^0(\mathcal{O}_{\mathbb{P}^2}(-d)) = 0$  for  $d > 0$ ,  $h^1(\mathcal{O}_{\mathbb{P}^2}(-d)) = 0$  for all  $d$ , and  $h^2(\mathcal{O}_{\mathbb{P}^2}(-d)) = h^0(\mathcal{O}_{\mathbb{P}^2}(d-3)) = \frac{(d-1)(d-2)}{2}$  for  $d \geq 3$ . Also adjunction gives

$$\omega_C \cong (K_{\mathbb{P}^2} \otimes \mathcal{O}_{\mathbb{P}^2}(C))|_C \cong \mathcal{O}_C(d-3).$$

For the third statement,  $\deg \mathcal{O}_C(1) = C \cdot H = d$  where  $H$  is a hyperplane class in  $\mathbb{P}^2$ , since  $C$  has degree  $d$  by definition.

**Exercise 1.378.** Give an example of a smooth curve of genus 2.

**Solution 1.379.** A standard example is the smooth projective completion of  $y^2 = x^5 - 1$ . More conceptually, any smooth double cover of  $\mathbb{P}^1$  branched at 6 points has genus 2. Another realization is as a smooth divisor of type  $(3, 2)$  on  $\mathbb{P}^1 \times \mathbb{P}^1$ , since such a curve has genus  $(3-1)(2-1) = 2$ .

**Exercise 1.380.** What is the genus of the projective completion of the affine curve defined by  $x^n - y^n = 1$ ? Use the degree-genus formula for plane curves, and then use Riemann–Hurwitz to check your answer.

**Solution 1.381.** The projective closure is  $X^n - Y^n + Z^n = 0 \subset \mathbb{P}^2$ , which is smooth if  $\text{char } k \nmid n$ . Since it is a smooth plane curve of degree  $n$ , its genus is

$$\frac{(n-1)(n-2)}{2}.$$

Alternatively, project to the  $y$ -coordinate:

$$\pi : C \rightarrow \mathbb{P}^1.$$

This has degree  $n$ . It is branched exactly over the  $n$  points with  $y^n = 1$ , and over each such point there is one totally ramified point of index  $n$ . So the total ramification is  $n(n-1)$ . Riemann–Hurwitz gives

$$2g - 2 = -2n + n(n-1),$$

hence

$$g = \frac{(n-1)(n-2)}{2}.$$

**Exercise 1.382.** Write down an explicit holomorphic differential on the smooth projective curve  $V(x^4 + y^4 + z^4) \subset \mathbb{P}^2$ .

**Solution 1.383.** *Let*

$$C = V(x^4 + y^4 + z^4) \subset \mathbb{P}^2.$$

*On the chart  $U_z = \{z \neq 0\}$ , write*

$$X = \frac{x}{z}, \quad Y = \frac{y}{z}.$$

*Then*

$$C \cap U_z = V(X^4 + Y^4 + 1),$$

*and we define*

$$\omega_z = \frac{dX}{4Y^3}.$$

*This is regular away from  $Y = 0$ . If  $Y = 0$ , then  $X \neq 0$ , and differentiating*

$$X^4 + Y^4 + 1 = 0$$

*gives*

$$4X^3 dX + 4Y^3 dY = 0,$$

*so*

$$\frac{dX}{4Y^3} = -\frac{dY}{4X^3},$$

*which is regular near  $Y = 0$ . Hence  $\omega_z$  is regular on  $C \cap U_z$ .*

*It remains to check regularity at infinity. On  $U_y = \{y \neq 0\}$ , write*

$$U = \frac{x}{y}, \quad V = \frac{z}{y}.$$

*On  $U_z \cap U_y$ ,*

$$X = \frac{U}{V}, \quad Y = \frac{1}{V}.$$

*Therefore*

$$dX = \frac{V dU - U dV}{V^2}, \quad Y^3 = \frac{1}{V^3}.$$

*Thus*

$$\omega_z = \frac{dX}{4Y^3} = \frac{V(V dU - U dV)}{4},$$

*which is regular on  $C \cap U_y$ .*

*Similarly, on  $U_x = \{x \neq 0\}$ , write*

$$R = \frac{y}{x}, \quad S = \frac{z}{x}.$$

*Using*

$$\omega_z = -\frac{dY}{4X^3},$$

*and noting that*

$$X = \frac{1}{S}, \quad Y = \frac{R}{S},$$

*we get*

$$dY = \frac{S dR - R dS}{S^2}, \quad X^3 = \frac{1}{S^3}.$$

### 1.8.4 Riemann–Roch theorem

Let  $R$  be a compact Riemann surface, and let  $L$  be a holomorphic line bundle on  $R$ . The Riemann–Roch theorem is the best possible answer to the question of finding the dimension of the space  $\Gamma(R; L)$  of holomorphic sections of a line bundle by topological methods. This question does not have a purely topological answer, as seen from the following.

**Example 1.384.** *Let  $g(R) > 0$ , fix a point  $y \in R$  and consider the family of divisors  $D_x = x - y$  on  $R$ , parametrized by  $x \in R$ . For  $x \neq y$ ,  $D_x$  is not principal, because a meromorphic function with a simple pole at  $x$  and no other poles would describe a degree one map  $R \rightarrow \mathbb{P}^1$ , which would have to be an isomorphism. However,  $D_y = 0$  has a nontrivial holomorphic section: the function 1. So the dimension of the space of holomorphic sections can jump in a family. Note that all line bundles  $\mathcal{O}(D_x)$  have degree zero, so they are topologically trivial. This means that no topological information will detect  $\dim \Gamma(R; L)$ .*

The topologically invariant quantity turns out to be

$$\chi(R; L) := \dim H^0(R; L) - \dim H^1(R; L).$$

**Definition 1.385.** *The space  $\mathcal{P}_x(V)$  of **principal parts** at  $x \in R$  with coefficients in a holomorphic vector bundle  $V$  is the quotient*

$$\mathcal{M}_x(V)/\mathcal{O}_x(V)$$

*of the space of germs at  $x$  of meromorphic sections of  $V$  by the subspace of germs of holomorphic sections. The space  $\mathcal{P}(R; V)$  of principal parts over  $R$  is the direct sum, over all points  $x \in R$ , of the  $\mathcal{P}_x(V)$ .*

In a local coordinate centred at  $x$ ,  $\mathcal{P}_x(V)$  is the space of negative Laurent polynomials with values in  $V$ . For later use, we note that  $\mathcal{P}(R; V)$  is the space of sections of a sheaf  $\mathcal{P}(V)$  over  $R$ ; however, in order to satisfy the sheaf condition for (possibly infinite) coverings, one must allow on a general open set  $U$  a distribution of principal parts over any discrete, but possibly infinite set. It is the compactness of  $R$  that forces the restriction to finitely many points, and limits us to the direct sum. There is a linear map induced by the morphism of sheaves  $\mathcal{M}(V) \rightarrow \mathcal{P}(V)$

$$p : \Gamma(R; \mathcal{M}(V)) \longrightarrow \Gamma(R; \mathcal{P}(V))$$

which assigns to each meromorphic section its principal parts. The kernel of  $p$  consists of (the sheaf of) holomorphic sections of  $V$ .

**Definition 1.386.** *The **first cohomology group**  $H^1(R; V)$  with coefficients in the (sheaf of holomorphic sections of the) vector bundle  $V$  is the ratio*

$$H^1(R; V) := \mathcal{P}(R; V)/\text{Im}(p),$$

*that is, principal parts modulo global meromorphic sections.*



In other words, we have a four-term exact sequence

$$0 \longrightarrow H^0(R; V) \longrightarrow \Gamma(R; \mathcal{M}(V)) \xrightarrow{p} \mathcal{P}(R; V) \longrightarrow H^1(R; V) \longrightarrow 0 \quad (2)$$

The 0 is justified by the fact that  $H^1(\mathcal{M}(V)) = 0$  where  $\mathcal{M}(V)$  is the sheaf of meromorphic sections of  $V$ , and  $\mathcal{P} = \mathcal{M}/\mathcal{O}$  is the quotient of the sheaf of meromorphic sections by the sheaf of holomorphic sections, so the long exact sequence in cohomology gives the above short exact sequence. The vanishing of  $H^1(\mathcal{M}(V))$  is a consequence of the following fact.

**Exercise 1.387.** *Show that, on a compact Hausdorff space  $X$ , sheaf cohomology commutes with filtered colimits: if  $\mathcal{S} = \varinjlim \mathcal{S}_n$ , then*

$$\varinjlim H^q(X; \mathcal{S}_n) = H^q(X; \mathcal{S}).$$

*Use this to prove that the sheaf of principal parts of meromorphic sections of a vector bundle on a Riemann surface has no  $H^1$  or higher cohomology.*

**Solution 1.388.** *Let  $X$  compact Hausdorff and  $\{\mathcal{S}_\alpha\}$  a filtered direct system of sheaves. Let  $\mathcal{S} = \varinjlim_\alpha \mathcal{S}_\alpha$  be the colimit in sheaves.*

*Take a finite open cover  $\mathfrak{U} = \{U_i\}_{i=1}^m$  of  $X$  (exists since  $X$  is compact). For any  $p$ ,*

$$C^p(\mathfrak{U}, \mathcal{S}) = \prod_{i_0 < \dots < i_p} \mathcal{S}(U_{i_0} \cap \dots \cap U_{i_p}).$$

*We will use a couple of lemmas and justify them at the end of the solution.*

**Lemma 1.389.** *Filtered colimits are computed stalkwise and commute with finite products.*

*The lemma implies that the Čech complex for  $\mathcal{S}$  is the filtered colimit of the Čech complexes for  $\mathcal{S}_\alpha$  since we have the following isomorphisms which commute with the differentials:*

$$\begin{aligned} C^p(\mathfrak{U}, \mathcal{S}) &= \prod_{i_0 < \dots < i_p} \mathcal{S}(U_{i_0} \cap \dots \cap U_{i_p}) = \prod_{i_0 < \dots < i_p} \varinjlim_\alpha \mathcal{S}_\alpha(U_{i_0} \cap \dots \cap U_{i_p}) \\ &\cong \varinjlim_\alpha \prod_{i_0 < \dots < i_p} \mathcal{S}_\alpha(U_{i_0} \cap \dots \cap U_{i_p}) = \varinjlim_\alpha C^p(\mathfrak{U}, \mathcal{S}_\alpha). \end{aligned}$$

**Lemma 1.390.** *Filtered colimits of abelian groups are exact, and in particular taking cohomology commutes with  $\varinjlim$ :*

$$\check{H}^q(\mathfrak{U}; \mathcal{S}) \cong \varinjlim_\alpha \check{H}^q(\mathfrak{U}; \mathcal{S}_\alpha).$$

*On a compact Hausdorff space (in particular, on a Riemann surface) Čech cohomology with respect to a good cover computes sheaf cohomology. Hence*

$$H^q(X; \mathcal{S}) \cong \varinjlim_\alpha H^q(X; \mathcal{S}_\alpha)$$

for all  $q$ .

For the second part, let  $R$  be a Riemann surface and  $V$  a holomorphic vector bundle. For each effective divisor  $D$  on  $R$ , let  $\mathcal{P}_D(V)$  be the sheaf of principal parts of meromorphic sections of  $V$  with poles bounded by  $D$ .

Observe that  $\mathcal{P}_D(V)$  is supported on the finite set  $\text{Supp}(D)$ , so it is a finite direct sum of skyscraper sheaves.

**Lemma 1.391.** *A sheaf on  $R$  with finite support has no higher cohomology. In particular, for  $q \geq 1$ ,*

$$H^q(R; \mathcal{P}_D(V)) = 0,$$

The full principal parts sheaf is the filtered colimit over all effective divisors:

$$\mathcal{P}(V) = \varinjlim_D \mathcal{P}_D(V),$$

directed by  $D \leq D'$  if  $D$  divides  $D'$ . Applying the result from the first part:

$$H^q(R; \mathcal{P}(V)) \cong \varinjlim_D H^q(R; \mathcal{P}_D(V)).$$

But each  $H^q(R; \mathcal{P}_D(V)) = 0$  for  $q \geq 1$ , so the colimit is 0. Therefore

$$H^q(R; \mathcal{P}(V)) = 0 \quad \text{for all } q \geq 1.$$

Now we justify the lemmas used above.

*Proof of Lemma 1.389.* Define a presheaf  $\mathcal{F}(U) := \varinjlim_{\alpha} \mathcal{S}_{\alpha}(U)$ . We claim  $\mathcal{F}$  is already a sheaf. Then  $\mathcal{F}$  is the colimit in the sheaf category, so for any open  $U$ , we have

$$\mathcal{S} := \varinjlim_{\alpha} \mathcal{S}_{\alpha} \quad \text{satisfies} \quad \mathcal{S}(U) = \varinjlim_{\alpha} \mathcal{S}_{\alpha}(U)$$

To show  $\mathcal{F}$  is a sheaf, take an open cover  $U = \bigcup_i U_i$ . For separatedness, suppose  $s \in \mathcal{F}(U)$  restricts to zero in every  $\mathcal{F}(U_i)$ . Represent  $s$  by some  $s_{\alpha} \in \mathcal{S}_{\alpha}(U)$ . Its restriction to  $\mathcal{F}(U_i)$  is the image of  $s_{\alpha}|_{U_i}$ . If that image is zero in the colimit, then for each  $i$  there is some stage  $\beta_i \geq \alpha$  with  $s_{\beta_i}|_{U_i} = 0$ . Since the system is filtered, find  $\gamma$  dominating all  $\beta_i$ . Then  $(s_{\gamma})|_{U_i} = 0$  for all  $i$ , so by the sheaf property of  $\mathcal{S}_{\gamma}$ ,  $s_{\gamma} = 0$ , hence  $s = 0$  in the colimit.

For gluing, suppose we have  $s_i \in \mathcal{F}(U_i)$  that agree on overlaps. Represent each  $s_i$  by some  $s_{i,\alpha_i} \in \mathcal{S}_{\alpha_i}(U_i)$ . Filteredness gives a stage  $\beta$  dominating all  $\alpha_i$ . Push all  $s_{i,\alpha_i}$  to  $\mathcal{S}_{\beta}$ ; they still agree on overlaps, so they glue to some  $s_{\beta} \in \mathcal{S}_{\beta}(U)$ . Its class in  $\mathcal{F}(U)$  glues the original  $s_i$ .

Therefore the presheaf colimit is a sheaf, and  $\mathcal{S}(U) = \varinjlim_{\alpha} \mathcal{S}_{\alpha}(U)$ . Thus filtered colimits are computed stalkwise, so it is enough to show that in the category of abelian groups filtered colimits commute with finite products.  $\square$

*Proof of Lemma 1.390.* Take a short exact sequence of direct systems

$$0 \rightarrow A_\alpha \xrightarrow{f_\alpha} B_\alpha \xrightarrow{g_\alpha} C_\alpha \rightarrow 0,$$

with all maps compatible. We need to show that

$$0 \rightarrow \varinjlim A_\alpha \rightarrow \varinjlim B_\alpha \rightarrow \varinjlim C_\alpha \rightarrow 0$$

is exact.

Left-exactness (injectivity at  $\varinjlim A_\alpha$ ) holds for any colimit: if an element becomes zero in  $\varinjlim B_\alpha$ , it is already zero after some stage, and then in  $A_\alpha$  by exactness there.

Surjectivity at  $\varinjlim C_\alpha$  is where filteredness is used. Let  $[c_\alpha] \in \varinjlim C_\alpha$ . Pick representative  $c_\alpha \in C_\alpha$ . Exactness at  $C_\alpha$  gives  $b_\alpha \in B_\alpha$  with  $g_\alpha(b_\alpha) = c_\alpha$ . Let  $[b_\alpha]$  be its class in  $\varinjlim B_\alpha$ ; its image is  $[c_\alpha]$ .

We need to check this does not depend on choices: if we change stage or lift, filteredness gives a common stage where the choices agree, because kernels and images are compatible. That is standard and uses only that the system is filtered.

Hence the right map is surjective and the colimit of a short exact sequence is exact.  $\square$

*Proof of Lemma 1.391.* Let  $\mathcal{F}$  be a sheaf on  $R$  with finite support  $\{p_1, \dots, p_n\}$ . Then

$$\mathcal{F} \cong \bigoplus_{i=1}^n (i_{p_i})_* \mathcal{F}_{p_i},$$

where  $i_{p_i} : \{p_i\} \hookrightarrow R$  is the inclusion. Each  $(i_{p_i})_* \mathcal{F}_{p_i}$  is a skyscraper sheaf supported at  $p_i$ .

Skyscraper sheaves are flasque. For  $\mathcal{G} = (i_x)_* A$ , where  $A$  is an abelian group and  $x \in R$ , we have for  $V \subset U$ , the restriction map  $\mathcal{G}(U) \rightarrow \mathcal{G}(V)$  is either:

$$\begin{cases} \text{id}_A & \text{if both contain } x \\ A \rightarrow 0 & \text{if } x \in U, x \notin V \\ 0 \rightarrow 0 & \text{otherwise} \end{cases}$$

In all cases it is surjective. Therefore every restriction map is surjective, so  $\mathcal{G}$  is flasque.

Finite direct sums of flasque sheaves are flasque since surjectivity is preserved under finite products. Flasque sheaves have vanishing higher cohomology. Since cohomology commutes with finite direct sums, we conclude that  $H^q(R; \mathcal{F}) = 0$  for  $q \geq 1$ .  $\square$

**Example 1.392** (Cohomology of  $\mathbb{P}^1$  via principal parts). *We compute  $H^1(\mathbb{P}^1, \mathcal{O}(-n \cdot 0))$  using principal parts. Write  $t$  for the coordinate on  $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$  and  $s = 1/t$  for a local coordinate at  $\infty$ .*

*Let  $\{a_i\}$  be finitely many points in  $\mathbb{A}^1$ , and prescribe principal parts*

$$\sum_{k=1}^{N_i} \frac{c_{i,k}}{(t - a_i)^k}$$

at each  $a_i$ . Then there exists a rational function

$$f_0(t) = \sum_i \sum_{k=1}^{N_i} \frac{c_{i,k}}{(t - a_i)^k}$$

realizing these principal parts. Thus principal parts away from  $\infty$  always lift to global meromorphic functions. If  $f$  and  $g$  are two rational functions with the same principal parts at all finite points, then  $f - g$  has no poles on  $\mathbb{A}^1$ , hence is a polynomial. Thus all such functions are of the form

$$f(t) = f_0(t) + P(t), \quad P(t) \in k[t].$$

Now we consider the principal part at  $\infty$ . For  $\mathcal{O}(-n) \cong \mathcal{O}(-n\infty)$ , the local condition at  $\infty$  is

$$\mathcal{O}(-n)_\infty = s^n \mathcal{O}_\infty,$$

so the space of principal parts at  $\infty$  is

$$\mathcal{P}_\infty(\mathcal{O}(-n)) = \mathcal{M}_\infty / s^n \mathcal{O}_\infty.$$

Thus a Laurent expansion

$$f(1/s) = \sum_{m \gg -\infty} a_m s^m$$

is remembered modulo  $s^n$ , so the principal part at  $\infty$  consists of the terms

$$a_{-1}s^{-1} + a_{-2}s^{-2} + \cdots + a_0 + a_1s + \cdots + a_{n-1}s^{n-1}.$$

A polynomial

$$P(t) = a_0 + a_1t + \cdots + a_dt^d$$

expands at  $\infty$  as

$$P(1/s) = a_0 + a_1s^{-1} + \cdots + a_ds^{-d}.$$

Thus adding a polynomial can change the coefficients of  $1, s^{-1}, s^{-2}, \dots$  but cannot affect the coefficients of  $s, s^2, \dots, s^{n-1}$ . Therefore, after fixing principal parts at finite points, the coefficients of  $s, s^2, \dots, s^{n-1}$  at  $\infty$  cannot be adjusted by any global meromorphic function. These give independent obstructions. Hence

$$\dim H^0(\mathbb{P}^1; \mathcal{O}(n)) = \begin{cases} n+1, & n \geq -1, \\ 0, & n < -1, \end{cases} \quad \dim H^1(\mathbb{P}^1; \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n < -1. \end{cases}$$

**Theorem 1.393** (Riemann–Roch theorem for vector bundles). *Let  $R$  be a compact Riemann surface of genus  $g$ , and let  $V$  be a holomorphic vector bundle of rank  $r$  and degree  $d$  on  $R$ . Then the Euler characteristic of  $V$  is given by*

$$\chi(R, V) = \dim H^0(R, V) - \dim H^1(R, V) = d + r(1 - g)$$

The following lemma has an elementary proof, but it also immediately follows from the fact that proper pushforward preserves coherence, and any finitely generated module over a regular local ring is free.

**Lemma 1.394.** *Let  $f : X \rightarrow Y$  be a branched cover of Riemann surfaces. If  $\mathcal{F}$  is a locally free  $\mathcal{O}_X$ -module, then  $f_*\mathcal{F}$  is a locally free  $\mathcal{O}_Y$ -module. The rank of  $f_*\mathcal{F}$  is  $n \cdot \text{rank } \mathcal{F}$ , where  $n$  is the degree of the cover.*

*Proof.* Suppose  $y$  is not a branch point. Then the fibre  $f^{-1}(y)$  consists of  $n$  distinct points  $x_1, \dots, x_n$  and there are neighbourhoods  $U$  of  $y$  and  $V_i$  of  $x_i$  such that

$$f^{-1}(U) = \bigsqcup_{i=1}^n V_i \quad \text{and} \quad f|_{V_i} : V_i \rightarrow U$$

is a biholomorphism.

Then

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U)) = \bigoplus_{i=1}^n \mathcal{F}(V_i)$$

Shrinking  $U$  if necessary, we may assume that  $\mathcal{F}$  is trivial on each  $V_i$ , so that

$$\mathcal{F}(V_i) \cong \mathcal{O}_X(V_i)^{\oplus r}$$

where  $r$  is the rank of  $\mathcal{F}$ . Since  $f|_{V_i}$  is a biholomorphism, we have

$$\mathcal{O}_X(V_i) \cong \mathcal{O}_Y(U)$$

Therefore,

$$(f_*\mathcal{F})(U) \cong \bigoplus_{i=1}^n \mathcal{O}_Y(U)^{\oplus r} \cong \mathcal{O}_Y(U)^{\oplus nr}$$

showing that  $f_*\mathcal{F}$  is locally free of rank  $nr$  near  $y$ .

Now suppose that  $y$  is a branch point. To show  $f_*\mathcal{F}$  is a vector bundle, we must show that its stalks are free  $\mathcal{O}_{Y,y}$ -modules. By definition, for any sheaf  $\mathcal{F}$  on  $X$  we have

$$(f_*\mathcal{F})_y = \varinjlim_{y \in \bar{U}} \mathcal{F}(f^{-1}(U)).$$

Given a finite map, the topology of the fiber decomposes the stalk into a direct sum over preimages:

$$(f_*\mathcal{F})_y \cong \bigoplus_{x_i \in f^{-1}(y)} \mathcal{F}_{x_i}$$

Note that  $\mathcal{F}_x$  is naturally an  $\mathcal{O}_{X,x}$ -module, and  $\mathcal{O}_{Y,y}$  acts on it via the pullback map

$$f^\# : \mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x}.$$

Thus  $(f_*\mathcal{F})_y$  is an  $\mathcal{O}_{Y,y}$ -module through  $f^\#$ , and thus we want to show that  $\mathcal{O}_X$  near  $x$  is a free module of rank  $n$  over  $\mathcal{O}_Y$  near  $y$ . We identify

$$\mathcal{O}_{Y,y} \cong \mathbb{C}\{z\}, \quad \mathcal{O}_{X,x} \cong \mathbb{C}\{w\}$$

Take any holomorphic function in  $w$ .

$$g(w) = \sum_{k=0}^{\infty} a_k w^k.$$

Grouping the terms according to their residue modulo  $n$ , we have

$$g(w) = h_0(z) + h_1(z)w + \cdots + h_{n-1}(z)w^{n-1},$$

Thus  $\mathbb{C}\{w\}$  is a free  $\mathbb{C}\{z\}$ -module of rank  $n$  with basis  $1, w, \dots, w^{n-1}$ . In other words,  $\mathcal{O}_{X,x}$  is a free  $\mathcal{O}_{Y,y}$ -module of rank  $n$ . This completes the proof.  $\square$

**Example 1.395** (Riemann-Roch on  $\mathbb{P}^1$ ). *We will directly verify the Riemann-Roch theorem for vector bundles on  $\mathbb{P}^1$ . Since every vector bundle on  $\mathbb{P}^1$  splits as a direct sum of line bundles by Grothendieck's theorem, it suffices to verify the theorem for line bundles ( $\chi$ , rank, and degree are additive under direct sum).*

For each  $n \in \mathbb{Z}$ , we have already computed

$$\dim H^0(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq -1, \\ 0, & n < -1, \end{cases} \quad \dim H^1(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n < -1, \end{cases}$$

so in all cases

$$\chi(\mathbb{P}^1, \mathcal{O}(n)) := \dim H^0(\mathbb{P}^1, \mathcal{O}(n)) - \dim H^1(\mathbb{P}^1, \mathcal{O}(n)) = n + 1.$$

is precisely the statement of Riemann-Roch for  $\mathbb{P}^1$ .

*Proof of Riemann-Roch for vector bundles.* We realize our surface  $R$  as an  $n$ -sheeted branched cover  $\pi : R \rightarrow \mathbb{P}^1$  for some  $n$ . We will prove the theorem on  $R$  by pushing  $V$  down to  $\mathbb{P}^1$  and using Riemann-Roch there. For  $\mathbb{P}^1$ , the theorem follows from Grothendieck's classification of vector bundles and the explicit calculation for line bundles.

Let  $B \in \text{Div}(R)$  be the branch divisor of the cover, the sum of branch points repeated according to the branching index. The degree of  $B$  is the total branching index  $b$ .

Consider the direct image bundle  $\pi_* V$  on  $\mathbb{P}^1$ , associated to the sheaf  $\pi_* \mathcal{O}(V)$ . I now make the following three claims:

1.  $H^i(R; V) \cong H^i(\mathbb{P}^1; \pi_* V)$  for  $i = 1, 2$ ;
2.  $\text{rank } \pi_* V = n \cdot \text{rank } V$ ;
3.  $\deg \pi_* V = \deg V - \frac{1}{2}b \cdot \text{rank } V$ ,

where  $b$  is the total branching index.

Claim (ii) follows from the previous lemma. (i) is clear for  $H^0$  but less so for  $H^1$ ; (iii) seems obscure. Granting the claims for now and writing  $r = \text{rank } V$ , we find

$$\chi(R; V) = \chi(\mathbb{P}^1; \pi_* V) = (\deg V - \frac{1}{2}br) + n \cdot r = \deg V + (1 - g) \cdot r,$$

using Riemann-Hurwitz, thus proving the Riemann-Roch theorem.

It remains to prove the claims. Actually, claim (i) follows from the two observations that

$$\mathcal{M}(\mathbb{P}^1; \pi_* V) = \mathcal{M}(R; V) \quad \text{and} \quad \mathcal{P}(\mathbb{P}^1; \pi_* V) = \mathcal{P}(R; V).$$

Both statements actually hold at the level of sheaves,  $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$  and similarly for  $\mathcal{P}$ , but we do not need that fact.

Where  $\pi$  is unramified, a holomorphic section is a tuple of  $r$ -vectors of holomorphic functions on each sheet, and a meromorphic section is a tuple of  $r$ -vectors of meromorphic functions on each sheet, so the equality of sheaves  $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$  and  $\pi_*\mathcal{O}(V) = \mathcal{O}(\pi_*V)$  is clear.

Likewise, at a branch point, in the local form  $w \mapsto z = w^n$ , we again have the decomposition of every holomorphic/meromorphic section, which after trivializing  $V$ , can be written as an  $r$ -vector of holomorphic/meromorphic functions in  $w$ . Grouping terms according to their residue modulo  $n$  again gives the desired equalities of sheaves. Thus again we have  $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$  and  $\pi_*\mathcal{O}(V) = \mathcal{O}(\pi_*V)$ . Quotienting gives the equality for principal parts.

Now claim (i) follows from the two exact sequences

$$\begin{aligned} 0 \rightarrow H^0(R, V) \rightarrow \Gamma(R, \mathcal{M}(V)) \rightarrow \Gamma(R, \mathcal{P}(V)) \rightarrow H^1(R, V) \rightarrow 0 \\ 0 \rightarrow H^0(\mathbb{P}^1, \pi_*V) \rightarrow \Gamma(\mathbb{P}^1, \mathcal{M}(\pi_*V)) \rightarrow \Gamma(\mathbb{P}^1, \mathcal{P}(\pi_*V)) \rightarrow H^1(\mathbb{P}^1, \pi_*V) \rightarrow 0. \end{aligned}$$

since we have identified the middle two terms in each sequence. The more sophisticated argument is to argue that  $\pi$  is affine and so  $R^i\pi_* = 0$  for  $i > 0$ , so that the Leray spectral sequence degenerates and gives the desired isomorphisms of cohomology groups.

The curious claim is (iii), which we now prove. For starters, observe that the fibre of  $\pi_*V$  at some  $z \in \mathbb{P}^1$  where  $R$  is not branched is the sum of the fibres of  $V$  at the points in  $\pi^{-1}(z)$ .

Recall that an elementary transformation of a vector bundle  $V$  at some point  $x \in R$  is

$$0 \rightarrow V' \rightarrow V \rightarrow S \rightarrow 0$$

where  $S$  is a skyscraper sheaf supported at  $x$  with fibre isomorphic to a quotient  $V_x/W$  for some subspace  $W \subset V_x$ . In particular the transformed bundle  $V'$  has the same rank as  $V$  but degree  $c = \text{codim } W$  less than that of  $V$ . The important fact to know that any vector bundle on a Riemann surface can be transformed to the trivial bundle  $\mathbb{C}^r$  by a sequence of elementary transformations.

An elementary transformation on  $V$  at some  $y \in \pi^{-1}(z)$  effects an elementary transformation on  $\pi_*V$  at  $z$ , with a subspace of the same codimension (the original  $S$ , plus the sum of the other fibres). So the degrees of the two bundles change by the same amount. Transforming all the way to the trivial bundle  $\mathcal{O}^r$  gives the relation

$$\deg \pi_*V - \deg V = \deg \pi_*\mathcal{O}^r - \deg \mathcal{O}^r = r \cdot \deg \pi_*\mathcal{O}.$$

So we need to show that  $\deg \pi_*\mathcal{O} = -b/2$ . We now consider the dual vector bundle  $(\pi_*\mathcal{O})^\vee$ , which has degree opposite that of  $\pi_*\mathcal{O}$ ; the desired equality is equivalent to

$$\deg ((\pi_*\mathcal{O})^\vee) = \deg \pi_*\mathcal{O} + b.$$

This equality, and with it the theorem, will follow from taking degrees in the isomorphism of Lemma 1.396 below. In particular the RHS becomes

$$\deg \pi_*(\mathcal{O}(B)) = \deg \mathcal{O}(B) + \text{rank}(\pi_*\mathcal{O}(B)) \cdot \deg \pi_*\mathcal{O} = b + 1 \cdot \deg \pi_*\mathcal{O}$$

as  $\square$

**Lemma 1.396.** *With notation as in the proof above,*

$$(\pi_*\mathcal{O})^\vee \cong \pi_*(\mathcal{O}(B)).$$

More generally,

$$(\pi_*(V))^\vee \cong \pi_*(V^\vee \otimes \mathcal{O}(B))$$

*Proof.* We first prove

$$(\pi_*\mathcal{O}_R)^\vee \cong \pi_*\mathcal{O}_R(B).$$

Let  $\mathcal{M}_R$  and  $\mathcal{M}_{\mathbb{P}^1}$  denote the sheaves of meromorphic functions. Since  $\pi$  is finite, there is a trace morphism

$$\mathrm{Tr}_\pi : \pi_*\mathcal{M}_R \rightarrow \mathcal{M}_{\mathbb{P}^1}.$$

We claim that

$$\mathrm{Tr}_\pi(\pi_*\mathcal{O}_R(B)) \subset \mathcal{O}_{\mathbb{P}^1}.$$

This is a local computation. Near a ramification point  $x \in R$  of index  $n$ , with  $z = \pi(x)$ , choose coordinates  $w$  at  $x$  and  $t$  at  $z$  such that

$$t = w^n.$$

Then locally

$$B = (n-1)x, \quad \mathcal{O}_R(B)_x = w^{-(n-1)}\mathbb{C}\{w\}.$$

Thus a local section of  $\mathcal{O}_R(B)$  has an expansion

$$\psi(w) = \sum_{m \geq -(n-1)} a_m w^m.$$

Its trace is

$$\mathrm{Tr}_\pi(\psi) = \sum_{j=0}^{n-1} \psi(\zeta^j w) = \sum_{m \geq -(n-1)} a_m \left( \sum_{j=0}^{n-1} \zeta^{jm} \right) w^m,$$

where  $\zeta = e^{2\pi i/n}$ . Since

$$\sum_{j=0}^{n-1} \zeta^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m, \end{cases}$$

only the terms with  $n \mid m$  survive. But the only multiple of  $n$  among the integers  $m \geq -(n-1)$  and  $m < 0$  is none, so no negative power survives. Hence  $\mathrm{Tr}_\pi(\psi)$  is holomorphic in  $t$ .

Therefore we obtain an  $\mathcal{O}_{\mathbb{P}^1}$ -bilinear pairing

$$\pi_*\mathcal{O}_R \times \pi_*\mathcal{O}_R(B) \rightarrow \mathcal{O}_{\mathbb{P}^1}, \quad (\varphi, \psi) \mapsto \mathrm{Tr}_\pi(\varphi\psi),$$

and hence a morphism

$$\pi_*\mathcal{O}_R(B) \rightarrow (\pi_*\mathcal{O}_R)^\vee.$$



To show that this is an isomorphism, it suffices to check stalkwise. At an unramified point, the map is obviously perfect. At a ramification point with local form  $t = w^n$ , the  $\mathcal{O}_{\mathbb{P}^1, z}$ -module  $\mathcal{O}_{R, x}$  has basis

$$1, w, \dots, w^{n-1},$$

while  $\mathcal{O}_R(B)_x$  has basis

$$1, w^{-1}, \dots, w^{-(n-1)}.$$

The pairing satisfies

$$\mathrm{Tr}_\pi(w^i \cdot w^{-j}) = \begin{cases} n, & i = j, \\ 0, & i \neq j, \end{cases}$$

so the matrix of the pairing is  $nI_n$ , hence invertible. Thus the pairing is perfect, and

$$(\pi_* \mathcal{O}_R)^\vee \cong \pi_* \mathcal{O}_R(B).$$

The same argument, after local trivialization of  $V$ , gives

$$(\pi_* V)^\vee \cong \pi_*(V^\vee \otimes \mathcal{O}_R(B)).$$

Indeed, the pairing is

$$\pi_* V \times \pi_*(V^\vee \otimes \mathcal{O}_R(B)) \rightarrow \mathcal{O}_{\mathbb{P}^1}, \quad (s, \lambda) \mapsto \mathrm{Tr}_\pi(\lambda(s)),$$

and in a local frame it is the tensor product of the scalar trace pairing above with the natural evaluation pairing  $V \times V^\vee \rightarrow \mathcal{O}_R$ .  $\square$

### 1.8.5 Serre duality on a curve

Let  $R$  be a compact Riemann surface,  $V$  vector bundle on  $R$ . We have the canonical bundle  $K_R = \Omega_R^1$  of holomorphic one-forms on  $R$ .

**Theorem 1.397** (Serre duality). *There is an isomorphism of vector spaces, functorial in  $R$ :*

$$\begin{aligned} H^0(R, V) &\cong H^1(R, K_R \otimes V^\vee)^\vee \\ H^1(R, V) &\cong H^0(R, K_R \otimes V^\vee)^\vee \end{aligned}$$

Functorial means that for  $f : R \rightarrow S$  a nonconstant map of compact Riemann surfaces, we have the following commutative diagram:

$$\begin{array}{ccc} H^0(R, V) & \xrightarrow{\cong} & H^1(R, K_R \otimes V^\vee)^\vee \\ \downarrow f^* & & \downarrow f^* \\ H^0(S, V) & \xrightarrow{\cong} & H^1(S, K_S \otimes V^\vee)^\vee \end{array}$$

We will say a little bit more about the map on  $H^1$ .

**Example 1.398.** We check the isomorphism for  $\mathcal{O}(n)$  on  $\mathbb{P}^1$  for  $n \geq -1$ . Move to  $\mathcal{O}(n\infty)$ . Check the residue pairing at  $z = 0$ . May assume the principal parts are at 0 and that meromorphic functions have only poles at 0. In this case

$$\begin{aligned} H^0(\mathbb{P}^1, \mathcal{O}(n\infty)) &= \text{span}\{1, z, z^2, \dots, z^n\} \\ H^1(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes \mathcal{O}(-n\infty)) &= \left\{ \frac{dz}{z}, \frac{dz}{z^2}, \dots, \frac{dz}{z^{n+1}} \right\} \end{aligned}$$

because by the principal parts argument  $dz/z^{n+2} = z^{-n}dz/z^n$  is holomorphic at  $\infty$ . The residue pairing is clear.

*Proof.* The global residue pairing

$$H^0(R, V) \times H^1(R, K_R \otimes V^*) \rightarrow \mathbb{C}$$

comes from a local residue pairing at a point  $x \in R$ .

$$\begin{aligned} \mathcal{O}(V)_x \otimes \mathcal{P}(K_R \otimes V^*)_x &\rightarrow \mathbb{C} \\ (\sigma, \tau) &\mapsto \text{Res}_x(\langle \sigma, \tau \rangle) \end{aligned}$$

In a local coordinate  $z$  near  $x$ , the pairing takes the form

$$z^n u \otimes \frac{v}{z^m} dz \mapsto \text{Res}_x \left( \frac{uv}{z^{m-n}} dz \right) = \begin{cases} u(x)v(x) & m = n + 1 \\ 0 & \text{otherwise} \end{cases}$$

Note that the holomorphic part of a differential has no residue so the map is defined on  $\mathcal{P}(K_R \otimes V^*)_x$ . The pairing is nondegenerate. The global residue pairing then is simply the sum of the local pairings over all points  $x \in R$ .

We will reduce the proof to the case  $R = \mathbb{P}^1$ . In this case, we have already explicitly verified the isomorphism. Using the affineness of the map  $f : R \rightarrow \mathbb{P}^1$ , we have  $H^i(R, V) = H^i(\mathbb{P}^1, f_*V)$  for  $i = 0, 1$ . We want to transport the residue pairing on  $\mathbb{P}^1$  along this isomorphism. Serre duality on  $\mathbb{P}^1$  for the bundle  $f_*V$  gives

$$H^0(\mathbb{P}^1, f_*V) \cong H^1(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes (f_*V)^\vee)^*.$$

So to match this with the group  $H^1(R, K_R \otimes V^\vee)^*$  we will construct an isomorphism of sheaves on  $\mathbb{P}^1$

$$f_*(K_R \otimes V^\vee) \cong K_{\mathbb{P}^1} \otimes (f_*V)^\vee. \quad (3)$$

Tensoring both sides by  $K_{\mathbb{P}^1}^{-1}$ , this is equivalent to

$$K_{\mathbb{P}^1}^{-1} \otimes f_*(K_R \otimes V^\vee) \cong (f_*V)^\vee. \quad (4)$$

By the projection formula,

$$K_{\mathbb{P}^1}^{-1} \otimes f_*(K_R \otimes V^\vee) \cong f_*(f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee),$$

so (4) is equivalent to the existence of an isomorphism

$$f_*(f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee) \cong (f_*V)^\vee. \quad (5)$$

By Lemma 1.396, we have

$$(f_*V)^\vee \cong f_*(V^\vee(B)),$$

where  $B$  is the branch divisor of  $f$ . Hence (5) will follow provided we can identify the bundles upstairs:

$$f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee \cong V^\vee(B). \quad (6)$$

Cancelling  $V^\vee$ , condition (6) is equivalent to the line-bundle identity

$$K_R \otimes f^*K_{\mathbb{P}^1}^{-1} \cong \mathcal{O}_R(B), \quad (7)$$

which is precisely the Riemann–Hurwitz formula in line-bundle form.

To check the compatibility of residue pairings on  $R$  and  $\mathbb{P}^1$ . We can use the isomorphisms

$$f_*\mathcal{P}_x(V) \cong \mathcal{P}_{f(x)}(f_*V),$$

and the discussion of differentials at branch points. Choose local coordinates in standard form and write the residue pairings in the Laurent basis.  $\square$

**Remark 1.399.** *Some other consequences of Riemann–Roch and Serre duality.*

1. *The degree of the canonical bundle  $K_R$  is  $2g - 2$  and there are  $g$  independent holomorphic differentials on  $R$ .*
2. *For all  $R$  with genus  $g \geq 1$ , for all  $p \in R$  there exists a nontrivial holomorphic differential not vanishing at  $p$ . If not, then all holomorphic differentials zeroes at  $p$  and so  $h^0(R, K_R(-p)) = g$ . This implies that  $h^0(R, \mathcal{O}(p)) = 2$  so there exists a nonconstant meromorphic function with a single simple pole at  $p$ . This function gives an isomorphism to  $\mathbb{P}^1$  and so  $g = 0$ , a contradiction.*
3. *There exists a map of degree  $g + 1$  from  $R$  to  $\mathbb{P}^1$ . This is because*

$$h^0(R, \mathcal{O}((g+1)p)) = h^1(R, K_R(-(g+1)p)) + (g+1) - g + 1 \geq 2$$

4. *Weierstrass gap theorem. For a point  $p \in R$ , there are exactly  $g$  positive integers  $n$  such that  $h^0(R, \mathcal{O}(np)) = h^0(R, \mathcal{O}((n-1)p))$ . These are called the Weierstrass gaps at  $p$ . The non-gap numbers are the pole orders of meromorphic functions with a single pole at  $p$ . The gap numbers are the pole orders of holomorphic differentials vanishing at  $p$ .*

### 1.8.6 Elliptic curves

**Definition** An **elliptic curve** over a field  $k$  is a smooth, projective, geometrically connected curve  $E$  of genus 1 together with a specified  $k$ -rational point  $O \in E(k)$ .

If  $\text{char}(k) \neq 2, 3$ , every elliptic curve admits a Weierstrass equation

$$E : y^2 = x^3 + ax + b$$

with discriminant

$$\Delta = -16(4a^3 + 27b^2) \neq 0.$$

**Proposition 1.400** (Weierstrass  $\wp$ -function). *Every elliptic curve  $E$  over  $\mathbb{C}$  can be realized as a quotient of  $\mathbb{C}$  by a lattice  $\Lambda \subset \mathbb{C}$ . The lattice  $\Lambda$  is the period lattice and is generated by two complex numbers  $\omega_1, \omega_2$  that are linearly independent over  $\mathbb{R}$ .  $\omega_1$  and  $\omega_2$  are called the periods of  $E$  and are defined by integrating the (unique up to scaling) holomorphic differential on  $E$  over a basis of the first homology group  $H_1(E, \mathbb{Z})$ .*

**Proposition 1.401.** *Being genus 1,  $E$  has a unique (up to scaling) holomorphic differential  $\omega$ . If we write  $E = \mathbb{C}/\Lambda$  as above, then  $\omega$  corresponds to the standard differential  $dz$  on  $\mathbb{C}$ . It is clearly nonvanishing and so it trivializes the canonical bundle  $K_E$ . In particular,  $K_E \cong \mathcal{O}_E$ .*

**Proposition 1.402.** *There are explicit meromorphic functions  $\wp(z)$  and  $\wp'(z)$  on  $E$  with the following properties:*

- $\wp(z)$  has a double pole at 0 and no other poles, and  $\wp'(z)$  has a triple pole at 0 and no other poles;
- $\wp(z)$  is an even function and  $\wp'(z)$  is an odd function
- $\wp(z)$  and  $\wp'(z)$  satisfy the Weierstrass equation

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$$

for some constants  $g_2, g_3$  determined by the lattice  $\Lambda$ ;

- Every meromorphic function on  $E$  can be expressed as a rational function in  $\wp(z)$  and  $\wp'(z)$ . Therefore the function field of  $E$  is generated by  $\wp(z)$  and  $\wp'(z)$  and can be presented as

$$\mathbb{C}(E) \cong \mathbb{C}(z)[w]/(w^2 - 4z^3 + g_2z + g_3)$$

**Proposition 1.403.** *Every degree 0 line bundle on  $E$  is of the form is linearly equivalent to  $\mathcal{O}_E(O - Q)$  for some  $Q \in E$ . The assignment  $Q \mapsto \mathcal{O}_E(O - Q)$  gives an isomorphism of groups*

$$E \cong \text{Pic}^0(E)$$

and the line bundle  $\mathcal{O}_E(P - Q)$  is trivial if and only if  $Q = P$ .

### 1.8.7 Hyperelliptic curves

**Definition 1.404.** *A compact Riemann surface  $R$  is hyperelliptic if there exists a degree 2 holomorphic map  $R \rightarrow \mathbb{P}^1$ .*

A standard family is  $y^2 = f(x)$ , where  $f(x)$  is squarefree of degree  $2g + 1$  or  $2g + 2$ . Its smooth projective model is hyperelliptic of genus  $g$ .

**Exercise 1.405.** *Prove that every compact Riemann surface of genus 2 is hyperelliptic*

**Solution 1.406.** *Being genus 2 means  $\dim H^0(R, K_R) = 2$ . Choose a basis  $\omega_1, \omega_2$ . Since  $\deg K_R = 2g - 2 = 2$ , any nonzero holomorphic differential has zero divisor of degree 2. We claim  $\omega_1, \omega_2$  have no common zero. If both vanished at  $p \in R$ , then*

$$H^0(R, K_R(-p)) \subset H^0(R, K_R)$$

*would still contain both  $\omega_1, \omega_2$ , so  $\dim H^0(R, K_R(-p)) \geq 2$ . But  $\deg K_R(-p) = 1$ , and a line bundle of degree 1 on a genus 2 curve cannot have two independent global sections (otherwise it would give a degree 1 map to  $\mathbb{P}^1$ , impossible for genus 2). Contradiction.*

*Hence  $f := \omega_1/\omega_2$  is a well-defined meromorphic function on  $R$ . Its poles are zeros of  $\omega_2$ , and its zeros are zeros of  $\omega_1$ . Each contributes total degree 2, so  $f$  has degree 2. Therefore  $f$  defines a holomorphic map*

$$R \rightarrow \mathbb{P}^1$$

*of degree 2. Thus  $R$  is hyperelliptic.*

If  $C$  is hyperelliptic, a degree-2 map  $\pi : C \rightarrow \mathbb{P}^1$  comes with a unique involution  $\iota : C \rightarrow C$  exchanging the two points in a general fiber. This is the **hyperelliptic involution**. It is unique. The quotient is

$$C/\langle \iota \rangle \cong \mathbb{P}^1;$$

and the fixed points of  $\iota$  are exactly the ramification points of  $\pi$ .

By Riemann–Hurwitz, if  $C$  has genus  $g$  and

$$\pi : C \rightarrow \mathbb{P}^1$$

has degree 2, then

$$2g - 2 = 2(-2) + \deg R,$$

so

$$\deg R = 2g + 2.$$

Hence there are  $2g+2$  ramification points counted with multiplicity. Over an algebraically closed field of characteristic  $\neq 2$ , a smooth hyperelliptic curve is branched at  $2g+2$  distinct points of  $\mathbb{P}^1$ . For a hyperelliptic curve, the ramification points of the double cover are the Weierstrass points. There are exactly  $2g + 2$  of them.

### 1.8.8 Genus 2 curves

Let  $C$  be a smooth projective curve of genus 2. Then  $C$  is hyperelliptic, and  $|K_C|$  is a  $g_2^1$  meaning a linear system of degree 2 and dimension 1. Moreover, the only nontrivial linear equivalences among effective divisors of degree 2 are those coming from  $|K_C|$ . Equivalently, if  $D, E$  are effective divisors of degree 2 with  $D \sim E$  and  $D \neq E$ , then  $D \sim K_C \sim E$ .

Indeed, if  $D, E$  are effective divisors of degree 2 with  $D \sim E$  and  $D \neq E$ , then  $|D|$  has dimension at least 1, so  $h^0(D) \geq 2$ . By Riemann–Roch,

$$h^0(D) - h^0(K_C - D) = 1.$$

Hence  $h^0(K_C - D) \geq 1$ . Since  $\deg(K_C - D) = 0$ , this forces  $K_C - D \sim 0$ , so  $D \sim K_C$ .

Now choose distinct points  $p, r, s \in C$  with  $r \neq \iota(p)$ , where  $\iota$  is the hyperelliptic involution, and set

$$L = \mathcal{O}_C(p + r - s).$$

Then  $\deg L = 1$ , so  $L$  is ample. If  $L$  were effective, then  $L \cong \mathcal{O}_C(x)$  for some  $x \in C$ , hence

$$p + r \sim s + x.$$

Since  $p + r \neq s + x$  as divisors, the previous paragraph implies  $p + r \sim K_C$ , so  $r = \iota(p)$ , a contradiction. Thus  $L$  is not effective.

### 1.8.9 Genus 4 curves and curves of every genus

As a consequence of the Riemann Hurwitz formula, we see that there are curves of every genus  $g$  over an algebraically closed field. In particular, take a branched cover of  $\mathbb{P}^1$  of degree 2 branched at  $2g + 2$  distinct points. Then

$$\chi(R) = 2\chi(\mathbb{P}^1) - \deg B$$

so

$$2 - 2g = 4 - (2g + 2)$$

so  $g$  is the genus of the cover. These curves are clearly hyperelliptic. We have already seen that every curve of genus 2 is hyperelliptic, basically because of the following fact.

**Proposition 1.407.** *If  $C$  is a curve of genus  $g \geq 1$  and  $C$  carries a degree 2 invertible sheaf with at least two independent global sections, then  $\mathcal{L}$  has exactly two independent global sections,  $\mathcal{L}$  is basepoint free, and the associated morphism  $C \rightarrow \mathbb{P}^1$  has degree 2. In particular, a curve of genus  $g \geq 1$  is hyperelliptic if and only if it has a  $g_2^1$ , that is a linear system of degree 2 and dimension 1.*

*Proof.* Suppose  $\mathcal{L}$  is a degree 2 line bundle on  $C$  with at least two independent global sections. Suppose, for contradiction, that  $h^0(C, \mathcal{L}) \geq 3$ . Choose three linearly independent sections  $s_0, s_1, s_2 \in H^0(C, \mathcal{L})$

Let  $B$  be the common base divisor of the linear system spanned by  $s_0, s_1, s_2$ . Then these sections define a base-point-free linear system on  $\mathcal{L}(-B)$ . Thus they give a morphism  $\varphi : C \rightarrow \mathbb{P}^2$  such that

$$\mathcal{L}(-B) \cong \varphi^* \mathcal{O}_{\mathbb{P}^2}(1).$$

Since  $s_0, s_1, s_2$  are linearly independent, the image  $\varphi(C)$  is not contained in a line. Hence

$$\deg \varphi(C) \geq 2.$$

Therefore

$$\deg \mathcal{L} - \deg B = \deg \mathcal{L}(-B) = \deg(\varphi) \deg(\varphi(C)) \geq 2.$$

But  $\deg \mathcal{L} = 2$ , so this forces

$$\deg B = 0, \quad \deg(\varphi) = 1, \quad \deg \varphi(C) = 2.$$

Thus  $B = 0$ , and  $\varphi$  is birational onto a plane conic. A plane conic is rational, so  $C$  is birational to  $\mathbb{P}^1$ . Since  $C$  is smooth and projective, this implies

$$C \cong \mathbb{P}^1,$$

contradicting  $g \geq 1$ . We conclude  $h^0(C, \mathcal{L}) = 2$ . Suppose  $p$  is a basepoint of  $\mathcal{L}$ . Then  $h^0(\mathcal{L}(-p)) = h^0(\mathcal{L})$  since every section of  $\mathcal{L}$  vanishes at  $p$ . Let  $\mathcal{M} = \mathcal{L}(-p)$ . Then  $\deg \mathcal{M} = 1$  and  $h^0(\mathcal{M}) = 2$ . Then  $\mathcal{M}$  is basepoint free, because if  $\mathcal{M}$  had a basepoint  $q$ ,  $\mathcal{M}(-q)$  would have the same number of global sections as  $\mathcal{M}$  and of degree 0, so  $\mathcal{M}(-q)$  would be trivial and would have  $H^0(\mathcal{M}(-q)) = 1$ , a contradiction. Thus  $\mathcal{M}$  is basepoint free and defines a morphism  $C \rightarrow \mathbb{P}^1$  of degree 1. This implies that  $C$  is rational, contradicting  $g \geq 1$ . We conclude that  $\mathcal{L}$  is basepoint free. Therefore if  $C$  has a  $g_2^1$ , then the a priori rational map  $C \rightarrow \mathbb{P}^1$  is in fact a morphism of degree 2 and so  $C$  is hyperelliptic.

Conversely, if  $C$  is hyperelliptic, the pullback of  $\mathcal{O}(1)$  along the degree 2 map  $C \rightarrow \mathbb{P}^1$  gives a line bundle  $\mathcal{L}$  of degree 2 with two independent global sections. We have to check that there are no other global sections. If there were, the morphism  $C \rightarrow \mathbb{P}^2$  given by these global sections would have image not contained in a line, since the global sections of  $\mathcal{L}$  were taken linearly independent. Thus the image would have degree at least 2. But then  $\mathcal{L} \cong \varphi^* \mathcal{O}(1)$  would have numerics  $2 = \deg L = \deg \varphi \cdot \deg(\varphi(C)) = 2 \cdot \deg(\varphi(C))$  so  $\deg(\varphi) = 1$ . This implies that  $\varphi(C)$  is birational onto a plane conic which contradicts  $g \geq 1$ .  $\square$

**Proposition 1.408.** *Assume  $g \geq 2$ . If  $\mathcal{L}$  corresponds to a hyperelliptic cover  $C \rightarrow \mathbb{P}^1$ , then  $\omega_C = \mathcal{L}^{\otimes(g-1)}$ .*

*Proof.* The composition  $C \rightarrow \mathbb{P}^1 \rightarrow \mathbb{P}^{g-1}$  identifies with  $\mathcal{L}^{\otimes(g-1)}$  and we are done if we can identify it with  $\omega_C$ . The pullback  $H^0(\mathbb{P}^{g-1}, \mathcal{O}(1)) \rightarrow H^0(C, \mathcal{L}^{\otimes(g-1)})$  is injective because if  $s$  is a nonzero section of  $\mathcal{O}(1)$  vanishing on the image of  $C$ , then  $\varphi(C)$  is contained in a hyperplane, but  $\varphi(C)$  is a rational normal curve of degree  $g-1$  so it is not contained in a hyperplane. Thus  $\mathcal{M} = \mathcal{L}^{\otimes(g-1)}$  has at least  $g$  independent global sections and degree  $2g-2$ . Riemann–Roch then implies that

$$h^0(C, \mathcal{M}) = g - 1 + h^0(C, \omega_C \otimes \mathcal{M}^{-1})$$

The left hand side is at least  $g$  and so  $h^0(C, \omega_C \otimes \mathcal{M}^{-1}) \geq 1$ . A degree 0 line bundle with a nonzero global section is trivial.  $\square$

Therefore, for hyperelliptic curves, the canonical map always factors as the hyperelliptic curve and then the Veronese embedding. For nonhyperelliptic curves, the canonical map in fact gives a closed embedding.

**Exercise 1.409.** *Show that if  $C$  is nonhyperelliptic and  $g \geq 2$ , then the canonical map  $C \rightarrow \mathbb{P}^{g-1}$  is a closed embedding.*

**Solution 1.410.** For any degree 2 effective divisor  $D$  consider the exact sequence

$$0 \rightarrow K_C(-D) \rightarrow K_C \rightarrow K_C|_D \rightarrow 0$$

The restriction map  $H^0(C, K_C) \rightarrow H^0(D, K_C|_D)$  is surjective if and only if

$$h^0(C, \omega_C) - h^0(C, \omega_C(-D)) = 2$$

This is because the image of the restriction map is at most 2-dimensional since  $D$  is length 2 scheme of dimension 0 and  $K_C|_D$  is a line bundle on  $D$ .

In turn, this is equivalent to  $h^0(C, K_C(-D)) = g - 2$ . Applying Riemann-Roch to  $K_C(-D)$ , the desired equality becomes equivalent to checking that  $h^0(C, \mathcal{O}_C(D)) = 1$ . If  $h^0(C, \mathcal{O}_C(D)) > 1$ , then  $\mathcal{O}_C(D)$  has two independent global sections and so  $C$  is hyperelliptic, a contradiction. Thus  $h^0(C, \mathcal{O}_C(D)) = 1$ , namely the constant section, and so we are done.

### 1.8.10 Jacobians

Recall the additive and multiplicative Cousin problems:

- Additive Cousin: Given a collection of principal parts on  $R$ , when does there exist a global meromorphic function with those principal parts?
- Multiplicative Cousin: Given a divisor  $D$  on  $R$ , when does there exist a global meromorphic function  $f$  with divisor  $(f) = D$ ?

We have seen that for  $g = 0$ , the additive Cousin problem always has a solution and the multiplicative Cousin problem has a solution if and only if the degree of the divisor is zero. For  $g = 1$ , the additive Cousin problem has a solution if and only if the sum of the residues is zero and the multiplicative Cousin problem has a solution if and only if the degree of the divisor is zero and the sum of the points (with multiplicity) modulo the period lattice is zero.

In general, the obstruction in the additive Cousin problem lies in  $H^1(R, \mathcal{O})$  because we defined it to be so:

$$H^1(R, \mathcal{O}) = \Gamma(R, \mathcal{P}) / \Gamma(R, \mathcal{M})$$

where  $\mathcal{P}$  is the sheaf of principal parts and  $\mathcal{M}$  is the sheaf of meromorphic functions. We proved the Serre duality isomorphism

$$H^1(R, \mathcal{O}) \cong H^0(R, K_R)^*$$

so the obstruction to solving the additive Cousin problem can be computed by pairing with holomorphic one-forms. In particular, a collection of principal parts  $\{p_i\}$  has a solution if and only if

$$\sum_i \text{Res}_{x_i}(p_i \omega) = 0$$

for all  $\omega \in H^0(R, K_R)$ . The obstruction in the multiplicative Cousin problem is encoded by the Abel-Jacobi map, which this section is devoted to describing.



### 1.8.11 Differentials on a curve

Recall that on a Riemann surface  $R$ , we have the Hodge star operator defined on all smooth differential 1-forms

$$* : \Omega_{sm}^1(R) \rightarrow \Omega_{sm}^1(R)$$

defined in local holomorphic coordinates  $z = x + iy$  by

$$*(f dx + g dy) = -g dx + f dy$$

so that

$$*(dz) = -i dz$$

$$*(d\bar{z}) = i d\bar{z}$$

The Hodge star operator satisfies

$$** = -1$$

on 1-forms. A differential form  $\omega$  is called **harmonic** if

$$d\omega = 0$$

$$d * \omega = 0$$

Recall that we had a decomposition of the square integrable smooth 1 forms on a compact Riemann surface  $R$

$$L^2\Omega_{sm}^1(R) = (dC^\infty(R)) \oplus H \oplus (*dC^\infty(R))$$

where  $H$  is the space of harmonic 1-forms. Moreover,  $H$  consists of smooth forms. Also recall that we defined an inner product on  $L^2\Omega_{sm}^1(R)$  by

$$\langle \alpha, \beta \rangle = \int_R \alpha \wedge * \beta$$

The decomposition is orthogonal with respect to this inner product in the sense that

$$\langle df, *dg \rangle = 0$$

for all smooth functions  $f, g$  on  $R$ .

Any closed differential form has  $2g$  periods given by integrating over the edges of the polygonal decomposition of  $R$ . The form is exact if and only if all its periods vanish.

1. In particular, the space of harmonic forms is at most  $2g$ -dimensional because it injects into the space of period vectors  $\mathbb{R}^{2g}$ .
2. There are  $g$  linearly independent holomorphic differentials on  $R$  and  $g$  linearly independent antiholomorphic differentials on  $R$ , where holomorphic and antiholomorphic are defined as the  $\pm i$  eigenspaces of the Hodge star operator  $*$ . Since eigenspaces don't intersect, we have

$$\dim H \geq 2g$$

Combining with the previous point, we have

$$\dim H = 2g$$

and the holomorphic and antiholomorphic differentials give bases for the eigenspaces.

Let  $a_1, \dots, a_g, b_1, \dots, b_g$  be the edges of the polygonal decomposition of  $R$ . Let  $\phi, \phi'$  be holomorphic differentials, and  $A_i = \int_{a_i} \phi$ ,  $B_i = \int_{b_i} \phi$  and similarly for  $\phi'$ . Using Stokes' theorem, we can compute

$$0 = \int_R \phi \wedge \phi' = \sum_{i=1}^g \left( \int_{a_i} \phi \int_{b_i} \phi' - \int_{b_i} \phi \int_{a_i} \phi' \right) = \sum_{i=1}^g (A_i B'_i - B_i A'_i)$$

where the second equality comes from writing  $\phi = dF$  (locally! there is monodromy) so that

$$\begin{aligned} \int_X \phi \wedge \phi' &= \int_P dF \wedge \phi' \\ &= \int_P d(F\phi') - F d\phi' \\ &= \oint_{\partial P} F \phi' \\ &= \sum_{i=1}^g \left( \int_{a_i} F \phi' + \int_{b_i} F \phi' - \int_{b_i^{-1}} F \phi' - \int_{a_i^{-1}} F \phi' \right) \\ &= \sum_{i=1}^g \left( \int_{a_i} F \phi' - \int_{a_i^{-1}} (F \phi') \right) + \sum_{i=1}^g \left( \int_{b_i} F \phi' - \int_{b_i^{-1}} (F \phi') \right) \end{aligned}$$

and the difference between integrating over  $a_i$  and  $a_i^{-1}$  is given by the period of  $\phi$  over the cycle, that is precisely  $A_i$ . Similarly for  $b_i$ . We also have the relation

$$\sum_i (A_i \overline{B_i} - B_i \overline{A_i}) = i \langle \phi, \phi' \rangle$$

We have the following consequences:

- For any nonzero holomorphic differential  $\phi$

$$\Im \left( \sum_i A_i \overline{B_i} \right) > 0$$

- A holomorphic differential with all periods real is zero. This is because

$$\sum_i (A_i \overline{B_i} - B_i \overline{A_i}) = 0 = i \langle \phi, \phi' \rangle$$

so  $\phi = 0$ .

- We can choose a basis of holomorphic differentials  $\phi_1, \dots, \phi_g$  such that

$$\int_{a_i} \phi_j = \delta_{ij}$$

Then the period matrix

$$\Pi_{ij} = \int_{b_i} \phi_j$$

is symmetric with positive definite imaginary part. In particular

$$[A|B] \rightarrow [I|\Pi]$$

where  $[A|B]$  is the old period matrix,  $\Pi$  is symmetric and  $\Im \Pi$  is positive definite.

- The periods form a lattice in  $\mathbb{C}^g$  of rank  $2g$ .

We are finally able to give a criterion for when a divisor is principal, completing the multiplicative Cousin problem.

**Theorem 1.411** (Abel's theorem). *A divisor  $D = \sum n_i p_i$  on a compact Riemann surface  $R$  is principal if and only if*

$$\deg D = 0$$

$$\sum n_i \int_{p_0}^{p_i} \vec{\Phi} \in \Lambda$$

for all  $j = 1, \dots, g$ , where  $\vec{\Phi} = (\phi_1, \dots, \phi_g)$  is a vector whose entries are a basis of holomorphic differentials normalized so that  $\int_{a_i} \phi_j = \delta_{ij}$  and  $\Lambda$  is the period lattice generated by the integrals of the  $\phi_j$  over the  $a_i$  and  $b_i$  cycles.

**Definition 1.412.** *The **Jacobian variety** of a compact Riemann surface  $R$  of genus  $g$  is the complex torus*

$$J(R) = \mathbb{C}^g / \Lambda$$

where  $\Lambda$  is the period lattice generated by the integrals of a basis of holomorphic differentials over the  $a_i$  and  $b_i$  cycles.

The Jacobian is independent of the choice of basis of holomorphic differentials and choice of symplectic basis of  $H_1(R, \mathbb{Z})$  up to isomorphism of complex tori. It is a compact complex manifold of dimension  $g$ , and an abelian group. One can show that every connected abelian group which is a complex manifold (and where the group structure is holomorphic) is a quotient of  $\mathbb{C}^g$  by a co-compact lattice. The specific properties of  $[A|B]$  ( $A^{-1}B$  is symmetric with positive definite imaginary part) impose a further restriction. They give rise to principally polarized abelian varieties.

We can give another interpretation of Abel's theorem.

**Theorem 1.413** (Abel's theorem). *The kernel of the Abel-Jacobi map*

$$\text{Div}^0(R) \rightarrow J(R)$$

$$D = \sum n_i p_i \mapsto \sum n_i \int_{p_0}^{p_i} \vec{\Phi}$$

is precisely the group of principal divisors on  $R$ . In particular, the Abel-Jacobi map descends to an isomorphism

$$\text{Pic}^0(R) \rightarrow J(R)$$

$J$  is isomorphic to the group  $\text{Div}^0(R) / \text{PDiv}(R)$  of isomorphism classes of degree-zero line bundles on  $R$ .

On line bundles, the group operation is the tensor product. Note that both sides have natural holomorphic structures: the divisor side can be built from symmetric powers of  $R$ , and the period map to  $J$  was given by integrating holomorphic forms, and is thus holomorphic: so we should suspect a stronger statement behind this.

Choose a point  $r \in R$ . On the product  $R \times J$ , there is the family of holomorphic line bundles over  $R$  parametrized by the points of  $J$ , with trivialized fibers over the point  $r$ . Without trivializing at  $r$ , each line bundle has a group  $\mathbb{C}^\times$  of automorphisms, and the family of line bundles is not uniquely determined: for instance, you can tensor it with your favorite line bundle over the base  $J$ , without changing the isomorphism type on any factor of  $R$ .

It turns out that these individual line bundles on  $R$  assemble to a holomorphic line bundle

$$\mathcal{P} \rightarrow R \times J,$$

the **universal** or **Poincaré** line bundle (of degree zero). These objects  $\mathcal{P}$  and  $J$  can be characterized by a universal property:

**Proposition 1.414.** *Fix  $r \in R$ . There exists a line bundle  $\mathcal{P}$  on  $R \times J$  together with a trivialization*

$$\mathcal{P}|_{\{r\} \times J} \cong \mathcal{O}_J$$

*such that for any complex space  $B$  and line bundle  $L$  on  $R \times B$  equipped with a trivialization along  $\{r\} \times B$ , there exists a unique morphism  $f_L : B \rightarrow J$  and a unique isomorphism*

$$(\mathrm{id}_R \times f_L)^* \mathcal{P} \cong L$$

*compatible with the given trivializations.*

For each degree  $d \geq 0$ , we have a holomorphic map

$$AJ_d : \mathrm{Sym}^d R \longrightarrow J_d,$$

defined by integrating the holomorphic differentials: choose an auxiliary base point  $r \in R$ , and then send a divisor  $D \in \mathrm{Sym}^d R$  to the sum of integrals along a chosen collection of paths from  $r$  to the points of  $D$ . A change of paths will change the integrals by an integral linear combination of periods, so the map to  $J_0$  is well defined; now use the divisor  $d\{r\}$  to identify  $J_0$  with  $J_d$ , and the resulting map to  $J_d$  is now also independent of your choice of  $r$ . The image of the divisor  $D$  is the isomorphism class of the line bundle  $\mathcal{O}(D)$ . In particular, the image of the Abel-Jacobi map is precisely the set of isomorphism classes of line bundles which admit a nontrivial holomorphic section (i.e., those line bundles which can be represented as  $\mathcal{O}(D)$  for some effective divisor  $D$ ).

**Remark 1.415** (Identifying the differential of  $AJ_d$ ). *First we identify the tangent space of  $J_d$  at a point  $[L]$  representing a line bundle  $L$  on  $R$ . Since  $J_d \cong J_0 \cong H^0(R, K_R)^* / H_1(R, \mathbb{Z})$ , we have*

$$T_{J_d, [L]} \cong H^0(R, K_R)^*.$$

Use the Abel-Jacobi map in holomorphic 1-form coordinates. Pick a basis  $\{\omega_1, \dots, \omega_g\}$  of  $H^0(R, K_R)$  and periods so that

$$AJ_1(p) = \left( \int_{r_0}^p \omega_1, \dots, \int_{r_0}^p \omega_g \right) \in J$$

For reduced  $D = p_1 + \dots + p_d$ , we have  $AJ_d(D) = \sum_{i=1}^d AJ_1(p_i)$ . Let  $v = (v_1, \dots, v_d) \in \bigoplus_i T_{p_i} R$  be tangent to  $\text{Sym}^d R$  at  $D$ . Vary  $p_i(t)$  with  $p'_i(0) = v_i$ . Then the derivative of an integral of a holomorphic form along a path with moving endpoint is just evaluation of the form at that endpoint against the velocity vector:

$$\frac{d}{dt} \bigg|_{t=0} \int_{r_0}^{p_i(t)} \omega_k = \omega_k(p_i)(v_i)$$

Summing over  $i$  gives the differential

$$d(AJ_d)_D : \bigoplus_{i=1}^d T_{p_i} R \longrightarrow H^0(R, K_R)^*$$

$$(v_1, \dots, v_d) \longmapsto \left( \omega \mapsto \sum_{i=1}^d \omega(p_i)(v_i) \right)$$

The situation is more delicate if  $D$  is not reduced. Choose a local coordinate  $z$  near each  $p_i$ . Write  $\omega = f(z) dz$ , so  $f(z)$  is holomorphic.

A tangent vector to  $\text{Sym}^d R$  at  $D$  corresponds to a choice of infinitesimal deformation of each  $m_i p_i$ , i.e. an element of  $H^0(\mathcal{O}_R(D)|_D) \cong \bigoplus_i \mathbb{C}[z]/(z^{m_i})$ . That means you specify the coefficients of  $(1, z, z^2, \dots, z^{m_i-1})$  at each  $p_i$ .

Then the differential of the Abel-Jacobi map acts by:

$$d(AJ_d)_D : \bigoplus_i \mathbb{C}[z]/(z^{m_i}) \longrightarrow H^0(K_R)^*$$

$$(v_{i,0}, v_{i,1}, \dots, v_{i,m_i-1})_i \mapsto \left( \omega \mapsto \sum_i \sum_{k=0}^{m_i-1} \frac{1}{k!} \omega^{(k)}(p_i) v_{i,k} \right)$$

where  $\omega^{(k)}(p_i)$  means the  $k$ -th derivative (the  $k$ -jet) of the local coefficient function  $f(z)$  of  $\omega = f(z) dz$ .

In general, there is a uniform way to describe the differential valid for all  $D$ : The differential

$$d(AJ_d)_D : H^0(\mathcal{O}_R(D)|_D) \rightarrow H^1(\mathcal{O}_R)$$

is given by the connecting homomorphism in the long exact sequence

$$0 \rightarrow \mathcal{O}_R \rightarrow \mathcal{O}_R(D) \rightarrow \mathcal{O}_R(D)|_D \rightarrow 0$$

The Abel Jacobi map will hold the key to Riemann's solution of the multiplicative Cousin problem, but we need to settle some of its properties first. The word generic in the next theorem means away from an analytic subspace of strictly lower dimension. Analytic subspaces are locally cut out by a collection of holomorphic functions; they are usually not submanifolds, but they are always stratified by locally closed submanifolds of decreasing dimensions.

**Theorem 1.416** (Properties of the Abel-Jacobi map). *Let  $AJ_d : \text{Sym}^d(R) \rightarrow J_d$  be the Abel-Jacobi map which sends a degree  $d$  effective divisor  $D$  on  $R$  to the isomorphism class of the line bundle  $\mathcal{O}(D)$ .*

1. *The fiber of  $AJ_d$  at a point  $j \in J$  representing a line bundle  $L$  on  $R$  is isomorphic to the projective space  $\mathbb{P}H^0(R; L)$ . (This is empty if  $h^0(R; L) = 0$ .)*
2. *For  $d \leq g$ , the map  $AJ_d$  is generically injective, and a local embedding in  $J_d$ .*
3. *For  $d = g$ , it is also surjective, and therefore a generic isomorphism.*
4. *For  $d \geq 2g - 1$ , the map  $AJ_d$  realizes  $\text{Sym}^d(R)$  as a holomorphic projective bundle over  $J_d$ .*

*Proof.*

1. A point in the fiber over  $j$  is an effective divisor  $D$  of degree  $d$  such that  $\mathcal{O}(D) \cong L$ . The set of such divisors is in bijection with the set of nonzero holomorphic sections of  $L$ , modulo scaling, which is precisely  $\mathbb{P}H^0(R; L)$ .
2. The fibers are connected. Local injectivity/isomorphism can be checked by computing the differential of  $AJ_d$  and checking that it is injective at generic points. The differential at  $(p_1 + \cdots + p_d) \in \text{Sym}^d R$  is given by the map

$$\begin{aligned} T_{p_1}R \oplus \cdots \oplus T_{p_d}R &\rightarrow T_{AJ_d(p_1+\cdots+p_d)}J_d \cong H^0(R, K_R)^* \\ (v_1, \dots, v_d) &\mapsto \left( \omega \mapsto \sum_{i=1}^d \omega(v_i) \right) \end{aligned}$$

The map fails to be injective if and only if there exist nonzero  $v_i \in T_{p_i}R$  such that

$$\sum_{i=1}^d \omega(v_i) = 0$$

for all  $\omega \in H^0(R, K_R)$ . This is equivalent to saying that there exists a holomorphic differential  $\omega$  vanishing at all  $p_i$ . By Riemann-Roch, for generic choices of  $p_i$ , there are no nontrivial holomorphic differentials vanishing at all  $p_i$  if  $d \leq g$ . Hence, the differential is injective at generic points, making  $AJ_d$  a local embedding.

3. The argument is the same as local injectivity.
4. By Riemann-Roch, for  $d \geq 2g - 1$ , we have

$$h^0(R, L) = d - g + 1$$

for all line bundles  $L$  of degree  $d$  because  $H^1(R, L) = 0$  by Serre duality. Therefore, the fibers of  $AJ_d$  are all projective spaces of dimension  $d - g$ . This gives  $\text{Sym}^d(R)$  the structure of a holomorphic projective bundle over  $J_d$ . The map is holomorphic submersion because the differential is surjective at all points, and moreover the map is proper since the domain and target are compact. Hence, by Ehresmann's fibration theorem, it is a holomorphic fiber bundle with fibers  $\mathbb{P}^{d-g}$ .  $\square$

**Remark 1.417** (Non-effective line bundles of nonnegative degree). *The discussion of the Abel Jacobi map shows that for  $g \geq 1$ , there exist line bundles of degree  $0 \leq d < g$  with no nontrivial holomorphic sections.*

Consider the Abel-Jacobi map

$$i_r : X \longrightarrow \mathrm{Pic}^0(X), \quad s \longmapsto \mathcal{O}_X(s - r).$$

Its image is a 1-dimensional subvariety of the  $g$ -dimensional torus  $\mathrm{Pic}^0(X)$  and hence cannot be all of  $\mathrm{Pic}^0(X)$ .

Pick a degree-zero line bundle  $T \in \mathrm{Pic}^0(X)$  not in  $i_r(X)$ . Set

$$L = \mathcal{O}_X(r) \otimes T \in \mathrm{Pic}^1(X).$$

Then  $\deg L = 1$ , but  $h^0(X, L) = 0$ . This is because if  $h^0(X, L) > 0$  then  $L \simeq \mathcal{O}_X(p)$  for some  $p \in X$ . Hence  $T \simeq L \otimes \mathcal{O}_X(-r) \simeq \mathcal{O}_X(p - r)$ , so  $T \in i_r(X)$ , contradicting the choice of  $T$ . Thus  $L$  is a non-effective class of nonnegative degree. More generally, for any  $0 \leq d < g$ , choose  $T \in \mathrm{Pic}^0(X) \setminus (W_d - D_0)$  and set  $L = \mathcal{O}_X(D_0) \otimes T$  with  $\deg D_0 = d$ ; then  $h^0(X, L) = 0$ .

### 1.8.12 Exercises

**Exercise 1.418.** Show that every vector bundle on  $\mathbb{P}^1$  is a direct sum of line bundles.

**Exercise 1.419.** Calculate  $\mathrm{Pic}$  of the nodal cubic and cuspidal cubic.

**Exercise 1.420.** Suppose  $f$  and  $g$  are (nice) homogeneous polynomials of degrees  $d$  and  $e$  in  $\mathbb{C}[X_0, \dots, X_n]$ . Let  $X$  be the subvariety of  $\mathbb{P}^n$  cut out by  $f$  and  $g$ . Calculate the cohomology of  $\mathcal{O}_X$ .

## 2 Lie groups and Lie algebras

### 2.1 Lie Groups and Algebraic Groups

The primary reference for this section is Kirillov's book *An Introduction to Lie Groups and Lie Algebras* chapters 2-7.

#### 2.1.1 Exponential Map and Campbell–Hausdorff Formula

**Proposition 2.1.** Let  $G$  be a Lie group,  $\mathfrak{g} = T_1G$ , and let  $x \in \mathfrak{g}$ . Then there exists a unique morphism of Lie groups

$$\gamma_x : \mathbb{R} \rightarrow G$$

such that

$$\dot{\gamma}_x(0) = x,$$

where the dot denotes differentiation with respect to  $t$ . The map  $\gamma_x$  will be called the one-parameter subgroup corresponding to  $x$ .

The uniqueness immediately implies that for any  $\lambda \in \mathbb{R}$

$$\gamma_x(\lambda t) = \gamma_{\lambda x}(t)$$

**Definition 2.2.** *Let  $G$  be a Lie group,  $\mathfrak{g} = T_1G$ . Then the **exponential map***

$$\exp: \mathfrak{g} \rightarrow G$$

*is defined by*

$$\exp(x) = \gamma_x(1),$$

*where  $\gamma_x(t)$  is the one-parameter subgroup with tangent vector at 1 equal to  $x$ .*

**Theorem 2.3.**

1.  $\exp(x) = 1 + x + \cdots$  (that is,  $\exp(0) = 1$  and  $\exp_*(0): \mathfrak{g} \rightarrow T_1G = \mathfrak{g}$  is the identity map).
2. The exponential map is a diffeomorphism of some neighborhood of 0 in  $\mathfrak{g}$  with a neighborhood of 1 in  $G$ . The local inverse map will be denoted by  $\log$ .
3.  $\exp((t+s)x) = \exp(tx)\exp(sx)$ .
4. For any morphism of Lie groups  $\varphi: G_1 \rightarrow G_2$  and any  $x \in \mathfrak{g}_1$ , we have

$$\exp(\varphi_*(x)) = \varphi(\exp(x)).$$

5. For any  $X \in G$ ,  $y \in \mathfrak{g}$ , we have

$$X \exp(y) X^{-1} = \exp(\text{Ad}_X(y)),$$

*where  $\text{Ad}$  is the adjoint action of  $G$  on  $\mathfrak{g}$  defined by (2.3).*

In general, the exponential map is not surjective. However, it can be shown that for compact Lie groups, the exponential map is surjective.

If  $x, y \in \mathfrak{g}$  are small enough, then  $\exp(x)\exp(y)$  is also close to 1, so it can be written as  $\exp(z)$  for some  $z \in \mathfrak{g}$ . The Campbell–Hausdorff formula gives an explicit expression for  $z$  in terms of  $x, y$  and the Lie bracket.

**Theorem 2.4** (Campbell–Hausdorff formula). *For small enough  $x, y \in \mathfrak{g}$  one has*

$$\exp(x)\exp(y) = \exp(\mu(x, y))$$

*for some  $\mathfrak{g}$ -valued function  $\mu(x, y)$  which is given by the following series which converges in some neighborhood of  $(0, 0)$ :*

$$\mu(x, y) = x + y + \sum_{n \geq 2} \mu_n(x, y), \tag{8}$$



where  $\mu_n(x, y)$  is a Lie polynomial in  $x, y$  of degree  $n$ , i.e. an expression consisting of commutators of  $x, y$ , their commutators, etc., of total degree  $n$  in  $x, y$ . This expression is universal: it does not depend on the Lie algebra  $\mathfrak{g}$  or on the choice of  $x, y$ .

It is possible to write the expression for  $\mu$  explicitly:

$$\mu(x, y) = x + y + \frac{1}{2}[x, y] + \frac{1}{12}([x, [x, y]] + [y, [y, x]]) + \cdots \quad (9)$$

**Corollary 2.5.** *The group operation in a connected Lie group  $G$  can be recovered from the commutator in  $\mathfrak{g} = T_1G$ . Indeed, locally the group law is determined by the Campbell–Hausdorff formula, and  $G$  is generated by a neighborhood of 1.*

**Exercise 2.6.** *Is the exponential map  $\exp : \mathfrak{sl}(2, \mathbb{R}) \rightarrow \mathrm{SL}(2, \mathbb{R})$  surjective?*

**Solution 2.7.** *No. The matrix  $\begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$  is in  $\mathrm{SL}(2, \mathbb{R})$  but is not in the image of the exponential map. This is because if this matrix had a logarithm, then the logarithm would have eigenvalues  $\log(-2) + 2\pi ik$  and  $\log(-\frac{1}{2}) + 2\pi im$  for some integers  $k, m$ . However, the logarithm being real, would have to have eigenvalues coming in complex conjugate pairs, which is not the case here.*

In general, the exponential map is surjective for compact connected Lie groups and for connected, simply connected nilpotent Lie groups. This is because for compact groups, every element is contained in a maximal torus, and the exponential map is clearly surjective on the torus. The map  $\exp : \mathfrak{t} \rightarrow T$  is a surjective group homomorphism with kernel given by the cocharacter lattice of  $G$  (by definition). This intermediate lattice determines the global form of  $G$  among connected compact groups with fixed semisimple Lie algebra. The simply connected form has the smallest exponential lattice  $\Lambda_G = Q^\vee$  and the adjoint form has the largest  $\Lambda_G = P^\vee$ .

### 2.1.2 Lie Group–Lie Algebra Correspondence

**Proposition 2.8.** *If  $G$  is a Lie group, then  $\mathrm{Lie}(G) = T_eG$  with Lie bracket induced by left-invariant vector fields. In particular, every  $v \in T_eG$  extends to a unique left-invariant vector field on  $G$ , and the bracket of two vector fields is defined by the action of first order differential operators. The bracket respects  $G$ -equivariance, so it induces a Lie algebra structure on  $T_eG$ . If  $f : G \rightarrow H$  is a Lie group homomorphism, then the differential at the identity*

$$(df)_e : \mathrm{Lie}(G) \rightarrow \mathrm{Lie}(H)$$

*is a Lie algebra homomorphism.*

**Proposition 2.9.** *Let  $G$  be connected, and let  $f_1, f_2 : G \rightarrow H$  be Lie group homomorphisms. If  $(df_1)_e = (df_2)_e$  then  $f_1 = f_2$ .*

*Proof.* For any  $X \in \mathrm{Lie}(G)$ , the one-parameter subgroup  $t \mapsto \exp_G(tX)$  is sent by  $f_i$  to

$$f_i(\exp_G(tX)) = \exp_H(t(df_i)_e(X)).$$

Since the differentials agree, these are equal for  $i = 1, 2$ . Hence

$$f_1(\exp_G(tX)) = f_2(\exp_G(tX))$$

for all  $t$ , so  $f_1$  and  $f_2$  agree on all one-parameter subgroups. Because  $G$  is connected, it is generated by any neighborhood of the identity, and in particular by exponentials from a small neighborhood in  $\text{Lie}(G)$ . Therefore  $f_1 = f_2$  on all of  $G$ .  $\square$

**Theorem 2.10** (Lie's second theorem). *Let  $G$  be a connected and simply connected Lie group with Lie algebra  $\mathfrak{g}$ , and let  $H$  be a Lie group with Lie algebra  $\mathfrak{h}$ . For every Lie algebra homomorphism  $\varphi : \mathfrak{g} \rightarrow \mathfrak{h}$  there exists a unique Lie group homomorphism  $\Phi : G \rightarrow H$  such that  $d\Phi_e = \varphi$ .*

The following examples show that the hypothesis in the previous theorem are necessary.

**Example 2.11.** *Let  $G_1 = S^1 = \mathbb{R}/\mathbb{Z}$ ,  $G_2 = \mathbb{R}$ . Then the Lie algebras are  $\mathfrak{g}_1 = \mathfrak{g}_2 = \mathbb{R}$  with zero bracket. Consider the identity map  $\mathfrak{g}_1 \rightarrow \mathfrak{g}_2$ . Then the corresponding morphism of Lie groups, if it exists, should be given by  $\theta \mapsto \theta$ ; on the other hand, it must also satisfy  $f(\mathbb{Z}) = \{0\}$ . Thus, this morphism of Lie algebras cannot be lifted to a morphism of Lie groups.*

**Example 2.12.** *Let  $G = T^2 = \mathbb{R}^2/\mathbb{Z}^2$ ; then  $\mathfrak{g} = \mathbb{R}^2$  with zero bracket. Choose an irrational number  $\alpha$  and consider the following subspace*

$$\mathfrak{h} = \{(t, \alpha t) \mid t \in \mathbb{R}\} \subset \mathfrak{g}.$$

*Since  $\mathfrak{g}$  is commutative,  $\mathfrak{h}$  is automatically a one-dimensional Lie subalgebra. However, there is no Lie subgroup corresponding to  $\mathfrak{h}$ : the image of  $\mathfrak{h}$  under the exponential map  $\mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$  is everywhere dense on the torus.*

**Theorem 2.13** (Universal covering Lie group). *If  $G$  is a connected Lie group, then its universal cover*

$$p : \tilde{G} \rightarrow G$$

*admits a unique Lie group structure such that  $p$  is a Lie group homomorphism. Moreover:*

1.  $\tilde{G}$  is connected and simply connected;
2.  $\text{Lie}(\tilde{G}) \cong \text{Lie}(G)$ ;
3.  $\ker(p)$  is discrete and central in  $\tilde{G}$ ;
4.  $\ker(p) \cong \pi_1(G)$ .

*Proof.* The nontrivial part of this theorem is that we can put a Lie group structure on the universal cover  $\tilde{G}$  of  $G$  such that the covering map  $p : \tilde{G} \rightarrow G$  is a homomorphism. Once we have this, all of the other statements follow from general covering space theory and the fact that  $p$  is a local diffeomorphism. In particular remark that the kernel will be discrete because  $p$  is locally injective.  $\square$

**Theorem 2.14** (Closed subgroup theorem, Lie's first theorem). *If  $H \subset G$  is a closed subgroup of a Lie group, then  $H$  is an embedded Lie subgroup and  $\text{Lie}(H) \subset \text{Lie}(G)$  is a Lie subalgebra.*

**Theorem 2.15** (Lie's third theorem). *Every finite-dimensional real Lie algebra is the Lie algebra of a Lie group. Equivalently, every such Lie algebra integrates to a connected simply connected Lie group.*

**Example 2.16.** *We give an example of a Lie algebra homomorphism that does not integrate to a Lie group homomorphism. Consider the Lie algebras  $\mathfrak{su}(2)$  and  $\mathfrak{so}(3)$ . The map  $\text{SU}(2) \rightarrow \text{SO}(3)$  is the universal double cover, with kernel  $\{\pm I\}$ , so it induces an isomorphism of Lie algebras  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ .*

*The inverse Lie algebra isomorphism  $\mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$  does not integrate to a Lie group homomorphism  $S^3 = \text{SO}(3) \rightarrow \text{SU}(2) = \mathbb{RP}^3$ , because if it did, it would split the covering  $\text{SU}(2) \rightarrow \text{SO}(3)$ . But the covering is nontrivial, so no splitting exists.*

### 2.1.3 Fundamental Group and Center

Let  $G$  be a connected Lie group, and let

$$p : \tilde{G} \rightarrow G$$

be its universal covering space. A basic theorem says that  $\tilde{G}$  carries a unique Lie group structure for which  $p$  is a Lie group homomorphism. Then  $\pi_1(G) \cong \ker(p)$  is a discrete subgroup of  $\tilde{G}$ . Moreover  $\ker(p) \subset Z(\tilde{G})$  because for any  $\gamma \in \ker(p)$ , its orbit under conjugation by  $\tilde{G}$  is connected because  $\tilde{G}$  is connected, and it is contained in the discrete set  $\ker(p)$ , so it must be a single point. Hence  $\gamma$  is central in  $\tilde{G}$ . Moreover, if  $G$  is semisimple adjoint, then by definition  $G = \text{Ad}(\tilde{G})$  where  $\text{Ad} : \tilde{G} \rightarrow \text{Aut}(\text{Lie}(\tilde{G}))$  is the adjoint representation. Then the kernel of the projection  $\tilde{G} \rightarrow G$  is precisely the kernel of the adjoint representation, which is the center  $Z(\tilde{G})$ . This is because if  $c \in Z(\tilde{G})$ , then  $\text{Ad}_c$  is the identity map on  $\text{Lie}(\tilde{G})$ , and thus  $c \in \ker(\text{Ad})$ . The other direction is immediate. Thus the projection  $\tilde{G} \rightarrow G$  can be identified with the quotient map  $\tilde{G} \rightarrow \tilde{G}/Z(\tilde{G})$ , so  $\pi_1(G) \cong Z(\tilde{G})$  in this case. This is because

In general one only has

$$\pi_1(G) \cong \ker(\tilde{G} \rightarrow G) \subset Z(\tilde{G}),$$

and for  $G$  connected semisimple, we get equality precisely when  $G$  is adjoint.

**Example 2.17.** *For  $G = S^1$  the universal cover is  $\tilde{G} = \mathbb{R}$  with covering map  $t \mapsto e^{2\pi it}$ . Then  $\pi_1(S^1) \cong \mathbb{Z}$  is only a proper discrete subgroup of the center  $Z(\mathbb{R}) = \mathbb{R}$ .*

For compact connected semisimple Lie groups, one can compute the center and the possible fundamental groups very explicitly from the root datum. If  $\tilde{G}$  is the simply connected form with root lattice  $Q$  and weight lattice  $P$  after fixing a maximal torus  $T$ , then

$$Z(\tilde{G}) \cong P/Q.$$

This follows from the sequence of following observations. Note that  $Z(\tilde{G}) \subset T$  because any  $z \in Z(\tilde{G})$  lies in some maximal torus  $T'$  of  $\tilde{G}$ , and all maximal tori are conjugate, so we can conjugate  $z$  into  $T$ . But  $z$  being central,  $z$  also lies in  $T$ .

Then  $t \in Z(\tilde{G})$  if and only if  $\alpha(t) = 1 \forall \alpha \in \Phi$ . This is because  $Z(\tilde{G})$  is the kernel of the adjoint representation, and the weights of the adjoint representation are precisely the roots. Hence

$$Z(\tilde{G}) = \{t \in \text{Hom}(P, \mathbb{C}^\times) : t|_Q = 1\} \cong \text{Hom}(P/Q, \mathbb{C}^\times).$$

where we are thinking of  $t \in T$  as a homomorphism  $P \rightarrow \mathbb{C}^\times$  by evaluating  $t$  on the weights. The quotient  $P/Q$  is finite because  $P$  and  $Q$  are full lattices of the same rank in the same real vector space  $\mathfrak{h}_{\mathbb{R}}^*$ , so  $Q \subset P$  has finite index.

**Example 2.18.** For type  $A_{n-1}$ :

$$G_{\text{sc}} = \text{SL}_n(\mathbb{C}), \quad G_{\text{ad}} = \text{PGL}_n(\mathbb{C}).$$

The fact that  $\text{SL}_n(\mathbb{C})$  is simply connected can be reduced to that of  $\text{SU}(n)$  by the polar decomposition, where the assertion follows from the long exact sequence of homotopy groups associated to the fibration  $\text{SU}(n) \rightarrow \text{SU}(n+1) \rightarrow S^{2n+1}$ , and induction on  $n$  starting from  $\text{SU}(1) = \{1\}$ . Being in the center of  $\text{SL}_n(\mathbb{C})$  is equivalent to being a scalar matrix. The determinant condition then implies that the scalar must be an  $n$ -th root of unity. Thus the center of the simply connected form is

$$Z(\text{SL}_n(\mathbb{C})) = \mu_n,$$

**Example 2.19.** For type  $B_n$ :

$$G_{\text{sc}} = \text{Spin}(2n+1), \quad G_{\text{ad}} = \text{SO}(2n+1),$$

We can show that  $\pi_1(\text{SO}(2n+1)) \cong \mathbb{Z}/2\mathbb{Z}$  by the long exact sequence of homotopy groups associated to the fibration  $\text{SO}(2n) \rightarrow \text{SO}(2n+1) \rightarrow S^{2n}$ . This sequence implies that  $\pi_1(\text{SO}(m)) \cong \pi_1(\text{SO}(3))$  for all  $m \geq 4$ , and then the identification  $\text{SO}(3) \cong \mathbb{RP}^3$  clearly gives  $\pi_1(\text{SO}(3)) \cong \mathbb{Z}/2\mathbb{Z}$ . Therefore we see that

$$Z(\text{Spin}(2n+1)) \cong \mathbb{Z}/2\mathbb{Z}.$$

**Example 2.20.** For type  $C_n$ :

$$G_{\text{sc}} = \text{Sp}_{2n}(\mathbb{C}), \quad G_{\text{ad}} = \mathbb{P}\text{Sp}_{2n}(\mathbb{C}),$$

and

$$Z(\text{Sp}_{2n}(\mathbb{C})) \cong \mathbb{Z}/2\mathbb{Z}.$$

**Example 2.21.** For type  $D_n$ :

$$G_{\text{sc}} = \text{Spin}(2n), \quad G_{\text{ad}} = \mathbb{P}\text{SO}(2n),$$

and

$$Z(\text{Spin}(2n)) \cong \begin{cases} \mathbb{Z}/4\mathbb{Z}, & n \text{ odd}, \\ \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, & n \text{ even}. \end{cases}$$

**Exercise 2.22.** Show that  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  is not a matrix group, i.e. it does not admit a faithful finite-dimensional representation.

**Solution 2.23.** The group  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  is a  $\mathbb{Z}$ -cover of  $\mathrm{SL}_2(\mathbb{R})$ . I claim that every finite-dimensional representation of  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  factors through  $\mathrm{SL}_2(\mathbb{R})$ , in particular the generator of the kernel  $\mathbb{Z}$  acts trivially. This claim will also imply that no finite sheeted cover of  $\mathrm{SL}_2(\mathbb{R})$  admits a faithful finite-dimensional representation.

Let  $V$  be a finite-dimensional representation of  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$ . Differentiating the action gives a Lie algebra representation of  $\mathfrak{sl}_2(\mathbb{R})$  on  $V$ . We can complexify this to a representation of  $\mathfrak{sl}_2(\mathbb{C})$  on  $V_{\mathbb{C}}$ . This representation integrates to a representation of  $\mathrm{SL}_2(\mathbb{C})$  on  $V_{\mathbb{C}}$  because  $\mathrm{SL}_2(\mathbb{C})$  is simply connected. Restricting this representation to  $\mathrm{SL}_2(\mathbb{R})$  gives a representation of  $\mathrm{SL}_2(\mathbb{R})$  on  $V_{\mathbb{C}}$ .

Now write  $V$  as a direct sum of irreducible  $\mathfrak{sl}_2(\mathbb{C})$ -representations. Each of these is of the form  $\mathrm{Sym}^n(\mathbb{C}^2)$  for some  $n \geq 0$ . Knowing this representation explicitly allows us to identify

$$\mathrm{Sym}^n(\mathbb{C}^2) \cong \mathrm{Sym}^n(\mathbb{R}^2) \otimes_{\mathbb{R}} \mathbb{C}$$

and clearly the action of  $\mathrm{SL}_2(\mathbb{R})$  on  $\mathrm{Sym}^n(\mathbb{C}^2)$  factors through  $\mathrm{Sym}^n(\mathbb{R}^2)$ , so we see that we get a representation of  $\mathrm{SL}_2(\mathbb{R})$  on  $V$ . Now we have two representations of  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  on  $V$ : the original one, and the one obtained by pulling back the representation of  $\mathrm{SL}_2(\mathbb{R})$  we just constructed. They have the same differential, so by uniqueness they must be the same. This shows that the original representation of  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  factors through  $\mathrm{SL}_2(\mathbb{R})$ , so the kernel  $\mathbb{Z}$  acts trivially. Hence no finite-dimensional representation of  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  is faithful.

#### 2.1.4 Compact Lie Groups, complexification, compact and split forms

Recall the classification of compact Lie groups.

- Let  $G$  be a compact Lie group. Let  $G_0$  be the identity component of  $G$ . Then there is a short exact sequence of Lie groups

$$1 \rightarrow G_0 \rightarrow G \rightarrow \pi_0(G) \rightarrow 1.$$

The component group  $\pi_0(G)$  is finite.

- Let  $G$  be a compact, connected Lie group. Then  $G$  has a universal cover  $\tilde{G}$ , and there is a short exact sequence

$$1 \rightarrow \pi_1(G) \rightarrow \tilde{G} \rightarrow G \rightarrow 1.$$

The fundamental group  $\pi_1(G)$  is an abelian group. If  $G$  is already compact semisimple then  $\pi_1(G)$  is finite. If we have torus factors then  $\pi_1(G)$  has free abelian parts. Moreover,  $\tilde{G}$  is the product of a simply connected compact Lie group and a torus.

- Let  $G$  be a compact, connected, simply connected Lie group. Then  $G$  is the product of a simply connected simple Lie groups.

- Let  $G$  be a compact, connected, simply connected simple Lie group. Then  $G$  is determined up to isomorphism by its Lie algebra  $\mathfrak{g}$ . The simple compact Lie algebras are classified by the Dynkin diagrams  $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4$ , and  $G_2$ .

**Definition 2.24.** Let  $\mathfrak{g}$  be a real Lie algebra. Its **complexification** is the complex Lie algebra

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{g} \oplus i\mathfrak{g}$$

with the commutator defined by

$$[x_1 + iy_1, x_2 + iy_2] = [x_1, x_2] - [y_1, y_2] + i([x_1, y_2] + [y_1, x_2])$$

for  $x_1, x_2, y_1, y_2 \in \mathfrak{g}$ . In particular, note that this bracket is  $\mathbb{C}$ -linear in both arguments, so  $\mathfrak{g}_{\mathbb{C}}$  is a complex Lie algebra. We say that  $\mathfrak{g}$  is a **real form** of  $\mathfrak{g}_{\mathbb{C}}$ .

**Example 2.25.** If  $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{R})$ , then  $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ . If  $\mathfrak{g} = \mathfrak{u}(n)$  then  $\mathfrak{g}_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$ . This follows from the fact that any complex matrix can be uniquely written as a sum of skew-hermitian (i.e., from  $\mathfrak{u}(n)$ ) and a hermitian (i.e.  $i\mathfrak{u}(n)$ ), namely

$$X = \frac{X - X^*}{2} + \frac{X + X^*}{2}$$

The Lie bracket also matches, because both sides use the commutator bracket and the map  $\mathfrak{u}(n) \oplus i\mathfrak{u}(n) \rightarrow \mathfrak{gl}(n, \mathbb{C})$  is compatible with the Lie bracket.

These notions can be extended to Lie groups. For simplicity, we only consider the case of connected groups.

**Definition 2.26.** Let  $G$  be a connected complex Lie group,  $\mathfrak{g} = \text{Lie}(G)$  and let  $K \subset G$  be a real Lie subgroup in  $G$  such that  $\mathfrak{k} = \text{Lie}(K)$  is a real form of  $\mathfrak{g}$ . Then  $K$  is called a real form of  $G$ .

It is true that if  $\mathfrak{g} = \text{Lie}(G)$  is the Lie algebra of a connected complex Lie group  $G$ , then every real form  $\mathfrak{k} \subset \mathfrak{g}$  can be obtained from a real form  $K \subset G$  of the Lie group.

Going in the opposite direction, from a real Lie group to a complex one, is more subtle: there are real Lie groups that cannot be obtained as real forms of a complex Lie group. It is still possible, however, to define a complexification  $G_{\mathbb{C}}$  for any real Lie group  $G$ ; however, in general  $G$  is not a subgroup of  $G_{\mathbb{C}}$ . It is defined as the complex Lie group which is initial among those receive a map from  $G$ . If  $G$  is connected and simply connected, one can consider  $\mathfrak{g}_{\mathbb{C}}$  and the simply connected complex Lie group  $G_{\mathbb{C}}$  integrating it. The hypothesis on  $G$  guarantee that we can integrate the map  $\mathfrak{g} \rightarrow \mathfrak{g}_{\mathbb{C}}$ .

**Example 2.27.** The group  $G = \text{SU}(n)$  is a compact real form of the complex group  $\text{SL}(n, \mathbb{C})$ .

Complexification defines an equivalence between connected compact real Lie groups and connected complex reductive Lie groups.

**Theorem 2.28** (Compact–reductive correspondence). *There is an equivalence between the following categories:*

- *Connected compact Lie groups (with smooth homomorphisms),*
- *Connected complex reductive algebraic groups (with morphisms of algebraic groups).*

*given as follows.*

1. *To a connected compact Lie group  $K$ , associate its complexification  $G = K_{\mathbb{C}}$ , which is a connected complex reductive algebraic group with Lie algebra*

$$\mathrm{Lie}(G) \cong \mathrm{Lie}(K) \otimes_{\mathbb{R}} \mathbb{C}.$$

2. *To a connected complex reductive algebraic group  $G$ , associate a maximal compact subgroup  $K \subset G$ , which is a compact real form of  $G$ , satisfying*

$$\mathrm{Lie}(K) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathrm{Lie}(G).$$

*These constructions are inverse to each other up to isomorphism.*

**Definition 2.29.** *A real form is **split** if it contains a Cartan subalgebra  $\mathfrak{h}$  such that every root of  $\mathfrak{g}_{\mathbb{C}}$  with respect to  $\mathfrak{h}_{\mathbb{C}}$  takes real values on  $\mathfrak{h}$ , equivalently if it contains a maximal abelian subalgebra consisting of semisimple elements of dimension equal to the complex rank.*

For a complex semisimple Lie algebra, both compact and split real forms exist, and are unique up to isomorphism.

**Theorem 2.30.** *Every complex semisimple Lie algebra admits:*

1. *a compact real form, unique up to inner automorphism;*
2. *a split real form, unique up to isomorphism.*

The compact form is the one with the most negative-definite Killing form. The split form is the one with the largest possible noncompact torus.

**Example 2.31.** *For  $\mathfrak{sl}_n(\mathbb{C})$ :*

$$\text{compact real form: } \mathfrak{su}(n), \quad \text{split real form: } \mathfrak{sl}_n(\mathbb{R}).$$

**Example 2.32.** *For  $\mathfrak{so}_{2n+1}(\mathbb{C})$ :*

$$\text{compact real form: } \mathfrak{so}(2n+1), \quad \text{split real form: } \mathfrak{so}(n+1, n).$$

**Example 2.33.** For  $\mathfrak{sp}_{2n}(\mathbb{C})$ :

$$\text{compact real form: } \mathfrak{sp}(n), \quad \text{split real form: } \mathfrak{sp}_{2n}(\mathbb{R}).$$

**Example 2.34.** For  $\mathfrak{so}_{2n}(\mathbb{C})$ :

$$\text{compact real form: } \mathfrak{so}(2n), \quad \text{split real form: } \mathfrak{so}(n, n).$$

One way to construct the compact real form is from a Chevalley basis. If

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha}$$

with root vectors  $X_{\alpha}$ , then the compact real form is spanned over  $\mathbb{R}$  by

$$iH_{\alpha}, \quad X_{\alpha} - X_{-\alpha}, \quad i(X_{\alpha} + X_{-\alpha}).$$

The split real form is spanned over  $\mathbb{R}$  by the same Chevalley basis without inserting the factors of  $i$ .

**Example 2.35.** For  $\mathfrak{sl}_2(\mathbb{C})$  we have the Chevalley basis  $E, F, H$  with  $[H, E] = 2E$ ,  $[H, F] = -2F$ , and  $[E, F] = H$ . The compact real form  $\mathfrak{su}(2)$  is spanned by the matrices

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

subject to the relations  $[A, B] = 2C$ ,  $[B, C] = 2A$ , and  $[C, A] = 2B$ . Note that the Cartan subalgebra is  $\mathbb{R}iH$  acts by imaginary eigenvalues on the root spaces, unlike the split form  $\mathfrak{sl}_2(\mathbb{R})$

$$\mathfrak{sl}_2(\mathbb{R}) = \mathfrak{k} \oplus \mathfrak{p} = \mathbb{R}(E - F) \oplus (\mathbb{R}H \oplus \mathbb{R}(E + F)).$$

**Exercise 2.36.** Show that  $\mathfrak{sl}_2(\mathbb{C})$  has exactly two nonisomorphic real forms: the compact form  $\mathfrak{su}(2)$  and the split form  $\mathfrak{sl}_2(\mathbb{R})$ .

*Proof.* Let  $\mathfrak{g}$  be a real form of  $\mathfrak{sl}_2(\mathbb{C})$ . The adjoint representation  $\mathfrak{g} \rightarrow \mathfrak{so}(\mathfrak{g}, \kappa)$  is injective because  $\mathfrak{g}$  is semisimple, and  $\mathfrak{so}(\mathfrak{g}, \kappa)$  is 3-dimensional, because after choosing a basis of  $\mathfrak{g}$  in which  $\kappa$  is given by an invertible symmetric matrix  $X$ , the invariance condition becomes those  $A \in \text{GL}(\mathfrak{g})$  such that  $A^T X + XA = 0$ . That is,  $SA$  is skew-symmetric, so  $A$  is determined by the skew-symmetric matrix  $SA$ , which has 3 independent entries.

This shows that  $\mathfrak{g} \rightarrow \mathfrak{so}(\mathfrak{g}, \kappa)$  is an isomorphism, so  $\mathfrak{g}$  is isomorphic to  $\mathfrak{so}(3)$  or  $\mathfrak{so}(2, 1)$ . The former is the compact form  $\mathfrak{su}(2)$  and the latter is the split form  $\mathfrak{sl}_2(\mathbb{R})$ . The first identification can be read off from the double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$  which is modeled on the unit quaternions acting on purely imaginary quaternions  $\cong \mathbb{R}^3$  by conjugation, which clearly preserves orientation.



The second identification follows from a direct calculation of the signature of the Killing form on  $\mathfrak{sl}_2(\mathbb{R})$ . On  $\mathfrak{sl}_n$  the Killing form is given by  $\kappa(X, Y) = 2n \operatorname{tr}(XY)$  which follows from direct calculation. Then

$$\begin{aligned}\kappa(H, H) &= 8, \\ \kappa(E, F) &= 4, \\ \kappa(E, E) &= \kappa(F, F) = 0\end{aligned}$$

and then the basis  $E - F$ ,  $H$ , and  $E + F$  is orthogonal with respect to  $\kappa$  and has signature  $(2, 1)$ .  $\square$

In general, real forms cannot be classified by the signature of the Killing form. The classification is given by Satake diagrams, which are Dynkin diagrams where some vertices are colored black and some are white, and some pairs of vertices are connected by an arrow.

### 2.1.5 Cartan Involution and Maximal Compact Subgroups

A central structural fact for real semisimple Lie groups is that they admit a canonical decomposition into a compact and a noncompact direction. This is encoded by a Cartan involution.

**Definition 2.37.** *Let  $\mathfrak{g}$  be a real semisimple Lie algebra. A **Cartan involution** is a Lie algebra automorphism*

$$\theta : \mathfrak{g} \rightarrow \mathfrak{g}$$

*such that  $\theta^2 = 1$  and the bilinear form*

$$B_\theta(X, Y) := -B(X, \theta Y)$$

*is positive definite, where  $B$  is the Killing form of  $\mathfrak{g}$ .*

Since  $\theta$  is an involution,  $\mathfrak{g}$  decomposes into its  $\pm 1$  eigenspaces:

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \quad \mathfrak{k} = \{X \in \mathfrak{g} : \theta(X) = X\}, \quad \mathfrak{p} = \{X \in \mathfrak{g} : \theta(X) = -X\}.$$

This is the **Cartan decomposition** of the Lie algebra. Since  $\theta$  is a Lie algebra automorphism, the bracket satisfies

$$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}.$$

The positivity condition implies that the Killing form has opposite signs on the two summands. Indeed, if  $X \in \mathfrak{k}$  then  $\theta X = X$ , so

$$B_\theta(X, X) = -B(X, X) < 0$$

whereas if  $Y \in \mathfrak{p}$  then  $\theta Y = -Y$ , so

$$B_\theta(Y, Y) = B(Y, Y) > 0$$

Then the compact real form of  $\mathfrak{g}$  is  $\mathfrak{k} \oplus i\mathfrak{p}$ , where the Killing form is clearly negative definite on both summands.

**Example 2.38.** Let  $\mathfrak{g} \subset M_n(\mathbb{R})$  be a real matrix Lie algebra closed under transpose, and every matrix is a sum of a symmetric and an antisymmetric matrix

$$X = \frac{X - X^*}{2} + \frac{X + X^*}{2}$$

where  $X^*$  is the conjugate transpose of  $X$ . Then letting  $\mathfrak{k}$  be the antisymmetric matrices and  $\mathfrak{p}$  be the symmetric matrices, we get a Cartan decomposition  $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$  and the compact real form is

$$\mathfrak{u}_0 = \mathfrak{k} \oplus i\mathfrak{p}$$

sits inside skew-Hermitian matrices  $\mathfrak{u}(n)$ . This is morally why  $\mathfrak{u}_0$  is the compact real form of  $\mathfrak{g}$ .

At the group level, if  $G$  is a connected semisimple Lie group with Lie algebra  $\mathfrak{g}$ , a Cartan involution on  $\mathfrak{g}$  integrates to a Lie group automorphism

$$\Theta : G \rightarrow G$$

with differential  $d\Theta_e = \theta$ . The fixed-point subgroup

$$K = G^\Theta = \{g \in G : \Theta(g) = g\}$$

has Lie algebra  $\mathfrak{k}$ . More precisely, the identity component of  $G^\Theta$  has Lie algebra  $\mathfrak{k}$ , and one often writes  $K$  for a maximal compact subgroup of  $G$  with this Lie algebra.

The key theorem is that every connected semisimple Lie group has maximal compact subgroups, and they are controlled by Cartan involutions.

**Theorem 2.39** (Cartan decomposition and maximal compact subgroups). *Let  $G$  be a connected semisimple Lie group with finite center. Then there exists a Cartan involution  $\Theta$  of  $G$ . If*

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

*is the corresponding eigenspace decomposition of  $d\Theta_e$ , and  $K = G^\Theta$ , then:*

1.  $K$  is a maximal compact subgroup of  $G$ ;
2. every maximal compact subgroup of  $G$  is conjugate to  $K$ ;
3. the map

$$K \times \mathfrak{p} \rightarrow G, \quad (k, X) \mapsto k \exp(X)$$

*is a diffeomorphism.*

Thus one may think of  $G$  as built from the compact group  $K$  and the vector space  $\mathfrak{p}$ . The decomposition is not a decomposition by subgroups, since  $\exp(\mathfrak{p})$  is generally not a subgroup, but it gives a very efficient global parametrization of  $G$ .

**Exercise 2.40** (Past Quals). Let  $G = \mathrm{SL}(n, \mathbb{R})$ . Show that  $\Theta(g) = (g^T)^{-1}$  is a Cartan involution of  $G$ , identify the corresponding decomposition  $\mathfrak{sl}_n(\mathbb{R}) = \mathfrak{k} \oplus \mathfrak{p}$  and deduce that  $\mathrm{SO}(n)$  is a maximal compact subgroup of  $\mathrm{SL}(n, \mathbb{R})$ .

**Solution 2.41.** Compute the differential. Writing  $g(t) = I + tX + O(t^2)$ , we have

$$\Theta(g(t)) = (g(t)^T)^{-1} = (I + tX^T + O(t^2))^{-1} = I - tX^T + O(t^2)$$

Hence  $d\Theta_e(X) = -X^T$  is the Cartan involution on  $\mathfrak{sl}_n(\mathbb{R})$ . We compute the eigenspaces.

$$\mathfrak{k} = \{X : -X^T = X\} = \{X : X^T = -X\} = \mathfrak{so}(n)$$

$$\mathfrak{p} = \{X : -X^T = -X\} = \{X : X^T = X\}$$

Thus

$$\mathfrak{sl}_n(\mathbb{R}) = \mathfrak{so}(n) \oplus \{X \in \mathfrak{sl}_n(\mathbb{R}) : X^T = X\}$$

Now we check the Cartan condition. The Killing form on  $\mathfrak{sl}_n(\mathbb{R})$  is a positive scalar multiple of  $(X, Y) \mapsto \mathrm{tr}(XY)$ . Therefore  $-B(X, \theta X)$  is a positive scalar multiple of  $-\mathrm{tr}(X(-X^T)) = \mathrm{tr}(XX^T)$ . But  $\mathrm{tr}(XX^T) = \sum_{i,j} x_{ij}^2 > 0$  for every nonzero matrix  $X$ . So  $B_\theta(X, X) = -B(X, \theta X)$  is positive definite, and  $\theta$  is a Cartan involution.

The fixed points of  $\Theta$  are the matrices satisfying  $g^T g = I$ , thus  $G^\Theta = \mathrm{SO}(n)$  is a maximal compact subgroup of  $\mathrm{SL}(n, \mathbb{R})$ .

### 2.1.6 Borel, Parabolic, and Levi Subgroups

Let  $G$  be a connected reductive algebraic group over an algebraically closed field  $k$ .

**Definition 2.42.** A **solvable group** is a group  $G$  such that the derived series

$$G^{(0)} = G, \quad G^{(n+1)} = [G^{(n)}, G^{(n)}]$$

eventually becomes trivial,  $[x, y]$  denoting the commutator  $xyx^{-1}y^{-1}$ . A **Borel subgroup** of  $G$  is a maximal connected solvable closed subgroup of  $G$ .

A fundamental fact is that Borel subgroups always exist, and they are all conjugate to each other. The basic example is  $G = \mathrm{GL}_n$ , where the subgroup  $B$  of invertible upper triangular matrices is a Borel subgroup. More generally, any conjugate of  $B$  is again a Borel subgroup, and in fact every Borel subgroup of  $\mathrm{GL}_n$  is of this form. In  $\mathrm{GL}_n$ , the quotient  $\mathrm{GL}_n/B$  identifies with the variety of complete flags

$$0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = k^n, \quad \dim V_i = i.$$

**Definition 2.43.** A **parabolic subgroup** of  $G$  is a closed subgroup  $P \subset G$  containing a Borel subgroup.

For  $G = \mathrm{GL}_n$ , the parabolic subgroups are exactly the stabilizers of partial flags. For example, the subgroup of block upper triangular matrices of type  $(n_1, \dots, n_r)$  is parabolic, and it stabilizes a flag

$$0 \subset V_{n_1} \subset V_{n_1+n_2} \subset \cdots \subset k^n.$$

**Definition 2.44.** If  $P$  is a parabolic subgroup, its **unipotent radical**  $R_u(P)$  is the largest connected normal unipotent subgroup of  $P$ . The quotient  $P/R_u(P)$  is the maximal reductive quotient of  $P$  and a Levi subgroup  $L \subset P$  is a choice of splitting of

$$1 \rightarrow R_u(P) \rightarrow P \rightarrow P/R_u(P) \rightarrow 1$$

The Levi subgroup  $L$  is not canonical, but it is unique up to conjugation by an element of  $R_u(P)$ . The decomposition

$$P = L \ltimes R_u(P).$$

is called a **Levi decomposition** of  $P$ .

For  $G = \mathrm{GL}_n$ , if  $P$  is the subgroup of block upper triangular matrices of type  $(n_1, \dots, n_r)$ , then

$$R_u(P) = \{\text{block upper triangular matrices with identity diagonal blocks}\},$$

and a Levi subgroup is

$$L \cong \mathrm{GL}_{n_1} \times \cdots \times \mathrm{GL}_{n_r},$$

Fix a maximal torus  $T \subset G$  and a Borel subgroup  $B \supset T$ . Then  $B$  determines a set of positive roots  $\Phi^+$  and a set of simple roots  $\Delta$ . With this choice, the standard parabolic subgroups containing  $B$  are classified by subsets  $\Theta \subset \Delta$ . For each such  $\Theta$ , one gets a parabolic subgroup  $P_\Theta$ .

Its Levi subgroup  $L_\Theta$  is the reductive subgroup generated by  $T$  together with the root subgroups corresponding to the roots in the span of  $\Theta$ . Its unipotent radical is generated by the positive root subgroups not belonging to that Levi factor.

### 2.1.7 Topology of Lie Groups

**Exercise 2.45.** Are there any Lie groups which are homeomorphic to  $S^2$ ?

**Solution 2.46.** Every Lie group is parallelizable, by picking a basis of the tangent space at the identity and translating it to every point.  $S^2$  is not parallelizable because if it were, by definition its tangent bundle would be trivial, and integrating the Euler class of the tangent bundle would give zero, contradicting the fact that the Euler characteristic of  $S^2$  is 2. Hence there are no Lie groups homeomorphic to  $S^2$ .

**Example 2.47.**  $S^3$  is a Lie group, namely  $\mathrm{SU}(2)$  the group of unit quaternions. A unit quaternion  $q$  acts on the purely imaginary quaternions  $V = \mathrm{Im}(\mathbb{H}) = \{bi + cj + dk\} \cong \mathbb{R}^3$  by conjugation  $v \mapsto qvq^{-1}$ . The operator  $T_q(v) = qvq^{-1}$  preserves lengths and angles, so it gives an element of  $\mathrm{SO}(V) \cong \mathrm{SO}(3)$ . Thus we get a homomorphism  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  whose kernel is the center of  $\mathrm{SU}(2)$ ,  $\{\pm 1\}$ , so it is the universal double cover.

## 2.2 Lie Algebras

### 2.2.1 Solvable and Nilpotent Lie Algebras

Let  $V$  be finite dimensional over an algebraically closed field. Let  $T$  be a linear operator on  $V$ .  $T$  has generalized eigenspaces  $V_\lambda$  for  $\lambda \in k$

$$V_\lambda = \{v \in V : (T - \lambda)^n v = 0 \text{ for some } n\}$$

**Theorem 2.48.**  *$V$  is the direct sum of its generalized eigenspaces:*

$$V = \bigoplus_{\lambda \in k} V_\lambda.$$

*There are polynomials  $p_\lambda(x) \in k[x]$  so that  $p_\lambda(T)$  is the projection onto  $V_\lambda$ . The dimension of  $V_\lambda$  is the multiplicity of  $\lambda$  as a root of the characteristic polynomial of  $T$ . The restriction of  $T$  to  $V_\lambda$  has the form  $\lambda + N$ , where  $N$  is nilpotent.*

We say that a linear operator  $T$  is **semisimple** if it is diagonalizable. We say that  $T$  is **nilpotent** if all of its eigenvalues are zero, or equivalently if  $T^n = 0$  for some  $n$ . The theorem is obtained by letting  $T_s = \sum_\lambda \lambda p_\lambda(T)$  and  $T_n = T - T_s$ .

**Theorem 2.49** (Jordan decomposition). *Let  $T$  be a linear operator on a finite-dimensional vector space  $V$  over an algebraically closed field. Then there exist unique linear operators  $T_s$  and  $T_n$  on  $V$  such that*

1.  $T = T_s + T_n$ ;
2.  $T_s$  is semisimple (diagonalizable);
3.  $T_n$  is nilpotent;
4.  $T_s$  and  $T_n$  commute.

**Theorem 2.50.** *Let  $k$  be algebraically closed, and let  $L$  be a linear operator on a finite-dimensional vector space  $V$ . Let  $M$  be a linear operator on  $V$ . Then  $M$  is killed by some power of  $\text{ad } L$  if and only if  $M$  preserves each generalized eigenspace of  $L$ .*

*Proof.* Let

$$V = \bigoplus_{\lambda \in \sigma(L)} V_\lambda$$

be the generalized eigenspace decomposition of  $L$ . Then

$$\text{End}(V) = \bigoplus_{\lambda, \mu} \text{Hom}(V_\lambda, V_\mu).$$

If  $T \in \text{Hom}(V_\lambda, V_\mu)$ , then on  $V_\lambda$  and  $V_\mu$  we may write

$$L|_{V_\lambda} = \lambda I + N_\lambda, \quad L|_{V_\mu} = \mu I + N_\mu,$$

where  $N_\lambda, N_\mu$  are nilpotent. Hence

$$(\operatorname{ad} L)(T) = LT - TL = (\mu - \lambda)T + N_\mu T - TN_\lambda.$$

The operator

$$T \mapsto N_\mu T - TN_\lambda$$

on  $\operatorname{Hom}(V_\lambda, V_\mu)$  is nilpotent. Therefore  $\operatorname{ad} L$  acts on  $\operatorname{Hom}(V_\lambda, V_\mu)$  with generalized eigenvalue  $\mu - \lambda$ . Thus the generalized eigenspaces of  $\operatorname{ad} L$  on  $\operatorname{End}(V)$  are indexed by differences  $\mu - \lambda$ . In particular,  $M$  is killed by some power of  $\operatorname{ad} L$  if and only if  $M$  has no component

$$V_\lambda \rightarrow V_\mu \quad (\lambda \neq \mu).$$

Equivalently,  $M$  preserves each generalized eigenspace  $V_\lambda$  of  $L$   $\square$ .

We say that a Lie algebra  $\mathfrak{n}$  is **nilpotent** if  $\operatorname{ad} x$  is nilpotent for every  $x \in \mathfrak{n}$ . We saw that for a given linear operator, we can write  $V$  as a direct sum of generalized eigenspaces, and the operator acts on each generalized eigenspace as a scalar plus a nilpotent operator. If  $\mathfrak{n}$  is nilpotent and  $V$  is a  $\mathfrak{n}$ -module, then we can write  $V$  as a direct sum of generalized eigenspaces for the action of  $\mathfrak{n}$ .

**Theorem 2.51.** *Let  $\mathfrak{g}$  be a nilpotent Lie algebra over  $k$ , let  $V$  be a finite-dimensional  $k$ -vector space, and let*

$$\rho : \mathfrak{g} \rightarrow \operatorname{End}(V)$$

*be a representation. Then the weight subspaces of  $\rho$  corresponding to distinct weights are linearly independent. If  $k$  is algebraically closed, and if  $\lambda_1, \dots, \lambda_r$  are all the distinct weights of  $\rho$ , then*

$$V = \bigoplus_{i=1}^r V_{\rho, \lambda_i}.$$

*Idea of proof.* The main point is to prove the decomposition when  $k$  is algebraically closed. We argue by induction on  $\dim V$ .

If every  $\rho(X)$  has only one eigenvalue, say  $\lambda(X)$ , then

$$V = V_{\rho, \lambda},$$

and there is nothing to prove.

Otherwise choose  $X_0 \in \mathfrak{g}$  such that  $\rho(X_0)$  has at least two distinct eigenvalues  $\mu_1, \dots, \mu_r$ . Then the generalized eigenspace decomposition for the single operator  $\rho(X_0)$  gives

$$V = \bigoplus_{i=1}^r V_{\rho(X_0), \mu_i}.$$

Each summand has smaller dimension than  $V$ .

The key point is that these summands are actually  $\mathfrak{g}$ -subrepresentations. Indeed, for any  $X \in \mathfrak{g}$ , since  $\mathfrak{g}$  is nilpotent, there is some  $p \geq 1$  such that

$$(\operatorname{ad} X_0)^p(X) = 0.$$

Applying the representation  $\rho$ , this gives

$$(\operatorname{ad} \rho(X_0))^p(\rho(X)) = 0.$$

By the linear algebra criterion for generalized eigenspaces, this implies that  $\rho(X)$  preserves each generalized eigenspace  $V_{\rho(X_0), \mu_i}$ . Therefore each  $V_{\rho(X_0), \mu_i}$  is a  $\mathfrak{g}$ -stable subspace.

Now apply the induction hypothesis to the restricted representation on each

$$V_i := V_{\rho(X_0), \mu_i}.$$

Each  $V_i$  decomposes as a direct sum of its weight spaces. Putting these decompositions together gives

$$V = \bigoplus_{\lambda} V_{\rho, \lambda}.$$

The linear independence of weight spaces corresponding to distinct weights follows from the same generalized-eigenspace separation: if  $\lambda \neq \mu$ , then for some  $X \in \mathfrak{g}$  one has  $\lambda(X) \neq \mu(X)$ , so the corresponding vectors lie in distinct generalized eigenspaces of  $\rho(X)$ .  $\square$

**Theorem 2.52.** *Let  $G$  be a connected nilpotent analytic group with Lie algebra  $\mathfrak{g}$ .*

1. *There is a polynomial map*

$$P : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

*such that*

$$\exp X \exp Y = \exp P(X, Y) \quad (X, Y \in \mathfrak{g}).$$

2. *Let  $\mathfrak{z} \subseteq \mathfrak{g}$  be the center, and define*

$$D := \{X \in \mathfrak{z} : \exp X = 1\}.$$

*Then  $D$  is a discrete additive subgroup of  $\mathfrak{g}$ , and the exponential map induces an analytic diffeomorphism*

$$\mathfrak{g}/D \xrightarrow{\sim} G.$$

*In particular,  $\exp : \mathfrak{g} \rightarrow G$  is surjective, and  $\mathfrak{g}$  is a covering manifold of  $G$  with covering map  $\exp$ .*

*Proof.* The Baker–Campbell–Hausdorff formula gives, for  $X, Y$  sufficiently close to 0,

$$\exp X \exp Y = \exp \left( X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \cdots \right).$$

Since  $\mathfrak{g}$  is nilpotent, all sufficiently long iterated brackets vanish. Hence the Baker–Campbell–Hausdorff series terminates, so it defines a polynomial map

$$P : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

such that

$$\exp X \exp Y = \exp P(X, Y)$$

near  $(0, 0)$ . Since both sides are analytic in  $X$  and  $Y$ , the identity holds for all  $X, Y \in \mathfrak{g}$ .

We next prove that  $\exp$  is surjective. Let

$$G' = \exp(\mathfrak{g}) \subseteq G.$$

The BCH formula implies that  $G'$  is closed under multiplication:

$$\exp X \exp Y = \exp P(X, Y) \in G'.$$

It is also closed under inverses, since

$$(\exp X)^{-1} = \exp(-X).$$

Thus  $G'$  is a subgroup of  $G$ . Since  $\exp$  is a local diffeomorphism near  $0 \in \mathfrak{g}$ , the subset  $G'$  contains an open neighborhood of the identity in  $G$ . Therefore  $G'$  is an open subgroup. But  $G$  is connected, so every open subgroup is also closed and must equal  $G$ . Hence  $\exp$  is surjective. Now consider the kernel of the exponential map,

$$\ker(\exp) = \{X \in \mathfrak{g} : \exp X = 1\}.$$

Since  $\exp$  is locally a diffeomorphism near 0, this kernel is discrete. Moreover, if  $\exp X = 1$ , then for every  $Y \in \mathfrak{g}$ ,

$$\exp X \exp Y \exp(-X) = \exp Y.$$

Infinitesimally, conjugation by  $\exp X$  acts as  $e^{\text{ad } X}$ . Thus  $e^{\text{ad } X}Y = Y$  for every  $Y \in \mathfrak{g}$ . Since  $\text{ad } X$  is nilpotent, the equality  $e^{\text{ad } X} = I$  implies

$$\text{ad } X = 0.$$

Therefore  $X \in \mathfrak{z}$ , and hence

$$\ker(\exp) = D = \{X \in \mathfrak{z} : \exp X = 1\}.$$

Thus  $\exp : \mathfrak{g} \rightarrow G$  is a surjective local diffeomorphism with discrete kernel  $D$ . Therefore it descends to an analytic diffeomorphism  $\mathfrak{g}/D \xrightarrow{\sim} G$ .  $\square$

If  $G$  is simply connected, then  $\mathfrak{g} \rightarrow G$  being the universal cover implies that  $D = 0$  and hence  $\exp : \mathfrak{g} \rightarrow G$  is a global diffeomorphism. Moreover, if  $H \subset G$  is an analytic subgroup with Lie algebra  $\mathfrak{h} \subset \mathfrak{g}$ , then  $H$  is closed, simply connected, and  $\exp : \mathfrak{h} \rightarrow H$  is a global diffeomorphism.

**Theorem 2.53.** *Let  $V$  be a finite-dimensional vector space over  $\mathbb{R}$  or  $\mathbb{C}$ , and set  $p = \dim V$ . Let  $\bar{\mathfrak{g}} \subseteq \mathfrak{gl}(V)$  be a Lie subalgebra consisting entirely of nilpotent endomorphisms of  $V$ , and let  $\bar{G} \subseteq GL(V)$  be the analytic subgroup with Lie algebra  $\bar{\mathfrak{g}}$ . Then  $\bar{\mathfrak{g}}$  is nilpotent, and  $\bar{G}$  is a simply connected unipotent algebraic subgroup of  $GL(V)$ .*

Moreover, for  $X \in \bar{\mathfrak{g}}$ ,

$$\exp X = I + \sum_{1 \leq s < p} \frac{X^s}{s!},$$



and for  $x \in \bar{G}$ ,

$$\log x = \sum_{1 \leq s < p} (-1)^{s-1} \frac{(x - I)^s}{s}.$$

In particular,  $\exp : \bar{\mathfrak{g}} \rightarrow \bar{G}$  and  $\log : \bar{G} \rightarrow \bar{\mathfrak{g}}$  are mutually inverse polynomial maps.

Two of the basic classes of Lie algebras are **solvable** and **nilpotent** Lie algebras. They may be viewed as the Lie-theoretic analogues of upper triangular and strictly upper triangular matrix algebras.

Given a Lie algebra  $\mathfrak{g}$ , its **derived series** is defined by

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \quad \mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^{(n+1)} = [\mathfrak{g}^{(n)}, \mathfrak{g}^{(n)}].$$

We say that  $\mathfrak{g}$  is **solvable** if  $\mathfrak{g}^{(n)} = 0$  for some  $n$ . Equivalently,  $\mathfrak{g}$  is solvable if and only if its derived subalgebra  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent.

The **lower central series** is

$$\mathfrak{g}_1 = \mathfrak{g}, \quad \mathfrak{g}_{n+1} = [\mathfrak{g}, \mathfrak{g}_n].$$

We say that  $\mathfrak{g}$  is **nilpotent** if  $\mathfrak{g}_n = 0$  for some  $n$ .

**Theorem 2.54** (Engel's theorem). *If every  $\text{ad}(x)$  is nilpotent, then  $\mathfrak{g}$  itself is nilpotent.*

Every nilpotent Lie algebra is solvable, but the converse is false. For example, the Lie algebra of strictly upper triangular matrices in  $\mathfrak{gl}_n(\mathbb{C})$  is nilpotent, while the Lie algebra of all upper triangular matrices is solvable but not nilpotent.

Solvable and nilpotent Lie algebras are the “building blocks” that appear inside general Lie algebras. Every finite-dimensional Lie algebra  $\mathfrak{g}$  has a largest solvable ideal, called the **radical**  $\text{rad}(\mathfrak{g})$ , and a largest nilpotent ideal, called the **nilradical**.

**Theorem 2.55** (Lie's theorem). *If  $\mathfrak{g}$  is a solvable Lie algebra over  $\mathbb{C}$  and  $V$  is a nonzero finite-dimensional  $\mathfrak{g}$ -module, then there exists a basis of  $V$  in which all operators from  $\mathfrak{g}$  are upper triangular.*

Equivalently,  $V$  contains a common eigenvector for  $\mathfrak{g}$ , namely pick a basis  $e_1, \dots, e_n$  of  $V$  such that the operators from  $\mathfrak{g}$  are upper triangular, and then  $e_1$  is a common eigenvector.

**Corollary 2.56.** *Every irreducible representation of a solvable Lie algebra is one-dimensional.*

Every finite-dimensional Lie algebra  $\mathfrak{g}$  over  $\mathbb{C}$  admits a decomposition

$$\mathfrak{g} \cong \text{rad}(\mathfrak{g}) \rtimes \mathfrak{s},$$

where  $\text{rad}(\mathfrak{g})$  is the maximal solvable ideal and  $\mathfrak{s}$  is a semisimple subalgebra (a Levi subalgebra). The radical is uniquely determined, while the Levi subalgebra is unique up

to conjugation. In particular, every Lie algebra is built from a semisimple Lie algebra by extension by a solvable one.

**Example 2.57.**

$$\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}$$

where  $\mathfrak{sl}_n$  is the semisimple part and  $\mathbb{C}$  is the center, which is solvable.

**Example 2.58.** Upper triangular matrices  $\mathfrak{b} \subset \mathfrak{gl}_n$  have no semisimple part, so  $\text{rad}(\mathfrak{b}) = \mathfrak{b}$  and  $\mathfrak{s} = 0$ .

## 2.2.2 Semisimple and Reductive Lie Algebras

**Definition 2.59.** A finite-dimensional Lie algebra  $\mathfrak{g}$  over  $\mathbb{C}$  is called **semisimple** if it has no nonzero solvable ideals. Equivalently, its radical  $\text{rad}(\mathfrak{g})$  is zero.

The fundamental examples are the classical matrix Lie algebras

$$\mathfrak{sl}_n(\mathbb{C}), \quad \mathfrak{so}_n(\mathbb{C}), \quad \mathfrak{sp}_{2n}(\mathbb{C}),$$

together with the five exceptional Lie algebras of types  $G_2, F_4, E_6, E_7, E_8$ .

**Theorem 2.60.** Let  $V$  be an irreducible representation of  $\mathfrak{g}$ . Then any  $h \in \text{rad}(\mathfrak{g})$  acts in  $V$  by scalar operators:

$$\rho(h) = \lambda(h) \text{id}.$$

Also, any  $h \in [\mathfrak{g}, \text{rad}(\mathfrak{g})]$  acts by zero.

*Proof.* By Lie's theorem, there is a common eigenvector in  $V$  for all  $h \in \text{rad}(\mathfrak{g})$ :

$$\rho(h)v = \lambda(h)v$$

for some  $\lambda : \text{rad}(\mathfrak{g}) \rightarrow \mathbb{C}$ . Define

$$V_\lambda = \{w \in V \mid \rho(h)w = \lambda(h)w \text{ for all } h \in \text{rad}(\mathfrak{g})\}.$$

Then the same argument as in the proof of Proposition 6.15 shows that for any  $x \in \mathfrak{g}$ , one has  $\rho(x)(V_\lambda) \subset V_\lambda$ . Thus,  $V_\lambda$  is a subrepresentation; since it is non-zero and  $V$  is irreducible, we must have  $V = V_\lambda$ , which proves the first statement of the theorem. The second statement immediately follows from the first one.  $\square$

From the point of view of representation theory, having non-zero elements which act by zero in any irreducible representation significantly complicates the theory. Thus, it is natural to consider a class of algebras for which

$$[\mathfrak{g}, \text{rad}(\mathfrak{g})] = 0.$$

Let  $\mathfrak{z} = \{x \in \mathfrak{g} : [x, y] = 0 \text{ for all } y \in \mathfrak{g}\}$  be the center of  $\mathfrak{g}$ .

**Definition 2.61.** A Lie algebra is called **reductive** if  $\text{rad}(\mathfrak{g}) = \mathfrak{z}(\mathfrak{g})$ . i.e. if  $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$  is semisimple.

Any semisimple Lie algebra is reductive. The model for a reductive Lie algebra which is not semisimple is  $\mathfrak{gl}_n(\mathbb{C}) = \mathfrak{sl}_n(\mathbb{C}) \oplus \mathbb{C}$ , where the center  $\mathbb{C}$  is the radical. In general, a reductive Lie algebra is a direct sum of a semisimple Lie algebra and an abelian Lie algebra.

**Proposition 2.62.** The following are equivalent for a finite-dimensional Lie algebra  $\mathfrak{g}$  over  $\mathbb{C}$ :

1.  $\mathfrak{g}$  is reductive;
2. every finite-dimensional representation of  $\mathfrak{g}$  is completely reducible;
3. the adjoint representation of  $\mathfrak{g}$  is completely reducible.
4.  $\mathfrak{g} = \mathfrak{z} \oplus \mathfrak{g}_{ss}$ , where  $\mathfrak{z}$  is abelian and  $\mathfrak{g}_{ss}$  is semisimple.

### 2.2.3 Killing Form and Complete Reducibility

Recall that the Killing form of a Lie algebra  $\mathfrak{g}$  is the symmetric bilinear form defined by

$$\kappa(x, y) = \text{tr}(\text{ad}(x) \text{ad}(y)).$$

The Killing form is invariant under the adjoint action, meaning that

$$\kappa([x, y], z) = \kappa(x, [y, z]) \quad \text{for all } x, y, z \in \mathfrak{g}.$$

This follows from the cyclicity of the trace  $\text{tr}(ABC) = \text{tr}(BCA)$ .

**Theorem 2.63.** Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra over  $\mathbb{C}$ . Then  $\mathfrak{g}$  is solvable if and only if  $\kappa(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]) = 0$ .

**Theorem 2.64.** A finite-dimensional Lie algebra is semisimple if and only if its Killing form is nondegenerate.

*Proof.* Suppose  $\text{rad } \mathfrak{g} \neq 0$ . Choose the last nonzero term in the derived series of  $\text{rad } \mathfrak{g}$ , say  $\mathfrak{a}$  which has the property that  $[\mathfrak{a}, \mathfrak{a}] = 0$ , i.e.  $\mathfrak{a}$  is abelian. Also  $[\mathfrak{g}, \mathfrak{a}] \subset \mathfrak{a}$  because each term in the derived series is an ideal (this is quickly verified by induction). Now we will show that  $\kappa(\mathfrak{g}, \mathfrak{a}) = 0$ . Observe that the operator  $\text{ad } x \text{ad } y$  where  $x \in \mathfrak{a}$  and  $y \in \mathfrak{g}$  carries  $\mathfrak{g}$  to  $\mathfrak{a}$ . Therefore its trace on  $\mathfrak{g}$  is the same as that of its trace to the restriction to  $\mathfrak{a}$ . This is elementary linear algebra, because in a basis adapted to  $\mathfrak{a} \subset \mathfrak{g}$ , the matrix is of block form

$$\begin{pmatrix} * & * \\ 0 & 0 \end{pmatrix}$$

But this operator is zero when restricted to  $\mathfrak{a}$ , because  $\text{ad } x \circ \text{ad } y = 0$  for  $x, y \in \mathfrak{a}$  since  $\mathfrak{a}$  is abelian. Thus the Killing form is singular if  $\text{rad } g \neq 0$ . Conversely, suppose that  $\kappa$  is degenerate. Then define its radical

$$\mathfrak{m} = \{x \in \mathfrak{g} : \kappa(x, y) = 0 \text{ for all } y \in \mathfrak{g}\}.$$

Then  $\mathfrak{m}$  is an ideal of  $\mathfrak{g}$  because  $\kappa([x, z], y) = \kappa(x, [z, y]) = 0$ . Because  $\mathfrak{m}$  is an ideal,  $\text{ad } X \circ \text{ad } Y$  carries  $\mathfrak{g}$  to  $\mathfrak{m}$  for any  $X, Y \in \mathfrak{m}$ . Therefore

$$\text{tr}_{\mathfrak{g}}(\text{ad } X \text{ ad } X') = \text{tr}_{\mathfrak{m}}(\text{ad}_{\mathfrak{m}} X \text{ ad}_{\mathfrak{m}} X')$$

But the left hand side is zero by definition of  $\mathfrak{m}$ , so the right hand side is also zero. Thus the Killing form of  $\mathfrak{m}$  is degenerate, so  $\mathfrak{m}$  is solvable. Thus  $\text{rad } g \neq 0$  and  $\mathfrak{g}$  is not semisimple.  $\square$

If  $\mathfrak{g}$  is semisimple, then  $\mathfrak{z}(\mathfrak{g}) = 0$  because the center is an abelian ideal. Thus, the adjoint representation of a semisimple Lie algebra is faithful. Moreover,  $\text{tr}(\text{ad } x) = 0$  for all  $x \in \mathfrak{g}$  because  $x$  can be written as a commutator and  $\text{tr}([A, B]) = 0$  for all  $A, B$ .

**Exercise 2.65** (Past Quals). *Let  $G$  be a connected Lie group whose Killing form  $B$  on  $\mathfrak{g}$  is negative definite. Show that  $G$  is compact.*

The general fact to know here is that a real Lie algebra with negative definite Killing form is the Lie algebra of a compact simply connected Lie group.

**Solution 2.66.** *The negative of  $B$  defines an inner product on  $\mathfrak{g}$  which is positive definite and  $\text{Ad}$ -invariant. Hence there is a map  $G \rightarrow O(\mathfrak{g})$  given by the adjoint representation. This map has kernel  $Z(G)$ , the center of  $G$ . The image of  $G$  is a subgroup of  $O(\mathfrak{g})$ , which is compact. Therefore, we must see that  $Z(G)$  is finite and the image is closed.*

Observe that the Lie algebra center is zero. Indeed, if  $X \in Z(\mathfrak{g})$ , then  $\text{ad}_X = 0$ , so for every  $Y \in \mathfrak{g}$ ,

$$B(X, Y) = \text{tr}(\text{ad}_X \text{ad}_Y) = 0.$$

Since  $B$  is nondegenerate, this implies  $X = 0$ . Hence  $Z(\mathfrak{g}) = 0$  and so we see that  $Z(G)$  is discrete. Also we immediately see that  $\mathfrak{g}$  is semisimple since Cartan's criterion says that the Killing form is nondegenerate if and only if  $\mathfrak{g}$  is semisimple if and only if the center of  $\mathfrak{g}$  is zero.

Next we show that the image is closed. Let  $H := \overline{\text{Ad}(G)} \subset O(\mathfrak{g})$ . Since  $O(\mathfrak{g})$  is compact,  $H$  is compact.

Now compare Lie algebras. The Lie algebra of  $\text{Ad}(G)$  is  $\text{ad}(\mathfrak{g}) \subset \mathfrak{so}(\mathfrak{g})$  because  $d(\text{Ad})_e = \text{ad}$ . Note the funny fact that the Lie algebra of  $O(n)$  is  $\mathfrak{so}(n)$ , because  $O(n)$  has two connected components, distinguished by  $\det = \pm 1$ .  $SO(n)$  is the identity component of  $O(n)$ .

The closure  $H$  has the same Lie algebra, so

$$\text{Lie}(H) = \text{ad}(\mathfrak{g}) = \text{Lie}(\text{Ad}(G))$$

Now  $\text{Ad}(G)$  is connected, because  $G$  is connected. In a Lie group, a connected Lie subgroup is determined by its Lie algebra. Since  $\text{Ad}(G)$  and  $H$  are connected Lie subgroups of  $O(\mathfrak{g})$  with the same Lie algebra, they must coincide so  $\text{Ad}(G) = H$  is closed. Hence  $G/Z(G) \cong \text{Ad}(G)$  is compact. We now have a covering map  $G \rightarrow G/Z(G) = \text{Ad}(G)$  with compact base and discrete fiber. We need to know that the fiber is in fact finite. We can prove this by reasoning about  $\pi_1$ .

$\pi_1(G)$  is abelian so  $\pi_1(G) \cong H_1(G, \mathbb{Z})$  by the Hurewicz map. Thus it is enough to show  $H_1(G, \mathbb{Z})$  is finite. On a compact connected Lie group, every de Rham cohomology class has a left-invariant representative. So it suffices to compute closed left-invariant 1-forms.

A left-invariant 1-form is the same as an element  $\alpha \in \mathfrak{g}^*$ . For left-invariant vector fields  $X, Y$ ,

$$d\alpha(X, Y) = -\alpha([X, Y]).$$

Thus  $\alpha$  is closed iff

$$\alpha([X, Y]) = 0 \quad \forall X, Y \in \mathfrak{g},$$

that is, iff  $\alpha$  vanishes on  $[\mathfrak{g}, \mathfrak{g}]$ . But  $\mathfrak{g}$  is semisimple, so  $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ . Hence the only closed left-invariant 1-form is 0. Therefore  $H^1(G, \mathbb{R}) = 0$ .

Since  $G$  is a compact manifold,  $H_1(G, \mathbb{Z})$  is a finitely generated abelian group. Also

$$H^1(G, \mathbb{R}) \cong \text{Hom}(H_1(G, \mathbb{Z}), \mathbb{R}).$$

Because  $H^1(G, \mathbb{R}) = 0$ , there are no nonzero homomorphisms

$$H_1(G, \mathbb{Z}) \rightarrow \mathbb{R}.$$

So  $H_1(G, \mathbb{Z})$  has no free  $\mathbb{Z}$ -part. Being finitely generated abelian, it must therefore be finite. Thus  $\pi_1(G) \cong H_1(G, \mathbb{Z})$  is finite.

**Theorem 2.67.** *Let  $\mathfrak{g}$  be a semisimple Lie algebra. If  $\mathfrak{h}$  is an ideal of  $\mathfrak{g}$ , then  $\mathfrak{h}^\perp$  is also an ideal,  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^\perp$ , and both  $\mathfrak{h}$  and  $\mathfrak{h}^\perp$  are semisimple. In particular,  $\mathfrak{g}$  is a direct sum of simple Lie algebras.*

*Proof.* We have the crucial fact that if  $\mathfrak{a}$  is an ideal of  $\mathfrak{g}$ , then  $\kappa_{\mathfrak{a}}(x, y) = \kappa_{\mathfrak{g}}(x, y)$  for  $x, y \in \mathfrak{a}$ . Applying this fact to the ideal  $\mathfrak{h} \cap \mathfrak{h}^\perp$ , we see that  $\mathfrak{h} \cap \mathfrak{h}^\perp$  is a solvable ideal of  $\mathfrak{g}$ , so it must be zero.  $\square$

This theorem implies that for semisimple  $\mathfrak{g}$ , we have  $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$  because otherwise,  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  would be nonzero, abelian, and semisimple, which is impossible because every abelian Lie algebra is solvable.

**Theorem 2.68.** *If  $\mathfrak{g}$  is semisimple, then every derivation  $D$  of  $\mathfrak{g}$  is inner, i.e. there is some  $x \in \mathfrak{g}$  such that  $D = \text{ad } x$ . Any*

*Proof.* Let  $D$  be a derivation of  $\mathfrak{g}$ . Let  $\lambda(Y) = \text{tr}(D \circ \text{ad } Y)$  be a linear functional, which by the nondegeneracy of  $\kappa$  can be written as  $\lambda(Y) = \kappa(X, Y)$  for some  $X \in \mathfrak{g}$ . We will

show that  $D = \text{ad } X$ . Put  $D' = D - \text{ad } X$ . Then  $D'$  is also a derivation with the property that  $\text{tr}(D' \circ \text{ad } Y) = 0$  for all  $Y$ . We will show that  $D' = 0$ , which will complete the proof.

Fix  $Z \in \mathfrak{g}$ . We have the identity since  $D'$  is a derivation,

$$\begin{aligned} [D', \text{ad } Z](W) &= D'([Z, W]) - [Z, D'(W)] \\ &= ([D'Z, W] + [Z, D'W]) - [Z, D'(W)] \\ &= [D'Z, W] = \text{ad}(D'Z)(W) \end{aligned}$$

Now we compute  $\text{tr}(\text{ad } Y [D', \text{ad } Z]) = \text{tr}(\text{ad } Y \text{ad}(D'Z)) = \kappa(Y, D'Z)$ . But at the same time

$$\begin{aligned} \text{tr}(\text{ad } Y \text{ad}(D'Z)) &= \text{tr}(\text{ad } Y [D', \text{ad } Z]) = \text{tr}(\text{ad } Y D' \text{ad } Z - \text{ad } Y \text{ad } Z D') \\ &= \text{tr}(D' \text{ad } Z \text{ad } Y) - \text{tr}(D' \text{ad } Y \text{ad } Z) \\ &= \text{tr}(D' \text{ad}[Z, Y]) \\ &= 0 \end{aligned}$$

□

The left hand side is equal to  $\kappa(Y, D'Z)$ , so  $\kappa(Y, D'Z) = 0$  for all  $Y$ , which implies  $D'Z = 0$ . Since  $Z$  was arbitrary, we have  $D' = 0$  and hence  $D = \text{ad } X$  is inner.

**Theorem 2.69.** *Let  $\text{Ad} : G \rightarrow \mathfrak{gl}(\mathfrak{g})$  be the adjoint representation of a connected semisimple Lie group  $G$ . Then  $\text{Ad}(G)$  is a closed subgroup of  $\mathfrak{gl}(\mathfrak{g})$ , and coincides with the connected component of the identity in  $\text{Aut}(\mathfrak{g})$ . Moreover  $\text{Aut}^0(\mathfrak{g}) = \text{Ad}(G) \subset \mathfrak{sl}(\mathfrak{g})$ .*

*Proof.* An element  $A \in \text{GL}(\mathfrak{g})$  is an automorphism of the Lie algebra  $\mathfrak{g}$  precisely if  $[Ax, Ay] = A[x, y]$  for all  $x, y \in \mathfrak{g}$ . Picking a basis for  $\mathfrak{g}$  expresses this condition in terms of polynomial equations in the entries of  $A$ . The Lie algebra of  $\text{Aut}(\mathfrak{g})$  are those infinitesimal automorphisms of  $\mathfrak{g}$ . Pick a curve  $A(t) \in \text{Aut}(\mathfrak{g})$  with  $A(0) = I$ . Writing  $A(t) = I + tX + O(t^2)$ , expanding the automorphism condition and comparing the  $t$ -linear terms gives  $[Xx, y] + [x, Xy] = X[x, y]$  for all  $x, y \in \mathfrak{g}$ . This is precisely the condition that  $X$  is a derivation of  $\mathfrak{g}$ . If  $\mathfrak{g}$  is semisimple, then every derivation is inner, so we see that

$$\text{Lie}(\text{Aut}(\mathfrak{g})) = \text{Der}(\mathfrak{g}) = \text{ad}(\mathfrak{g})$$

Therefore we have  $\text{Ad}(G) \subset \text{Aut}^0(\mathfrak{g})$  and both are connected analytic Lie groups with the same Lie algebra, so they must coincide. Finally,  $\text{Ad}(G) \subset \mathfrak{sl}(\mathfrak{g})$  because  $\text{tr}(\text{ad } x) = 0$  for all  $x \in \mathfrak{g}$ . □

**Theorem 2.70** (Whitehead lemmas). *Let  $\mathfrak{g}$  be a semisimple Lie algebra. Then*

$$H^1(\mathfrak{g}, V) = H^2(\mathfrak{g}, V) = 0$$

*for every finite-dimensional  $\mathfrak{g}$ -module  $V$ .*

*Proof.* One checks the vanishing of  $H^1$  directly and we omit the  $H^2$  calculation. The statement amounts to saying that for every linear map  $f : \mathfrak{g} \rightarrow V$  satisfying the 1-cocycle condition

$$f([x, y]) = x \cdot f(y) - y \cdot f(x)$$

then there is some  $v \in V$  such that  $f(x) = x \cdot v$  for all  $x \in \mathfrak{g}$ . Write  $V = V_0 \oplus V_1$  where  $V_0$  is the generalized 0 eigenspace for the action of  $c$  and  $V_1$  is the sum of the other generalized eigenspaces, where  $c$  is the Casimir element of  $U(\mathfrak{g})$ . These are  $\mathfrak{g}$ -submodules of  $V$  because  $c$  is central. Since cohomology is additive, it is enough to prove the result for  $V = V_0$  where the Casimir acts nilpotently, or  $V = V_1$  where the Casimir acts invertibly. In the first case,  $v = 0$  does the trick. In the second case, let  $e_i$  be a basis of  $\mathfrak{g}$  and let  $e^i$  be the dual basis with respect to the Killing form. Let  $w = \sum_i e_i \cdot f(e^i)$ . Then

$$xw = \sum_i e_i x f(e^i) + \sum_i [x, e_i] f(e^i)$$

Note that

$$[x, e_i] = \sum_j \kappa([x, e_i], e^j) e_j = \sum_j -\kappa(e_i, [x, e^j]) e_j$$

using the expansion into the dual basis, along with invariance of the Killing form. Therefore

$$\sum_i [x, e_i] f(e^i) = - \sum_{i,j} \kappa([x, e^j], e_i) e_j f(e^i)$$

Swapping the order of summation and applying the linearity of  $f$  gives

$$\sum_i [x, e_i] f(e^i) = - \sum_j e_j f([x, e^j])$$

The 1-cocycle condition says that  $f([x, e^j]) = x f(e^j) - e^j f(x)$ , so we have

$$- \sum_j e_j f([x, e^j]) = - \sum_j e_j (x f(e^j) - e^j f(x))$$

Therefore the second term is  $-\sum_i e_i x f(e^i) + \sum_i e_i e^i f(x)$ . Cancelling, we find that

$$xw = \sum_i e_i e^i f(x)$$

so the vector  $v = w/c$  does the trick.  $\square$

**Theorem 2.71** (Complete reducibility). *If  $\mathfrak{g}$  is a semisimple Lie algebra, then every finite-dimensional representation of  $\mathfrak{g}$  is completely reducible.*

*Proof.* Let  $V$  be a finite-dimensional representation of  $\mathfrak{g}$ , and let  $W \subset V$  be a subrepresentation. We will show that there is a complementary subrepresentation  $W' \subset V$  such that  $V = W \oplus W'$ . Consider the short exact sequence of  $\mathfrak{g}$ -modules

$$0 \rightarrow W \rightarrow V \rightarrow V/W \rightarrow 0.$$

This corresponds to an element of  $H^1(\mathfrak{g}, \text{Hom}(V/W, W))$ . By the first Whitehead lemma, this cohomology group vanishes, so the extension splits and we have  $V = W \oplus W'$  for some subrepresentation  $W'$ . This proves complete reducibility.  $\square$

### 2.2.4 Cartan Subalgebras

Our next step would be considering not just one semisimple element but a family of commuting semisimple elements.

**Definition 2.72.** A subalgebra  $\mathfrak{h} \subset \mathfrak{g}$  is called **toroidal** if it is commutative and consists of semisimple elements.

**Theorem 2.73.** Let  $\mathfrak{h} \subset \mathfrak{g}$  be a toroidal subalgebra. Then

1.  $\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$ , where  $\mathfrak{g}_\alpha$  is a common eigenspace for all operators  $\text{ad } h$ ,  $h \in \mathfrak{h}$ , with eigenvalue  $\alpha$ :

$$\text{ad } h \cdot x = \langle \alpha, h \rangle x, \quad h \in \mathfrak{h}, \quad x \in \mathfrak{g}_\alpha.$$

In particular,  $\mathfrak{h} \subset \mathfrak{g}_0$ .

2.  $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ .
3. If  $\alpha + \beta \neq 0$ , then  $\mathfrak{g}_\alpha$  and  $\mathfrak{g}_\beta$  are orthogonal with respect to the Killing form  $K$ .
4. For any  $\alpha$ , the Killing form gives a non-degenerate pairing

$$\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \rightarrow \mathbb{C}.$$

In particular, the restriction of  $K$  to  $\mathfrak{g}_0$  is non-degenerate.

Our next goal is to produce as large a toroidal subalgebra in  $\mathfrak{g}$  as possible. The standard way of formalizing this is as follows.

**Definition 2.74.** Let  $\mathfrak{g}$  be a semisimple Lie algebra. A **Cartan subalgebra**  $\mathfrak{h} \subset \mathfrak{g}$  is a toroidal subalgebra which coincides with its centralizer:

$$C(\mathfrak{h}) = \{x \in \mathfrak{g} \mid [x, \mathfrak{h}] = 0\} = \mathfrak{h}.$$

**Example 2.75.** Let  $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$  and let  $\mathfrak{h}$  be the set of diagonal matrices with trace 0. Then  $\mathfrak{h}$  is a Cartan subalgebra. Indeed, it is obviously commutative, and every diagonal element is semisimple. On the other hand, choose  $h \in \mathfrak{h}$  to be a diagonal matrix with distinct eigenvalues. By a well-known result of linear algebra, if  $[x, h] = 0$ , and  $h$  has distinct eigenvalues, then any eigenvector of  $h$  is also an eigenvector of  $x$ , thus  $x$  must also be diagonal. Hence  $C(\mathfrak{h}) = \mathfrak{h}$ .

We still need to prove existence of Cartan subalgebras.

**Theorem 2.76.** Let  $\mathfrak{h} \subset \mathfrak{g}$  be a maximal toroidal subalgebra, i.e. a toroidal subalgebra which is not properly contained in any other toroidal subalgebra. Then  $\mathfrak{h}$  is a Cartan subalgebra.



### 2.2.5 Root Systems and Root Space Decomposition

**Definition 2.77.** An **abstract root system** is a finite set of elements  $R \subset E \setminus \{0\}$ , where  $E$  is a real vector space with a positive definite inner product  $(\cdot, \cdot)$ , such that the following properties hold:

(R1)  $R$  generates  $E$  as a vector space.

(R2) For any two roots  $\alpha, \beta$ , the number

$$n_{\alpha\beta} = \frac{2(\alpha, \beta)}{(\beta, \beta)} \quad (10)$$

is an integer.

(R3) Let  $s_\alpha : E \rightarrow E$  be defined by

$$s_\alpha(\lambda) = \lambda - \frac{2(\alpha, \lambda)}{(\alpha, \alpha)} \alpha. \quad (11)$$

Then for any roots  $\alpha, \beta$ , one has  $s_\alpha(\beta) \in R$ .

The number  $r = \dim E$  is called the **rank** of  $R$ .

If, in addition,  $R$  satisfies the following property:

(R4) If  $\alpha, c\alpha$  are both roots, then  $c = \pm 1$ ,

then  $R$  is called a **reduced root system**.

**Theorem 2.78.** Let  $\mathfrak{g}$  be a semisimple Lie algebra and  $\mathfrak{h} \subset \mathfrak{g}$  a Cartan subalgebra. Then the set of nonzero roots  $\Phi \subset \mathfrak{h}^*$  forms a reduced root system.

There is a complete classification of reduced root systems, which is given by the following list:

- $A_n = \{e_i - e_j \mid 1 \leq i \neq j \leq n+1\}$ , where  $e_i$  is the  $i$ -th standard basis vector in  $\mathbb{R}^{n+1}$ .
- $B_n = \{\pm e_i, \pm e_i \pm e_j \mid 1 \leq i < j \leq n\}$ .
- $C_n = \{\pm 2e_i, \pm e_i \pm e_j \mid 1 \leq i < j \leq n\}$ .
- $D_n = \{\pm e_i \pm e_j \mid 1 \leq i < j \leq n\}$ .
- $G_2 = \{\pm e_i, \pm e_i \pm e_j, \pm 2e_i \pm e_j \mid 1 \leq i < j \leq 3\}$ .
- $F_4 = \{\pm e_i, \pm e_i \pm e_j, \pm \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4) \mid 1 \leq i < j \leq 4\}$ .
- $E_6 = \{\pm e_i \pm e_j, \pm \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4 \pm e_5 \pm e_6) \mid 1 \leq i < j \leq 6\}$ .
- $E_7 = \{\pm e_i \pm e_j, \pm \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4 \pm e_5 \pm e_6 \pm e_7) \mid 1 \leq i < j \leq 7\}$ .

- $E_8 = \{\pm e_i \pm e_j, \pm \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4 \pm e_5 \pm e_6 \pm e_7 \pm e_8) \mid 1 \leq i < j \leq 8\}.$

Root systems are more succinctly described by their Dynkin diagrams, which are graphs with vertices corresponding to simple roots and edges encoding the angles between them. This information also coincides with the Cartan matrix of the root system, which encodes the integers  $n_{\alpha\beta}$  defined in (10) for simple roots  $\alpha, \beta$ .

### 2.2.6 Examples: $\mathfrak{sl}_n, \mathfrak{so}_n, \mathfrak{sp}_{2n}, \mathfrak{u}_n$

## 2.3 Representation Theory

A representation of a Lie group  $G$  is a homomorphism  $G \rightarrow \mathrm{GL}(V)$  for some vector space  $V$ . A representation of a Lie algebra  $\mathfrak{g}$  is a homomorphism  $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$  for some vector space  $V$ . A representation of a Lie group  $G$  induces a representation of its Lie algebra  $\mathfrak{g}$ , and if  $G$  is connected and simply connected, then the representation of  $\mathfrak{g}$  determines the representation of  $G$ .

We will focus on finite dimensional representations over the complex numbers. It turns out that to study these, there is no harm in then passing to complexification.

**Lemma 2.79.** *Let  $\mathfrak{g}$  be a real Lie algebra, and  $\mathfrak{g}_{\mathbb{C}}$  its complexification as defined in Definition 3.51. Then any complex representation of  $\mathfrak{g}$  has a unique structure of representation of  $\mathfrak{g}_{\mathbb{C}}$ , and*

$$\mathrm{Hom}_{\mathfrak{g}}(V, W) = \mathrm{Hom}_{\mathfrak{g}_{\mathbb{C}}}(V, W).$$

*In other words, the categories of complex representations of  $\mathfrak{g}$  and  $\mathfrak{g}_{\mathbb{C}}$  are equivalent.*

**Example 2.80.** *The categories of representations of  $\mathrm{SL}(2, \mathbb{C})$ ,  $\mathrm{SU}(2)$ ,  $\mathfrak{sl}(2, \mathbb{C})$  and  $\mathfrak{su}(2)$  are all equivalent.*

This, in particular, allows us to reduce the problem of studying representations of a non-compact Lie group  $\mathrm{SL}(2, \mathbb{C})$  to the study of representations of a compact Lie group  $\mathrm{SU}(2)$ . This is useful because, as we will show below, representation theory of compact Lie groups is especially nice.

### 2.3.1 Representation theory of compact Lie groups

**Definition 2.81.** *A complex representation of a Lie group  $K$  is called **unitary** if there is a  $K$ -invariant positive-definite Hermitian form on the representation space.*

**Theorem 2.82.** *Finite-dimensional representations of a compact Lie group  $K$  are semisimple, i.e. they decompose as direct sums of irreducible representations. Moreover, every irreducible representation of  $K$  is unitary.*

*Proof.* The first statement follows from induction on the dimension of the representation. The second statement follows from the existence of the Haar measure on  $K$ : given any representation, we can average any positive-definite Hermitian form over the group to get a  $K$ -invariant positive-definite Hermitian form.  $\square$

**Theorem 2.83.**

1. Let  $V, W$  be non-isomorphic irreducible representations of  $G$ . Choose bases  $v_i \in V$ ,  $i = 1, \dots, n$  and  $w_a \in W$ ,  $a = 1, \dots, m$ . Then for any  $i, j, a, b$ , the matrix coefficients  $\rho_{ij}^V(g)$ ,  $\rho_{ab}^W(g)$  are orthogonal:

$$(\rho_{ij}^V(g), \rho_{ab}^W(g)) = 0,$$

where  $(\ , \ )$  is the inner product on  $C^\infty(G, \mathbb{C})$  given by

$$(f_1, f_2) = \int_G f_1(g) \overline{f_2(g)} dg. \quad (12)$$

2. Let  $V$  be an irreducible representation of  $G$  and let  $v_i \in V$  be an orthonormal basis with respect to a  $G$ -invariant inner product (which exists by Theorem 4.38). Then the matrix coefficients  $\rho_{ij}^V(g)$  are pairwise orthogonal, and each has norm squared  $\frac{1}{\dim V}$ :

$$(\rho_{ij}^V(g), \rho_{kl}^V(g)) = \frac{\delta_{ik} \delta_{jl}}{\dim V}. \quad (13)$$

*Proof.* The proof is based on the following easy lemma.  $\square$

**Lemma 2.84.**

1. Let  $V, W$  be non-isomorphic irreducible representations of  $G$  and  $f : V \rightarrow W$  a linear map. Then

$$\int_G g f g^{-1} dg = 0.$$

2. If  $f : V \rightarrow V$  is linear, then

$$\int_G g f g^{-1} dg = \frac{\text{tr}(f)}{\dim V} \text{id}$$

So irreducible representations give us a way of constructing an orthonormal set of functions on the group. Unfortunately, they depend on the choice of basis. However, there is one particular combination of matrix coefficients that does not depend on the choice of basis.

**Definition 2.85.** A character of a representation  $V$  is the function on the group defined by

$$\chi_V(g) = \text{tr } \rho(g) = \sum_i \rho_{ii}^V(g).$$

It is immediate from the definition that the character does not depend on the choice of basis in  $V$ . It also has a number of other properties, listed below; proof of them is left to the reader as an exercise.

The orthogonality relation for matrix coefficients immediately implies the following result for the characters.

**Theorem 2.86.**

1. Let  $V, W$  be non-isomorphic complex irreducible representations of a compact real Lie group  $G$ . Then the characters  $\chi_V, \chi_W$  are orthogonal with respect to inner product (12):

$$(\chi_V, \chi_W) = 0.$$

2. For any irreducible representation  $V$ ,

$$(\chi_V, \chi_V) = 1.$$

In particular  $V$  is irreducible if and only if  $(\chi_V, \chi_V) = 1$  and the multiplicity of  $V_i$  in  $V$  can be obtained by pairing  $(\chi_{V_i}, \chi_V)$ .

The statement of Peter Weyl theorem can be understood as a coordinate free formulation of the orthonormality of characters.

**Theorem 2.87** (Peter-Weyl). *The map*

$$m : \bigoplus_V V \otimes V^\vee \rightarrow C^\infty(G, \mathbb{C})$$

which takes  $v \times v^\vee \mapsto (g \cdot v, v^\vee)$  is an injective morphism of  $G$ -bimodules which preserves the inner product, if we define the inner product on the left hand side to be

$$((v \otimes v^\vee ee), (w \otimes w^\vee)) = \frac{1}{\dim V} (v^\vee, w^\vee ee)(v, w)$$

Moreover, if we complete the LHS with respect to this metric, then  $m$  induces an isomorphism with the  $L^2$  integrable functions on  $G$ .

**2.3.2 Universal Enveloping Algebra and PBW Theorem**

Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$ . The **tensor algebra** of  $\mathfrak{g}$  is

$$T(\mathfrak{g}) = \bigoplus_{n \geq 0} \mathfrak{g}^{\otimes n}.$$

**Definition 2.88.** *The **universal enveloping algebra** of  $\mathfrak{g}$  is*

$$U(\mathfrak{g}) := T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] \mid x, y \in \mathfrak{g} \rangle.$$

The algebra  $U(\mathfrak{g})$  is associative and comes equipped with a Lie algebra map

$$\iota : \mathfrak{g} \rightarrow U(\mathfrak{g}),$$

such that any Lie algebra map  $\mathfrak{g} \rightarrow A$  into an associative algebra factors uniquely through  $U(\mathfrak{g})$ . The bracket on  $U(\mathfrak{g})$  is  $[x, y] = xy - yx$ , and the relations in the definition of  $U(\mathfrak{g})$  ensure that  $\iota$  is a Lie algebra homomorphism.

Define an increasing filtration on  $U(\mathfrak{g})$  by

$$F^n U(\mathfrak{g}) := \text{image of } \bigoplus_{i \leq n} \mathfrak{g}^{\otimes i} \subset T(\mathfrak{g}).$$

**Theorem 2.89** (Poincaré–Birkhoff–Witt). *Let  $\mathfrak{g}$  be a Lie algebra with ordered basis  $\{x_i\}_{i \in I}$ . Then the monomials*

$$x_{i_1} x_{i_2} \cdots x_{i_n}, \quad i_1 \leq i_2 \leq \cdots \leq i_n,$$

*form a basis of  $U(\mathfrak{g})$  as a vector space. Equivalently, there is a natural isomorphism of graded algebras*

$$\text{gr } U(\mathfrak{g}) \cong S(\mathfrak{g}),$$

*where  $S(\mathfrak{g})$  is the symmetric algebra.*

### 2.3.3 Verma Modules and Highest Weight Modules

An element  $\lambda$  of  $\mathfrak{h}^*$  is called an **integral weight** if  $(\lambda|\alpha^\vee) \in \mathbb{Z}$  for all  $\alpha \in \Phi^+$ , or equivalently, for all simple roots  $\alpha$ . The integral weights form a lattice

$$\Lambda = \{x \in V | (\alpha^\vee|x) \in \mathbb{Z} \text{ for } \alpha \in \Phi^+\},$$

called the **weight lattice**. We call  $\lambda \in \mathfrak{h}^*$  **dominant** if  $(\lambda|\alpha^\vee) \geq 0$  for all  $\alpha \in \Phi^+$  (or equivalently for simple roots  $\alpha$ ). We call  $\lambda$  **strongly dominant** if  $(\lambda|\alpha^\vee) > 0$ . Thus the special vector  $\rho$  is a strongly dominant integral weight.

Here are a couple of important properties of the Weyl group action. Let  $V$  be the  $\mathbb{R}$ -span of  $\Phi$  in  $\mathfrak{h}^*$ . The inner product  $(|)$  makes  $V$  into a Euclidean space, and  $\mathfrak{h}^* = V + iV$ . The set

$$C^+ = \{x \in V | (\alpha^\vee|x) > 0 \text{ for } \alpha \in \Phi^+\}$$

is called the **positive Weyl chamber**. The dominant weights are the ones in  $C^+$ .

**Proposition 2.90** (Fundamental domain). *The positive Weyl chamber is a fundamental domain for the action of the Weyl group: if  $x \in V$  there is a unique element of  $C^+$  in the  $W$  orbit of  $x$ .*

Recall that there is a partial order (known as dominance order)  $\succeq$  on  $\mathfrak{h}^*$  defined by

$$\lambda \succeq \mu \iff \lambda - \mu = \sum_{\alpha \in \Phi^+} n_\alpha \alpha \text{ with } n_\alpha \in \mathbb{Z}_{\geq 0}.$$

**Proposition 2.91.** *Let  $\lambda$  be a dominant, integral weight and let  $w \in W$ . Then  $\lambda \succeq w\lambda$ .*

Let  $V$  be a  $\mathfrak{g}$ -module. For  $\lambda \in \mathfrak{h}^*$  we denote the **weight space**

$$V_\lambda = \{v \in V \mid h \cdot v = \lambda(h)v \text{ for } h \in \mathfrak{h}\}.$$

We will say that  $V$  is  **$\mathfrak{h}$ -diagonalizable** if  $V$  is the algebraic direct sum of the  $V_\lambda$ .

**Proposition 2.92.** *If  $V$  is  $\mathfrak{h}$ -diagonalizable, then so is any submodule or quotient module.*

*Proof.* Let  $U \subset V$  be a submodule. We must show that an element of  $U$  may be expressed as a finite linear sum of  $u_\lambda \in U_\lambda$ . Since  $V$  has a weight space decomposition, we may write  $u$  as a sum of  $u_\lambda \in V_\lambda$ , and the problem is then to show that  $u_\lambda \in U$ . There exist a finite number of  $\lambda_i$  such that

$$u = \sum_{i=1}^m u_{\lambda_i},$$

and we choose  $h \in \mathfrak{h}$  such that the values  $\lambda_i(h)$  are all distinct. Then for  $j = 0, \dots, m-1$

$$h^j \cdot u = \sum \lambda_i(h)^j u_{\lambda_i} \in \mathfrak{h}.$$

The  $m \times m$  matrix  $\{\lambda_i(h)^j\}$  is invertible since its determinant is a Vandermonde determinant. Consider the vectors  $u, h \cdot u, h^2 \cdot u, \dots, h^{m-1} \cdot u$ . Each is in  $U$ , and together they form a linear system:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1(h) & \lambda_2(h) & \cdots & \lambda_m(h) \\ \lambda_1(h)^2 & \lambda_2(h)^2 & \cdots & \lambda_m(h)^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1(h)^{m-1} & \lambda_2(h)^{m-1} & \cdots & \lambda_m(h)^{m-1} \end{bmatrix} \begin{bmatrix} u_{\lambda_1} \\ u_{\lambda_2} \\ \vdots \\ u_{\lambda_m} \end{bmatrix} = \begin{bmatrix} u \\ h \cdot u \\ \vdots \\ h^{m-1} \cdot u \end{bmatrix}.$$

This is a Vandermonde system. The matrix is invertible because the  $\lambda_i(h)$  are distinct. Applying the inverse to this shows that each  $u_{\lambda_i} \in U$ , as required.

This proves that a submodule of a  $\mathfrak{h}$ -diagonalizable module is diagonalizable. It follows that the same is true for quotient modules, with  $(V/U)_\lambda = V_\lambda/U_\lambda$ .  $\square$

We will work exclusively with diagonalizable modules with  $\dim(V_\lambda) < \infty$  for all  $\lambda \in \mathfrak{h}^*$ . We will define the **support**  $\text{supp}(V) = \{\lambda \in \mathfrak{h}^* \mid V_\lambda \neq 0\}$ .

Let  $U(\mathfrak{g})$  be the universal enveloping algebra. Let  $\mathfrak{n}^+$  be the nilpotent subalgebra of  $\mathfrak{g}$  generated by the  $\mathfrak{g}_\alpha$  ( $\alpha \in \Phi^+$ ), and let  $\mathfrak{n}^-$  be the subalgebra generated by the  $\mathfrak{g}_\alpha$  with  $\alpha \in \Phi^-$ . Then clearly we have the **triangular decomposition**

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+.$$

**Lemma 2.93.** *We have  $U(\mathfrak{g}) \cong U(\mathfrak{n}^-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}^+)$  in the sense that the multiplication map*

$$U(\mathfrak{n}^-) \times U(\mathfrak{h}) \times U(\mathfrak{n}^+) \longrightarrow U(\mathfrak{g})$$

*induces a vector space isomorphism  $U(\mathfrak{n}^-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}^+) \longrightarrow U(\mathfrak{g})$ .*

*Proof.* This follows from the Poincaré-Birkhoff-Witt theorem (PBW) together with the triangular decomposition. Namely, if  $\{x_i\}$  is a basis for  $\mathfrak{g}$ , then PBW asserts that a basis for  $U(\mathfrak{g})$  consists of all elements of the form

$$x_1^{k_1} \cdots x_d^{k_d}, \quad 0 \leq k_i \in \mathbb{Z}.$$

Now we take the basis in a particular way, where its first  $\frac{1}{2}|\Phi|$  elements are a basis for  $\mathfrak{n}^-$ , the next  $\ell$  elements are a basis for  $\mathfrak{h}$ , and the last  $\frac{1}{2}|\Phi|$  elements are a basis for  $\mathfrak{n}^+$ . Then the element  $x_1^{k_1} \cdots x_d^{k_d}$  factors uniquely as a product  $abc$  where  $a$  runs through a basis of  $U(\mathfrak{n}^-)$ ,  $b$  runs through a basis of  $U(\mathfrak{h})$  and  $c$  runs through a basis of  $U(\mathfrak{n}^+)$ .  $\square$

We will call a vector  $v \in V$  a **highest weight vector** of weight  $\lambda$  if  $v \in V_\lambda$  and if  $x_\alpha v = 0$  for  $\alpha \in \Phi^+$ . (Humphreys calls such  $v$  a **maximal vector**.) We will call  $V$  a **highest weight module** of weight  $\lambda$  if it is generated by a highest weight vector  $v \in V_\lambda$ . (Humphreys calls a highest weight module a **standard cyclic module**.)

**Proposition 2.94.** *Suppose that  $v \in V$  is a highest weight vector. Then the  $\mathfrak{g}$ -submodule  $U(\mathfrak{g})v$  generated by  $v$  equals  $U(\mathfrak{n}^-)v$ . The weight space  $V_\mu = 0$  unless  $\mu \preceq \lambda$ . We have  $\dim(V_\lambda) = 1$ .*

*Proof.* We note that any element of  $U(\mathfrak{n}^+)$  may be written as a constant times an element of the left ideal  $U(\mathfrak{n}^+)\mathfrak{n}^+$ , but this ideal annihilates  $v$ , so  $U(\mathfrak{n}^+)v = \mathbb{C}v$ . Similarly  $U(\mathfrak{h})v = \mathbb{C}v$  since  $v \in V_\lambda$ . By Lemma 4,

$$U(\mathfrak{g}) = U(\mathfrak{n}^-)U(\mathfrak{h})U(\mathfrak{n}^+)v = U(\mathfrak{n}^-)v.$$

Consider the basis  $\{x_{-\alpha}\}$  ( $\alpha \in \Phi^+$ ) of  $\mathfrak{n}^-$  with  $x_{-\alpha} \in \mathfrak{g}_{-\alpha}$ . Using a fixed order on  $\Phi^+$ , the elements  $\prod_{\alpha \in \Phi^+} x_{-\alpha}^{k_\alpha}$  are a PBW basis of  $U(\mathfrak{n}^-)$ . Since  $x_{-\alpha}$  maps  $V_\mu$  to  $V_{\mu-\alpha}$ ,

$$\prod_{\alpha \in \Phi^+} x_{-\alpha}^{k_\alpha} v \in V_\mu, \quad \mu = \lambda - \sum_{\alpha \in \Phi^+} k_\alpha \alpha,$$

so  $\mu \preceq \lambda$ . Unless all  $k_\alpha = 0$ ,  $\mu$  is strictly  $\prec \lambda$ , so  $V_\lambda$  is one-dimensional.  $\square$

**Proposition 2.95.** *Let  $V$  be a highest weight module with highest weight  $\lambda$ . A submodule  $U$  of  $V$  is proper if and only if  $U \cap V_\lambda = 0$ .*

*Proof.* Since  $\dim(V_\lambda) = 1$ , if  $U \cap V_\lambda \neq 0$  then  $V_\lambda \subseteq U$  and then since  $V_\lambda$  generates  $V$ , it is clear that  $U = V$ . On the other hand if  $U \cap V_\lambda = 0$  then clearly  $U$  is proper.  $\square$

**Proposition 2.96.** *Let  $V$  be a highest weight module with highest weight  $\lambda$ . Then  $V$  has a unique maximal proper submodule. Moreover  $V$  has a unique irreducible quotient.*

*Proof.* Let  $\Sigma$  be the set of proper submodules of  $V$ , and let

$$W = \sum_{U \in \Sigma} U.$$

By Proposition 3 each  $U \in \Sigma$  is diagonalizable, so evidently for  $\mu \in \mathfrak{h}^*$

$$W_\mu = \sum_{U \in \Sigma} U_\mu.$$

We apply this with  $\mu = \lambda$ . Since  $U \in \Sigma$  is proper,  $U_\lambda = 0$  by the previous proposition, and so  $W_\lambda = 0$ . This shows that  $W$  is proper. We have proved that  $W$  is the unique maximal proper submodule of  $V$ , and consequently  $V/W$  is the unique irreducible quotient.  $\square$

**Theorem 2.97.** *Let  $\lambda \in V^*$ . There is a highest weight module  $M = M(\lambda)$  with highest weight vector  $m \in M_\lambda$  with the following universal property. If  $V$  is another highest weight module with highest weight  $\lambda$  and if  $v \in V_\lambda$ , then there is a unique  $\mathfrak{g}$ -module homomorphism  $M \rightarrow V$  mapping  $m \mapsto v$ . The map  $\xi \mapsto \xi \cdot v$  is vector space isomorphism  $U(\mathfrak{n}^-) \rightarrow M$ .*

*Proof.* Note that since  $\mathfrak{h}$  normalizes  $\mathfrak{n}^+$ ,  $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}^+$  is a subalgebra of  $\mathfrak{g}$ , the “Borel subalgebra.” As in Lemma 4,  $U(\mathfrak{g}) \cong U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$ , that is, the multiplication map  $U(\mathfrak{n}^-) \times U(\mathfrak{b}) \rightarrow U(\mathfrak{g})$  induces a vector space isomorphism  $U(\mathfrak{n}^-) \otimes U(\mathfrak{b}) \rightarrow U(\mathfrak{g})$ . This result is a simple consequence of this fact.

To elaborate, regarding  $\mathbb{C}$  as a one-dimensional abelian Lie algebra, we have a Lie algebra homomorphism  $\theta_\lambda : \mathfrak{b} \rightarrow \mathbb{C}$  that maps  $H \in \mathfrak{h}$  to  $\lambda(H)$ , and  $\mathfrak{n}^+$  to zero. Thus let  $H_1, \dots, H_\ell$  be a basis of  $\mathfrak{h}$  and  $x_\alpha$  ( $\alpha \in \Phi^+$ ) be a basis of  $\mathfrak{n}^+$ . By the PBW theorem, the elements

$$H_1^{k_1} \cdots H_\ell^{k_\ell} \prod_{\alpha \in \Phi^+} x_\alpha^{k_\alpha}$$

with  $k_i$  and  $k_\alpha$  nonnegative integers are a basis for  $U(\mathfrak{b})$ . It is understood that in the product  $\prod x_\alpha^{k_\alpha}$  the roots  $\alpha \in \Phi^+$  are taken in a fixed definite order. We then have

$$\theta_\lambda \left( H_1^{k_1} \cdots H_\ell^{k_\ell} \prod_{\alpha} x_\alpha^{k_\alpha} \right) = \begin{cases} \prod \lambda(H_i)^{k_i} & \text{if all } k_\alpha = 0, \\ 0 & \text{if any } k_\alpha > 0. \end{cases}$$

Now let  $J_\psi$  be the left ideal generated by  $\xi - \theta_\lambda(\xi)$  for  $\xi \in \mathfrak{b}$ . Let

$$M = M(\lambda) = U(\mathfrak{g})/J_\psi,$$

and let  $m$  be the image of  $1 \in U(\mathfrak{g})$  in  $M(\lambda)$ .

It is clear from the PBW theorem that  $Hv = \lambda(H)m$  for  $H \in \mathfrak{h}$ , while  $\mathfrak{n}^+v = 0$ , and moreover from  $U(\mathfrak{g}) \cong U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$ , it is clear that every element of  $M(\lambda)$  may be written uniquely as  $\eta \cdot v$  for  $\eta \in U(\mathfrak{n}^-)$ .

Now let us verify the universal property. Let  $V$  be a highest weight module with weight  $\lambda$ , and let  $v_\lambda \in V_\lambda$  be a generator. Then we have a surjective  $U(\mathfrak{g})$ -module homomorphism

$$U(\mathfrak{g}) \rightarrow V, \quad \xi \mapsto \xi \cdot v_\lambda,$$

and since  $\beta \cdot v = \theta_\lambda(\beta)v$  for  $\beta \in \mathfrak{b}$ ,  $J_\psi$  is in the kernel. Thus the map factors uniquely through  $U(\mathfrak{g})/J_\psi = M(\lambda)$ .  $\square$

**Corollary 2.98.** *Let  $\lambda \in \mathfrak{h}^*$ . Up to isomorphism,  $\mathfrak{g}$  has a unique irreducible highest weight module  $L(\lambda)$  with highest weight  $\lambda$ .*

*Proof.* Every highest weight module is a quotient of  $M(\lambda)$ . Since  $M(\lambda)$  has a unique irreducible quotient, there is a unique irreducible highest weight module.  $\square$



**Remark 2.99.** *The irreducible quotient  $L(\lambda)$  might be finite or infinite dimensional. Recall that  $\lambda$  is called **integral** if  $\langle \alpha^\vee, \lambda \rangle \in \mathbb{Z}$  for all coroots  $\alpha^\vee$ , and **dominant** if  $\langle \alpha^\vee, \lambda \rangle \geq 0$ . If  $\lambda$  is a dominant integral weight, then  $L(\lambda)$  is finite-dimensional. On the other hand if  $\lambda$  is not integral,  $L(\lambda)$  will be infinite dimensional, and unless  $\langle \alpha^\vee, \lambda \rangle \in \mathbb{Z}$  for some coroot  $\alpha^\vee$ , we will actually have  $M(\lambda)$  irreducible, and  $L(\lambda) = M(\lambda)$ .*

**Theorem 2.100** (Classification of finite-dimensional irreducible modules). *Let  $V$  be a finite dimensional irreducible module. Then  $V \cong L(\lambda)$  where  $\lambda$  is a dominant integral weight. Conversely, if  $\lambda$  is a dominant integral weight, then  $L(\lambda)$  is finite-dimensional.*

*Proof.* Assume that  $V$  is finite-dimensional. Choose a vector  $v \in V_\lambda$  where  $\lambda$  is a weight of  $V$  that is maximal with respect to  $\succ$ . If  $\alpha \in \Phi^+$  then  $x_\alpha v \in V_{\lambda+\alpha}$  so  $x_\alpha v = 0$ . Therefore  $v$  is a highest weight vector. Then  $V = U(\mathfrak{g})v$  since  $V$  is irreducible. We have proved that  $V$  is a highest weight module; it is irreducible so  $V \cong L(\lambda)$ .

To show that  $\lambda$  is a dominant integral weight, let  $\alpha$  be a simple positive root. The restriction of  $V$  to the  $\mathfrak{sl}_2$  spanned by  $x_\alpha, x_{-\alpha}$  and  $h_\alpha$  is finite-dimensional, and  $x_\alpha v = 0$ . From the classification of finite-dimensional  $\mathfrak{sl}_2$ -modules, this means that  $(\alpha^\vee | \lambda) = \lambda(h_\alpha) \in \mathbb{Z}$  is a nonnegative integer. Therefore  $\lambda$  is dominant and integral.

We will omit the slightly tedious proof of the converse, that if  $\lambda$  is a dominant integral weight then  $L(\lambda)$  is finite-dimensional. For a proof of this see Kac, Lemma 10.1.  $\square$

**Corollary 2.101** (Weyl). *For  $V$  an irreducible finite-dimensional  $\mathfrak{g}$ -module, the highest weight  $\lambda$  is a dominant integral weight, and*

$$V \longleftrightarrow \lambda$$

*is a bijection between the irreducible highest weight modules and the dominant integral weights.*

Now let  $\mathfrak{g}$  be finite dimensional. The Casimir element of the universal enveloping algebra  $U(\mathfrak{g})$  may be defined as follows. Let  $\{\gamma_i\}$  be a basis of  $\mathfrak{g}$  and  $\{\gamma^i\}$  the dual basis with respect to the Killing form, so  $\kappa(\gamma_i, \gamma^j) = \delta_{ij}$ . Then

$$c_{\mathfrak{g}} = \sum_{i=1}^{\dim(\mathfrak{g})} \gamma_i \gamma^i$$

**Proposition 2.102.**  *$c_{\mathfrak{g}}$  is independent of the choice of basis  $\{\gamma_i\}$ . It lies in the center of  $U(\mathfrak{g})$ .*

*Proof.* For any finite-dimensional vector space  $V$ , there is a canonical iso  $V \otimes V^* \cong \text{End}(V)$ ,  $v \otimes \phi \mapsto (w \mapsto \phi(w)v)$ . Under this isomorphism, the element  $\sum_i v_i \otimes \phi_i$ , where  $\{v_i\}$  is a basis and  $\{\phi_i\}$  the dual basis, corresponds to the identity map on  $V$ . This element is independent of the chosen basis (it's just the coordinate expression of the identity endomorphism). Now take  $V = \mathfrak{g}$  with the Killing form  $\kappa$ . So the canonical element

$\sum_i \gamma_i \otimes \gamma^i \in \mathfrak{g} \otimes \mathfrak{g}$  corresponds to the identity operator  $\text{id}_{\mathfrak{g}}$ . Finally push  $\Omega$  into  $U(\mathfrak{g})$  using multiplication

$$c_{\mathfrak{g}} = m(\Omega) = \sum_i \gamma_i \gamma^i$$

This shows that  $c_{\mathfrak{g}}$  is independent of the choice of basis. To see that  $c_{\mathfrak{g}}$  is central, let  $x \in \mathfrak{g}$ . Then

Write

$$[x, \gamma_i] = \sum_j a_{ij} \gamma_j.$$

Use ad-invariance of  $\kappa$ :

$$0 = \kappa([x, \gamma_i], \gamma^k) + \kappa(\gamma_i, [x, \gamma^k]) = a_{ik} + \kappa(\gamma_i, [x, \gamma^k]).$$

If we expand  $[x, \gamma^k] = \sum_j b_{kj} \gamma^j$ , the relation above gives

$$b_{ki} = -a_{ik},$$

i.e.

$$[x, \gamma^k] = - \sum_i a_{ik} \gamma^i.$$

Now compute in  $U(\mathfrak{g})$ :

$$\begin{aligned} [x, c_{\mathfrak{g}}] &= \sum_i [x, \gamma_i] \gamma^i + \sum_i \gamma_i [x, \gamma^i] \quad \text{by the Leibniz rule} \\ &= \sum_{i,j} a_{ij} \gamma_j \gamma^i - \sum_{i,k} a_{ki} \gamma_i \gamma^k. \end{aligned}$$

so  $[x, c_{\mathfrak{g}}] = 0$ .  $\square$

**Proposition 2.103.** *Let  $h_i$  be a basis of  $\mathfrak{h}$  and let  $h^i$  be the dual basis of  $\mathfrak{h}$  with respect to the Killing form, so  $\kappa(h_i, h^j) = \delta_{ij}$ . Then if  $\lambda, \mu \in \mathfrak{h}^*$  we have*

$$(\lambda|\mu) = \sum_i \lambda(h^i) \mu(h_i).$$

*Proof.* Recall that the defining property of  $t_{\mu}$  is that

$$\kappa(t_{\mu}, h) = \mu(h) \quad \text{for all } h \in \mathfrak{h}.$$

First let us show that

$$t_{\mu} = \sum_i \mu(h_i) h^i. \tag{14}$$

To check this, we pair both sides with  $h_j$ . We have

$$\kappa(t_{\mu}, h_j) = \mu(h_j) = \kappa\left(\sum_i \mu(h_i) h^i, h_j\right) = \mu(h_j)$$

is exactly the defining property of  $t_{\mu}$ . Since the  $h_j$  span  $\mathfrak{h}$  and  $\kappa$  restricted to  $\mathfrak{h}$  is nondegenerate, this proves (14).

Now (14) implies

$$(\lambda|\mu) = \kappa(t_\lambda, t_\mu) = \sum_i \mu(h_i) \kappa(t_\lambda, h^i) = \sum_i \mu(h_i) \lambda(h^i).$$

again using the defining property of  $t_\mu$ .  $\square$

**Proposition 2.104.** *Let  $V$  be a highest weight module with highest weight  $\lambda$ . Then the Casimir element  $c_{\mathfrak{g}}$  acts by the scalar  $|\lambda + \rho|^2 - |\rho|^2$  on  $V$ .*

*Proof.* Since  $c_{\mathfrak{g}}$  is central in  $U(\mathfrak{g})$  it commutes with the action of  $\mathfrak{g}$  on any module  $V$ . Because  $V$  is generated by a highest weight vector  $v_\lambda \in V_\lambda$ , it is sufficient to show that

$$c_{\mathfrak{g}}v = (|\lambda + \rho|^2 - |\rho|^2)v$$

We need to choose dual bases of  $\mathfrak{g}$  with respect to the Killing form. For one basis, we choose a basis  $h_i$  of  $\mathfrak{h} = \mathfrak{g}_0$ , and vectors  $x_\alpha \in \mathfrak{g}_\alpha$ .

Now we describe the dual basis. We know that the Killing form is nondegenerate on  $\mathfrak{h}$ , so we find  $h^i$  such that  $\kappa(h_i, h^j) = \delta_{ij}$ . Then we define another set of representatives  $y_\alpha \in \mathfrak{g}_\alpha$  so that

$$y_\alpha = \frac{x_\alpha}{\kappa(x_\alpha, x_{-\alpha})}$$

so that

$$\kappa(x_\alpha, y_{-\beta}) = \delta_{\alpha\beta}$$

Thus we have dual bases  $\{h_i, x_\alpha\}$  and  $\{h^i, y_{-\alpha}\}$ . Then

$$c_{\mathfrak{g}} = \sum_i h_i h^i + \sum_{\alpha \in \Phi} x_\alpha y_{-\alpha}.$$

We want to rewrite this slightly. We write this as

$$c_{\mathfrak{g}} = \sum_i h_i h^i + \sum_{\alpha \in \Phi^+} x_\alpha y_{-\alpha} + \sum_{\alpha \in \Phi^+} x_{-\alpha} y_\alpha.$$

Now observe that  $[x_\alpha, y_{-\alpha}] = t_\alpha$ . Certainly  $[x_\alpha, y_{-\alpha}] \in \mathfrak{h}$  so write it as  $t_\alpha$ . Then for  $h \in \mathfrak{h}$ , we check that  $\kappa([x_\alpha, y_{-\alpha}], h) = \alpha(h)$ .

$$\kappa([x_\alpha, y_{-\alpha}], h) = \kappa(x_\alpha, [y_{-\alpha}, h])$$

But since  $h$  acts on the root vector  $y_{-\alpha}$  by  $[h, y_{-\alpha}] = -\alpha(h)y_{-\alpha}$ , we get  $[y_{-\alpha}, h] = \alpha(h)y_{-\alpha}$ . So  $\kappa([x_\alpha, y_{-\alpha}], h) = \kappa(x_\alpha, \alpha(h)y_{-\alpha}) = \alpha(h)\kappa(x_\alpha, y_{-\alpha})$ .

And by construction of  $y_{-\alpha}$ ,  $\kappa(x_\alpha, y_{-\alpha}) = 1$ . Thus  $\kappa([x_\alpha, y_{-\alpha}], h) = \alpha(h)$ .

That is exactly the defining property of  $t_\alpha$ . Hence  $[x_\alpha, y_{-\alpha}] = t_\alpha$ .

So in the enveloping algebra

$$x_\alpha y_{-\alpha} = t_\alpha + y_{-\alpha} x_\alpha.$$

Thus

$$c_{\mathfrak{g}} = \sum_i h_i h^i + \sum_{\alpha \in \Phi^+} t_\alpha + \sum_{\alpha \in \Phi^+} (y_{-\alpha} x_\alpha + x_{-\alpha} y_\alpha).$$

Since  $v_\lambda$  is a highest weight vector it is annihilated by  $x_\alpha$  and  $y_\alpha$  when  $\alpha \in \Phi^+$ . On the other hand,  $Hv_\lambda = \lambda(H)v_\lambda$  for  $H \in \mathfrak{h}$ , and so

$$c_{\mathfrak{g}} v_\lambda = \sum_i \lambda(h_i) \lambda(h^i) + \sum_{\alpha \in \Phi^+} \lambda(t_\alpha).$$

The first expression equals  $(\lambda|\lambda)$  by the previous proposition, while

$$\sum_{\alpha \in \Phi^+} \lambda(t_\alpha) = \sum_{\alpha \in \Phi^+} \langle \lambda, \alpha \rangle = 2(\lambda|\rho).$$

Thus

$$c_{\mathfrak{g}} v_\lambda = ((\lambda|\lambda) + 2(\lambda|\rho)) v_\lambda = ((\lambda + \rho|\lambda + \rho) - (\rho|\rho)) v_\lambda,$$

as desired.  $\square$

### 2.3.4 Category $\mathcal{O}$ and the Weyl character formula

We will now prove the Weyl character formula following Kac. It will be useful to work in the following category of representations, Category  $\mathcal{O}$ , introduced by Bernstein, Gelfand and Gelfand.

**Definition 2.105.** *A module is in Category  $\mathcal{O}$  if it is  $\mathfrak{h}$ -diagonalizable with finite dimensional weight spaces  $V_\lambda$ , such that there exists a finite set of weights  $\{\lambda_1, \dots, \lambda_N\}$  such that  $V_\mu = 0$  unless  $\mu \preceq \lambda_i$  for some  $i$ .*

This category contains all highest weight modules, is closed under finite direct sums, and it contains all submodules and quotient modules of a Category  $\mathcal{O}$  module. In particular it is an abelian category with enough projectives and injectives and has a good homological theory. The Verma modules  $M(\lambda)$  may or may not be irreducible. We will say a module  $V$  is a **subquotient** of a module  $W$  if there are submodules  $U \supset Q$  of  $W$  such that  $U/Q \cong V$ . Thus either a submodule or a quotient module is a subquotient.

**Proposition 2.106.** *Suppose that  $V$  is a highest weight module with weight  $\mu$  and  $V$  is a subquotient of  $M(\lambda)$ . Then*

$$|\lambda + \rho|^2 = |\mu + \rho|^2.$$

*Proof.* Since  $c$  commutes with the action of  $\mathfrak{g}$  it must act as a scalar on  $M(\lambda)$ , and we computed that scalar to be  $|\lambda + \rho|^2 - |\rho|^2$ . So it acts by the same scalar on any submodule,

quotient module or subquotient. Also  $c$  acts by the scalar  $|\mu + \rho|^2 - |\rho|^2$  on any highest weight module  $V$  with highest weight  $\lambda$ , so

$$|\lambda + \rho|^2 - |\rho|^2 = |\mu + \rho|^2 - |\rho|^2.$$

as desired.  $\square$

**Definition 2.107.** Let  $V$  be a module in Category  $\mathcal{O}$ . We define the **character** of  $V$  to be the formal expression

$$\chi_V = \sum_{\lambda} \dim(V_{\lambda}) e^{\lambda}$$

where  $e^{\lambda}$  is a formal symbol for  $\lambda \in \mathfrak{h}^*$ .

**Proposition 2.108** (Character of Verma modules). The character of  $M(\lambda)$  is

$$e^{\lambda} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})^{-1}.$$

*Proof.* Let  $v$  be the highest weight vector. We recall from Theorem 8 that the map

$$\xi \mapsto \xi \cdot v$$

from  $U(\mathfrak{n}^-)$  to  $M(\lambda)$  is a vector space isomorphism. So by the PBW theorem a basis of  $M(\lambda)$  consists of the vectors

$$\left( \prod_{\alpha \in \Phi^+} x_{-\alpha}^{k_{\alpha}} \right) v, \quad k_{\alpha} \geq 0,$$

where the positive roots  $\Phi^+$  are taken in some fixed definite order. The weight of this vector is

$$\lambda - \sum_{\alpha \in \Phi^+} k_{\alpha} \alpha,$$

so

$$\chi_V = e^{\lambda} \prod_{\alpha \in \Phi^+} e^{-k_{\alpha} \alpha} = e^{\lambda} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})^{-1}.$$

as desired.  $\square$

**Definition 2.109.** Let  $V$  be a module in Category  $\mathcal{O}$ . A nonzero vector  $v \in V$  is called **primitive** if there exists a proper submodule  $U \subset V$  such that  $v \notin U$  but  $x_{\alpha} v \in U$  for all  $\alpha \in \Phi^+$  (or equivalently, for all simple roots). We can take  $U = 0$ , so if  $x_{\alpha} v = 0$  then  $v$  is primitive. In other words, a highest weight vector is a primitive vector. More generally,  $v$  being primitive means that the image of  $v$  in  $V/U$  is a highest weight vector for some proper submodule  $U$  of  $V$ . We will call  $\mu$  a **primitive weight** if  $V_{\mu}$  contains a primitive vector.

A primitive vector is like a “hidden” highest weight vector, but visible only in a quotient.

**Proposition 2.110.** *Let  $V$  be a module in Category  $\mathcal{O}$ . Then  $V$  is generated by its primitive vectors.*

*Proof.* If not, consider the submodule  $U$  generated by the primitive vectors. Then  $Q = V/U$  would be a nonzero submodule. If we choose a nonzero vector in  $Q$  whose weight is maximal with respect to  $\preceq$ , then its preimage in  $V$  would be a primitive vector, which is a contradiction.  $\square$

**Proposition 2.111.** *Let  $V$  be a module in Category  $\mathcal{O}$ . Assume that  $V$  has only a finite number of weights. Then  $V$  has finite length. That is, it has a composition series*

$$V = V_m \supset V_{m-1} \supset \cdots \supset V_0 = 0$$

*such that each quotient  $V_i/V_{i-1}$  is irreducible, isomorphic to  $L(\mu)$ , where  $\mu$  is a primitive weight of  $V$ . (The quotients  $V_i/V_{i-1}$  are called composition factors, and they are independent of the composition series, by the Jordan–Hölder theorem.)*

*Proof.* We argue by induction on the number of linearly independent primitive vectors.

Choose a primitive weight  $\mu$  that is maximal with respect to  $\preceq$ . Then clearly a primitive vector  $v$  of weight  $\mu$  must be a highest weight vector, so  $W = U(\mathfrak{g}) \cdot v = U(\mathfrak{n}^-)v$  is a highest weight module. It has a maximal submodule  $W'$  and the quotient  $Q = W/W'$  is irreducible. Both  $V/W$  and  $W'$  have fewer independent primitive vectors than  $V$  (note that there are finitely many weights and each weight space is finite dimensional since we are in Category  $\mathcal{O}$ ), so by induction they have finite length. Since  $V/W$ ,  $W'$  and the irreducible quotient  $W/W'$  all have finite length, it follows that  $V$  has finite length.  $\square$

**Proposition 2.112** (Character of irreducible highest weight modules). *Let  $\lambda \in \mathfrak{h}^*$ . Then the character  $\chi_{L(\lambda)}$  is of the form*

$$\chi_{L(\lambda)} = \sum_{\substack{\mu \preceq \lambda \\ |\mu + \rho|^2 = |\lambda + \rho|^2}} c_\mu \chi_{M(\mu)} \quad (15)$$

where  $c_\lambda = 1$ .

*Proof.* The weight  $\mu$  of a primitive vector must satisfy  $\mu \preceq \lambda$  and  $|\mu + \rho|^2 = |\lambda + \rho|^2$ .

Since the inner product is positive definite, this implies that there are only a finite number of possible weights for primitive vectors (because  $|\mu + \rho|^2 = |\lambda + \rho|^2$  cuts out a sphere, and the lattice of weights intersected with that sphere is finite).  $M(\mu)$  has finite length because only finitely many irreducibles can appear as factors, and modules in Category  $\mathcal{O}$  have finite dimensional weight spaces. Also the composition factors of  $M(\mu)$  must be  $L(\nu)$  where  $|\nu + \rho|^2 = |\mu + \rho|^2 = |\lambda + \rho|^2$ . This is because every composition factor is a highest weight module and every irreducible highest weight module is of the form  $L(\nu)$ .

Let  $d(\mu, \nu)$  be the multiplicity of such  $L(\nu)$ . Then

$$\chi_{M(\mu)} = \sum_{\substack{\nu \preceq \mu \\ |\nu + \rho|^2 = |\lambda + \rho|^2}} d(\mu, \nu) \chi_{L(\nu)}.$$

Now the matrix  $d(\mu, \nu)$  indexed by pairs  $\mu, \nu$  is triangular since  $d(\mu, \mu) = 1$  and  $d(\mu, \nu) = 0$  unless  $\nu \preceq \mu$ . So it is invertible and we may write

$$\chi_{L(\mu)} = \sum_{\substack{\nu \preceq \mu \\ |\nu + \rho|^2 = |\lambda + \rho|^2}} d'(\mu, \nu) \chi_{M(\nu)}.$$

Applying this to  $\mu = \lambda$  gives (2).  $\square$

We will define the **Weyl denominator**

$$\Delta = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha}).$$

**Lemma 2.113.** *Let  $w \in W$  (the Weyl group). Then*

$$w(\Delta) = \text{sgn}(w)\Delta.$$

*Proof.* It is sufficient to check this if  $w = s_{\alpha_i}$  is a simple reflection. We recall that  $s_{\alpha_i}$  maps the simple root  $\alpha_i$  to  $-\alpha_i$  and it permutes the remaining positive roots. Moreover  $s_{\alpha_i}(\rho) = \rho - \alpha_i$ . So if we write

$$\Delta = e^\rho (1 - e^{-\alpha_i}) \prod_{\substack{\alpha \in \Phi^+ \\ \alpha \neq \alpha_i}} (1 - e^{-\alpha}),$$

then  $s_i$  maps  $e^\rho(1 - e^{-\alpha_i})$  to

$$e^{\rho - \alpha_i}(1 - e^{\alpha_i}) = -e^\rho(1 - e^{-\alpha_i}),$$

and it fixes the product. Hence  $s_i(\Delta) = -\Delta$ .  $\square$

**Theorem 2.114** (Weyl Character Formula). *Let  $V$  be a finite-dimensional irreducible representation of  $\mathfrak{g}$ . Thus by Theorem 2.100 there is a dominant integral weight  $\lambda$  such that  $V = L(\lambda)$ . We have*

$$\chi_V = \Delta^{-1} \sum_{w \in W} \text{sgn}(w) e^{w(\lambda + \rho)}$$

where  $W$  is the Weyl group and

$$\Delta = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha}).$$

The following argument is due to Kac, improving the proof of BGG. As an application, Kac extended the applicability of the Weyl character formula for characters of integrable representations of infinite-dimensional Kac-Moody Lie algebras. We will discuss this in more detail in the section, following Chapter 10 of Kac [?].

*Proof.* Using Proposition 2.108 we may rewrite (2) in the form (note that the  $\rho$  in the exponent comes from dividing by  $\Delta$ )

$$\chi_{L(\lambda)} = \sum_{\substack{\mu \preceq \lambda \\ |\mu+\rho|^2 = |\lambda+\rho|^2}} c_\mu e^{\mu+\rho} \Delta^{-1}$$

It may be simpler to write this as

$$\chi_{L(\lambda)} = \sum_{\mu \in P^+} c_\mu e^{\mu+\rho} \Delta^{-1}$$

and remember that  $c_\mu = 0$  unless  $\mu \preceq \lambda$  and  $|\mu+\rho|^2 = |\lambda+\rho|^2$ . We claim that if  $w \in W$ , then

$$c_\mu = \text{sgn}(w) c_{w \circ \mu}. \quad (16)$$

Indeed, since  $\chi_{L(\lambda)}$  is invariant under the action of  $W$ , and since  $w(\Delta) = \text{sgn}(w)\Delta$ , we have an identity

$$\sum_{\mu \in P^+} c_\mu e^{\mu+\rho} \Delta^{-1} = \sum_{\mu \in P^+} \text{sgn}(w) c_\mu e^{w(\mu+\rho)} \Delta^{-1}$$

and comparing the coefficients of  $e^{w \circ \mu} = e^{w(\mu+\rho)-\rho}$  on both sides of this equation gives (2).

We know that  $c_\lambda = 1$ , since this is part of Proposition 2.112. So by (2), we will have terms corresponding to  $\mu$  of the form  $w \circ \lambda$  and the sum of these terms is

$$\Delta^{-1} \sum_{w \in W} c_{w \circ \lambda} e^{w(\lambda+\rho)-\rho} e^\rho = \Delta^{-1} \sum_{w \in W} \text{sgn}(w) e^{w(\lambda+\rho)}.$$

This is the right hand side of the Weyl character formula, so our task is to show that these are the **only terms**. That is, we must show that  $c_\mu = 0$  unless  $\mu$  is of the form  $w \circ \lambda$  for some  $w \in W$ .

Therefore we start with  $\mu$  such that  $c_\mu \neq 0$ . By Proposition 2.90, there exists  $w \in W$  such that  $w(\mu+\rho)$  is dominant. Let  $\nu = w \circ \mu = w(\mu+\rho) - \rho$ . We will show that  $\nu = \lambda$ . In any case by (2),  $c_\nu \neq 0$  and so  $\nu \preceq \lambda$  and  $|\lambda+\rho|^2 = |\nu+\rho|^2$ . We write

$$\lambda - \nu = \sum_{\alpha \in \Phi^+} k_\alpha \alpha,$$

where since  $\nu \preceq \lambda$  we have  $k_\alpha \geq 0$ . We note the identity, for  $a, b \in \mathfrak{h}^*$ :

$$|a|^2 - |b|^2 = (a+b|a-b).$$

We apply this and learn that

$$|\lambda+\rho|^2 - |\nu+\rho|^2 = \left( \lambda + \nu + 2\rho \mid \sum_{\alpha \in \Phi^+} k_\alpha \alpha \right).$$

Now  $\lambda$  and  $\nu + \rho = w(\mu + \rho)$  are both dominant, so  $\lambda + \nu + 2\rho$  is **strongly dominant** meaning

$$(\alpha^\vee \mid \lambda + \nu + 2\rho) > 0$$

for all positive roots  $\alpha$ . So  $|\lambda+\rho|^2 = |\nu+\rho|^2$  implies that  $k_\alpha = 0$  for all  $\alpha$  and so  $\nu = \lambda$ .

□



The asymptotics of Weyl's character formula imply the Weyl dimension formula.

**Theorem 2.115** (Weyl Dimension Formula). *Let  $V$  be a finite-dimensional irreducible representation of  $\mathfrak{g}$  with highest weight  $\lambda$ . Then the dimension of  $V$  is given by*

$$\dim V = \prod_{\alpha \in \Phi^+} \frac{\langle \lambda + \rho, \alpha^\vee \rangle}{\langle \rho, \alpha^\vee \rangle}.$$

*Proof.* Note that  $\dim V(\lambda) = \chi_\lambda(1)$  but we cannot just plug in 1 into the Weyl character formula we would get  $0/0$ . Instead we look at the first nonzero homogeneous term in the Taylor expansion of the numerator and denominator. It turns out that the numerator and denominator both vanish to order  $|\Phi^+|$  at 1. Their first nonzero homogeneous terms are alternating polynomials, hence scalar multiples of  $\prod_{\alpha \in \Phi^+} \langle \alpha, t \rangle$ . Computing the scalar in the numerator and denominator gives

$$\sum_{w \in W} \operatorname{sgn}(w) e^{\langle w\mu, t \rangle} = C \left( \prod_{\alpha \in \Phi^+} \langle \mu, \alpha^\vee \rangle \right) \left( \prod_{\alpha \in \Phi^+} \langle \alpha, t \rangle \right) + \text{higher order terms}$$

with the same constant  $C$  for all  $\mu$ . Taking  $\mu = \lambda + \rho$  upstairs and  $\mu = \rho$  downstairs, the common factor cancels, and one gets

$$\dim V(\lambda) = \prod_{\alpha \in \Phi^+} \frac{\langle \lambda + \rho, \alpha^\vee \rangle}{\langle \rho, \alpha^\vee \rangle}$$

□

### 2.3.5 Finite-Dimensional Representations

**Exercise 2.116.** *Let  $V$  be a finite-dimensional vector space over  $\mathbb{C}$  equipped with a nondegenerate symmetric bilinear form  $B$ . Show that  $O(V, B)$  acts irreducibly on  $V$ .*

We present an elementary solution and then another solution using representation theory of Lie algebras.

**Solution 2.117.** *Take a nonzero  $O(V, B)$ -invariant subspace  $0 \neq W \subseteq V$ . Pick a nonzero vector  $w = (a_1, \dots, a_n) \in W$ . Some coordinate  $a_i$  is nonzero.*

*Now  $O(V, B)$  contains the sign-change matrix*

$$s_i = \operatorname{diag}(1, \dots, 1, -1, 1, \dots, 1),$$

*with the  $-1$  in the  $i$ -th place. Since  $s_i \in O(V, B)$  and  $W$  is  $O(V, B)$ -stable,  $s_i w \in W$ . Therefore*

$$w - s_i w = 2a_i e_i \in W.$$

*Since  $a_i \neq 0$ , this gives  $e_i \in W$ . Next,  $O(V, B)$  contains permutation matrices, so for any  $j$ , there is  $p_{ij} \in O(V, B)$  with  $p_{ij} e_i = e_j$ . Because  $W$  is  $O(V, B)$ -stable and  $e_i \in W$ , we get  $e_j \in W$  for all  $j$ . Hence all standard basis vectors lie in  $W$ , so  $W = V$ .*

**Solution 2.118.** It is enough to show that  $V$  is  $SO(V, B)$ -irreducible. This is equivalent to showing that  $V$  is  $\mathfrak{so}(V, B)$ -irreducible because  $SO(V, B)$  is connected and the Lie algebra of  $SO(V, B)$  is  $\mathfrak{so}(V, B)$ . Choose the standard Cartan subalgebra  $\mathfrak{h} \subset \mathfrak{g}$  consisting of diagonal matrices of the form

$$\text{diag}(t_1, \dots, t_m, -t_1, \dots, -t_m)$$

in type  $D_m$ , and

$$\text{diag}(t_1, \dots, t_m, 0, -t_1, \dots, -t_m)$$

in type  $B_m$ . Let  $\varepsilon_i \in \mathfrak{h}^*$  be given by  $\varepsilon_i(h) = t_i$ .

With respect to the standard basis, the defining representation  $V$  has weight decomposition

$$V = \bigoplus_{\lambda} V_{\lambda},$$

where the weights are

$$\{\pm\varepsilon_1, \dots, \pm\varepsilon_m\} \quad \text{in type } D_m,$$

and

$$\{\pm\varepsilon_1, \dots, \pm\varepsilon_m, 0\} \quad \text{in type } B_m.$$

Moreover, each weight space is one-dimensional.

For the standard choice of positive roots,  $V$  has a highest weight vector  $e_1$  of weight  $\varepsilon_1$ . Let  $W \subseteq V$  be the  $\mathfrak{g}$ -subrepresentation generated by  $e_1$ . Then  $W$  is a nonzero subrepresentation of  $V$ , so it suffices to show that  $W = V$ . For this, note that the negative root spaces move  $e_1$  to all the other weight vectors in  $V$ :

- root vectors for roots  $-(\varepsilon_1 - \varepsilon_j)$  send  $e_1$  to a nonzero multiple of  $e_j$  for  $j > 1$ ;
- root vectors for roots  $-(\varepsilon_1 + \varepsilon_j)$  send  $e_1$  to a nonzero multiple of  $e_{-j}$ ;
- in type  $B_m$ , a root vector for the root  $-\varepsilon_1$  sends  $e_1$  to a nonzero multiple of  $e_0$ .

General theory says that if a representation has a unique highest weight vector up to scaling, then the representation is irreducible. We have shown that  $e_1$  is a highest weight vector and that the subrepresentation it generates contains all the other weight vectors, so it must be all of  $V$ . Hence  $V$  is irreducible.

**Exercise 2.119.** Explain the difference between  $SO(5, \mathbb{R})$  and the compact symplectic group  $\text{Sp}(2)$ . Describe the root diagrams. Find the 4 and 5-dimensional representations. Are they representations of the groups or just the Lie algebras?

**Solution 2.120.** Recall that

$$\begin{aligned} SO(5, \mathbb{R}) &= \{A \in M_5(\mathbb{R}) : A^t A = I, \det A = 1\} \\ \text{Sp}(2) &= \{A \in M_4(\mathbb{C}) : A^t J A = J\} \end{aligned}$$

where  $J$  is the standard symplectic form on  $\mathbb{C}^4$ . Both are compact, connected Lie groups of dimension 10 and rank 2. The fact to know is  $\text{Sp}(2) \cong \text{Spin}(5)$  and  $SO(5) = \text{Spin}(5)/\{\pm 1\}$

so that  $\mathrm{Sp}(2)$  is the simply connected double cover of  $\mathrm{SO}(5)$ . Then they clearly have the same Lie algebra  $\mathfrak{so}(5)$ . The complexification is type  $B_2 = C_2$ :

$$\mathfrak{g}_{\mathbb{C}} \cong \mathfrak{so}_5(\mathbb{C}) \cong \mathfrak{sp}_4(\mathbb{C})$$

and the Lie algebras correspond to the compact real form:

$$\mathfrak{g}_0 \cong \mathfrak{so}(5, \mathbb{R}) \cong \mathfrak{sp}(2)$$

Root systems are defined after choosing a Cartan subalgebra of the complexification  $\mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$  so even though the Lie algebra is real, the root diagram is always the complex one.

Being rank 2 we have two fundamental weights  $\omega_1$  and  $\omega_2$ . The 5-dimensional representation is the one with highest weight  $\omega_1$ , and the 4-dimensional representation is the one with highest weight  $\omega_2$ . In contrast, the 5-dimensional representation is the defining representation of  $\mathrm{SO}(5)$ , and so  $\mathrm{SO}(5)$  acts on it (and of course does  $\mathrm{Sp}(2)$ ).

The 4-dimensional representation is the defining representation of  $\mathrm{Sp}(2)$ , in particular the inclusion of  $\mathrm{Sp}(2) \subset \mathrm{GL}(4)$  differentiates to  $\mathfrak{sp}(2) \rightarrow \mathfrak{gl}(4)$ . However it does not descend to a representation of  $\mathrm{SO}(5)$  because the central element of  $\mathrm{Sp}(2)$  does not act trivially on it.

How do we identify how the center acts? The action of the center is determined by the highest weight modulo the root lattice, in particular recall that

$$Z(G) \cong P/Q$$

where  $P$  is the weight lattice and  $Q$  is the root lattice. For  $\mathrm{Spin}(5)$ , the center is  $\mathbb{Z}/2$ . The 5-dimensional representation has highest weight in the root lattice, so the center acts trivially and it descends to  $\mathrm{SO}(5)$ . The 4-dimensional representation has highest weight not in the root lattice, so the nontrivial central element acts by  $-1$ , and it does not descend. To see this, recall that to check if something is in the root lattice, it simply must pair integrally against all of the fundamental coweights.

We start with the Cartan matrix of type  $B_2$ :

$$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$$

Inverting the Cartan matrix and taking columns tells us how to write the fundamental coweights in terms of the coroots:

$$\begin{aligned} \omega_1^\vee &= \alpha_1^\vee + \frac{1}{2} \alpha_2^\vee \\ \omega_2^\vee &= 1 \cdot \alpha_1^\vee + 1 \cdot \alpha_2^\vee \end{aligned}$$

Then pairing with the fundamental weight simply extracts the coefficient of the corresponding simple coroot and so we see that  $\omega_1$  is in the root lattice but  $\omega_2$  is not. To describe the root diagram, we can recover the lengths from the off diagonal entries of the Cartan matrix

$$\frac{(\alpha_1, \alpha_1)}{(\alpha_2, \alpha_2)} = \frac{A_{12}}{A_{21}} = \frac{-2}{-1} = 2$$

so  $\alpha_1$  is the long root and  $\alpha_2$  is the short root. The angle between them is given by

$$\cos \theta = \frac{(\alpha_1, \alpha_2)}{\sqrt{(\alpha_1, \alpha_1)(\alpha_2, \alpha_2)}} = \frac{A_{12}}{\sqrt{A_{12}A_{21}}} = -\frac{1}{\sqrt{2}}$$

so  $\theta = 135^\circ$ . We can choose coordinates consistent with the above:

$$\alpha_1 = e_1 - e_2 \quad (\text{long}), \quad \alpha_2 = e_2 \quad (\text{short}).$$

Generate all roots by reflections, we get  $\Phi = \{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}$ . So the diagram is 4 short roots along the axes, and 4 long roots along the diagonals. It becomes clear by the reflections that  $W = D_4$  is the dihedral group of order 8. The root diagram is also consistent with the fact that  $\dim \mathfrak{g}$  is 10.

**Exercise 2.121.** Let  $\mathfrak{g}$  be the adjoint representation of  $\mathfrak{sl}_3(\mathbb{C})$ . Decompose the second tensor power of  $\mathfrak{g}$  into irreducible representations of  $\mathfrak{sl}_3(\mathbb{C})$ .

**Solution 2.122.** Let  $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ , viewed as the adjoint representation, so

$$\mathfrak{g} \cong V(\omega_1 + \omega_2).$$

We decompose

$$\mathfrak{g} \otimes \mathfrak{g} = S^2 \mathfrak{g} \oplus \Lambda^2 \mathfrak{g}.$$

The highest weight of  $\mathfrak{g}$  is  $\omega_1 + \omega_2$ , so the highest weight of  $\mathfrak{g} \otimes \mathfrak{g}$  is  $2\omega_1 + 2\omega_2$ . Hence  $V(2\omega_1 + 2\omega_2)$  occurs, and since the highest weight vector tensors with itself symmetrically, it lies in  $S^2 \mathfrak{g}$ . Recall by Weyl's formula that the dimension of  $V(a\omega_1 + b\omega_2)$  is given by the formula

$$\frac{(a+1)(b+1)(a+b+2)}{2}$$

The Killing form gives a nonzero invariant symmetric bilinear form on  $\mathfrak{g}$ , so  $V(0) \subset S^2 \mathfrak{g}$ . For  $\mathfrak{sl}_3$  there is also a nonzero equivariant symmetric map

$$S^2 \mathfrak{g} \rightarrow \mathfrak{g},$$

which may seem not so obvious at first, but it comes from the most obvious bilinear operation on matrices, that is multiplication  $(X, Y) \mapsto XY$ . However, this is not symmetric and does not land in  $\mathfrak{sl}_n$ . So fix both defects in the most obvious way by symmetrizing and removing the trace part:

$$XY + YX - c \operatorname{tr}(XY)I.$$

This is a nonzero equivariant symmetric bilinear map  $S^2 \mathfrak{g} \rightarrow \mathfrak{g}$ , and the image of the highest weight vector is a nonzero highest weight vector, so  $V(\omega_1 + \omega_2) \subset S^2 \mathfrak{g}$ .

Now  $\dim S^2 \mathfrak{g} = \frac{8 \cdot 9}{2} = 36$ , while  $27 + 8 + 1 = 36$ . Therefore

$$S^2 \mathfrak{g} \cong V(2\omega_1 + 2\omega_2) \oplus V(\omega_1 + \omega_2) \oplus V(0)$$

Next,

$$\dim \Lambda^2 \mathfrak{g} = \frac{8 \cdot 7}{2} = 28.$$

The Lie bracket gives a nonzero equivariant map

$$\Lambda^2 \mathfrak{g} \rightarrow \mathfrak{g},$$

so

$$V(\omega_1 + \omega_2) \subset \Lambda^2 \mathfrak{g}.$$

This accounts for one copy of  $V(\omega_1 + \omega_2)$  of dimension 8, leaving 20 dimensions. The remaining highest weights are  $3\omega_1$  and  $3\omega_2$ , and

$$\dim V(3\omega_1) = \dim V(3\omega_2) = 10.$$

Hence

$$\Lambda^2 \mathfrak{g} \cong V(3\omega_1) \oplus V(3\omega_2) \oplus V(\omega_1 + \omega_2).$$

To see that  $V(3\omega_1)$  and  $V(3\omega_2)$  occur in  $\Lambda^2 \mathfrak{g}$ , let

$$\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C}) \cong V(\omega_1 + \omega_2),$$

and let  $e_{\alpha_1}, e_{\alpha_2}, e_{\alpha_1+\alpha_2}$  be root vectors for the positive roots. Since the highest weight of the adjoint representation is

$$\theta = \alpha_1 + \alpha_2 = \omega_1 + \omega_2,$$

the vectors

$$e_{\alpha_1+\alpha_2} \wedge e_{\alpha_1}, \quad e_{\alpha_1+\alpha_2} \wedge e_{\alpha_2}$$

have weights

$$\theta + \alpha_1 = 2\alpha_1 + \alpha_2 = 3\omega_1, \quad \theta + \alpha_2 = \alpha_1 + 2\alpha_2 = 3\omega_2.$$

A direct check shows that both are annihilated by the positive root operators, so they are highest weight vectors. Therefore

$$V(3\omega_1) \subset \Lambda^2 \mathfrak{g}, \quad V(3\omega_2) \subset \Lambda^2 \mathfrak{g}.$$

**Exercise 2.123.** When is the trivial representation of a Lie algebra the only one-dimensional representation?

**Solution 2.124.** The answer is precisely when  $\mathfrak{g}$  is perfect, meaning  $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ . Every one dimensional representation of  $\mathfrak{g}$  is given by a Lie algebra homomorphism  $\mathfrak{g} \rightarrow \mathbb{C}$ , but  $\mathbb{C}$  is abelian, so such a homomorphism factors through the abelianization  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ . Thus any 1 dimensional rep is given by a choice of linear functional on  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ . If  $\mathfrak{g}$  is perfect, then  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] = 0$ , so there is only the trivial representation. Conversely, if  $\mathfrak{g}$  is not perfect, then  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  is nonzero, so there are nontrivial one-dimensional representations.

Note that  $\mathfrak{g}$  is semisimple implies that it is perfect. A counterexample to the converse is  $\mathfrak{g} = \mathfrak{sl}_2 \ltimes \mathbb{C}^2$ , where  $\mathbb{C}^2$  is the standard  $\mathfrak{sl}_2$ -module. Then  $\mathfrak{g}$  is not semisimple, but  $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ . It is not semisimple because it has an ideal  $\mathbb{C}^2$  which is abelian, and in particular solvable. Recall that a Lie algebra is semisimple if it has no nonzero solvable ideals, so  $\mathfrak{g}$  is not semisimple.

### 2.3.6 Flag Variety and Borel–Weil–Bott Theorem

**Theorem 2.125.** *If  $G$  is semisimple and simply connected, then the forgetful functor from the groupoid of equivariant line bundles on  $X$  to the groupoid of line bundles on  $X$  is an equivalence of categories.*

*Proof.* The forgetful functor is faithful by construction. To see that it is full, we must show that every isomorphism  $\mathcal{L} \rightarrow \mathcal{L}'$  of equivariant line bundles is actually  $G$ -equivariant. To specify such an isomorphism is to specify a nowhere vanishing section of the equivariant line bundle  $\mathcal{L}^{-1} \otimes \mathcal{L}'$ . This line bundle is trivial, hence  $G$  acts by a character on its space of global sections. Since  $G$  is semisimple, this character must be trivial, and any global section is invariant.

It remains to show that the forgetful functor is essentially surjective. In other words, we must show that any line bundle  $\mathcal{L}$  on  $X$  admits an action of  $G$ , compatible with the action of  $G$  on  $X$ . Let  $a : G \times X \rightarrow X$  classify the action of  $G$  on  $X$ , and let

$$\pi_1 : G \times X \rightarrow G, \quad \pi_2 : G \times X \rightarrow X$$

be the projections. Let  $\mathcal{M} = a^*\mathcal{L} \otimes \pi_2^*\mathcal{L}^{-1}$ . Since  $X$  is simply connected (for example, by the Bruhat decomposition), its Picard variety is trivial, so the fibers of a connected family of line bundles on  $X$  are all isomorphic. In particular,  $\mathcal{M}$  is trivial on each fiber of  $\pi_1$ , hence  $\mathcal{M} \simeq \pi_1^*\mathcal{N}$  for some line bundle  $\mathcal{N}$  on  $G$ .

Since  $H^2(G, \mathbb{Z}) = 0$ , any line bundle on  $G$  is topologically trivial. As  $G$  is Stein, such a line bundle is holomorphically trivial. Thus  $\mathcal{M}$  is trivial, so  $a^*\mathcal{L} \simeq \pi_2^*\mathcal{L}$ . Consequently there exists a projective action of  $G$  on  $\mathcal{L}$  compatible with its action on  $X$ . This corresponds to a central extension  $\tilde{G}$  of  $G$  by  $\mathbb{C}^\times$ . Since  $G$  is simply connected, this extension splits. Hence  $G$  acts on  $\mathcal{L}$ .  $\square$

Equivariant line bundles on  $X$  are classified by one-dimensional representations of  $B$ . Let  $U$  be the unipotent radical of  $B$  and  $T$  a maximal torus, so that  $B = TU$ . Then such representations correspond to characters of  $T$ . This is because  $U$  is unipotent, so as a variety it is isomorphic to  $\mathbb{C}^n$ , and any algebraic group homomorphism  $\phi : \mathbb{C}^n \rightarrow \mathbb{C}^\times$  must be constant. To see this, one can differentiate at the identity:  $d\phi : \mathbb{C}^n \rightarrow \mathbb{C}$ , but  $\mathbb{C}^n$  is nilpotent as a Lie algebra (coming from  $U$ ), while  $\mathbb{C}$  is abelian with no nilpotents compatible with multiplicative structure. The only possibility is  $d\phi = 0$ , hence  $\phi = 1$  and since  $\chi$  is trivial on  $U$ , it factors through  $B \twoheadrightarrow B/U \cong T$ .

Let  $\Lambda$  denote the character group of  $T$ . For each  $\lambda \in \Lambda$ , let  $\mathcal{L}_\lambda$  denote the corresponding equivariant line bundle  $\mathcal{L}_\lambda = G \times_B \mathbb{C}_\lambda$ .

**Theorem 2.126.** *If  $-\lambda$  is not dominant, then  $H^0(X, \mathcal{L}_\lambda) = 0$ . If  $-\lambda$  is dominant, then  $H^0(X, \mathcal{L}_\lambda)$  is an irreducible representation of  $G$  with lowest weight  $\lambda$ .*

*Proof.* Let  $V = H^0(X, \mathcal{L}_\lambda)$ . Identify  $V$  with the space of holomorphic functions  $v$  on  $G$  satisfying

$$v(gb) = \lambda(b)v(g), \quad b \in B.$$

Then  $G$  acts by left translation:  $(gv)(x) = v(gx)$ .

Let  $U'$  be the unipotent radical of the opposite Borel. Let  $V_0 \subset V$  be the subspace of vectors invariant under  $U'$ . Then  $V_0$  consists of functions satisfying  $v(ux) = v(x)$  for all  $u \in U'$ . Since  $U'B$  is dense in  $G$ ,  $V_0$  is at most one-dimensional. If nonzero, it is generated by a function  $v$  with  $v(1) = 1$ .

For  $t \in T$ ,

$$(tv)(1) = v(t) = \lambda(t)v(1),$$

so  $tv = \lambda(t)v$ , and hence  $V$  is irreducible with lowest weight  $\lambda$ . This is only possible if  $-\lambda$  is dominant.

If  $-\lambda$  is dominant, one constructs such a function  $v$  using the Bruhat decomposition and checks that it extends holomorphically. To construct the function  $v$ , let  $B' = U'T$  be a Borel subgroup opposite to  $B$ , and consider the Bruhat decomposition of  $G$  into double cosets

$$G = \bigsqcup_{w \in W} B'wB.$$

The *big cell* is  $U'B$ , which is biholomorphic to  $U' \times B$ . Hence the condition

$$v(ub) = \lambda(b)$$

determines  $v$  uniquely on  $U'B$ .

We must show that this holomorphic function extends to all of  $G$ . Since  $v$  is algebraic on  $U'B$ , it suffices to show that  $v$  has no poles along Bruhat cells of codimension 1.

Let  $C$  be such a cell, corresponding to a simple root  $\alpha$ . If  $v$  had a pole along  $C$ , then since  $v$  has no zeros on  $U'B$ , it would approach infinity near every point of  $C$ . Recall that at the generic point  $\eta_C$ , the local ring is a DVR, so we can talk about the order of vanishing of  $v$  at  $\eta_C$ . The order of vanishing is negative if and only if  $v$  has a pole along  $C$ . However, this is not quite the same as saying that  $v$  approaches infinity near every point of  $C$ , because the generic point  $\eta_C$  does not necessarily know about all of the closed points. However it does guarantee that there is an open dense subset of  $C$  such that  $v$  approaches infinity near every point of this subset. Write

$$C \cup U'B = SB,$$

where  $S$  is the subgroup generated by  $U'$  and the root subgroup corresponding to  $\alpha$ . Let  $G'$  denote the universal cover of a Levi factor of  $S$  whose maximal torus is contained in  $T$ . Then  $G' \simeq \mathrm{SL}_2(\mathbb{C})$ .

Pulling everything back to  $G'$ , we reduce to proving the statement for  $G = G'$ .

Now identify  $G$  with  $\mathrm{SL}_2(\mathbb{C})$ ,  $B$  with upper triangular matrices, and  $U'$  with lower triangular unipotent matrices. Identifying  $T$  with diagonal matrices, the character  $\lambda$  is given by

$$\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \mapsto t^k$$

for some integer  $k$ . On the big cell, the function  $v$  is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a^k$$

This function extends holomorphically to all of  $G$  if and only if  $k \geq 0$ , i.e. if and only if  $-\lambda$  is dominant.  $\square$

Let  $X = \mathbb{P}^1$ . Let  $K$  be the canonical bundle, so  $\deg K = -2$ . For an equivariant line bundle  $\mathcal{O}(n)$ , we have  $\deg(\mathcal{O}(n) \otimes K^{\otimes n+1}) = -n - 2$ . By Serre duality,

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \cong H^1(\mathbb{P}^1, \mathcal{O}(n) \otimes K^{\otimes n+1})^*.$$

More generally, for a  $\mathbb{P}^1$ -bundle  $\pi : E \rightarrow S$  with relative canonical bundle  $K$ , and a line bundle  $\mathcal{L}$  of degree  $n \geq -1$  on fibers, there are natural isomorphisms

$$R^i \pi_* \mathcal{L} \cong R^{i+1} \pi_* (\mathcal{L} \otimes K^{\otimes n+1}).$$

**Theorem 2.127.** *Let  $\pi : E \rightarrow S$  be a  $\mathbb{P}^1$ -bundle with relative canonical bundle  $K$ , and let  $\mathcal{L}$  be a line bundle on  $E$  of degree  $n \geq -1$  on fibers. Then*

$$H^i(E, \mathcal{L}) \cong H^{i+1}(E, \mathcal{L} \otimes K^{\otimes n+1}).$$

Let  $\rho$  be half the sum of positive roots and set  $\mathcal{L}^\lambda = \mathcal{L}_{\rho-\lambda}$ .

**Lemma 2.128.** *Let  $\alpha$  be a simple root and suppose  $\langle \alpha^\vee, \lambda \rangle \geq 0$ . Then there is a canonical isomorphism*

$$H^i(X, \mathcal{L}^\lambda) \cong H^{i+1}(X, \mathcal{L}^{w_\alpha(\lambda)}),$$

where  $w_\alpha$  is the simple reflection.

*Proof.* Let  $P$  be the minimal parabolic corresponding to  $\alpha$ . Then  $X$  is a  $\mathbb{P}^1$ -bundle over  $G/P$ , and the result follows from the previous theorem.  $\square$

**Theorem 2.129** (Borel–Weil–Bott). *Let  $\lambda \in \Lambda$ . If  $\lambda$  is not regular, then  $H^i(X, \mathcal{L}^\lambda) = 0$  for all  $i$ . If  $\lambda$  is regular, then there exists a unique  $i(\lambda)$  such that  $H^{i(\lambda)}(X, \mathcal{L}^\lambda)$  is nonzero, and is dual to the irreducible representation of highest weight  $w(\lambda) - \rho$ , where  $w(\lambda)$  is dominant.*

### 2.3.7 Coadjoint Orbits

**Exercise 2.130.** *Talk about coadjoint orbits of  $\mathfrak{sl}_2(\mathbb{C})$ . What is the adjoint representation of a simple Lie algebra? Is it irreducible? Is the adjoint representation (of a Lie algebra) isomorphic to its dual? What about an arbitrary irreducible representation of a Lie algebra? Calculate the orbits of the Lie group adjoint representation  $\mathrm{SL}_2 \rightarrow \mathrm{GL}(\mathfrak{sl}_2)$ .*"



**Solution 2.131.** *A basic piece of general theory is the following. Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . The group  $G$  acts on  $\mathfrak{g}$  by the adjoint action*

$$\mathrm{Ad}(g)(X) = gXg^{-1},$$

*and on  $\mathfrak{g}^*$  by the coadjoint action*

$$\mathrm{Ad}^*(g)(\lambda)(X) = \lambda(\mathrm{Ad}(g^{-1})X).$$

*For any  $\lambda \in \mathfrak{g}^*$ , its coadjoint orbit is*

$$\mathcal{O}_\lambda = G \cdot \lambda \cong G / \mathrm{Stab}_G(\lambda).$$

*When  $\mathfrak{g}$  carries a nondegenerate  $G$ -invariant bilinear form, one may identify  $\mathfrak{g}$  with  $\mathfrak{g}^*$ , and then adjoint and coadjoint orbits become the same objects. This is what happens for semisimple Lie algebras: the Killing form identifies  $\mathfrak{g} \simeq \mathfrak{g}^*$ , so every statement about coadjoint orbits may be reformulated as a statement about adjoint orbits. Coadjoint orbits are more natural to study because they carry a canonical symplectic structure.*

*Now let*

$$G = \mathrm{SL}_2(\mathbb{C}), \quad \mathfrak{g} = \mathfrak{sl}_2(\mathbb{C}),$$

*and let*

$$T = \left\{ \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} : t \in \mathbb{C}^\times \right\} \subset \mathrm{SL}_2(\mathbb{C})$$

*be the diagonal maximal torus. It is enough to find an element of  $\mathfrak{g}$  whose stabilizer under the adjoint action is  $T$ . Take*

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathfrak{sl}_2(\mathbb{C}).$$

*We claim that  $\mathrm{Stab}_G(h) = T$ . Indeed, if*

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{C})$$

*satisfies  $\mathrm{Ad}(g)(h) = h$ , then  $ghg^{-1} = h$ , equivalently  $gh = hg$ . But*

$$gh = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \quad hg = \begin{pmatrix} a & b \\ -c & -d \end{pmatrix},$$

*so  $gh = hg$  forces  $b = c = 0$ . Hence  $g$  is diagonal, and since  $\det(g) = 1$ , we get  $g \in T$ . Therefore  $\mathrm{Stab}_G(h) = T$*

*More generally, we can characterize all the coadjoint orbits of  $\mathfrak{sl}_2(\mathbb{C})$ . Every element of  $\mathfrak{sl}_2(\mathbb{C})$  has the form*

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

*The determinant is conjugation-invariant, it helps classify adjoint orbits.*

*The orbits are:*

1. the zero orbit  $\{0\}$ ;
2. the unique nonzero nilpotent orbit, represented by

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

The stabilizer of  $e$  is the subgroup  $B$  of upper triangular matrices, so the orbit is  $G/B \cong \mathbb{P}^1$ .

3. the semisimple orbits, represented by

$$\lambda h = \begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix}, \quad \lambda \in \mathbb{C}^\times.$$

In general, for a simple Lie algebra  $\mathfrak{g}$ , the adjoint representation is always irreducible. Moreover, it is always self-dual, the isomorphism being given by the Killing form. However, for semisimple Lie algebras, the adjoint representation may be reducible, with one irreducible summand for each simple factor.

In general, if  $\mu$  is a weight of an irreducible representation  $V(\lambda)$ , then  $-\mu$  is a weight of the dual representation  $V(\lambda)^*$ , occurring with the same multiplicity. In particular, if  $\lambda$  is the highest weight of  $V(\lambda)$ , then  $w_0(\lambda)$  is the lowest weight of  $V(\lambda)$ , and so  $-w_0(\lambda)$  is the highest weight of  $V(\lambda)^*$ . Hence  $V(\lambda)$  is self-dual if and only if  $\lambda = -w_0(\lambda)$ .

**Theorem 2.132** (Jordan decomposition). *Let  $\mathfrak{g}$  be a complex semisimple Lie algebra and let  $x \in \mathfrak{g}$ . Then there exist unique elements  $x_s, x_n \in \mathfrak{g}$  such that*

$$x = x_s + x_n, \quad [x_s, x_n] = 0,$$

where  $x_s$  is semisimple and  $x_n$  is nilpotent. Moreover,  $x_s$  and  $x_n$  are polynomials in  $x$ , and the decomposition is functorial under Lie algebra homomorphisms. In particular,

$$(\text{Ad}(g)x)_s = \text{Ad}(g)(x_s), \quad (\text{Ad}(g)x)_n = \text{Ad}(g)(x_n).$$

**Theorem 2.133.** *Let  $\mathfrak{g}$  be a complex semisimple Lie algebra.*

1. *There are only finitely many nilpotent adjoint orbits in  $\mathfrak{g}$ .*
2. *If  $\mathcal{O} \subset \mathfrak{g}$  is a nilpotent orbit and  $t \in \mathbb{C}^\times$ , then*

$$t\mathcal{O} = \mathcal{O}.$$

*In particular, the closure  $\overline{\mathcal{O}}$  is a cone.*

**Remark 2.134.** *If  $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$ , then the nilpotent orbits are classified by partitions of  $n$ , because the eigenvalues of a nilpotent matrix are all zero, so the only invariant is the sizes of the Jordan blocks. In this case, it is clear that there are only finitely many nilpotent orbits, and that they are cones. However, for an arbitrary semisimple Lie algebra, the classification of nilpotent orbits is more complicated, and it is not a priori clear that there are only finitely many of them, or that they are cones.*

**Remark 2.135.** *If  $x \in \mathfrak{g}$  is nilpotent, then by Jacobson-Morozov there exists an  $\mathfrak{sl}_2$ -triple  $(x, h, y)$ . Then*

$$\mathrm{Ad}(\exp(th))(x) = e^{2t}x.$$

*So every nonzero scalar multiple of  $x$  lies in the same adjoint orbit. That is exactly the statement that the nilpotent orbit is a cone under scaling.*

**Theorem 2.136** (Jacobson–Morozov). *Let  $\mathfrak{g}$  be a finite-dimensional semisimple Lie algebra over a field of characteristic 0. If  $e \in \mathfrak{g}$  is nilpotent, then there exist elements  $h, f \in \mathfrak{g}$  such that  $[h, e] = 2e$ ,  $[h, f] = -2f$ ,  $[e, f] = h$ .*

### 3 Algebraic Topology

#### 3.1 Fundamental groups

**Theorem 3.1** (Van Kampen’s theorem). *Let  $X$  be the union of path connected open subsets  $X_\alpha$  such that  $X_\alpha \cap X_\beta$  is path connected for all  $\alpha, \beta$ . Then there is a surjection*

$$*_\alpha \pi_1(X_\alpha) \twoheadrightarrow \pi_1(X)$$

*whose kernel is the normal subgroup generated by elements of the form  $i_{\alpha\beta}(\gamma)i_{\beta\alpha}(\gamma)^{-1}$ , where  $\gamma \in \pi_1(X_\alpha \cap X_\beta)$  and  $i_{\alpha\beta} : \pi_1(X_\alpha \cap X_\beta) \rightarrow \pi_1(X_\alpha)$  is the map induced by inclusion.*

A priori, loops in  $X$  can move through different  $X_\alpha$ , but if the cover is nice enough, we can subdivide the loop into pieces that lie in a single  $X_\alpha$ . This gives a surjection from the free product of the  $\pi_1(X_\alpha)$  to  $\pi_1(X)$ . However, if we have a loop that lies in  $X_\alpha \cap X_\beta$ , then it can be viewed as a loop in  $X_\alpha$  or in  $X_\beta$ , and these two views must be identified in  $\pi_1(X)$ . This gives the relations that generate the kernel of the surjection.

Note that the subgroup generated by the elements  $i_{\alpha\beta}(\gamma)i_{\beta\alpha}(\gamma)^{-1}$  is not necessarily normal, so we must take the normal closure to get the kernel.

**Corollary 3.2.** *If  $X$  is obtained from  $Y$  by attaching a 2-cell along a loop  $\gamma : S^1 \rightarrow Y$ , then  $\pi_1(X)$  is the quotient of  $\pi_1(Y)$  by the normal subgroup generated by  $[\gamma]$ .*

The same argument shows that attaching  $n$ -cells for  $n \geq 3$  does not change the fundamental group.

*Proof.* Take the open cover  $A = Y \cup_\gamma \{z \in D^2 \mid |z| > 1/3\}$  and  $B = \{z \in D^2 \mid |z| < 2/3\}$ . Then  $A$  is homotopy equivalent to  $Y$ ,  $B$  is contractible, and  $A \cap B$  is homotopy equivalent to  $S^1$ . We get

$$\pi_1(X) \cong \pi_1(Y)/\mathbb{Z}$$

where the generator of  $\pi_1(S^1)$  maps to  $[\gamma]$  in  $\pi_1(Y)$ .  $\square$

**Example 3.3.** *The genus  $g$  oriented surface  $M_g$  has a CW structure with one 0-cell,  $2g$  1-cells, and one 2-cell. The attaching map of the 2-cell is given by the loop*

$\gamma = [a_1, b_1] \cdots [a_g, b_g]$ , where  $a_i, b_i$  are the loops corresponding to the 1-cells. In particular,  $\pi_1(M_g)$  is generated by  $a_i, b_i$  with a single relation  $\prod_{i=1}^g [a_i, b_i] = 1$ .

**Example 3.4.** *If we attach a 2 cell to the wedge sum of  $g$  circles by the word  $a_1^2 \cdots a_g^2$ , we obtain a nonorientable surface  $N_g$ . For  $g = 1$ , we get  $\mathbb{RP}^2$ , and for  $g = 2$  we get the Klein bottle, which can also be presented by the word  $abab^{-1}$ .*

### 3.2 Covering spaces

A covering space of a topological space  $X$  is a topological space  $\tilde{X}$  together with a continuous surjective map  $p : \tilde{X} \rightarrow X$  such that for every  $x \in X$ , there exists an open neighborhood  $U$  of  $x$  such that  $p^{-1}(U)$  is a disjoint union of open sets in  $\tilde{X}$ , each of which is homeomorphic to  $U$  via  $p$ .

**Proposition 3.5.** *Covering spaces have the homotopy lifting property, that is if  $p : \tilde{X} \rightarrow X$  is a covering space, and  $f : Y \times [0, 1] \rightarrow X$  is a homotopy, and  $\tilde{f}_0 : Y \rightarrow \tilde{X}$  is a lift of  $f_0 = f(-, 0)$ , then there exists a unique homotopy  $\tilde{f} : Y \times [0, 1] \rightarrow \tilde{X}$  such that  $p \circ \tilde{f} = f$  and  $\tilde{f}(-, 0) = \tilde{f}_0$ .*

**Theorem 3.6.** *The map  $p_* : \pi_1(\tilde{X}) \rightarrow \pi_1(X)$  induced by a covering space  $p : \tilde{X} \rightarrow X$  is an injective homomorphism. The image of  $p_*$  is a subgroup of  $\pi_1(X)$  consisting of those loops in  $X$  that lift to loops in  $\tilde{X}$ . The index of  $p_*(\pi_1(\tilde{X}))$  in  $\pi_1(X)$  is equal to the number of sheets of the covering space.*

**Theorem 3.7.** *Let  $p : (\tilde{X}, x_0) \rightarrow (X, x)$  be a covering space. Let  $(Y, y)$  be path connected and locally path connected. Then  $f : (Y, y) \rightarrow (X, x)$  lifts to a map  $\tilde{f} : (Y, y) \rightarrow (\tilde{X}, x_0)$  if and only if  $f_*(\pi_1(Y, y)) \subset p_*(\pi_1(\tilde{X}, x_0))$ .*

**Theorem 3.8.** *Let  $p : (\tilde{X}, x_0) \rightarrow (X, x)$  be a covering space. Then if  $Y$  is connected and two lifts of a map  $f : (Y, y) \rightarrow (X, x)$  agree at a point, they are equal.*

Having established the basic theory of covering spaces, we can now classify them. Let  $X$  be a path connected, locally path connected, and semilocally simply connected space. Then there is a universal cover  $\tilde{X}$  of  $X$ , characterized by the property that  $\tilde{X}$  is simply connected. Every other covering space of  $X$  is a quotient of  $\tilde{X}$ .

The universal cover  $\tilde{X}$  is modeled on the set of homotopy classes of paths in  $X$  starting at  $x$ . The map  $p : \tilde{X} \rightarrow X$  sends a path to its endpoint. A consequence of the lifting criterion is that the universal cover  $\tilde{X}$  is a covering space of any other path connected covering space of  $X$ .

**Theorem 3.9.** *Let  $X$  be path-connected, locally path-connected, and semilocally simply connected. Then there is a bijection between the set of basepoint-preserving isomorphism*

classes of path-connected covering spaces

$$p : (\widetilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$$

and the set of subgroups of  $\pi_1(X, x_0)$ , obtained by associating to such a covering space the subgroup

$$p_* (\pi_1(\widetilde{X}, \tilde{x}_0)) \subseteq \pi_1(X, x_0).$$

If basepoints are ignored, this correspondence gives a bijection between isomorphism classes of path-connected covering spaces

$$p : \widetilde{X} \rightarrow X$$

and conjugacy classes of subgroups of  $\pi_1(X, x_0)$ .

Covering spaces can be represented by permutations. Let  $[\gamma] \in \pi_1(X, x_0)$  be a loop based at  $x_0$ . Then for each point  $\tilde{x}_0 \in p^{-1}(x_0)$ , there is a unique lift  $\tilde{\gamma}$  of  $\gamma$  starting at  $\tilde{x}_0$ . The endpoint of  $\tilde{\gamma}$  is another point in  $p^{-1}(x_0)$ , and this gives a permutation of the fiber  $p^{-1}(x_0)$ . This defines a homomorphism from  $\pi_1(X, x_0)$  to the symmetric group on the fiber  $p^{-1}(x_0)$ , and the image of this homomorphism is called the monodromy group of the covering space. The cover is connected if and only if the monodromy group acts transitively on the fiber. Relabeling the sheets changes the permutation by conjugation. Thus we have the following.

**Theorem 3.10.** *Let  $X$  be path-connected, locally path-connected, and semilocally simply connected, with basepoint  $x_0$ . Isomorphism classes of connected  $n$ -sheeted covering spaces  $p : \widetilde{X} \rightarrow X$  are equivalent to transitive permutation representations  $\rho : \pi_1(X, x_0) \rightarrow S_n$  up to conjugation by  $S_n$ .*

**Definition 3.11.** *The group of **deck transformations** of a covering space  $p : \widetilde{X} \rightarrow X$  is the group of homeomorphisms  $f : \widetilde{X} \rightarrow \widetilde{X}$  such that  $p \circ f = p$ . The covering space is called **Galois** if the group of deck transformations acts transitively on the fibers of  $p$ .*

The covering spaces  $\mathbb{R} \rightarrow S^1$  and  $S^1 \rightarrow S^1$  given by  $z \mapsto z^n$  are Galois, with deck transformation groups  $\mathbb{Z}$  and  $\mathbb{Z}/n\mathbb{Z}$  respectively. Galois covering spaces are in correspondence with normal subgroups of  $\pi_1(X, x_0)$ , and the deck transformation group is isomorphic to the quotient of  $\pi_1(X, x_0)$  by the corresponding normal subgroup.

**Proposition 3.12.** *Let  $p : (\widetilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$  be a covering space. Let*

$$H = p_*(\pi_1(\widetilde{X}, \tilde{x}_0)) \subseteq \pi_1(X, x_0)$$

*be the stabilizer in  $\pi_1(X, x_0)$  of the chosen lift  $\tilde{x}_0$  of  $x_0$ . Then*

1. *the covering space is Galois if and only if  $H$  is a normal subgroup of  $\pi_1(X, x_0)$ .*
2.  *$\text{Aut}(p) \cong N_{\pi_1(X, x_0)}(H)/H$ , where  $N_{\pi_1(X, x_0)}(H)$  is the normalizer of  $H$  in  $\pi_1(X, x_0)$ .*

*In particular, if  $H$  is normal, then  $\text{Aut}(p) \cong \pi_1(X, x_0)/H$ .*

*Proof.* We need to think about loops in  $X$  which move the chosen lift  $\tilde{x}_0$  to another point of the fiber in a way that comes from a deck transformation. Recall that if  $\tilde{x}_1$  is another point in the fiber, then  $p_*(\pi_1(\tilde{X}, \tilde{x}_1))$  is a conjugate of  $H$  in  $\pi_1(X, x_0)$ , and the conjugating element is given by the projection of any path  $[\gamma]$  from  $\tilde{x}_0$  to  $\tilde{x}_1$ . If there is a deck transformation sending  $\tilde{x}_0$  to  $\tilde{x}_1$ , then  $\tau_*$  induces an isomorphism from  $\pi_1(\tilde{X}, \tilde{x}_0)$  to  $\pi_1(\tilde{X}, \tilde{x}_1)$ . The identity  $p \circ \tau = p$  implies that  $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = p_*(\pi_1(\tilde{X}, \tilde{x}_1))$ , so we see that we must have

$$[\gamma]H[\gamma]^{-1} = H$$

In particular  $[\gamma]$  must lie in the normalizer of  $H$ . The lifting criterion gives the converse, so there is a deck transformation taking  $\tilde{x}_0$  to  $\tilde{x}_1$  if and only if  $H_{\tilde{x}_0} = H_{\tilde{x}_1}$ .

Given  $[\gamma] \in N(H)$  we can construct a deck transformation  $\tau_{[\gamma]}$  as follows. Lift  $\gamma$  to a path starting at  $x_0$ . Since  $[\gamma]$  normalizes  $H$ , the endpoint of this lift is a point  $\tilde{x}_1$  in the fiber such that  $p_*(\pi_1(\tilde{X}, \tilde{x}_1)) = H$ . Hence, by the lifting criterion, there is a deck transformation  $\tau_{[\gamma]}$  taking  $\tilde{x}_0$  to  $\tilde{x}_1$ . The map  $[\gamma] \mapsto \tau_{[\gamma]}$  is a homomorphism from  $N(H)$  to  $\text{Aut}(p)$ . It is clearly surjective because given  $\tau \in \text{Aut}(p)$ , we can take any path from  $\tilde{x}_0$  to  $\tau(\tilde{x}_0)$ , and the projection of this path is a loop in  $X$  whose class lies in  $N(H)$  and maps to  $\tau$ . The kernel of this homomorphism is exactly  $H$  because if  $[\gamma]$  induces the identity deck transformation, then the lift of  $\gamma$  starting at  $\tilde{x}_0$  is a loop, so  $[\gamma] \in H$ .  $\square$

**Exercise 3.13.** Find all 2-sheeted covering spaces of  $S^1 \vee S^1$ .

**Solution 3.14.** The fundamental group of  $S^1 \vee S^1$  is the free group on two generators, say  $a$  and  $b$ . A 2-sheeted covering space is determined by the action of  $\pi_1(S^1 \vee S^1)$  on the fiber of the cover, which is a set of two points. Therefore, we simply have to classify homomorphisms  $F_2 \rightarrow S_2$  with transitive image, up to conjugation by  $S_2$ . There are three such homomorphisms

$$(a, b) \mapsto (1, 1)$$

$$(a, b) \mapsto (0, 1)$$

$$(a, b) \mapsto (1, 0)$$

**Exercise 3.15.** Find all 3-sheeted covering spaces of  $S^1 \vee S^1$ .

**Solution 3.16.** Homomorphisms from  $F_2$  to  $S_3$  are determined by the images of the generators  $a$  and  $b$ . Thus we need to classify pairs of elements of  $S_3$  that generate a transitive subgroup, up to simultaneous conjugation by  $S_3$ . There are two transitive subgroups of  $S_3$ , namely  $S_3$  itself and  $A_3 \cong \mathbb{Z}/3$ .

Consider the cyclic monodromy cases where  $\langle \sigma, \tau \rangle = A_3$ . Since  $A_3 = \{e, (123), (132)\}$  and conjugation by a transposition swaps  $(123)$  and  $(132)$ , the nonzero pairs in  $A_3^2$  modulo this symmetry give 4 cases:

$$(e, (123)), ((123), e), ((123), (123)), ((123), (132))$$

These are normal covers, because the stabilizer of a single sheet, say labeled 1, is the subgroup of  $A_3$  that fixes 1, which is  $H = \{e\}$ , and  $H$  is normal in  $A_3$ .

Now suppose  $\langle \sigma, \tau \rangle = S_3$ . Then there are three more simultaneous-conjugacy classes:

$$((12), (23)), ((12), (123)), ((123), (12))$$

These are genuinely non-normal covers, because the stabilizer of a single sheet, say labeled 1, is the subgroup of  $S_3$  that fixes 1, which is  $H = \{e, (23)\}$ , and  $H$  is not normal in  $S_3$ . Note that a surjective homomorphism  $F_2 \rightarrow S_3$  carries normal subgroups to normal subgroups, and preimages of normal subgroups are normal.

### 3.3 Homology and cohomology

We begin with a discussion of cellular homology. The degree of a map  $f : S^n \rightarrow S^n$  is the integer  $d$  such that  $f_* : H_n(S^n) \rightarrow H_n(S^n)$  is multiplication by  $d$ .

Let  $X$  be a CW complex. Define a chain complex  $C_\bullet(X)$  by letting  $C_n(X)$  be the free abelian group generated by the  $n$ -cells of  $X$ , and differential  $d_n : C_n(X) \rightarrow C_{n-1}(X)$  be given by the degree of the attaching map of each  $n$ -cell.

$$d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta} e_\beta^{n-1}$$

where  $d_{\alpha\beta}$  is the degree of the following map of  $S^{n-1}$ :

$$S^{n-1} \xrightarrow{\text{attaching map}} X^{n-1} \rightarrow X^{n-1}/X^{n-2} \cong \bigvee_\gamma S^{n-1} \rightarrow S^{n-1}$$

where the last map collapses all spheres except the one corresponding to  $e_\beta^{n-1}$ . The differential obeys  $d^2 = 0$  because it is derived from the long exact sequence of the triple  $(X^n, X^{n-1}, X^{n-2})$ .

**Theorem 3.17.** *Let  $(X, A)$  be a CW pair. Then there is a long exact sequence in homology*

$$\cdots \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A) \rightarrow \cdots$$

Moreover the quotient map  $X \rightarrow X/A$  induces an isomorphism

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$$

**Theorem 3.18 (Excision).** *For subspaces  $A, B$  whose interiors cover  $X$ , the inclusion  $(B, B \cap A) \rightarrow (X, A)$  induces an isomorphism in homology*

$$H_n(B, B \cap A) \cong H_n(X, A).$$

In particular, the statement applies to CW complexes  $X$  covered by two subcomplexes  $A$  and  $B$ .

**Example 3.19.** *Consider the genus  $g$  oriented surface  $M_g$ . We can give  $M_g$  a CW structure with one 0-cell,  $2g$  1-cells, and one 2-cell. The cellular chain complex is*

$$0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z}^{2g} \xrightarrow{0} \mathbb{Z} \rightarrow 0$$

The differential  $d_2 : C_2(M_g) \rightarrow C_1(M_g)$  is zero because the coefficient of a 1-cell in  $d_2(e^2)$  is the total exponent of that generator in the attaching word, which is zero because the commutator  $[a_i, b_i]$  has total exponent zero for both  $a_i$  and  $b_i$ . The differential  $d_1 : C_1(M_g) \rightarrow C_0(M_g)$  is zero because each 1-cell has both endpoints at the single 0-cell. Hence the homology groups are

$$\begin{aligned} H_0(M_g) &\cong \mathbb{Z} \\ H_1(M_g) &\cong \mathbb{Z}^{2g} \\ H_2(M_g) &\cong \mathbb{Z} \end{aligned}$$

**Example 3.20.** We consider the homology of  $\mathbb{RP}^n$ , which as a CW complex has one cell in each dimension from 0 to  $n$ . The cellular chain complex is

$$0 \rightarrow \mathbb{Z}e^n \rightarrow \mathbb{Z}e^{n-1} \rightarrow \cdots \rightarrow \mathbb{Z}e^1 \rightarrow \mathbb{Z} \rightarrow 0$$

where the degrees of the differentials are given by studying the attaching maps

$$S^{k-1} \rightarrow \mathbb{RP}^{k-1} \rightarrow \mathbb{RP}^{k-1}/\mathbb{RP}^{k-2} \cong S^{k-1}$$

This map identifies with the antipodal map on  $S^{k-1}$ , which has degree  $(-1)^k$ . This can be seen by embedding  $S^{k-1}$  in  $\mathbb{R}^k$  and noting that the antipodal map is given by multiplication by  $-1$ , which has determinant  $(-1)^k$ . Hence the cellular chain complex is

$$0 \rightarrow e^n \xrightarrow{(-1)^n} \mathbb{Z}e^{n-1} \xrightarrow{(-1)^{n-1}} \cdots \xrightarrow{-1} \mathbb{Z}e^1 \xrightarrow{1} \mathbb{Z} \rightarrow 0$$

The homology groups are

$$\begin{aligned} H_k(\mathbb{RP}^n) &\cong \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}/2\mathbb{Z} & 0 < k < n \text{ odd} \\ 0 & \text{otherwise} \end{cases} \\ H_n(\mathbb{RP}^n) &\cong \begin{cases} \mathbb{Z} & n \text{ odd} \\ 0 & n \text{ even} \end{cases} \end{aligned}$$

**Theorem 3.21** (Mayer–Vietoris). Suppose that  $X = A \cup B$  where the interiors of  $A$  and  $B$  cover  $X$ . Then there is a long exact sequence in homology

$$\cdots \rightarrow H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(X) \rightarrow H_{n-1}(A \cap B) \rightarrow \cdots$$

where the map

$$\begin{aligned} H_n(A \cap B) &\rightarrow H_n(A) \oplus H_n(B) \\ x &\mapsto (i_{A*}(x), -i_{B*}(x)) \end{aligned}$$

is induced by inclusion and

$$\begin{aligned} H_n(A) \oplus H_n(B) &\rightarrow H_n(X) \\ (a, b) &\mapsto a + b \end{aligned}$$



There is an identical Mayer–Vietoris sequence in reduced homology.

**Exercise 3.22.** Compute  $H_i(S^n \setminus X)$  for  $X = S^k \vee S^\ell$  and  $X = S^k \amalg S^\ell$ .

**Solution 3.23.** If  $X = S^k \vee S^\ell$ , then we can write  $U = S^n \setminus S^k$  and  $V = S^n \setminus S^\ell$ . Then  $U \cap V = S^n \setminus X$  and  $U \cup V = S^n \setminus * \cong \mathbb{R}^n$ . The Mayer–Vietoris sequence for the cover  $U \cup V$  of  $S^n \setminus X$  gives

$$\cdots \rightarrow H_i(U \cap V) \rightarrow H_i(U) \oplus H_i(V) \rightarrow H_i(S^n \setminus X) \rightarrow H_{i-1}(U \cap V) \rightarrow \cdots$$

Since  $U \cup V$  is contractible, the map  $H_i(U) \oplus H_i(V) \rightarrow H_i(S^n \setminus X)$  is an isomorphism for  $i \geq 1$ . In degree  $i = 0$ ,  $S^n \setminus X$  is path connected, so  $H_0(S^n \setminus X) \cong \mathbb{Z}$ . We can identify  $H_i(U) = H_i(S^n \setminus S^k) \cong H_i(S^{n-k-1})$  and  $H_i(V) = H_i(S^n \setminus S^\ell) \cong H_i(S^{n-\ell-1})$  because  $S^n \setminus S^k$  is homotopy equivalent to  $S^{n-k-1}$ .

Put  $S^k$  as the equator  $\{(x, 0)\} \subset S^n \subset \mathbb{R}^{k+1} \oplus \mathbb{R}^{n-k}$ . In the complement, the normal coordinate  $y$  is nonzero. The map  $(x, y) \mapsto (0, y/|y|)$  is a deformation retraction, obtained by shrinking the  $x$ -coordinate to zero and renormalizing. Thus the complement retracts onto the sphere of normal directions  $S^{n-k-1}$ .

Hence we have that  $H_i(S^n \setminus X)$  is  $\mathbb{Z}$  in degrees 0,  $n - k - 1$ , and  $n - \ell - 1$ , and is zero otherwise.

Let  $G$  be an abelian group. The cohomology with coefficients in  $G$  is defined by dualizing the chain complex  $C_\bullet(X)$  which defines homology. The failure of the homology of the two chain complexes to agree is measured by the functor  $\text{Ext}_{\mathbb{Z}}^1(-, G)$ .

**Theorem 3.24** (Universal coefficient theorem for cohomology). *Let  $X$  be a topological space and let  $G$  be an abelian group. Then there is a short exact sequence*

$$0 \rightarrow \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X), G) \rightarrow H^n(X; G) \rightarrow \text{Hom}_{\mathbb{Z}}(H_n(X), G) \rightarrow 0$$

*which splits, but not naturally.*

The second map  $h$  in the sequence is given by evaluating a cohomology class on homology classes, i.e.  $[\alpha] \mapsto (x \mapsto \alpha(x))$ . The first map is given by the following construction. The kernel of  $h$  is those cohomology classes  $[\alpha]$  whose cocycle representative  $\alpha \in C^n(X; G)$  has the following property. Restrict  $\alpha : Z_n(X) \rightarrow G$ . Since  $\alpha$  is a cocycle,  $\alpha$  vanishes on  $B_n(X)$ , so  $\alpha$  induces a homomorphism from  $Z_n(X)/B_n(X) = H_n(X)$  to  $G$ . The kernel of  $h$  are those  $[\alpha]$  so that the induced map  $\alpha : H_n(X) \rightarrow G$  is zero. Now we can identify  $\text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X), G)$  with the kernel of  $h$ . Being in the kernel of  $h$  means that  $\alpha : Z_n(X) \rightarrow G$  is 0. There is a short exact sequence

$$0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$$

Since  $\varphi$  vanishes on  $Z_n$ , it factors uniquely through the quotient  $C_n/Z_n \cong B_{n-1}$ . So there is a unique homomorphism  $f : B_{n-1} \rightarrow G$  such that  $\varphi = f \circ d_n$ . We have to divide

by coboundaries, which can be identified with  $\text{im}(\text{Hom}_{\mathbb{Z}}(C_{n-1}, G) \rightarrow \text{Hom}_{\mathbb{Z}}(B_{n-1}, G))$ . Altogether we have

$$\ker h \cong \frac{\text{Hom}(B_{n-1}, G)}{\text{im}(\text{Hom}(Z_{n-1}, G) \rightarrow \text{Hom}(B_{n-1}, G))} \cong \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X), G)$$

The last identification follows from applying  $\text{Hom}(-, G)$  to the short exact sequence

$$0 \rightarrow B_{n-1} \rightarrow Z_{n-1} \rightarrow H_{n-1} \rightarrow 0$$

**Proposition 3.25** (Facts about  $\text{Ext}_{\mathbb{Z}}^1$ ). *Let  $A$  and  $B$  be abelian groups. Then*

1.  $\text{Ext}_{\mathbb{Z}}^1(A, B) = 0$  if  $A$  is free or  $B$  is injective.
2.  $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, B) \cong B/nB$
3. In particular,  $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ .

### 3.3.1 Cup product

We are now prepared to introduce an important structure on cohomology, namely the cup product. It is defined on cochains by the formula

$$(\alpha \cup \beta)(\sigma) = \alpha(\sigma|_{[0, \dots, p]}) \cdot \beta(\sigma|_{[p, \dots, p+q]})$$

where  $\sigma$  is a singular simplex of dimension  $p + q$ , and  $\alpha$  and  $\beta$  are cochains of degree  $p$  and  $q$  respectively. The cup product satisfies the relation

$$\delta(\alpha \cup \beta) = \delta\alpha \cup \beta + (-1)^p \alpha \cup \delta\beta \quad (17)$$

This formula implies that the cup product of two cocycles is a cocycle, and the cup product of a coboundary with any cochain is a coboundary. Hence the cup product descends to cohomology, giving  $H^\bullet(X; G)$  the structure of a graded ring. Moreover the cup product respects pullback, in the sense that if  $f : X \rightarrow Y$  is a continuous map, then  $f^*(\alpha \cup \beta) = f^*(\alpha) \cup f^*(\beta)$ .

**Proposition 3.26.** *The cup product on cohomology is graded commutative, in the sense that if  $\alpha \in H^p(X; G)$  and  $\beta \in H^q(X; G)$ , then  $\alpha \cup \beta = (-1)^{pq} \beta \cup \alpha$ .*

The external cup product map

$$\begin{aligned} H^*(X; R) \times H^*(Y; R) &\rightarrow H^*(X \times Y; R) \\ (\alpha, \beta) &\mapsto \pi_X^*(\alpha) \cup \pi_Y^*(\beta) \end{aligned}$$

is bilinear and therefore induces a homomorphism of  $R$ -modules

$$H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$$

If we define the multiplication on the left hand side by the formula

$$(\alpha_1 \otimes \beta_1)(\alpha_2 \otimes \beta_2) = (-1)^{|\beta_1||\alpha_2|}(\alpha_1 \cup \alpha_2) \otimes (\beta_1 \cup \beta_2)$$

then the external cup product map is a homomorphism of graded rings. The Künneth formula gives a criterion for when this map is an isomorphism.

**Theorem 3.27** (Künneth formula). *The external cup product map*

$$H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$$

*is an isomorphism if  $X$  and  $Y$  are CW complexes and  $H^*(X; R)$  is a finitely generated free  $R$ -module in each degree.*

In fact, the CW approximation theorem implies that we can remove the CW hypothesis. However the freeness is indispensable.

### 3.4 Poincare duality

Let  $M$  be an  $n$ -manifold. A local orientation of  $M$  with coefficients in a ring  $R$  at a point  $x \in M$  is a choice of generator of the  $R$ -module  $H_n(M, M \setminus \{x\}; R) \cong H_{n-1}(S^{n-1}; R)$ . An orientation of  $M$  with coefficients in  $R$  is an assignment  $x \mapsto o_x$  of a local orientation to each point  $x \in M$  such that for every  $x \in M$ , there is an open neighborhood  $U$  of  $x$  such that the local orientations  $o_y$  for  $y \in U$  are compatible in the following sense. For every  $y, z \in U$ , the local orientations  $o_y$  and  $o_z$  are compatible if they induce the same generator of  $H_n(U, U \setminus \{y, z\}; R)$ .

In other words, an orientation of  $M$  is a section of the orientation sheaf  $\mathcal{O}_M$  of  $M$ , which is the locally constant sheaf on  $M$  whose stalk at  $x$  is  $H_n(M, M \setminus \{x\}; R)$ . If  $R = \mathbb{Z}$  then the total space of this sheaf is a double sheeted cover of  $M$ ,

**Proposition 3.28.** *Let  $R = \mathbb{Z}$ .  $M$  is orientable if and only if the orientation sheaf  $\mathcal{O}_M$  is trivial, i.e. admits a global section, if and only if  $\tilde{M} \rightarrow M$  has two connected components. In particular, this is the case if  $M$  is simply connected or  $\pi_1(M)$  has no subgroup of index 2.*

In general, one has the canonical map

$$\begin{aligned} H_n(M; R) &\rightarrow \Gamma(M, \mathcal{O}_M) \\ [\alpha] &\mapsto (x \mapsto \alpha(o_x)) \end{aligned}$$

The following theorem asserts that this map is an isomorphism when  $M$  is closed. If moreover  $M$  is connected, then a section of a local system is determined by its value at a single point, provided that value is invariant under monodromy. If  $M$  is  $R$ -orientable, then the monodromy is trivial and any nonzero value at a single point gives a global section. In this case, we can identify  $\Gamma(M, \mathcal{O}_M)$  with  $R$ , and the isomorphism  $H^n(M; R) \cong \Gamma(M, \mathcal{O}_M)$  gives an isomorphism  $H^n(M; R) \cong R$ . If  $M$  is not  $R$ -orientable, then the monodromy is nontrivial and acts by multiplication by  $-1$  on the fiber of  $\mathcal{O}_M$ . In this case, the invariant sections of  $\mathcal{O}_M$  are those  $r \in R$  such that  $-r = r$ , the 2-torsion elements of  $R$ . In summary we have the following theorem:

**Theorem 3.29.** *Let  $M$  be a connected closed  $n$  manifold. If  $M$  is  $R$ -orientable, then  $H_n(M; R) \cong R$  and if  $M$  is not  $R$ -orientable, then  $H_n(M; R) \cong \{r \in R : 2r = 0\}$ .*

The Poincaré duality theorem asserts for a closed connected  $R$ -orientable manifold  $M$  that there is a  $H^*(M; R)$ -module structure on  $H_*(M; R)$  which induces isomorphisms  $H^k(M; R) \cong H_{n-k}(M; R)$  for all  $k$ . The cap product  $C^p(X; G) \times C_q(X; G) \rightarrow C_{q-p}(X; G)$  is defined on cochains by the formula

$$\alpha \cap \sigma = \alpha(\sigma|_{[v_0, \dots, v_p]}) \cdot \sigma|_{[v_p, \dots, v_q]}$$

where  $\sigma$  is a singular simplex of dimension  $q$  and  $\alpha$  is a cochain of degree  $p$ . The cap product descends to homology and cohomology, thereby inducing

$$\cap : H^p(X; G) \times H_q(X; G) \rightarrow H_{q-p}(X; G)$$

functorial in the sense that

$$f_*(\alpha \cap x) = f^*(\alpha) \cap f_*(x)$$

for any continuous map  $f : X \rightarrow Y$ .

**Theorem 3.30** (Poincaré duality). *Let  $M$  be a closed connected  $R$ -orientable  $n$ -manifold. Then capping with a generator of  $H_n(M; R)$  gives an isomorphism of  $H^*(M; R)$ -modules  $H^k(M; R) \cong H_{n-k}(M; R)$  for all  $k$ .*

Let  $M$  be connected. There is a natural evaluation pairing

$$H^k(M; R) \times H_k(M; R) \rightarrow H_0(M; R) \cong R$$

given by evaluating a cohomology class on a homology class. Note that  $H^0(M; R) \cong R$  is generated by the class of the constant function 1. If  $M$  is a closed oriented  $n$ -manifold and  $B \subset M$  is an oriented codimension- $k$  submanifold, then  $\text{PD}([B]) \in H^k(X; \mathbb{Z})$  is characterized by the property that for every  $k$ -cycle  $A \subset X$ ,

$$\langle \text{PD}([B]), [A] \rangle = [B] \cdot [A]$$

where the right side is the signed intersection number, assuming  $A$  and  $B$  are transverse.

**Example 3.31.** *Recall that the additive cohomology of  $\mathbb{RP}^n$  with  $\mathbb{Z}/2\mathbb{Z}$  coefficients is  $\mathbb{Z}/2\mathbb{Z}$  in each degree from 0 to  $n$ . Let  $x \in H^1(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z})$  be the generator. We will show that  $x^k$  is the generator of  $H^k(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z})$  for each  $k$ . Let  $H \subset \mathbb{RP}^n$  be a projective hyperplane, so  $H \cong \mathbb{RP}^{n-1}$ . The class  $x \in H^1(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z})$  may be identified with the Poincaré dual of  $H$ .*

*Let  $x^k \in H^k(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z})$  be the  $k$ -fold cup product of  $x$  with itself. Then  $x^k = m\eta_k$  where  $\eta_k$  is the generator of  $H^k(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z})$  and  $m \in \mathbb{Z}/2\mathbb{Z}$ . We want to show that  $m = 1$ . We can calculate  $m$  using the intersection pairing. We have*

$$\begin{aligned} m &= \langle x^k, [H] \rangle = \# \{ H \cap H \cap \cdots \cap H \} \pmod{2} \\ &= 1 \end{aligned}$$

**Example 3.32.** The cup product allows us to distinguish  $\mathbb{CP}^2$  from  $S^2 \vee S^4$ . The cohomology groups of both spaces are  $\mathbb{Z}$  in degrees 0, 2, and 4, and are zero otherwise. However

$$\begin{aligned} H^*(\mathbb{CP}^2) &\cong \mathbb{Z}[x]/(x^3) \\ H^*(S^2 \vee S^4) &\cong \mathbb{Z}[y, z]/(y^2, yz, z^2) \end{aligned}$$

where  $\deg y = 2$  and  $\deg z = 4$ . The key difference is that  $y^2 = 0$  because  $y^2$  is a degree 4 class that is supported on  $S^2$ , of which there are none.

**Theorem 3.33.** Let  $X$  be a CW complex, and let  $X^n$  denote its  $n$ -skeleton. Then for any coefficient group  $A$ ,

$$H_k(X; A) \cong \varinjlim_n H_k(X^n; A).$$

Moreover, there is a natural Milnor exact sequence

$$0 \rightarrow \varprojlim_n^1 H^{k-1}(X^n; A) \rightarrow H^k(X; A) \rightarrow \varprojlim_n H^k(X^n; A) \rightarrow 0.$$

For the skeletal filtration of a CW complex, the inverse system  $\{H^k(X^n; A)\}_n$  is eventually constant, so  $\varprojlim_n^1 = 0$ . Hence

$$H^k(X; A) \cong \varprojlim_n H^k(X^n; A).$$

*Proof.* The key reason for stabilization is that the inclusion  $X^n \hookrightarrow X$  induces

$$H^k(X; A) \xrightarrow{\cong} H^k(X^n; A)$$

whenever  $n > k$ . This is because the relative cohomology group  $H^k(X, X^n; A)$  vanishes for  $n > k$ . This is because  $X/X^n$  has no cells in dimensions  $\leq n$ , so  $H^k(X/X^n; A) = 0$  for  $n > k$ .

For fixed  $k$ , once the skeleton is past dimension  $k$ , the cohomology group no longer changes. Thus the inverse system is eventually constant, and the possible  $\varprojlim_n^1$  obstruction disappears.  $\square$

**Example 3.34.** Consider the infinite-dimensional complex projective space  $\mathbb{CP}^\infty$ . It has the CW structure with one cell in each even dimension. The cohomology groups are  $\mathbb{Z}$  in even degrees and zero in odd degrees. This can be seen directly from the cellular chain complex, or from the skeleton filtration and the Milnor exact sequence.

To compute the cup product, let  $x \in H^2(\mathbb{CP}^\infty)$  be the generator. We will show that  $x^k$  is the generator of  $H^{2k}(\mathbb{CP}^\infty)$  for each  $k$ . Let  $N > k$  and pullback  $x$  to  $H^2(\mathbb{CP}^N)$ . Pullback respects cup product, and we have already shown that  $x^k$  is the generator of  $H^{2k}(\mathbb{CP}^N)$ . Since  $N > k$ , the pullback  $H^{2k}(\mathbb{CP}^\infty) \rightarrow H^{2k}(\mathbb{CP}^N)$  is an isomorphism, so  $x^k$  is the generator of  $H^{2k}(\mathbb{CP}^\infty)$  as well.

### 3.5 Higher homotopy groups

**Definition 3.35.** *The  $n$ -th homotopy group of a pointed space  $(X, x)$  is the set of homotopy classes of maps  $S^n \rightarrow X$  sending the base point to  $x$ . We denote it by  $\pi_n(X, x)$  or simply  $\pi_n(X)$  if  $X$  is path connected.*

The group law is given by writing  $S^n \rightarrow S^n \vee S^n$  as the pinch map and then composing with the wedge of two maps  $S^n \rightarrow X$ .  $\pi_n(X, x)$  can also be modeled by the set of homotopy classes of maps  $I^n \rightarrow X$  sending the boundary to  $x$ . In this model, the group law can be defined by concatenating along the 1st coordinate. The homotopy groups are abelian for  $n \geq 2$ .

There is an action of  $\pi_1(X)$  on  $\pi_n(X)$  for  $n \geq 2$  by dragging the base point around loops. We say that  $X$  is  **$n$ -simple** if the action of  $\pi_1(X)$  on  $\pi_n(X)$  is trivial, and  $X$  is **simple** if it is  $n$ -simple for all  $n$ .

**Proposition 3.36.** *A covering space projection  $p : \tilde{X} \rightarrow X$  induces an isomorphism on  $\pi_n$  for  $n \geq 2$ .*

*Proof.* The map is surjective because we can lift maps  $S^n \rightarrow X$  to  $\tilde{X}$  because  $S^n$  is simply connected. The map is injective because if a map  $S^n \rightarrow \tilde{X}$  becomes nullhomotopic in  $X$ , then we can lift the nullhomotopy to  $\tilde{X}$  and get a nullhomotopy there.  $\square$

Moreover, if  $\tilde{X}$  is the universal cover of  $X$ , then  $\pi_1(X)$  acts on  $\tilde{X}$  by deck transformations and this action induces the action of  $\pi_1(X)$  on  $\pi_n(\tilde{X}) \cong \pi_n(X)$ .

**Proposition 3.37.** *A product of path connected spaces  $X_\alpha$  has homotopy groups given by the product of the homotopy groups of the factors, i.e.  $\pi_n(\prod X_\alpha) \cong \prod \pi_n(X_\alpha)$ .*

*Proof.* This follows immediately from the universal property of products.  $\square$

Another useful notion is the relative homotopy groups  $\pi_n(X, A, x)$  for a pair of spaces  $A \subset X$ .

**Definition 3.38.** *Let  $x \in A \subset X$  be a pointed pair of spaces. The  $n$ -th relative homotopy group  $\pi_n(X, A, x)$  is the set of homotopy classes of maps  $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x)$  where*

$$J^{n-1} = \overline{\partial I^n \setminus (I^{n-1} \times \{1\})}$$

*with homotopies through such maps. Equivalently, it is homotopy classes of maps*

$$(D^n, S^{n-1}, *) \rightarrow (X, A, x)$$

*modulo homotopies through such maps.*

**Theorem 3.39.** *There is a long exact sequence of homotopy groups*

$$\cdots \rightarrow \pi_n(A) \rightarrow \pi_n(X) \rightarrow \pi_n(X, A) \rightarrow \pi_{n-1}(A) \rightarrow \cdots$$

coming from the inclusion of pairs. The boundary map is the restriction of a map  $D^n \rightarrow X$  to the boundary  $S^{n-1} \rightarrow A$ .

**Theorem 3.40** (Whitehead's theorem). *A map  $f : X \rightarrow Y$  between CW complexes is a homotopy equivalence if and only if it induces isomorphisms on all homotopy groups.*

Note that two CW complexes may have the same homotopy groups but not be homotopy equivalent. In particular, having a map inducing isomorphisms on homotopy groups is a much stronger condition than just having isomorphic homotopy groups. One situation where the latter implies the former is when all homotopy groups vanish, in which case the inclusion of a point is a homotopy equivalence.

**Example 3.41.** *Consider  $X = \mathbb{RP}^2$  and  $Y = S^2 \times \mathbb{RP}^\infty$ . These both have fundamental group  $\mathbb{Z}/2\mathbb{Z}$ . Their higher homotopy groups are isomorphic since their universal covers  $S^2$  and  $S^2 \times S^\infty$  are homotopy equivalent,  $S^\infty$  being contractible. But  $\mathbb{RP}^2$  and  $S^2 \times \mathbb{RP}^\infty$  are not homotopy equivalent since their homology groups are vastly different,  $S^2 \times \mathbb{RP}^\infty$  having nonvanishing homology in infinitely many dimensions since it retracts onto  $\mathbb{RP}^\infty$ . Recall that retracts induce injections on cohomology.*

**Example 3.42.** *Another pair of CW complexes that are not homotopy equivalent but have isomorphic homotopy groups is  $S^2$  and  $S^3 \times \mathbb{CP}^\infty$ . Indeed,  $\mathbb{CP}^\infty$  is an Eilenberg–MacLane space  $K(\mathbb{Z}, 2)$ , so*

$$\pi_n(\mathbb{CP}^\infty) = \begin{cases} \mathbb{Z}, & n = 2, \\ 0, & n \neq 2. \end{cases}$$

Therefore

$$\pi_n(S^3 \times \mathbb{CP}^\infty) \cong \pi_n(S^3) \oplus \pi_n(\mathbb{CP}^\infty)$$

for all  $n \geq 1$ , and hence

$$\pi_n(S^3 \times \mathbb{CP}^\infty) \cong \begin{cases} \mathbb{Z}, & n = 2, \\ \pi_n(S^3), & n \geq 3. \end{cases}$$

Now consider the Hopf fibration

$$S^1 \longrightarrow S^3 \longrightarrow S^2.$$

Its long exact sequence in homotopy implies that

$$\pi_n(S^3) \cong \pi_n(S^2) \quad \text{for all } n \geq 3,$$

since  $\pi_n(S^1) = 0$  for  $n \geq 2$ . Also,

$$\pi_2(S^2) \cong \mathbb{Z}, \quad \pi_1(S^2) = 0.$$

Since  $S^3 \times \mathbb{CP}^\infty$  is simply connected as well, it follows that

$$\pi_n(S^3 \times \mathbb{CP}^\infty) \cong \pi_n(S^2) \quad \text{for all } n \geq 1.$$

However, these spaces are not homotopy equivalent. For example,

$$H_4(S^2) = 0,$$

while

$$H_4(S^3 \times \mathbb{C}P^\infty) \cong H_4(\mathbb{C}P^\infty) \cong \mathbb{Z},$$

so their homology groups are not isomorphic. Therefore  $S^2$  and  $S^3 \times \mathbb{C}P^\infty$  cannot be homotopy equivalent.

A map  $f : X \rightarrow Y$  is **cellular** if it sends the  $n$ -skeleton of  $X$  to the  $n$ -skeleton of  $Y$  for all  $n$ .

**Theorem 3.43.** *Any map  $f : X \rightarrow Y$  between CW complexes is homotopic to a cellular map. If  $f$  is already cellular on a subcomplex  $A \subset X$ , then the homotopy can be taken to be relative to  $A$ .*

It immediately follows that  $\pi_k(S^n) = 0$  for  $k < n$  because the  $k$ -skeleton of  $S^n$  is a point.

**Definition 3.44.** *A map  $f : X \rightarrow Y$  is a weak homotopy equivalence if it induces isomorphisms on all homotopy groups.*

Whitehead's theorem reformulated says that a weak homotopy equivalence between CW complexes is a homotopy equivalence. However, a weak homotopy equivalence between general spaces need not be a homotopy equivalence. For example, there exist spaces which have trivial homotopy groups but are not contractible. For these spaces, the constant map to a point is a weak homotopy equivalence but not a homotopy equivalence.

**Example 3.45.** *Consider the quasi-circle  $Y$  which is a closed subspace of  $\mathbb{R}^2$  obtained by taking a portion of the graph of  $y = \sin(1/x)$ , the segment  $[-1, 1]$  in the  $y$ -axis, and connecting them with a curve. This space comes with a map  $p : Y \rightarrow S^1$  which collapses the sine curve and the interval to a point. Then this space has trivial homotopy groups. Any map from  $S^n$  to this space is nullhomotopic because the image is compact, so it must meet the sine-curve in finitely many oscillations. This will clearly be nullhomotopic.*

*However, the quasi-circle is not contractible because if it were, then  $p$  would be nullhomotopic and we could lift  $p$  to the universal cover  $\mathbb{R}$  of  $S^1$ . The lifted map  $\tilde{p} : Y \rightarrow \mathbb{R}$  in fact factors through  $Y/I = S^1$  because the quotient map collapses  $I$  to a point. Then this gives a section of the universal cover  $\mathbb{R} \rightarrow S^1$ , which is impossible.*

**Definition 3.46.** *Let  $Y$  be a topological space. A CW-approximation of  $Y$  is a CW complex  $X$  together with a map  $f : X \rightarrow Y$  such that for every CW complex  $Z$ , the induced map  $f_* : [Z, X] \rightarrow [Z, Y]$  is a bijection, where  $[-, -]$  denotes the set of homotopy classes of maps.*

Taking  $Z = S^n$  for all  $n$  shows that a CW approximation is a weak homotopy equivalence.



Conversely, if  $f : X \rightarrow Y$  is a weak homotopy equivalence and  $X$  is a CW complex, then  $f$  is a CW approximation by Whitehead's theorem.

**Theorem 3.47.** *Every topological space  $Y$  admits a CW approximation  $X \rightarrow Y$ . If  $X$  is path connected,  $X$  can be chosen to have a single 0 cell, with all other cells attached by basepoint-preserving maps.*

There is also a relative version of CW approximation.

**Theorem 3.48** (Relative CW approximation). *Let  $(X, A)$  be a pair of spaces with  $A$  a CW complex (or more generally a subspace). Then there exists a relative CW complex  $(Z, A)$  and a map of pairs*

$$f : (Z, A) \longrightarrow (X, A)$$

*such that:*

1.  $f|_A = \text{id}_A$ ,
2.  $f$  is a weak homotopy equivalence of pairs, i.e.

$$\pi_k(Z, A) \xrightarrow{\cong} \pi_k(X, A) \quad \text{for all } k.$$

*If  $(X, A)$  is  $n$ -connected, i.e.*

$$\pi_k(X, A) = 0 \quad \text{for all } k \leq n,$$

*then there exists a relative CW complex  $(Z, A)$  and a weak equivalence of pairs*

$$(Z, A) \xrightarrow{\cong} (X, A)$$

*such that  $(Z, A)$  is obtained from  $A$  by attaching no cells in dimensions  $\leq n$ . Equivalently, the first relative cells occur in dimension  $n + 1$ .*

**Theorem 3.49** (Homotopy excision). *Let  $X$  and  $Y$  be subcomplexes of a CW complex  $Z = X \cup Y$ , and let*

$$A = X \cap Y.$$

*Assume that the pairs  $(X, A)$  and  $(Y, A)$  are  $m$ -connected and  $n$ -connected respectively. Then inclusion of pairs induces a map*

$$\pi_k(X, A) \longrightarrow \pi_k(X \cup Y, Y).$$

*The map is an isomorphism for  $k < m + n$  and a surjection for  $k = m + n$ .*

**Corollary 3.50.** *Let  $Y = CA$  be the cone on  $A$  as in the previous setup. Note that the long exact sequence of the pair  $(Y, A)$  shows that  $(Y, A)$  is  $n + 1$ -connected if  $A$  is  $n$ -connected. Then  $X \cup_A CA = X/A$  and since  $CA$  is contractible, we get*

$$\pi_k(X \cup_A CA, CA) \cong \pi_k(X/A)$$

For  $(X, A)$   $m$ -connected and  $A$   $(n-1)$ -connected, homotopy excision gives a comparison map

$$\pi_k(X, A) \rightarrow \pi_k(X/A)$$

which is an isomorphism for  $k < m + n$  and a surjection for  $k = m + n$ .

**Corollary 3.51** (Freudenthal suspension theorem). *Let  $Z$  be an  $n$ -connected space. Then the suspension map  $\pi_k(Z) \rightarrow \pi_{k+1}(\Sigma Z)$  is an isomorphism for  $k < 2n$  and a surjection for  $k = 2n$ .*

*Proof.* Write the suspension

$$\Sigma Z = CZ \cup_X CZ$$

as the union of two cones on  $Z$  glued along  $Z$ . Now apply homotopy excision to  $X = CZ$  and  $Y = CZ$ . Then  $A = Z$  and the pairs  $(X, A)$  and  $(Y, A)$  are  $n+1$ -connected by the long exact sequence in homotopy.  $\square$

**Theorem 3.52** (Hurewicz theorem). *Let  $X$  be a path connected space. For each  $n$  there is a map*

$$\pi_n(X) \rightarrow H_n(X)$$

*which takes  $[f : S^n \rightarrow X]$  to  $f_*([S^n])$ . If  $X$  is  $(n-1)$ -connected, then this map is an isomorphism in degree  $n$ , zero in degree less than  $n$ , and a surjection in degree  $n+1$ .*

The Hurewicz theorem recovers the Hopf theorem from differential topology which says that if  $M$  is an  $n$ -dimensional compact connected oriented manifold and  $f, g$  are two continuous maps  $M \rightarrow S^n$ , then the degree of  $f$  equals the degree of  $g$  if and only if  $f$  and  $g$  are homotopic.

This result can be established traditionally by understanding the degree of a smooth map as the signed count of the preimages of a regular value, where a preimage has a positive sign if the derivative is orientation preserving and a negative sign if the derivative is orientation reversing.

**Corollary 3.53.**  $\pi_n(S^n) \cong \mathbb{Z}$  for all  $n$ .

*Proof.* The Hurewicz map  $\pi_n(S^n) \rightarrow H_n(S^n)$  is an isomorphism because  $S^n$  is  $(n-1)$ -connected. Since  $H_n(S^n) \cong \mathbb{Z}$ , we get  $\pi_n(S^n) \cong \mathbb{Z}$  as well.  $\square$

**Example 3.54.** *The Hopf fibration  $S^1 \rightarrow S^3 \rightarrow S^2$  can be constructed as follows. We can identify  $S^3$  with the unit quaternions, and  $S^2$  with the unit imaginary quaternions. Then the map  $S^3 \rightarrow S^2$  is given by  $q \mapsto qi q^{-1}$ . The fiber over  $i$  is the subgroup of unit quaternions that commute with  $i$ , which is precisely the unit complex numbers, i.e.  $S^1$ .*

The long exact sequence in homotopy associated to this fibration gives  $\pi_n(S^3) \cong \pi_n(S^2)$  for all  $n \geq 3$ .

In particular  $\pi_3(S^2) \cong \mathbb{Z}$  is generated by the Hopf map. An application of the Freudenthal suspension theorem shows that  $\pi_3(S^2) \rightarrow \pi_4(S^3)$  is a surjection, and in fact indeed  $\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$  is generated by the suspension of the Hopf map.

**Example 3.55.** We can calculate all of the homotopy groups of  $\mathbb{CP}^\infty$ . The Hopf fibrations  $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$  give rise to the infinite Hopf fibration  $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ . Since  $S^\infty$  is contractible, the long exact sequence in homotopy associated to this fibration gives  $\pi_n(\mathbb{CP}^\infty) = 0$  for  $n \geq 3$  and  $\pi_2(\mathbb{CP}^\infty) \cong \pi_1(S^1) \cong \mathbb{Z}$ . Clearly  $\pi_1(\mathbb{CP}^\infty) = 0$  as well, so we have computed all of the homotopy groups of  $\mathbb{CP}^\infty$ .

The previous computation shows that  $\mathbb{CP}^\infty$  is an Eilenberg–MacLane space  $K(\mathbb{Z}, 2)$ .

**Definition 3.56.** Let  $G$  be a group and  $n \geq 1$ . An Eilenberg–MacLane space of type  $K(G, n)$  is a path connected space  $X$  such that  $\pi_n(X) \cong G$  and  $\pi_k(X) = 0$  for all  $k \neq n$ . For  $n \geq 2$ , the group  $G$  must be abelian since  $\pi_n(X)$  is abelian for  $n \geq 2$ .

**Definition 3.57.** An  $H$ -space is a space  $X$  and a point  $e \in X$  together with a continuous map  $\mu : X \times X \rightarrow X$  such that  $\mu(x, -)$  and  $\mu(-, x)$  are homotopic to  $x$  through  $(X, e)$ -based homotopies for all  $x \in X$ .

**Theorem 3.58.** Eilenberg–MacLane spaces of type  $K(G, n)$  exist for all groups  $G$  and all  $n \geq 1$ .

*Proof.* We first discuss existence. For  $n = 1$ , let  $G$  be any group. Choose a connected CW complex  $X$  with

$$\pi_1(X) \cong G.$$

For example, take a wedge of circles with one circle for each generator of  $G$ , and attach 2-cells realizing the relations. This constructs a connected CW complex with fundamental group  $G$ .

The higher homotopy groups of  $X$  may be nonzero. We kill them inductively by attaching cells. Suppose we have constructed a CW complex  $X_r$  with

$$\pi_1(X_r) \cong G$$

and such that  $\pi_k(X_r) = 0$  for  $2 \leq k < r$ . To kill  $\pi_r(X_r)$ , attach  $(r + 1)$ -cells along representatives

$$S^r \rightarrow X_r$$

of elements of  $\pi_r(X_r)$ . Attaching an  $(r + 1)$ -cell along a map  $S^r \rightarrow X_r$  makes that class nullhomotopic. This is because if  $X' = X_r \cup_f D^{r+1}$  where  $f : S^r \rightarrow X_r$  is the attaching map, then  $f : S^r \rightarrow X'$  extends across the disk  $D^{r+1}$ , so  $f$  becomes nullhomotopic in  $X'$ , i.e.  $i_*(f) = 0$ .

Attaching enough such cells kills all of  $\pi_r(X_r)$  without changing  $\pi_1$  or the already-vanishing lower homotopy groups. This is because the relative CW complex  $(X', X_r)$  has no cells in dimensions  $\leq r$ . Continuing inductively gives a CW complex  $K(G, 1)$  with

$$\pi_1(K(G, 1)) \cong G, \quad \pi_k(K(G, 1)) = 0 \quad \text{for } k \geq 2.$$

For  $n \geq 2$ , the group  $G$  must be abelian. One starts with a CW complex whose only first nonzero homotopy group is  $G$  in degree  $n$ . For example, if  $G$  is free abelian, take a product or wedge of  $n$ -spheres and then arrange the desired group in  $\pi_n$ . For a general abelian group, one can build a Moore space  $M(G, n)$  satisfying

$$\widetilde{H}_k(M(G, n); \mathbb{Z}) = 0 \quad \text{for } k \neq n, \quad \widetilde{H}_n(M(G, n); \mathbb{Z}) \cong G.$$

By the Hurewicz theorem, if  $M(G, n)$  is chosen  $(n-1)$ -connected, then

$$\pi_n(M(G, n)) \cong H_n(M(G, n)) \cong G.$$

The higher homotopy groups may again be nonzero. Attach cells of dimension  $> n+1$  inductively to kill

$$\pi_{n+1}, \pi_{n+2}, \dots$$

without changing  $\pi_n$ . This produces a CW complex  $K(G, n)$  with

$$\pi_n(K(G, n)) \cong G, \quad \pi_k(K(G, n)) = 0 \quad \text{for } k \neq n \quad \square$$

**Theorem 3.59.**  $K(G, n)$  represents cohomology in the sense that for any CW complex  $X$ ,

$$[X, K(G, n)] \cong H^n(X; G)$$

$K(G, n)$  is unique up to homotopy equivalence. The addition law on cohomology gives  $K(G, n)$  the structure of an  $H$ -space.

*Proof.* Since  $K(G, n)$  has no homotopy below degree  $n$ , any map on  $X^{n-1}$  to  $K(G, n)$  is nullhomotopic. To extend over an  $n$ -cell  $e^n$  having fixed the boundary map  $S^{n-1} \rightarrow K(G, n)$ , is to choose a map  $S^n \rightarrow K(G, n)$ , which is the same as choosing an element of  $\pi_n(K(G, n))$ . Having fixed the boundary  $g : S^{n-1} \rightarrow K(G, n)$ , two extensions  $f, f' : D^n \rightarrow K(G, n)$  determine a map  $S^n \rightarrow K(G, n)$  by gluing  $f$  and  $f'$  along the boundary. This gives an element  $[f - f'] \in \pi_n(K(G, n)) \cong G$ . So choices of extensions form a torsor for  $\pi_n(Y)$ .

It remains to explain uniqueness. Let  $X$  and  $Y$  be two  $K(G, n)$  spaces. By obstruction theory, maps from a CW complex into a  $K(G, n)$  are classified by degree- $n$  cohomology:

$$[X, K(G, n)] \cong H^n(X; G).$$

Equivalently, a map  $f : X \rightarrow K(G, n)$  is determined up to homotopy by the pullback of the universal class

$$u \in H^n(K(G, n); G).$$

Now suppose  $X$  and  $Y$  are both  $K(G, n)$  spaces. Choose an isomorphism

$$\varphi : \pi_n(X) \xrightarrow{\cong} \pi_n(Y).$$

Since  $Y$  is a  $K(G, n)$ , choosing a map  $X \rightarrow Y$  is the same as choosing a class in  $H^n(X; \pi_n(Y))$ . On the other hand, since  $X$  is also a  $K(G, n)$ , the Hurewicz theorem and universal coefficients identify the relevant degree- $n$  cohomology with

$$H^n(X; \pi_n(Y)) \cong \text{Hom}(\pi_n(X), \pi_n(Y)).$$

Under this identification, choose the class corresponding to  $\varphi$ . Obstruction theory gives a map  $f : X \rightarrow Y$  whose induced map on  $\pi_n$  is precisely  $\varphi$ .

Since  $X$  and  $Y$  have no other nonzero homotopy groups,  $f$  induces isomorphisms on all homotopy groups. Therefore  $f$  is a weak homotopy equivalence, and by Whitehead's theorem it is a homotopy equivalence.

Let  $u \in H^n(K(G, n); G)$  be the universal class, corresponding to the identity map under the isomorphism  $[K(G, n), K(G, n)] \cong H^n(K(G, n); G)$ . On the product  $K(G, n) \times K(G, n)$ , we have the class  $p_1^*u + p_2^*u$  where  $p_1$  and  $p_2$  are the projections. Their sum is a class  $p_1^*u + p_2^*u$  in  $H^n(K(G, n) \times K(G, n); G)$  and is classified by a map  $\mu : K(G, n) \times K(G, n) \rightarrow K(G, n)$ . Now for any space  $X$ , given two maps  $f, g : X \rightarrow K(G, n)$  their sum is defined by

$$X \xrightarrow{(f, g)} K(G, n) \times K(G, n) \xrightarrow{\mu} K(G, n).$$

Then

$$(\mu \circ (f, g))^*u = (f, g)^*\mu^*u = (f, g)^*(p_1^*u + p_2^*u) = f^*u + g^*u$$

In particular  $\mu$  is defined so that  $[f] + [g] = [\mu \circ (f, g)]$ .  $\square$

**Example 3.60.** Recall that the standard model for a contractible space with a free  $S^1$  action is  $ES^1 \cong S^\infty$ . The quotient  $BS^1 = ES^1/S^1$  is modeled by  $\mathbb{CP}^\infty$  and has the property that it classifies  $S^1$ -bundles in the sense that for any space  $X$ , there is a natural bijection

$$[X, \mathbb{CP}^\infty] \cong \{\text{isomorphism classes of } S^1\text{-bundles on } X\}.$$

In particular, topological line bundles on  $X$  are classified by maps  $X \rightarrow \mathbb{CP}^\infty$ , which are determined up to homotopy by their first Chern class  $f^*u \in H^2(X; \mathbb{Z})$  where  $u$  is the universal class in  $H^2(\mathbb{CP}^\infty; \mathbb{Z})$  and  $f : X \rightarrow \mathbb{CP}^\infty$  is the classifying map of the line bundle. In terms of the  $H$ -space structure on  $\mathbb{CP}^\infty$ , the tensor product of line bundles corresponds to the addition of maps  $X \rightarrow \mathbb{CP}^\infty$  under the group law on  $\mathbb{CP}^\infty$ .

### 3.6 Exercises

**Example 3.61.** We compute  $\pi_1$  and  $\pi_2$ , homology, and cohomology of  $X = S^1 \vee S^2$ . By Van Kampen's theorem,  $\pi_1(X) \cong \pi_1(S^1) * \pi_1(S^2) \cong \mathbb{Z}$ . The universal cover of  $S^1 \vee S^2$  is a wedge of countably many copies of  $S^2$  and therefore  $\pi_2(S^1 \vee S^2)$  is a countable direct sum of copies of  $\mathbb{Z}$ . There is an action of  $\pi_1(X) = \mathbb{Z}$  acting by deck transformations on the universal cover, and this induces an action of  $\mathbb{Z}$  on  $\pi_2(X)$  by translating the summands.

Reduced homology of a wedge is the direct sum of reduced homologies, so therefore

$$H_k(X) \cong H_k(S^1) \oplus H_k(S^2)$$

for  $k \geq 1$ . By the universal coefficient theorem, since the homology is free abelian, the cohomology agrees with the homology. The cup product is trivial because the cohomology classes are supported on different wedge summands.

**Example 3.62.** We write down the cohomology ring of the torus  $T^2$  and  $S^2 \vee S^1 \vee S^1$ . Writing  $T^2 = S^1 \times S^1$ , the Künneth formula gives

$$H^*(T^2) \cong H^*(S^1) \otimes H^*(S^1)$$

Write  $\alpha$  for the pullback of the generator of  $H^1(S^1)$  along the first projection and  $\beta$  for the pullback of the generator of  $H^1(S^1)$  along the second projection. Then  $\alpha^2 = (\eta \otimes 1)^2 = \eta^2 \otimes 1 = 0$  and similarly  $\beta^2 = 0$ . However,  $\alpha\beta = (\eta \otimes 1)(1 \otimes \eta) = \eta \otimes \eta$ , and  $\beta\alpha = (1 \otimes \eta)(\eta \otimes 1) = -\eta \otimes \eta$

$$H^*(T^2) \cong \Lambda_{\mathbb{Z}}(\alpha, \beta)$$

the exterior algebra on two generators in degree 1. However, the cohomology ring of  $S^2 \vee S^1 \vee S^1$  is isomorphic to  $H^*(S^2) \oplus H^*(S^1) \oplus H^*(S^1)$  as a graded abelian group, and the cup product is trivial because the cohomology classes are supported on different wedge summands.

**Example 3.63.** We compute the homology and cohomology of  $\mathrm{SO}(3)$ . The double cover  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  induced by letting  $\mathrm{SU}(2)$  act on the imaginary quaternions by conjugation is a universal cover and identifies  $\mathrm{SO}(3)$  with  $\mathbb{RP}^3$ . Therefore

$$H_k(\mathrm{SO}(3)) \cong H_k(\mathbb{RP}^3) \cong \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/2\mathbb{Z}, & k = 1, \\ 0, & k = 2, \\ \mathbb{Z}, & k = 3, \\ 0, & k > 3. \end{cases}$$

An application of the universal coefficient theorem gives

$$H^k(\mathrm{SO}(3)) \cong H^k(\mathbb{RP}^3) \cong \begin{cases} \mathbb{Z}, & k = 0, 3, \\ 0, & k = 1, \\ \mathbb{Z}/2\mathbb{Z}, & k = 2, \\ 0, & k > 3. \end{cases}$$

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